**Supplementary Materials for**

**Comorbidities in Juvenile-onset Rheumatic Diseases: a systematic review and meta-analysis**

**Shall be considered in conjunction with the**

**Thesis submitted in accordance with the requirements of the University of Liverpool**

**for the degree of Doctor of Philosophy**

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## Supplementary Table S4.1: Prevalence and incidence rates of comorbidities in JIA, jSLE and JDM patients presenting in childhood or apparent in adulthood

### Childhood / JIA

| Study | Gender | Age in yrs | Onset age, yrs | Age diagnosis, yrs | Disease duration | Follow up period for the study | Comorbidity | Sample size, N (patients/ controls) | Prevalence | Incidence (per 1000 person-yrs) | Ethnicity /Race |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Demirkaya, et al, 2011, Turkey[1] | F: 354 | Mean 11.84±4.66 yrs | | | Mean 4.35 ±3.25 yrs | Mean 7.6±4.4 yrs | | 634 | | | |
| | | | | | | | Macrophage Activation Syndrome | 14 | 0.022082 | 2.905528806 | |
| | | | | | | | Infections | 14 | 0.022082 | 2.905528806 | |
| | | | | | | | Uveitis | 62 | 0.0977918 | 12.86734186 | |
| Jain, et al, 2001, India[2] | F: 10 | Mean 8.7± 3.2 yrs | | | Mean 16.4 ± 5.23 months | | | 50 | | | |
| | | | | | | | Keratoconjunctivitis sicca | 7 | 0.14 | | |
| Rangel, et al, 2003, UK | F:20 | Mean 15.2±2.4 yrs | Mean 5.5±3.6 yrs | | Mean 9.8 ± 3.8 yrs | | | 30 | | | Caucasian 27; Asian (SE) 3 |
| | | | | | | | Major depressive episode | 2/27 | 0.0740741 | | |
| | | | | | | | Over-anxious disorder/generalized anxiety disorder | 3/27 | 0.1111111 | | |
| | | | | | | | Dysthymic disorder | 2/27 | 0.0740741 | | |
| | | | | | | | Night terrors | 1 | 0.0333333 | | |
| | | | | | | | Oppositional defiant disorder | 1 | 0.0333333 | | |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Saurenman, et al, 2007, Canada[26] | F: 711 | Mean 13.8 yrs (1.2 to 34.1) | Mean 6.9 yrs (0.5-17.8 yrs) | Mean 6.9 yrs (Range 0.08 to 25.7) | Mean 6.9 yrs (Range 0.08 to 25.7) | | 1081 | | | |
| | | | | | | Uveitis | 142 | 0.1313599 | 19.03765971 | |
| | | | | | | Cataracts | 33 | 0.0305273 | 4.424244862 | |
| | | | | | | Synechiae | 31 | 0.0286772 | 4.15610881 | |
| | | | | | | Glaucoma | 22 | 0.0203515 | 2.949496575 | |
| | | | | | | Band keratopathy | 20 | 0.0185014 | 2.681360522 | |
| | | | | | | Macular oedema | 7 | 0.0064755 | 0.938476183 | |
| | | | | | | Legal blindness | 10 | 0.0092507 | 1.340680261 | |
| | | | | | | Impaired vision | 6 | 0.0055504 | 0.804408157 | |
| Chabra, et al, 2020, Canada[27] | F: 142 | Median 16.9 yrs (IQR 12.9 - 18.0) | | | Median 5.6 yrs (IQR 3.3-8.3) | | 247 | | | |
| | | | | | | Uveitis | 24 | 0.097166 | 17.35106998 | |
| | | | | | | Band keratopathy | 1 | 0.0040486 | 0.722961249 | |
| | | | | | | Decreased visual acuity | 1 | 0.0040486 | 0.722961249 | |
| | | | | | | Cataract | 1 | 0.0040486 | 0.722961249 | |
| | | | | | | Micrognathia or facial asymmetry | 11 | 0.0445344 | 7.952573742 | |
| | | | | | | Leg-length discrepancy | 12 | 0.048583 | 8.675534991 | |
| Beukelman, et al, 2012, USA[28] | F: 5,421 | Mean 9.7±4.4 yrs | | | Median 1.1 yrs (IQR 0.5-2.2) | | 8479 | | | White 4402; Black 1448; Latino 1721; other/unknown 932 |
| | | | | | | Upper respiratory tract | 110 | 0.0129732 | 11.79384361 | |
| | | | | | | Pneumonia | 87 | 0.0102606 | 9.327858131 | |
| | | | | | | Bacteraemia/septicaemia | 67 | 0.0079019 | 7.183522928 | |
| | | | | | | Urinary tract/pyelonephritis | 65 | 0.007666 | 6.969089408 | |
| | | | | | | Skin and soft tissue | 44 | 0.0051893 | 4.717537445 | |
| | | | | | | Abdominal abscess | 20 | 0.0023588 | 2.144335202 | |
| | | | | | | Gastroenteritis | 30 | 0.0035382 | 3.216502804 | |
| | | | | | | Diabetes mellitus | 56 | 0.0066046 | 6.004138567 | |
| | | | | | | Asthma | 669 | 0.0789008 | 71.72801252 | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| van Rossum, et al, 2007, Netherlands[29] | F: 41 | Median 18 yrs (range 10-25) | | Median 10.7 yrs (range 8-23) | Median 9 yrs (IQR 8-9 yrs) | | 61 | | |
| | | | | | | Uveitis | 12 | 0.1967213 | 21.8579235 |
| Gerloni, et al, 2008, Italy[30] | F:124 | Mean 17.1 yrs SD 9.2 yrs (range 1.9-49.8) | Mean 6.4 yrs SD 4.8 (Range 0.5-16.8) | Mean 10.7 yrs SD 8.6 (range 1.8-49.8) | | | 163 | | |
| | | | | | | Flare or new onset of chronic iridocyclitis | 7 | 0.0429448 | |
| | | | | | | Gonadal effects | 6 | 0.0368098 | |
| | | | | | | Hypertension | 8 | 0.0490798 | |
| | | | | | | Infective events | 41 (split below) | | |
| | | | | | | Recurrent URI or UTI or mycotic vaginitis | 21 | 0.1288344 | |
| | | | | | | Viral thyroiditis | 1 | 0.006135 | |
| | | | | | | Varicella | 4 | 0.0245399 | |
| | | | | | | EBV infections | 3 | 0.0184049 | |
| | | | | | | Single URI | 3 | 0.0184049 | |
| | | | | | | Rotavirus gastro-enteritis | 3 | 0.0184049 | |
| | | | | | | Helicobacter gastritis | 1 | 0.006135 | |
| | | | | | | Scarlet fever | 1 | 0.006135 | |
| | | | | | | Hip prosthetic infection | 1 | 0.006135 | |
| | | | | | | CMV pneumonia complicated by MAS | 1 | 0.006135 | |
| | | | | | | Pneumonia | 1 | 0.006135 | |
| | | | | | | Impetigo | 1 | 0.006135 | |
| | | | | | | Neuropsychiatric manifestations | 53 (split below) | | |
| | | | | | | Headache | 10 | 0.0613497 | |
| | | | | | | Severe headache (dose-dependent) | 4 | 0.0245399 | |
| | | | | | | Psychoses | 2 | 0.0122699 | |
| | | | | | | Depression | 3 | 0.0184049 | |

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|---|---|---|---|---|---|---|
| | | | | | Panic attacks | 2 | 0.0122699 |
| | | | | | Anxiety | 4 | 0.0245399 |
| | | | | | Anorexia nervosa | 1 | 0.006135 |
| | | | | | Nervousness/hyperactivity | 7 | 0.0429448 |
| | | | | | Memory and concentration deficit | 1 | 0.006135 |
| | | | | | Aggressiveness | 3 | 0.0184049 |
| | | | | | Fatigue | 5 | 0.0306748 |
| | | | | | Hyperalgesia | 1 | 0.006135 |
| | | | | | Pain amplification syndrome | 1 | 0.006135 |
| | | | | | Sleepiness | 2 | 0.0122699 |
| | | | | | Insomnia | 1 | 0.006135 |
| | | | | | Vertigo | 3 | 0.0184049 |
| | | | | | Paraesthesia | 1 | 0.006135 |
| | | | | | Hypoglossal paralysis | 1 | 0.006135 |
| | | | | | Neuroalgodistrofy | 1 | 0.006135 |
| Klotsche, et al, 2016, Germany[31] | F: 1,537 | Mean 11.3 yrs (SD 4.6) | Mean 7.8 yrs (SD 3.7) | Mean 3.5 yrs (SD 3.6) | | 2263 | |
| | | | | | Malignancies | 8 (split below) | |
| | | | | | Acute lymphoblastic leukaemia | 2 | 0.0008838 |
| | | | | | Thyroid gland carcinoma | 1 | 0.0004419 |
| | | | | | Yolk sac carcinoma | 1 | 0.0004419 |
| | | | | | Non-Hodgkin’s lymphoma | 1 | 0.0004419 |
| | | | | | Hodgkin’s lymphoma | 2 | 0.0008838 |
| | | | | | Oligodendroglioma | 1 | 0.0004419 |
| | | | | | Uveitis | 361 | 0.1595228 |
| | | | | | inflammatory bowel diseases (IBD) | 14 (split below) | |
| | | | | | Colitis | 4 | 0.0017676 |
| | | | | | Crohn’s disease | 10 | 0.0044189 |
| | | | | | Infections* | 471 (split below) | |
| | | | | | Bacterial | 29 | 0.0128148 |

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|---|---|---|---|---|---|---|---|
| | | | | Viral | 107 (split below) | | |
| | | | | Epstein Barr infection | 8 | 0.0035351 | |
| | | | | Varicella zoster infection | 34 | 0.0150243 | |
| | | | | Chickenpox | 16 | 0.0070703 | |
| | | | | Herpes Zoster | 18 | 0.007954 | |
| | | | | HPV | 2 | 0.0008838 | |
| Beukelman, et al, 2013, USA[32] | F: 5,421 | Median 10.1 yrs (IQR 5.9-13.3) | Median 1.2 yrs (IQR 0.6-2.4) | | 8503 | | White 4,402; Black 1,448; Latino 1,721; Other/unknown 932 |
| | | | | Infections | | (300/100,000 [95% CI 216-406] yrs split below) | |
| | | | | Herpes zoster | | 225 /100,000 yrs (95% CI 154–318) | |
| | | | | Salmonella | | 86/100,000 yrs (95% CI 44–150) | |
| | | | | Coccidioides | | 35/100,000 yrs (95% CI 11–81) | |
| | | | | Nontuberculous Mycobacterium | | 21/100,000 yrs (95% CI 4–61) | |
| | | | | Nocardia | | 7/100,000 yrs (95% CI 0.2-39) | |
| | | | | Pneumocystis | | 7/100,000 yrs (95% CI 0.2-39) | |
| | | | | Toxoplasma | | 7/100,000 yrs (95% CI 0.2-39) | |
| | | | | Blastomyces | | 0/100,000 yrs (95% CI 0.2-39) | |

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|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Histoplasma | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | JC virus | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | Cryptococcus | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | Tuberculosis | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | Legionella | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | Listeria | | | 0/100,000 yrs (95% CI 0-36) |
| | | | | | | Aspergillus | | | 0/100,000 yrs (95% CI 0-36) |
| Schenck, et al, 2018, Germany[33] | F: 8,043 | Mean 11.6 yrs | | Mean 4.2 yrs | | | 12,269 | | |
| | | | | | | Type 1 Diabetes | 58 | 0.0047274 | |
| | | | | | | Type 2 Diabetes | 2 | 0.000163 | |
| | | | | | | Mature onset diabetes of the young | 2 | 0.000163 | |
| Angeles-Han, et al, 2015, USA[34] | F: 208 | | Median 6.4 (2.9-11.7) yrs | Median 3.4, range (1.8 – 5.9) yrs | Mean 15.1± 12 months | | 287 | | Caucasian 221, African American 36, Others 30 |
| | | | | | | Uveitis | 52 | 0.1811847 | 11.9989847 |
| | | | | | | Cataracts | 14 | 0.0487805 | 3.230495881 |
| | | | | | | Glaucoma | 7 | 0.0243902 | 1.615247941 |
| | | | | | | Synechiae | 12 | 0.0418118 | 2.76899647 |

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|---|---|---|---|---|---|---|---|---|
| | | | | | Band keratopathy | 11 | 0.0383275 | 2.538246764 | |
| | | | | | Cystoid macular oedema | 7 | 0.0243902 | 1.615247941 | |
| | | | | | Cataract extraction | 3 | 0.010453 | 0.692249117 | |
| Thiele, et al, 2021, Germany[35] | F: 2,490 | Median 12.3, IQR (8.6-15.2) | | | | 3258 | | | Caucasian 95%, African and Asian 5% |
| | | | | | Pneumonia | 30 | 0.0092081 | | |
| | | | | | Herpes zoster | 34 | 0.0104359 | | |
| | | | | | Varicella | 23 | 0.0070595 | | |
| | | | | | SAE infection | 108 | 0.0331492 | | |
| | | | | | Infections | 813 (split below) | | | |
| | | | | | Upper respiratory tract infection | 412 | 0.1264579 | | |
| | | | | | Otitis media | 28 | 0.0085942 | | |
| | | | | | Urinary tract infection | 32 | 0.009822 | | |
| | | | | | Tonsillitis | 67 | 0.0205648 | | |
| | | | | | Gastroenteritis | 63 | 0.019337 | | |
| | | | | | Streptococcal infection | 20 | 0.0061387 | | |
| Kearsley-Fleet, et al, 2022, multiple countries[36] | F: 2,014 | Mean, 6.5 (SD 4.6) Median 14, IQR (10–17) yrs | Median 1, IQR (0, 4) yrs, N=2894 | Median 5, IQR (3, 9) yrs, N=2926 | | 2693 | | | |
| | | | | | MAS: macrophage activation syndrome (systemic JIA only) | 37/250 | 0.148 | 29.6 | |
| | | | | | Varicella | 822/2238 | 0.3672922 | 73.45844504 | |
| | | | | | Tuberculosis | 17 | 0.0063127 | 1.262532492 | |
| | | | | | Uveitis | 556 | 0.2064612 | 41.29223914 | |
| | F: 1,046 | Mean 13.2 (SD 6.1) Median 22, IQR (19–25) yrs | Median 3, IQR (1, 7) yrs, N=1531 | Median 14, IQR (7, 18) yrs, N=1514 | | 1541 | | | |

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|---|---|---|---|---|---|---|---|---|---|
| | | | | | Uveitis | 238 | 0.1544452 | 11.03179753 | |
| | | | | | MAS: macrophage activation syndrome (systemic JIA only) | 5/82 | 0.0609756 | 4.355400697 | |
| | | | | | Varicella | 127/871 | 0.1458094 | 10.41495818 | |
| | | | | | Tuberculosis | 2/1541 | 0.0012979 | 0.092704181 | |
| | F : 2,476 | Mean 6.5 SD (4.5) Median 13, IQR (9–17) yrs | Median 3, IQR (1–6) yrs | Median 6, IQR (3–9) yrs | | 3562 | | | |
| | | | | | Uveitis | 676/3484 | 0.1940299 | 32.33830846 | |
| | | | | | MAS: macrophage activation syndrome (systemic JIA only) | 62/366 | 0.1693989 | 28.23315118 | |
| | | | | | Varicella | 1166/2312 | 0.5043253 | 84.05420992 | |
| | | | | | Tuberculosis | 54/3006 | 0.0179641 | 2.994011976 | |
| Krause, et al, 2017, USA[37] | F: 57 | Mean 8.6 (±5.1) yrs at index | Median 1.3, range (0.2–4.0) yrs | Mean 6.3 (±4.4) yrs | | 89 | | | Caucasian 75, Hispanic 2, Black 4, Asian 3, Other 1, Unknown 4 |
| | | | | | Joint surgeries | 6 | 0.0674157 | 10.70090958 | |
| | | | | | Non-joint surgeries | 14 | 0.1573034 | 24.96878901 | |
| | | | | | Non-surgical hospitalisations (due to infection, JIA management etc.) | 12 | 0.1348315 | 21.40181915 | |
| | | | | | Hospitalisations for depression | 5 | 0.0561798 | 8.917424648 | |
| | | | | | Pregnancies | 9/28 | 0.3214286 | 51.02040816 | |
| | | | | | Depression | 7 | 0.0786517 | 12.48439451 | |
| Lee, et al, 2020, USA[38] | F: 10,233 | Mean 13.86 ±3.44 yrs | | Mean 2.68 yrs SD 2.34 | | 15210 | | | |
| | | | | | Type 1 diabetes | 18 | 0.0011834 | 44/100,000 person yrs (95% CI 28-70) | |
| | | | | | Type 2 diabetes | 27 | 0.0017751 | 66/100,000 person yrs | |

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| | | | | |
|---|---|---|---|---|
| | | | | (95% CI 45-96) |
| | | Hyperlipidaemia | 191.646 | 0.0126 | 4.701492537 |
| | | Hypertension | 158.184 | 0.0104 | 3.880597015 |
| | | Obesity | 305.721 | 0.0201 | 7.5 |
| Horneff, et al, 2022, Germany[39] | aged 2 to 15 yrs | | 546 | | |
| | | Allergic rhinitis | 76 | 0.1391941 | |
| | | Predominantly allergic asthma | 15 | 0.0274725 | |
| | | Amyloidosis | <5 | <5/546 | |
| | | Anaemia | 15 | 0.0274725 | |
| | | Phobic anxiety disorders | 17 | 0.0311355 | |
| | | Essential hypertension | <5 | <5/546 | |
| | | Atopic dermatitis | 80 | 0.1465201 | |
| | | Persistent somatoform pain disorder | 46 | 0.0842491 | |
| | | Ulcerative colitis | <5 | <5/546 | |
| | | Depression | 21 | 0.0384615 | |
| | | Diabetes mellitus | 5 | 0.0091575 | |
| | | Iron deficiency | 8 | 0.014652 | |
| | | Post-viral fatigue syndrome | 0 | 0 | |
| | | Fibromyalgia | 13 | 0.0238095 | |
| | | Autoimmune thyroiditis | 6 | 0.010989 | |
| | | Thyrotoxicosis | 0 | 0 | |
| | | Hypothyroidism | 8 | 0.014652 | |
| | | Lack of expected normal physiological development | 26 | 0.047619 | |
| | | Crohn’s disease | 12 | 0.021978 | |
| | | Migraine | 14 | 0.025641 | |
| | | Kidney disease (chronic) | <5 | <5/546 | |
| | | Psoriasis | 21 | 0.0384615 | |
| | | Sicca syndrome | 5 | 0.0091575 | |
| | | Uveitis | 59 | 0.1080586 | |

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|---|---|---|---|---|---|---|---|
| | | | | | 849 | | |
| | | | | Allergic rhinitis | 94 | 0.1107185 | |
| | | | | Predominantly allergic asthma | 25 | 0.0294464 | |
| | | | | Amyloidosis | 11 | 0.0129564 | |
| | | | | Anaemia | 24 | 0.0282686 | |
| | | | | Phobic anxiety disorders | 30 | 0.0353357 | |
| | | | | Essential (primary) hypertension | <5 | <5/849 | |
| | | | | Atopic dermatitis | 105 | 0.1236749 | |
| | | | | Persistent somatoform pain disorder | 71 | 0.0836278 | |
| | | | | Ulcerative colitis | <5 | <5/849 | |
| | | | | Depression | 19 | 0.0223793 | |
| | | | | Diabetes mellitus | 7 | 0.008245 | |
| | | | | Iron deficiency | 7 | 0.008245 | |
| | | | | Post-viral fatigue syndrome | <5 | <5/849 | |
| | | | | Fibromyalgia | 9 | 0.0106007 | |
| | | | | Autoimmune thyroiditis | 13 | 0.0153121 | |
| | | | | Thyrotoxicosis | <5 | <5/849 | |
| | | | | Hypothyroidism | 18 | 0.0212014 | |
| | | | | Lack of expected normal physiological development | 31 | 0.0365135 | |
| | | | | Crohn‘s disease | <5 | <5/849 | |
| | | | | Migraine | 29 | 0.0341578 | |
| | | | | Kidney disease (chronic) | 0 | 0 | |
| | | | | Osteoporosis | 5 | 0.0058893 | |
| | | | | Psoriasis | 39 | 0.0459364 | |
| | | | | Sicca syndrome | <5 | <5/849 | |
| | | | | Uveitis | 90 | 0.1060071 | |
| Bolt, IB. et al, 2008, Switzerland[40] | F: 174 | ≤18 yrs (see comments) | Mean 6.49, range (0.67–15.67) yrs | Mean 4.82, range (0.08–16.58) yrs | 265 | | |
| | | | | Synechiae | 10 | 0.0377358 | 7.829014327 |
| | | | | Cataracts | 9 | 0.0339623 | 7.046112894 |

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|---|---|---|---|---|---|---|
| | | | Band keratopathy | 5 | 0.0188679 | 3.914507164 |
| | | | Glaucoma | 3 | 0.0113208 | 2.348704298 |
| | | | Macular oedema | 2 | 0.0075472 | 1.565802865 |
| | | | Uveitis | 35 | 0.1320755 | 27.40155014 |
| | | | Impaired visual acuity | 1 | 0.0037736 | 0.782901433 |
| | | | Legal blindness in one eye | 2 | 0.0075472 | 1.565802865 |
| Simon, et al, 2020, USA[41] | | < 18 yrs | | | | |
| | F: 6,173 | Mean 10.6±4.5 yrs | | 8633 | | |
| | | | Multiple sclerosis | 1 | 0.0001158 | |
| | | | Asthma | 777 | 0.0001158 | |
| | | | Anxiety | 295 | 0.0001158 | |
| | | | Depression | 208 | 0.0001158 | |
| | | | Diabetes mellitus | 92 | 0.0001158 | |
| | | | Anaemia | 498 | 0.0001158 | |
| | | | Hypertension | 2 | 0.0001158 | |
| | | | Hyperlipidaemia | 174 | 0.0001158 | |
| | | | Malignancy | 56 | 0.0001158 | |
| | F: 6,587 | Mean 10.5±4.5 yrs | | 9155 | | |
| | | | Multiple sclerosis | 1 | 0.0001092 | |
| | | | Asthma | 887 | 0.0001092 | |
| | | | Anxiety | 317 | 0.0001092 | |
| | | | Depression | 214 | 0.0001092 | |
| | | | Diabetes mellitus | 84 | 0.0001092 | |
| | | | Anaemia | 479 | 0.0001092 | |
| | | | Hypertension | 6 | 0.0001092 | |
| | | | Hyperlipidaemia | 202 | 0.0001092 | |
| | | | Malignancy | 49 | 0.0001092 | |

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|---|---|---|---|---|---|---|---|---|
| Schulz, C. et al, 2022, Germany[42] | F 333 | Mean 6.8 yrs (range: 2-22) | Median 6.0 yrs (<1-16.4) | | Median 4.4 yrs (range: 0.9–14.1) | | | |
| | | | | | | 499 | | Europe 413, Asia 31, Turkish 28 , Africa Arab 2, mixed countries of origin 30 and unknown 23 |
| | | | | | | Hypothyroidism | 6 | 0.012024 |
| | | | | | | Autoimmune thyroiditis | 8 | 0.0160321 |
| | | | | | | Other thyroid disorder | 2 | 0.004008 |
| | | | | | | Coeliac disease | 3 | 0.006012 |
| | | | | | | Scleroderma | 1 | 0.002004 |
| | | | | | | Overlap syndrome | 1 | 0.002004 |
| Swart, et al, 2018, Italy[43] | F: 5,584 | ≤ 18 yrs | Median 5.4 yrs (IQR : 2.4–10.0) | Med 6.2 (IQR 2.8–10.9) yrs | Med 5.3 (IQR 2.7–8.8) yrs | | 8274 | |
| | | | | | | Infections | 1523 (split below) | |
| | | | | | | Serious/targeted infections (Epstein– Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections) | 674 | 0.08146 |
| | | | | | | Tuberculosis | 27 | 0.0032632 |
| | | | | | | Other infections | 822 | 0.0993474 |
| | | | | | | Macrophage activation syndrome | 75 | 0.0090645 |
| | | | | | | Autoimmune disease | 50 (split below) | |
| | | | | | | Inflammatory bowel disease (IBD) | 21 | 0.0025381 |
| | | | | | | Multiple sclerosis | 2 | 0.0002417 |
| | | | | | | Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder | 18 | 0.0021755 |
| | | | | | | Malignancies | 16 (split below) | |
| | | | | | | Leukaemia’s | 3 | 0.0003626 |
| | | | | | | Lymphomas | 2 | 0.0002417 |

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| | | | | | |
|---|---|---|---|---|---|
| | | | Hematopoietic neoplasms (excluding leukaemia and lymphoma) | 1 | 0.0001209 |
| | | | Neoplasm (other) | 10 | 0.0012086 |
| | | | Gastrointestinal (GI) ulcer/GI bleed/GI perforation | 17 | 0.0020546 |
| | | | Pregnancy | 9 | 0.0010877 |
| | | | Congestive heart failure | 1 | 0.0001209 |
| F: 2,670 | Median 7.6 yrs (IQR 3.2–11.7) | Med 6.1 (IQR 3.5–9.5) yrs | | 3990 | |
| | | | Infections | 1509 (split below) | |
| | | | Serious/targeted infections (Epstein– Barr virus, cytomegalovirus, papilloma virus, herpes zoster primary and reactivation, and opportunistic infections) | 171 | 0.0428571 |
| | | | Tuberculosis | 0 | 0 |
| | | | Other infections | 1338 | 0.3353383 |
| | | | Macrophage activation syndrome | 11 | 0.0027569 |
| | | | Autoimmune disease | 50 (split below) | |
| | | | Inflammatory bowel disease (IBD) | 23 | 0.0057644 |
| | | | Multiple sclerosis | 0 | 0 |
| | | | Other autoimmune diseases excluding IBD, uveitis, and demyelinisation disorder | 24 | 0.006015 |
| | | | Malignancies | 13 (split below) | |
| | | | Leukaemia | 2 | 0.0005013 |
| | | | Lymphoma | 5 | 0.0012531 |
| | | | Hematopoietic neoplasms (excluding leukaemia and lymphoma) | 2 | 0.0005013 |
| | | | Neoplasm (other) | 4 | 0.0010025 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | Gastrointestinal (GI) ulcer/GI bleed/GI perforation | 4 | 0.0010025 | |
| | | | | Pregnancy | 7 | 0.0017544 | |
| | | | | Congestive heart failure | 0 | 0 | |
| Giancane, et al, 2019, Italy[44] | F: 189 | Med 3.1 yrs (IQR 1.8–5.8) | Med 7.7 yrs (IQR 6.1–10.2) | | 239 | | |
| | | | | Ocular | 21 | 0.0878661 | |
| | | | | Muscular atrophy | 19 | 0.0794979 | |
| | | | | Osteoporosis | 3 | 0.0125523 | |
| | | | | Subcutaneous atrophy | 9 | 0.0376569 | |
| | | | | Growth failure | 19 | 0.0794979 | |
| | | | | Pubertal delay | 5 | 0.0209205 | |
| | | | | Avascular necrosis | 0 | 0 | |
| | | | | Amyloidosis | 0 | 0 | |
| | | | | Diabetes | 0 | 0 | |
| | F: 219 | Med 2.5 yrs (IQR 1.6–4.7) | Med 6.9 yrs (IQR 5.4–9.7) | | 269 | | |
| | | | | Ocular | 7 | 0.0260223 | |
| | | | | Muscular atrophy | 8 | 0.0297398 | |
| | | | | Osteoporosis | 1 | 0.0037175 | |
| | | | | Avascular necrosis | 1 | 0.0037175 | |
| | | | | Subcutaneous atrophy | 6 | 0.0223048 | |
| | | | | Growth failure | 1 | 0.0037175 | |
| | | | | Diabetes | 0 | 0 | |
| | | | | Amyloidosis | 0 | 0 | |
| Grassi, et al, 2007, Italy[45] | F: 244 | mean 4.9 ± 3.6 yrs | mean 7.6 ± 5.6 yrs | | 309 | | |
| | | | | Synechiae | 12 | 0.038835 | 5.109862034 |
| | | | | Band keratopathy | 8 | 0.02589 | 3.406574689 |
| | | | | Cataract | 6 | 0.0194175 | 2.554931017 |
| | | | | Glaucoma | 2 | 0.0064725 | 0.851643672 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Retinal detachment | 1 | 0.0032362 | 0.425821836 |
| | | | | | Iris atrophy | 1 | 0.0032362 | 0.425821836 |
| | | | | | Blindness* | 4 | 0.012945 | 1.703287345 |
| | | | | | Uveitis | 62 | 0.2006472 | 26.40095384 |
| | | | | | Severe visual loss | 8 | 0.02589 | 3.406574689 |
| Horneff, et al, 2016, Germany[46] | | | | 13,198 observation yrs | | 3695 | | |
| | | | | | Suspected Malignancies | 12 (split below) | | |
| | | | | | Hodgkin's lymphoma | 2 | 0.0005413 | 4.10117E-05 |
| | | | | | non-Hodgkin's lymphoma | 1 | 0.0002706 | 2.05058E-05 |
| | | | | | acute lymphatic leukaemia (ALL) | 2 | 0.0005413 | 4.10117E-05 |
| | | | | | lymphoproliferative disorder | 1 | 0.0002706 | 2.05058E-05 |
| | | | | | thyroid carcinoma | 1 | 0.0002706 | 2.05058E-05 |
| | | | | | yolk sac carcinoma | 1 | 0.0002706 | 2.05058E-05 |
| | | | | | anaplastic ependymoma | 1 | 0.0002706 | 2.05058E-05 |
| Lovell, DJ. et al, 2021, USA[47] | F: 867 | Mean 6.6 yrs (SD 4.5) | Mean 8.0 (SD 4.8) | | | 1332 | | |
| | | | | | Autoimmune diseases | 1085 (split below) | | |
| | | | | | Addison disease | 11 | 0.0082583 | |
| | | | | | Autoimmune/Hashimoto's thyroiditis | 16 | 0.012012 | |
| | | | | | Graves' disease | 1 | 0.0007508 | |
| | | | | | Morphoea | 12 | 0.009009 | |
| | | | | | Multiple sclerosis | 1 | 0.0007508 | |
| | | | | | Takayasu arteritis | 1 | 0.0007508 | |
| | | | | | Type 1 diabetes mellitus | 17 | 0.0127628 | |
| | | | | | Vitiligo | 5 | 0.0037538 | |
| | | | | | Myasthenia Gravis | 0 | 0 | |
| | | | | | Pernicious Anaemia | 0 | 0 | |
| | | | | | Primary biliary cirrhosis | 0 | 0 | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Associated conditions | | 0 | |
| | | | | | Autoimmune disease not elsewhere classified | 8 | 0.006006 | |
| | | | | | Chronic glomerulonephritis | 1 | 0.0007508 | |
| | | | | | Chronic urticaria | 49 | 0.0367868 | |
| | | | | | Episcleritis/scleritis | 5 | 0.0037538 | |
| | | | | | Erythema nodosum | 1 | 0.0007508 | |
| | | | | | Haemolytic anaemia | 1 | 0.0007508 | |
| | | | | | Immune thrombocytopenia purpura | 2 | 0.0015015 | |
| | | | | | Leukocytoclastic vasculitis | 2 | 0.0015015 | |
| | | | | | Pulmonary fibrosis/interstitial lung disease | 15 | 0.0112613 | |
| | | | | | Raynaud's syndrome | 19 | 0.0142643 | |
| | | | | | Sjögren's syndrome/sicca syndrome | 3 | 0.0022523 | |
| | | | | | Thrombocytopenia purpura | 1 | 0.0007508 | |
| | | | | | Vasculitis | 8 | 0.006006 | |
| Barthel, et al, 2015, Germany[48] | F: 2,073 | Mean 6.1 ± 3.9 yrs | | 8389 patient-yrs | | 3071 | | |
| | | | | | Inflammatory bowel disease | 11 (split below) | | |
| | | | | | Crohn Disease | 8 | 0.002605 | 0.000310527 |
| | | | | | Ulcerative colitis | 3 | 0.0009769 | 0.000116448 |
| Sahin, et al, 2021, Turkey[49] | F: 289 | | 8.8 ± 4.7 yrs (mean) (range 9 months to 17 yrs) | 4.0 ± 1.9 yrs (mean) | | 500 | | |
| | | | | | Uveitis | 34 | 0.068 | |
| | | | | | Osteoporosis | 8 | 0.016 | |
| | | | | | Delayed puberty | 6 | 0.012 | |
| | | | | | Autoimmune Thyroiditis | 4 | 0.008 | |
| | | | | | Inflammatory Bowel Disease (IBD) | 2 | 0.004 | |
| | | | | | Depression | 2 | 0.004 | |
| | | | | | Hypertension | 1 | 0.002 | |
| | | | | | Amyloidosis | 1 | 0.002 | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | familial Mediterranean fever (FMF) | 63 | 0.126 |
| | | | | | | Psoriasis | 15 | 0.03 |
| | | | | | | Cataract | 4 | 0.008 |
| | | | | | | Band keratopathy | 1 | 0.002 |
| | | | | | | Posterior synechia | 1 | 0.002 |
| Pohjankoski, et al, 2010, Finland[50] | F: 283 | | | Mean 7 (1–15) yrs | | | 417 | |
| | | | | | | Uveitis | 91 | 0.2182254 |
| | | | | | | Diabetes mellitus type 1 | 9 | 0.0215827 |
| | | | | | | Coeliac disease | 3 | 0.0071942 |
| | | | | | | Hypothyroidism | 3 | 0.0071942 |
| Cakan, et al, 2020, Turkey[51] | F: 202 | | | | Mean 29.2±18.0 months | | 405 | |
| | | | | | | Uveitis | 27 | 0.0666667 |
| Cosickic, et al, 2017, Bosnia and Herzegovina[52] | F: 61 | Med 7.1 rs (IQR, 2.3 - 10.6 yrs) | | | | | 97 | 2.283105023 |
| | | | | | | Synechia | 3 | 0.0309278 |
| | | | | | | Band keratopathy | 2 | 0.0206186 |
| | | | | | | Cataracts | 2 | 0.0206186 |
| | | | | | | Glaucoma | 1 | 0.0103093 |
| | | | | | | Uveitis | 14 | 0.1443299 |
| Berthold, et al, 2022, Sweden[53] | F: 433 | | | Med 8.7 yrs ((IQR) 3.7 – 12.7 yrs) | | | 640 | |
| | | | | | | Depressive disorder | 93 | 0.1453125 |
| | | | | | | Anxiety disorder | 111 | 0.1734375 |
| Lien, et al, 2003, Norway[54] | F: 80 | Mean 17.0 ±1.8 yrs | Mean 2.8±1.4 yrs | Mean 14.2 ±1.4 yrs | | | 105 | |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Osteopenia/Low total-body BMC | 42/103 | 0.407767 |
| | | | | | Osteoporosis (BMD Z score≤-2.5) | 5/103 | 0.0485437 |
| Saraux, et al, 2015, France[55] | F: 18 | Mean 13.5 yrs (SD 5.84) | | | | 26 | |
| | | | | | Infections | 3 | 0.1153846 |
| | | | | | Bacterial infection | 3 | 0.1153846 |
| | | | | | Viral infection | 1 | 0.0384615 |
| Alberdi-Saugstrup, et al, 2017, Nordic countries[56] | F: 135 | | Median 6.0 (IQR 2.8-10.0) | | | 193 | |
| | | | | | Uveitis | 30 | 0.1554404 |
| Amine, et al, 2011, Morrocco[57] | F: 32 | Mean 11 ± SD 3.3 yrs range [2-16] | Mean 7 ±SD 3.4 range [1–15] | Mean 3 ± SD 2.4 yrs range [1-12] | | 58 | |
| | | | | | Obesity | 13 | 0.2241379 |
| **Adulthood / JIA** | | | | | | | |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Smitherman, et al, 2022, USA[58] | F: 144 | Mean 20±1.4 yrs | CARRA (or any other) JIA registry, Age onset will have childhood onset disease. | | | | 187 | | | White 147; Hispanic 23; Black 9; Other* 8 |
| | | | | | | Self-reported symptoms of pain or discomfort | 122 | 0.6524064 | | |
| | | | | | | Self-reported symptoms of anxiety or depression | 84 | 0.4491979 | | |
| Mena-Vazquez, et al, 2022, Spain[14] | F: 35 | Mean 25 yrs +/- 6.1 | | 161.4 months (103.6 - 220.4 IQR) | 24 months | | 52 | | | Caucasian 50 |
| | | | | | | Depression | 13 | 0.25 | 125 | |
| | | | | | | Arterial hypertension | 1 | 0.0192308 | 9.615384615 | |
| | | | | | | Diabetes melllitus | 1 | 0.0192308 | 9.615384615 | |
| | | | | | | Dyslipidaemia | 4 | 0.0769231 | 38.46153846 | |
| | | | | | | Obesity (BMI ≥30) | 7 | 0.1346154 | 67.30769231 | |
| | | | | | | Uveitis | 11 | 0.2115385 | 105.7692308 | |
| Raab, et al, 2013, Germany[59] | F: 196 | Mean 19.7 yrs (SD 2.8) | | Mean 10.6 yrs (since symptom onset) 5.6 SD | | | 344 | | | |
| | | | | | | Cardiovascular diseases | 34 (split below) | | | |
| | | | | | | Hypertension | 25 | 0.0726744 | 6.856077227 | |
| | | | | | | Cardiac insufficiency | 1 | 0.002907 | 0.274243089 | |
| | | | | | | Pulmonary diseases | 28 (split below) | | | |

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| | | | |
|---|---|---|---|
| Asthma | 26 | 0.0755814 | 7.130320316 |
| Gastrointestinal diseases | 14 (split below) | | |
| Crohn's disease | 4 | 0.0116279 | 1.096972356 |
| Ulcerative colitis | 3 | 0.0087209 | 0.822729267 |
| Gastrointestinal ulcer | 1 | 0.002907 | 0.274243089 |
| Liver and gallbladder diseases | 3 | 0.0087209 | 0.822729267 |
| Diseases of the kidneys or the Urinary tract | 16 (split below) | | |
| Renal insufficiency | 1 | 0.002907 | 0.274243089 |
| Kidney stone | 4 | 0.0116279 | 1.096972356 |
| Amyloidosis | 3 | 0.0087209 | 0.822729267 |
| Gynaecological diseases | 12 | 0.0348837 | 3.290917069 |
| Haematological diseases | 23 (split below) | | |
| Anaemia | 11 | 0.0319767 | 3.01667398 |
| Iron deficiency | 6 | 0.0174419 | 1.645458534 |
| Allergic diseases | 84 (split below) | | |
| Allergic rhinitis | 50 | 0.1453488 | 13.71215445 |
| House dust mite allergy | 16 | 0.0465116 | 4.387889425 |
| Thyroid/endocrine diseases | 23 (split below) | | |
| Hypothyroidism | 7 | 0.0203488 | 1.919701624 |
| Hashimoto thyroiditis | 4 | 0.0116279 | 1.096972356 |
| Diabetes mellitus | 3 | 0.0087209 | 0.822729267 |
| Skin diseases | 67 (split below) | | |
| Psoriasis | 22 | 0.0639535 | 6.03334796 |
| Atopic dermatitis | 30 | 0.0872093 | 8.227292672 |
| Eye diseases | 80 (split below) | | |
| Uveitis | 59 | 0.1715116 | 16.18034226 |
| Cataract | 3 | 0.0087209 | 0.822729267 |
| Glaucoma | 9 | 0.0261628 | 2.468187802 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Other eye diseases | 8 (split below) | |
| | | | | | | Benign eye tumour | 1 | 0.002907 | 0.274243089 |
| | | | | | | Episcleritis | 1 | 0.002907 | 0.274243089 |
| | | | | | | Scleritis | 2 | 0.005814 | 0.548486178 |
| | | | | | | Sicca syndrome/ sicca symptoms | 3 | 0.0087209 | 0.822729267 |
| | | | | | | Chronic conjunctivitis | 1 | 0.002907 | 0.274243089 |
| | | | | | | Astigmatism | 1 | 0.002907 | 0.274243089 |
| | | | | | | Neurologic diseases | 41 (split below) | |
| | | | | | | Migraine | 30 | 0.0872093 | 8.227292672 |
| | | | | | | Epilepsy | 5 | 0.0145349 | 1.371215445 |
| | | | | | | Psychiatric disorders | 32 (split below) | |
| | | | | | | Depression | 17 | 0.0494186 | 4.662132514 |
| | | | | | | Anxiety disorder | 7 | 0.0203488 | 1.919701624 |
| | | | | | | Other comorbidities | 30 (split below) | |
| | | | | | | Osteoporosis | 11 | 0.0319767 | 3.01667398 |
| Aggarwal, et al, 2006, India[60] | M: 30 | 22.6 ± 5.12 yrs | Mean 11.37 ± 2.67 yrs | | Mean 11.27 ± 5.39 yrs | | 30 | | |
| | | | | | | Osteopenia (Lumbar Spine) | 14 | 0.4666667 | 41.40786749 |
| | | | | | | Osteopenia (Hip) | 15 | 0.5 | 44.36557232 |
| | | | | | | Osteopenia (Distal third of forearm) | 15 | 0.5 | 44.36557232 |
| | | | | | | Osteoporosis (Lumbar Spine) | 8 | 0.2666667 | 23.66163857 |
| | | | | | | Osteoporosis (Hip) | 7 | 0.2333333 | 20.70393375 |
| | | | | | | Osteoporosis (Distal third of forearm) | 12 | 0.4 | 35.49245785 |
| | | | | | | Joint deformities | 5 | 0.1666667 | 14.78852411 |
| Arvidsson, et al, 2010, Norway[61] | F: 44 | Mean 35.2 yrs +/- 3.6 (27.9-44.4) | Mean 5.5 yrs +/- 3.1 (0.8-14.5 yrs) | | Mean 29.7 yrs +/- 3.3 yrs (24.4-38.0) | Mean 27 yrs | 60 | | |
| | | | | | | Craniofacial growth disturbance | 34 | 0.5666667 | 20.98765432 |
| | | | | | | Micrognathia | 16 | 0.2666667 | 9.87654321 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Packham, et al, 2002, UK[62] | F: 176 | Mean 35.4 yr (range 19-78, SD = 11.1) | Mean 7.1 yr (range 0.8–15.9, S.D.=4.5 ) | Mean 28.3 yr (range 8–73, S.D.=10. 8) | | 246 | | | Asian 1, Afro-Caribbean 1, Caucasian descent 244 |
| | | | | | Uveitis | 22% | 0.22 | 7.77385159 | |
| | | | | | Glaucoma | 4.10% | 0.041 | 1.448763251 | |
| | | | | | Amyloidosis | 8.90% | 0.089 | 3.144876325 | |
| | | | | | Cataracts | 10.20% | 0.102 | 3.604240283 | |
| | | | | | Required eye surgery | 10.60% | 0.106 | 3.745583039 | |
| | | | | | Sicca syndrome | 15.00% | 0.15 | 5.300353357 | |
| | | | | | At least one major prosthetic joint replacement | 51.2% (split below) | | | |
| | | | | | Hip prosthetic | 43.90% | 0.439 | 15.51236749 | |
| | | | | | Revision hip prostheses | 15.40% | 0.154 | 5.441696113 | |
| | | | | | Knee prosthetic | 27.60% | 0.276 | 9.752650177 | |
| | | | | | Shoulder prosthetic | 4.90% | 0.049 | 1.731448763 | |
| | | | | | Elbow prosthetic | 3.30% | 0.033 | 1.166077739 | |
| | | | | | Wrist prosthetic | 1.60% | 0.016 | 0.565371025 | |
| | | | | | Ankle prosthetic | 1.60% | 0.016 | 0.565371025 | |
| Mars, et al, 2019, Finland[63] | F: 92 | Mean 32.4 ± 13.4 yrs | | | | 119 | | | |
| | | | | | Cardiovascular diseases* | 17 | 0.1428571 | | |
| | | | | | Diabetes | 5 | 0.0420168 | | |
| | | | | | Eye disorders | 16 | 0.1344538 | | |
| | | | | | Gastrointestinal disorders | 11 | 0.092437 | | |
| | | | | | Infections | 27 | 0.2268908 | | |
| | | | | | Malignancies | 2 | 0.0168067 | | |
| | | | | | Mental disorders | 12 | 0.1008403 | | |
| | | | | | Neurological disorders | 18 | 0.1512605 | | |
| | | | | | Other musculoskeletal disorders | 38 | 0.3193277 | | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Rheumatic diseases | 76 | 0.6386555 | |
| | | | | | | Skin disorders | 19 | 0.1596639 | |
| Dimopoulou, et al, 2017, Greece[64] | F: 72 | Mean 25 yrs SD 5.9, median 24 yrs range 8-41 yrs | Median 7 yrs Mean 7.7 yrs SD 4 (range 1-15.5) | Median 17 yrs range 5-31.5 | Mean 17.2 yrs SD 6.7 yrs | | 102 | | |
| | | | | | | Uveitis | 11 | 0.1078431 | 6.26994984 |
| | | | | | | Prosthetic arthroplasty | 9 | 0.0882353 | 5.12995896 |
| | | | | | | Mandibular correction | 1 | 0.0098039 | 0.56999544 |
| | | | | | | Arthrodesis of the radiocarpal joint | 1 | 0.0098039 | 0.56999544 |
| | | | | | | Hands or foot small joint correction | 4 | 0.0392157 | 2.27998176 |
| | | | | | | Blindness in one eye | 2 | 0.0196078 | 1.13999088 |
| | | | | | | Leg length discrepancy ≥0.5cm | 37 | 0.3627451 | 21.08983128 |
| | | | | | | Mandibular dysfunction (such as reduced mouth opening, frontal facial asymmetry or retrognathia) | 29.40% | 0.0028824 | 0.167578659 |
| Rebane, et al, 2022, Finland[65] | F: 62 | Mean 23 yrs (range 16-23) | | | 17 yrs | | 97 | | |
| | | | | | | Coeliac disease | 1 | 0.0103093 | 0.606428138 |
| | | | | | | Crohn's disease | 2 | 0.0206186 | 1.212856277 |
| | | | | | | Irritable bowel syndrome (IBS) | 2 | 0.0206186 | 1.212856277 |
| | | | | | | Hypothyroidism | 1 | 0.0103093 | 0.606428138 |
| | | | | | | Depression | 2 | 0.0206186 | 1.212856277 |
| | | | | | | Panic disorder | 2 | 0.0206186 | 1.212856277 |
| | | | | | | Uveitis | 4 | 0.0412371 | 2.425712553 |
| | | | | | | Ulcerative colitis | 0 | 0 | 0 |
| Minden, et al, 2019, Germany[66] | F: 464 | Mean 23.4±3.9 yrs | | Mean 14.3±6.1 yrs | Mean 9.1±3.7 yrs | | 701 | | |
| | | | | | | Arthroplasty | 36 | 0.0513552 | 5.643429324 |
| | | | | | | Eye surgery | 22 | 0.0313837 | 3.448762365 |
| | | | | | | Uveitis | 63 | 0.0898716 | 9.876001317 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | Synovectomy | 112 | 0.1597718 | 17.55733567 | |
| Smith, et al, 2019, USA[67] | F: 170 | Mean 30.55±0.39 yrs (Age at estimated due date) | Mean 20.32±7.32 yrs | | 170 | | | White 145; African American 3; Asian/Pacific islander 2; Native America 3; Other 1 |
| | | | | Comorbid pregestational conditions | | | | |
| | | | | Thyroid dysfunction | 13 | 0.0764706 | | |
| | | | | Hypertension | 21 | 0.1235294 | | |
| | | | | Diabetes Mellitus (Type 1 and 2) | 0 | 0 | | |
| | | | | Other autoimmune diseases | | | | |
| | | | | Inflammatory bowel disease | 11 | 0.0647059 | | |
| | | | | Sjögren's syndrome or UCTD | 3 | 0.0176471 | | |
| | | | | Antiphospholipid syndrome | 1 | 0.0058824 | | |
| | | | | Psoriasis | 11 | 0.0647059 | | |
| | | | | Lupus | 4 | 0.0235294 | | |
| | | | | Obstetric complications | | | | |
| | | | | Preeclampsia | 15 | 0.0882353 | | |
| | | | | PIH without preeclampsia | 10 | 0.0588235 | | |
| | | | | Placenta previa and/or abruption | 4 | 0.0235294 | | |
| | | | | Preterm labour | 25 | 0.1470588 | | |
| | | | | GDM | 13 | 0.0764706 | | |
| | | | | Fever during pregnancy | 47 | 0.2764706 | | |

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|---|---|---|---|---|---|---|---|
| | | | | | Caesarean section | 70 | 0.4117647 |
| Barth, et al, 2016, Germany[68] | F: 1,617 | range (18–73) yrs | Median 8, range (0–15) yrs IQR (4-11) | Median 27; range (3–70) yrs IQR (15-40) | | 2592 | |
| | | | | | Pain/discomfort | (split below) | |
| | | | | | Moderate problems | 1395 | 0.5381944 |
| | | | | | Severe problems | 94 | 0.0362654 |
| | | | | | Anxiety/depression: | (split below) | |
| | | | | | Moderate problems | 624 | 0.2407407 |
| | | | | | Severe problems | 54 | 0.0208333 |
| | | | | | Severe disability | 783 | 0.3020833 |
| Simon, et al, 2020, USA[41] | | ≥ 18 yrs | | 1 year | | | |
| | F: 4,506 | Mean 37.0 ±16.5 yrs | | | | 5789 (split below) | |
| | | | | | Multiple sclerosis | 20 | 0.0034548 | 3.454828122 |
| | | | | | Asthma | 472 | 0.0815339 | 81.53394369 |
| | | | | | Anxiety | 555 | 0.0958715 | 95.87148039 |
| | | | | | Depression | 653 | 0.1128001 | 112.8001382 |
| | | | | | Diabetes mellitus | 440 | 0.0760062 | 76.00621869 |
| | | | | | Anaemia | 948 | 0.1637589 | 163.758853 |
| | | | | | Hypertension | 103 | 0.0177924 | 17.79236483 |
| | | | | | Hyperlipidaemia | 1162 | 0.2007255 | 200.7255139 |
| | | | | | Malignancy | 278 | 0.0480221 | 48.0221109 |
| | F: 4,315 | Mean 34.6 ±14.7 yrs | | | | 5638 (split below) | |
| | | | | | Multiple sclerosis | 19 | 0.00337 | 3.369989358 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | Asthma | 486 | 0.0862008 | 86.20078042 |
| | | | | | | Anxiety | 549 | 0.097375 | 97.37495566 |
| | | | | | | Depression | 622 | 0.1103228 | 110.3228095 |
| | | | | | | Diabetes mellitus | 371 | 0.0658035 | 65.80347641 |
| | | | | | | Anaemia | 655 | 0.1161759 | 116.1759489 |
| | | | | | | Hypertension | 103 | 0.0182689 | 18.26888968 |
| | | | | | | Hyperlipidaemia | 1023 | 0.1814473 | 181.4473217 |
| | | | | | | Malignancy | 162 | 0.0287336 | 28.73359347 |
| Packham and Hall, 2003, UK[69] | F: 177 | Mean 35.4±11.1 yrs (19-78) | Mean 7.1±4.2 yrs (range 0.8-15.9) | Mean 28.7±10.8 yrs (range 8-73) | | | 177 | | |
| | | | | | | Primary ovarian failure | 9 (split below) | | |
| | | | | | | Primary ovarian failure (unrelated to medication use) | 6 | 0.0338983 | 1.181125613 |
| Ma, et al, 2022, Taiwan[70] | F: 1,155 | 0-30 yrs | | | 249 593 months | | 2791 | | |
| | | | | | | Obstructive sleep apnoea (OSA) | 31 | 0.0111071 | 0.64 per 10000 person yrs 95% CI 0.5-0.82) |
| | | | | | | Allergic rhinitis | 401 | 0.1436761 | 0.006907699 |
| | | | | | | Obesity | 19 | 0.0068076 | 0.000327297 |
| | | | | | | Osteoporosis | 7 | 0.0025081 | 0.000120583 |
| | | | | | | Disorders of mineral metabolism | 3 | 0.0010749 | 5.16785E-05 |
| | | | | | | Hypocalcaemia/hypomagnesemia | 0 | 0 | 0 |
| Zhang-Jian, et al, 2020, Taiwan[71] | F: 549 | Age at pregnancy is 24.4 ± 3.4 yrs | | | | | 549 | | |
| | | | | | | Pregnancy outcomes | | | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Stillbirth | 4 | 0.007286 | | |
| | | | | | Foetal distress | 32 | 0.0582878 | | |
| | | | | | Foetal abnormalities, any | 38 | 0.0692168 | | |
| | | | | | Preeclampsia | 17 | 0.0309654 | | |
| | | | | | Gestational diabetes | 24 | 0.0437158 | | |
| | | | | | Pregnancy-related hypertension | 24 | 0.0437158 | | |
| | | | | | Antepartum haemorrhage | 55 | 0.1001821 | | |
| | | | | | Severe postpartum haemorrhage | 25 | 0.0455373 | | |
| | | | | | Chorioamnionitis | 5 | 0.0091075 | | |
| | | | | | Caesarean delivery | 155 | 0.2823315 | | |
| | | | | | Low birth weight <2500g | 64 | 0.1165756 | | |
| | | | | | Prematurity <37 weeks | 49 | 0.0892532 | | |
| Rypdal, et al, 2021, USA[72] | F: 297 | ≥19 yrs | Mean 5.7 IQR 2.6-9.8 | 17.6 yrs IQR (16.7-18.4) | | 434 | | | |
| | | | | | Uveitis | 96 | 0.2211982 | 12.56807708 | |
| | | | | | At least 1 ocular complication | 31/80 | 0.3875 | 22.01704545 | |
| | | | | | Cataract | 25/80 | 0.3125 | 17.75568182 | |
| | | | | | Glaucoma | 22/80 | 0.275 | 15.625 | |
| | | | | | Synechiae | 14/80 | 0.175 | 9.943181818 | |
| | | | | | Macular oedema | 8/80 | 0.1 | 5.681818182 | |
| | | | | | Band keratopathy | 7/80 | 0.0875 | 4.971590909 | |
| | | | | | Epiretinal membrane | 3/80 | 0.0375 | 2.130681818 | |
| | | | | | Hypotony | 3/80 | 0.0375 | 2.130681818 | |
| | | | | | Phthisis | 3/80 | 0.0375 | 2.130681818 | |
| | | | | | Blindness <3/60 - binocular | 2/80 | 0.025 | 1.420454545 | |
| | | | | | Blindness <3/60 - monocular | 3/80 | 0.0375 | 2.130681818 | |
| **Childhood / jSLE** | | | | | | | | | |
| Davis, et al, 2023, USA[73] | F: 571 | 18.4±3.7 yrs (10-24 yrs) | | | | 650 | | | White 367; Black 110; |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | Depression | 117 | 0.18 | Hispanic 91; Asian 41; Unknown 41 |
| | | | | Anxiety | 78 | 0.12 | |
| | | | | Adjustment disorder | 65 | 0.1 | |
| | | | | Seizure/stroke disorder | 52 | 0.08 | |
| Garf, et al, 2015, Egypt[74] | F: 49 | Mean 13.1±3.0 yrs | Mean 3.9±3.0 yrs (range 0.2-12 yrs) | | 70 | | |
| | | | | Vitamin D deficiency | 28 | 0.4 | |
| | | | | Vitamin D insufficiency (≤30/ng/ml) | 42 | 0.6 | |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Paim-Marques, et al, 2019, Brazil[75] | F: 61 | Mean 17.9±3.07 yrs | Mean 13.56 ± 2.60 yrs (corneal verticillata) 12.63 ± 2.9 (no corneal verticillata) | | Mean 5.06 ± 2.04 yrs (with corneal verticillata) | | 76 | | | |
| | | | | | | Visual acuity reduction | 25 | 0.3289474 | 65.00936135 | |
| | | | | | | Posterior subcapsular cataracts | 4 | 0.0526316 | 10.40149782 | |
| | | | | | | Keratitis | 40 | 0.5263158 | 104.0149782 | |
| | | | | | | Crystalline abnormalities | 1 | 0.0131579 | 2.600374454 | |
| | | | | | | Corneal vortex keratopathy | 16 | 0.2105263 | 41.60599126 | |
| Silva, et al, 2015, Brazil[76] | F: 732 | Median 14.5 (6–20) yrs, IFI; 17.0 (2–25.9) non-IFI | Median 12.4 (range 5.1–17.2) yrs, IFI; 11.8 (0.2–17.8) non IFI | | Median 1.0 (range 0–7.3) yrs, IFI; 4.7 (range 0–23.4) non- IFI | | 852 | | | White 592/830 |
| | | | | | | Invasive Fungal Infections (Proven) | 22 (split as follows) | | | |
| | | | | | | Candidiasis | 13 | 0.0152582 | | |
| | | | | | | Aspergillosis | 6 | 0.0070423 | | |
| | | | | | | Cryptococcosis | 2 | 0.0023474 | | |
| | | | | | | Histoplasmosis | 1 | 0.0011737 | | |
| | | | | | | Arterial hypertension | 119/843 | 0.1411625 | | |
| | | | | | | Acute renal failure | 37/841 | 0.0439952 | | |
| | | | | | | Chronic renal failure | 27/842 | 0.0320665 | | |
| | | | | | | Renal replacement Therapy | 28/842 | 0.0332542 | | |
| Knight, et al, 2016, USA[77] | F: 805 | Mean 14.7 SD 2.0 (10-18 yrs) | | | | | 970 | | | White 141; African American |
| | | | | | | Depression diagnosis | 181 | 0.1865979 | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Anxiety | 72 | 0.0742268 | | 410; Latino 265; Other 154 |
| | | | | Acute stress/adjustment | 60 | 0.0618557 | | |
| | | | | Other psychiatric disorder | 175 | 0.1804124 | | |
| | | | | Seizures/stroke | 155 | 0.1597938 | | |
| Das Chagas Medeiros, et al, 2016, Brazil[78] | F: 52 | Mean 15.5±2.0 yrs | Mean 10.4±6.4 yrs | | 60 | | | |
| | | | | Antiphospholipid syndrome | 3/49 | 0.0612245 | 5.886970173 | |
| | | | | Hypothyroidism | 1/37 | 0.027027 | 2.598752599 | |
| | | | | Thrombosis | 7/60 (split below) | | | |
| | | | | Deep vein thrombosis | 1 | 0.0166667 | 1.602564103 | |
| | | | | Mesenteric artery thrombosis | 1 | 0.0166667 | 1.602564103 | |
| | | | | CVA | 1 | 0.0166667 | 1.602564103 | |
| | | | | Pulmonary embolus | 1 | 0.0166667 | 1.602564103 | |
| | | | | Portal vein thrombosis | 1 | 0.0166667 | 1.602564103 | |
| | | | | Renal vein thrombosis | 1 | 0.0166667 | 1.602564103 | |
| | | | | DVT + PE | 1 | 0.0166667 | 1.602564103 | |
| | | | | Neoplasia | 1/58 | 0.0172414 | 1.657824934 | |
| | | | | hypertension | 27/60 | 0.45 | 43.26923077 | |
| | | | | Cardiovascular diseases | 3/58 (split below) | | | |
| | | | | Congestive heart failure | | | | |
| | | | | Coronary artery disease | | | | |
| | | | | Aortic and mitral valve disorder | | | | |
| | | | | Diabetes | 0/60 | 0 | 0 | |
| | | | | Seizure | 7/60 | 0.1166667 | 11.21794872 | |
| | | | | Psychosis | 2/60 | 0.0333333 | 3.205128205 | |
| Frittoli, et al, 2016, Brazil[79] | F: 38 | Mean 14.5±2.84 yrs | | | 41 | | | |
| | | | | Cognitive dysfunction | 17 | 0.4146341 | | |
| | | | | Anxiety symptoms | 8 | 0.195122 | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Holland , et al, 2018, USA[80] | F: 869 | Mean 12.2±3.1 yrs | Mean 3.81±2.98 yrs | Mean 3.8 yrs | Depression symptoms | 1 1048 | 0.0243902 | |
| | | | | | Ocular | | | |
| | | | | | Cataracts | 4.20% | 4.008E-05 | 0.010546404 |
| | | | | | Retinal change | 1.70% | 1.622E-05 | 0.004268783 |
| | | | | | Neuropsychiatric | | | |
| | | | | | Cognitive impairment | 8.40% | 8.015E-05 | 0.021092808 |
| | | | | | Seizures requiring therapy | 3.90% | 3.721E-05 | 0.00979309 |
| | | | | | Cerebrovascular accident | 2.9%/0.4% | 0.0069179 | 1.820510245 |
| | | | | | Cranial/peripheral neuropathy | 2.40% | 2.29E-05 | 0.006026517 |
| | | | | | Transverse myelitis | 0.40% | 3.817E-06 | 0.001004419 |
| | | | | | Renal | | | |
| | | | | | End-stage renal disease | 1.30% | 1.24E-05 | 0.003264363 |
| | | | | | Pulmonary | | | |
| | | | | | Pulmonary hypertension | 0.40% | 3.817E-06 | 0.001004419 |
| | | | | | Pleural fibrosis | 1.30% | 1.24E-05 | 0.003264363 |
| | | | | | Shrinking lung | 0.80% | 7.634E-06 | 0.002008839 |
| | | | | | Pulmonary fibrosis | 0.90% | 8.588E-06 | 0.002259944 |
| | | | | | Pulmonary infarction | 0% | 0 | 0 |
| | | | | | Cardiovascular | | | |
| | | | | | Angina/artery bypass | 0.10% | 9.542E-07 | 0.000251105 |
| | | | | | Myocardial infarction | 0.1 / 0.1 | 0.0009542 | 0.251104861 |
| | | | | | Cardiomyopathy | 0.90% | 8.588E-06 | 0.002259944 |
| | | | | | Valvular disease | 0.90% | 8.588E-06 | 0.002259944 |
| | | | | | Pericarditis/pericardiectomy | 1.60% | 1.527E-05 | 0.004017678 |
| | | | | | Peripheral vascular | | | |
| | | | | | Claudication for 6 months | 0.30% | 2.863E-06 | 0.000753315 |
| | | | | | minor tissue loss | 2.40% | 2.29E-05 | 0.006026517 |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | significant tissue loss | 0.7/0.3% | 0.0022265 | 0.585911343 |
| | | | | | Venous thrombus swelling, ulcer/stasis | 2.50% | 2.385E-05 | 0.006277622 |
| | | | | | Gastrointestinal | | | |
| | | | | | Bowel infarction or resection | 1.3 / 0.3% | 0.0041349 | 1.088121066 |
| | | | | | Mesenteric insufficiency | 0.10% | 9.542E-07 | 0.000251105 |
| | | | | | Peritonitis | 0.20% | 1.908E-06 | 0.00050221 |
| | | | | | Stricture/upper GI tract surgery ever | 0.40% | 3.817E-06 | 0.001004419 |
| | | | | | Pancreatic insufficiency | 0.50% | 4.771E-06 | 0.001255524 |
| | | | | | Musculoskeletal | | | |
| | | | | | Muscle atrophy/weakness | 7.90% | 7.538E-05 | 0.019837284 |
| | | | | | Deforming/erosive arthritis | 4.70% | 4.485E-05 | 0.011801928 |
| | | | | | Osteoporosis with fracture | 2.60% | 2.481E-05 | 0.006528726 |
| | | | | | Avascular necrosis | 2.7 / 1.0 | 0.0025763 | 0.677983126 |
| | | | | | Osteomyelitis | 0.40% | 3.817E-06 | 0.001004419 |
| | | | | | Ruptured tendons | 0.10% | 9.542E-07 | 0.000251105 |
| | | | | | Skin | | | |
| | | | | | Scarring chronic alopecia | 9.10% | 8.683E-05 | 0.022850542 |
| | | | | | Extensive scar/panniculum | 1.70% | 1.622E-05 | 0.004268783 |
| | | | | | Skin ulceration for 6 months | 2.70% | 2.576E-05 | 0.006779831 |
| | | | | | Premature gonadal failure | 2% | 1.908E-05 | 0.005022097 |
| | | | | | Diabetes | 0.90% | 8.588E-06 | 0.002259944 |
| | | | | | Malignancy | 0.20% | 1.908E-06 | 0.00050221 |
| Gad, et al, 2013, Egypt[81] | F: 17 | Mean 12.90±2.673 yrs (8-16 yrs) | | Mean 4.40±1.875 yrs | | 20 | | |
| | | | | | Sensorineural hearing loss (SNHL) | 5 | 0.25 | |
| | | | | | Mild Conductive hearing loss (CHL) | 2 | 0.1 | |
| Jongvilaikasem and Rianthavorn, 2021, Thailand[82] | F: 87 | Mean 11.3 ± 2.4 yrs | | 26 months | | 97 | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | Short final height | 22 | 0.2268041 | 104.6788263 | |
| | | | | Mild CKD | 88 | 0.9072165 | 418.7153053 | |
| | | | | Advanced CKD | 9 | 0.0927835 | 42.82315623 | |
| AlAhmed, et al, 2020, USA[83] | F: 306 | Mean 11.39±3.8 yrs | Mean 12.04±3.62 year (Age first seen by paediatric rheumatologist) | | 386 | | | Hispanic 102; Non-Hispanic 283, White 216, African American 105, Asia 33, Other 13 |
| | | | | Cerebrovascular accident | 2 | 0.0051813 | | |
| | | | | Cranial nerve disease | 1 | 0.0025907 | | |
| | | | | Renal dialysis | 3 | 0.007772 | | |
| | | | | Arterial thrombosis | 3 | 0.007772 | | |
| | | | | Venous thrombosis | 10 | 0.0259067 | | |
| | | | | Avascular necrosis | 5 | 0.0129534 | | |
| | | | | Autoimmune thyroiditis | 10 | 0.0259067 | | |
| | | | | Coeliac disease | 1 | 0.0025907 | | |
| | | | | T1DM | 1 | 0.0025907 | | |
| | | | | Other autoimmune disorder | 22 | 0.0569948 | | |
| | | | | Pulmonary hypertension | 0 | 0 | | |
| | | | | Coronary heart disease | 0 | 0 | | |
| | | | | Renal transplant | 0 | 0 | | |
| Karve, S. et al, 2012, USA[12] | | 0-15 yrs | 18 months | | 71 | | | |

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| | | | |
|---|---|---|---|
| Asthma | 8 | 0.1126761 | 75.11737089 |
| Atopic dermatitis | 2 | 0.028169 | 18.77934272 |
| Allergic rhinitis | 6 | 0.084507 | 56.33802817 |
| General diabetes | 1 | 0.0140845 | 9.389671362 |
| Rheumatoid arthritis | 14 | 0.1971831 | 131.4553991 |
| Multiple sclerosis | 1 | 0.0140845 | 9.389671362 |
| Thyroiditis | 1 | 0.0140845 | 9.389671362 |
| Fracture in post-index period | 4 | 0.056338 | 37.55868545 |
| Psoriasis | 0 | 0 | 0 |
| Type 1 diabetes | 0 | 0 | 0 |
| Other CNS demyelinating disease | 0 | 0 | 0 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Vitiligo | 0 | 0 | 0 |
| | | | | | | Grave's disease | 0 | 0 | 0 |
| Ye, Q., et al, 2021, China[84] | F: 82 | Median age 169.02 months (range 31.00-220.53) | 133.83, range (27.70–202.83) months | | 28.00, range (0–144.00) months | | 107 | | |
| | | | | | | Cataracts | 21 | 0.1962617 | |
| | | | | | | Steroid-induced osteonecrosis of femoral head | 3 | 0.0280374 | |
| | | | | | | Osteoporosis | 5 | 0.046729 | |
| | | | | | | Short stature | 8 | 0.0747664 | |
| | | | | | | Obesity | 7 | 0.0654206 | |
| | | | | | | Diabetes | 6 | 0.0560748 | |
| | | | | | | HDL resistance | 1 | 0.0093458 | |
| | | | | | | Upper respiratory tract infection | 11 | 0.1028037 | |
| | | | | | | Pneumonia | 13 | 0.1214953 | |
| | | | | | | EB virus infection | 15 | 0.1401869 | |
| | | | | | | Progressive systemic sclerosis | 1 | 0.0093458 | |
| | | | | | | Secondary Sjogren | 6 | 0.0560748 | |
| | | | | | | Juvenile rheumatoid arthritis | 1 | 0.0093458 | |
| | | | | | | Antiphospholipid syndrome | 2 | 0.0186916 | |
| Cann, et al, 2022, Australia[85] | F: 37 | | | *12.5 yrs (mean) 13 yrs (median)* | Mean 4.3 yrs , IQR (2.5–6.1) yrs | | 42 | | | Caucasian 16, Indigenous Australian 11, Asian 10, Other 5 |
| | | | | | | Cataract | 2 | 0.047619 | 11.07419712 | |
| | | | | | | retinal change | 2 | 0.047619 | 11.07419712 | |
| | | | | | | CVA | 3 | 0.0714286 | 16.61129568 | |
| | | | | | | Significant tissue loss | 1 | 0.0238095 | 5.53709856 | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Venous thrombosis | 1 | 0.0238095 | 5.53709856 |
| | | | | | Muscle weakness | 1 | 0.0238095 | 5.53709856 |
| | | | | | AVN | 1 | 0.0238095 | 5.53709856 |
| | | | | | Scarring alopecia | 2 | 0.047619 | 11.07419712 |
| | | | | | Diabetes | 1 | 0.0238095 | 5.53709856 |
| | | | | | Growth failure | 3 | 0.0714286 | 16.61129568 |
| | | | | | Delayed puberty | 1 | 0.0238095 | 5.53709856 |
| | | | | | Premature gonadal failure | 0 | 0 | 0 |
| | | | | | Malignancy | 0 | 0 | 0 |
| Medhat, et al, 2020, Egypt[86] | F: 86 | Mean 12.7 ±2.6 yrs (range 6–16) | Mean 9.9±2.5, range (4–15) yrs | Median 24, IQR (12–48) months | | 106 | | |
| | | | | | Osteoporosis | 6 | 0.0566038 | |
| | | | | | Hypothyroidism | 2 | 0.0188679 | |
| | | | | | Diabetes mellitus | 4 | 0.0377358 | |
| | | | | | Hypertension | 31 | 0.2924528 | |
| | | | | | Hepatitis C virus infection | 5 | 0.0471698 | |
| | | | | | Chronic kidney disease | 5 | 0.0471698 | |
| | | | | | End-stage renal disease | 1 | 0.009434 | |
| | | | | | Avascular necrosis | 1 | 0.009434 | |
| | | | | | Sicca manifestations | 2 | 0.0188679 | |
| | | | | | Ischaemic heart disease | 0 | 0 | |
| | | | | | CVA | 0 | 0 | |
| | | | | | Malignancy | 0 | 0 | |
| Aggarwal, et al, 2018, India[87] | F: 250 | ≤ 18 yrs | Median 14 yrs range (1-17) | Median 3.5 yrs, Range (0.25-28 yrs) (248 patients); 25 patients were not followed up | | 273 | | |
| | | | | | Cushingoid habitus | 35 | 0.1282051 | 36.63003663 |
| | | | | | Growth retardation | 32 | 0.1172161 | 33.4903192 |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Avascular necrosis | 9 | 0.032967 | 9.419152276 | |
| | | | | | | Cataract | 8 | 0.029304 | 8.372579801 | |
| | | | | | | Osteopenia | 10 | 0.03663 | 10.46572475 | |
| | | | | | | Infection | 72 | 0.2637363 | 75.35321821 | |
| | | | | | | Hypertension | 42 | 0.1538462 | 43.95604396 | |
| | | | | | | DVT | 8 | 0.029304 | 8.372579801 | |
| | | | | | | Pulmonary thromboembolism | 2 | 0.007326 | 2.09314495 | |
| | | | | | | Stroke | 3 | 0.010989 | 3.139717425 | |
| | | | | | | Renal vein thrombosis | 1 | 0.003663 | 1.046572475 | |
| Al-Mayouf, et al, 2008, Saudi Arabia[88] | F: 76 | F 14.4±3.9 yrs, male 15.8±4.4 yrs (mean) | | Mean 9.3±2.8 yrs(F); 10±2.5 yrs (M) | F 66.5 ± 39.2 months, Male 76.2 ± 56.4 months, median 5 yrs | F 60.5 ± 38 months, Male 69.1 ± 57.5 months | 89 | | | |
| | | | | | | Diabetes | 1 | 0.011236 | 0.185718265 | |
| | | | | | | Malignancy | 0 | 0 | 0 | |
| | | | | | | Dialysis + transplant (ESRD) | 5 | 0.0561798 | 0.928591327 | |
| | | | | | | Premature gonadal failure | 0 | 0 | 0 | |
| | | | | | | Transverse myelitis | 2 | 0.0224719 | 0.371436531 | |
| | | | | | | Pyoderma gangrenosum | 1 | 0.011236 | 0.185718265 | |
| | | | | | | Pancreatic pseudocyst | 1 | 0.011236 | 0.185718265 | |
| Costagliola, et al, 2018, Italy[89] | F: 24 | | Mean 14.6 ± 1.6 yrs | | | Mean 14.17 ± 8.04 yrs (1 patient excluded) | 25 | | | Caucasian 24, Hispanic 1 |
| | | | | | | Renal damage (including nephrotic syndrome or ESRD) | 9 | 0.36 | 25.40578687 | |

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| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | Vascular involvement (including DVT or ischaemic arterial lesions) | 7 | 0.28 | 19.76005646 | |
| | | | | | Nervous system involvement (with ischemic stroke, cranial neuropathies and cerebral vasculitis) | 4 | 0.16 | 11.29146083 | |
| | | | | | Cutaneous damage (chronic non-scarring alopecia) | 3 | 0.12 | 8.468595625 | |
| | | | | | Myositis | 1 | 0.04 | 2.822865208 | |
| | | | | | Acute myocardial infarction | 1 | 0.04 | 2.822865208 | |
| | | | | | Pulmonary (restrictive syndrome and pulmonary heart) | 1 | 0.04 | 2.822865208 | |
| | | | | | CMV reactivation | 3 | 0.12 | 8.468595625 | |
| | | | | | Herpes zoster reactivation | 2 | 0.08 | 5.645730416 | |
| | | | | | Severe EBV infection | 1 | 0.04 | 2.822865208 | |
| | | | | | Mycobacterium tuberculosis infection | 1 | 0.04 | 2.822865208 | |
| | | | | | ESRD | 3 | 0.12 | 8.468595625 | |
| | | | | | Requiring dialysis | 1 | 0.04 | 2.822865208 | |
| | | | | | Multi-infarct encephalopathy | 1 | 0.04 | 2.822865208 | |
| Torrente-Segarra, et al, 2017, Spain[90] | F: 435 | Mean 31.5±10.5 Median 30 range [24-28] yrs Childhood | Mean 16.6 ± 6.3 yrs, median 16 [13-18] | Mean 10.7±7.2 yrs | | 484 | | | Caucasian 441 |
| | | | | | Articular | | | | |
| | | | | | Erosive arthritis | 40 | 0.0826446 | | |
| | | | | | Atrophy, muscle weakness | 15 | 0.0309917 | | |
| | | | | | Tendon rupture | 5 | 0.0103306 | | |
| | | | | | Osteoporosis | 15 | 0.0309917 | | |
| | | | | | Fibromyalgia | 11 | 0.0227273 | | |
| | | | | | Avascular necrosis | 1.6 ± 0.6 2 [1-2] | | | |
| | | | | | Pulmonary embolism | 11 | 0.0227273 | | |
| | | | | | Lung fibrosis | 6 | 0.0123967 | | |

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| | | |
|---|---|---|
| Pulmonary hypertension | 8 | 0.0165289 |
| Alveolar haemorrhage | 6 | 0.0123967 |
| Shrinking lung | 4 | 0.0082645 |
| Pleural fibrosis | 1 | 0.0020661 |
| Pulmonary infarction | 1 | 0.0020661 |
| Vascular disease | 25 | 0.0516529 |
| Valvular dysfunction | 19 | 0.0392562 |
| Libman-Sachs Endocarditis | 8 | 0.0165289 |
| Cardiomyopathy | 7 | 0.0144628 |
| Angina or cardiac bypass | 6 | 0.0123967 |
| Myocarditis | 3 | 0.0061983 |
| Number of heart attacks | 1.1 ± 0.0<br>1.0 [1-1] | |
| Small tissue loss | 22 | 0.0454545 |
| Major tissue loss | 3 | 0.0061983 |
| Venous thrombosis | 3 | 0.0061983 |
| Intermittent claudication >6 months | 1 | 0.0020661 |
| Renal terminal failure | 27 | 0.0557851 |
| Seizures (requiring therapy for 6 months) | 46 | 0.0950413 |
| Cognitive impairment/ Psychosis | 21 | 0.0433884 |
| Neuropathy | 13 | 0.0268595 |
| Transverse myelitis | 3 | 0.0061983 |
| Number of strokes | 1.2±0.6<br>1[1-1] | |
| Cataract | 31 | 0.0640496 |
| Retinopathy | 22 | 0.0454545 |
| Amenorrhoea <40 | 22 (51.2%) | 0.0454545 |
| Dyslipidaemia | 104 | 0.214876 |
| High blood pressure | 100 | 0.2066116 |
| Autoimmune thyroiditis | 31 | 0.0640496 |
| Alcoholism | 14 | 0.0289256 |
| Congestive heart failure | 8 | 0.0165289 |
| Diabetes mellitus with organ involvement | 8 | 0.0165289 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Diabetes mellitus without organ involvement | 0 | 0 | |
| | | | | | | Chronic obstructive pulmonary disease | 7 | 0.0144628 | |
| | | | | | | Peripheral arterial disease | 3 | 0.0061983 | |
| | | | | | | Malignancy Solid without metastasis | 15 | 0.0309917 | |
| | | | | | | Malignancy Solid with metastasis | 0 | 0 | |
| | | | | | | Leukaemia | 1 | 0.0020661 | |
| | | | | | | Lymphoma | 2 | 0.0041322 | |
| | | | | | | Dialysis | 23 | 0.0475207 | |
| | | | | | | Kidney transplantation | 15 | 0.0309917 | |
| | | | | | | Splenectomy | 14 | 0.0289256 | |
| Ponin, et al, 2022, Thailand[91] | F: 87 | Mean 20.6 ± 3.0 yrs | Mean 12.1 ± 2.3 yrs | Mean 5.5 ± 2.7 yrs | | | 106 | | |
| | | | | | | Growth impairment | 25 | 0.2358491 | |
| Avar Aydin, et al, 2018, USA[92] | F: 87 | | Mean yrs 13.2±3.1 | | Median 29.5 months (IQR 6.0–55.2) | | 110 | | Caucasian 54; African American 39; Asian 8; Other 9 |
| | | | | | | Hypertension | 22 | 0.2 | 81.3559322 |
| | | | | | | Obesity | 19 | 0.1727273 | 70.26194145 |
| | | | | | | Dyslipidaemia | 14 | 0.1272727 | 51.77195686 |
| Ahluwalia, et al, 2014, India[93] | F: 17 | | | | Mean 82 months (range 9–210 months) | | 21 (27 patients initially, 6 lost to follow up) | | |
| | | | | | | Thrombosis | 3 (split below) | | |
| | | | | | | Cerebral infarct (death) | 1 | 0.047619 | 6.968641115 |
| | | | | | | Pulmonary thromboembolism (death) | 1 | 0.047619 | 6.968641115 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Spinal infarct | 1 | 0.047619 | 6.968641115 |
| Sinicato, et al, 2017, Brazil[94] | F: 37 | 16.7 ± 4.0 (mean) | | Mean 4.8 ± 4.1 yrs | | 41 | | |
| | | | | | iglyceridemia | 11 | 0.268 | |
| | | | | | Insulin resistance | 2 | 0.049 | |
| | | | | | Arterial hypertension | 4 | 0.097 | |
| | | | | | Albumin Exretion | 5 | 0.122 | |
| | | | | | high LDL-c levels | 19 | 0.463 | |
| | | | | | hypercholesterolemia | 8 | 0.195 | |
| Cabral, et al, 2013, Portugal[95] | F: 46 | | Mean 12.6±4.04 yrs Median 13.5 range (1-17) | Mean 5.5±5.4 yrs Median 4 range (0-24) | | 56 | | |
| | | | | | Cataract | 4 | 0.0714286 | 12.98701299 |
| | | | | | Epilepsy | 3 | 0.0535714 | 9.74025974 |
| | | | | | Obesity | 25 | 0.4464286 | 81.16883117 |
| | | | | | Cardiac failure | 2 | 0.0357143 | 6.493506494 |
| | | | | | Arterial hypertension | 5 | 0.0892857 | 16.23376623 |
| | | | | | Avascular necrosis | 1 | 0.0178571 | 3.246753247 |
| | | | | | Growth delay | 3 | 0.0535714 | 9.74025974 |
| | | | | | Osteoporosis | 5 | 0.0892857 | 16.23376623 |
| | | | | | Gastrointestinal tract bleeding | 1 | 0.0178571 | 3.246753247 |
| | | | | | Macrophage activation syndrome | 2 | 0.0357143 | 6.493506494 |
| | | | | | Infection | 6 (split below) | | |
| | | | | | Pneumonia | 2 | 0.0357143 | 6.493506494 |
| | | | | | Sepsis | 1 | 0.0178571 | 3.246753247 |
| | | | | | Bacterial endocarditis | 1 | 0.0178571 | 3.246753247 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Cellulitis | 3 | 0.0535714 | 9.74025974 | |
| | | | | | Cognitive impairment | 4 | 0.0714286 | 12.98701299 | |
| | | | | | End stage renal disease | 2 | 0.0357143 | 6.493506494 | |
| | | | | | Vascular thrombosis | 5 (split below) | | | |
| | | | | | Cerebrovascular (thrombosis) | 4 | 0.0714286 | 12.98701299 | |
| | | | | | Cerebral hemorrhage | 2 | 0.0357143 | 6.493506494 | |
| | | | | | Myocardium infarct | 1 | 0.0178571 | 3.246753247 | |
| | | | | | Epilepsy | 3 | 0.0535714 | 9.74025974 | |
| Bernatsky, et al, 2017, North America[96] | F: 978 | Mean 13±3.3 yrs | | Mean 7.6 yrs, SD 6.4 yrs, IQ 3.4-10.0 yrs/8839 PY | | 1168 | | | |
| | | | | | Non-Hodgkin lymphoma | 3 | 0.0025685 | 0.337959625 | |
| | | | | | Head and neck squamous cell carcinomas (tongue and nasopharynx) | 2 | 0.0017123 | 0.225306417 | |
| | | | | | Glioblastoma | 1 | 0.0008562 | 0.112653208 | |
| | | | | | Thyroid cancer (papillary adenoma) | 1 | 0.0008562 | 0.112653208 | |
| | | | | | Breast cancer (infiltrating duct carcinoma) | 1 | 0.0008562 | 0.112653208 | |
| | | | | | Adenocarcinoma not otherwise specified | 1 | 0.0008562 | 0.112653208 | |
| | | | | | Cancers whose morphology and histology codes were indeterminate | 3 | 0.0025685 | 0.337959625 | |
| Yang, et al, 2015, Canada[97] | F: 30 (AVN), F: 82 (non-AVN) | | Mean 13.8 ± 2.5 (AVN), 13.8 ± 2.4 (non-AVN) | Mean 6.89 ± 5.39 yrs (AVN), 5.72 ± 5.09 yrs (non-AVN) | | 617 | | | AVN: South Asian 10; *Other Asian 9; Black 9; White 8; Other 1. Non-AVN: South Asian 10, |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | | | | Other Asian 30, Black 12, WhIte 40, Hispanic 1 Other 13, Unknown 5 |
| | | | | | Symptomatic Avascular Necrosis (AVN) | 37 | 0.0599676 | 8.703568228 | |
| Lin, et al, 2018, Taiwan[98] | | | Mean 12.9±3.12 yrs (range 6-18 yrs) | mean 6 yrs | | 35 | | | |
| | | | | | Vitamin D deficiency | 28 | 0.8 | 133.3333333 | |
| Kahwage, et al, 2017, Brazil[99] | F: 6/7 (uveitis) | ≤25 yrs, Median 10.4 range (8.8 -14.5) (Uveitis) | | | | 852 | | | |
| | | | | | Uveitis | 7 | 0.008216 | | |
| | | | | | Cataracts & bilateral irreversible blindness | 1 | 0.0011737 | | |
| | | | | | Unilateral blindness | 1 | 0.0011737 | | |
| Lai, et al, 2022, China[100] | F: 95 | Age at enrolment - Girls mean 14.4 (SD 3.4), Boys mean 13.9 (SD 5.0) | | Girls mean 6.6 (SD 4.1) Boys mean 5.2 (SD 4.0) | | 115 | | | |
| | | | | | Hypertension | 22 | 0.1913043 | | |
| | | | | | Chronic kidney disease | 14 | 0.1217391 | | |
| Adulthood / jSLE | | | | | | | | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Al Hamzi, et al, 2014, Saudi Arabia[101] | F: 45 | Mean 23.6±4 yrs | Mean 10.3±6 yrs | Mean 15±4 yrs | | 48 | | |
| | | | | | Growth retardation/Short stature | 24 | 0.5 | |
| | | | | | Delayed puberty | 13 | 0.2708333 | |
| | | | | | Renal disease | 12 (as split below) | | |
| | | | | | Progressive renal disease and required dialysis | 7 | 0.1458333 | |
| | | | | | Renal transplant | 5 | 0.1041667 | |
| | | | | | Neuropsychiatric | 9 | 0.1875 | |
| | | | | | Cardiovascular | 6 | 0.125 | |
| | | | | | Musculoskeletal | 9 | 0.1875 | |
| | | | | | Skin | 2 | 0.0416667 | |
| | | | | | Seizure disorder | 6 | 0.125 | |
| | | | | | Chorea | 3 | 0.0625 | |
| | | | | | Cerebrovascular accident | 3 | 0.0625 | |
| | | | | | Miscarriage | 4 | 0.0833333 | |
| Tanigava, et al, 2022, Brazil[102] | F: 65 | Median 21 yrs (IQR 20-26) | Median 14 IQR 12-16 | Median 9 yrs IQR 6-14 | | 69 | | White 35; Black 31; Asian 3 |

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| | | |
|---|---|---|
| Mental disorders | 16 | 0.2318841 |
| Obesity | 14 | 0.2028986 |
| Thyroid diseases | 7 | 0.1014493 |
| Fibromyalgia | 6 | 0.0869565 |
| Antiphospholipid syndrome | 56 | 0.8115942 |
| Osteoporosis and osteoporotic fracture | 5 | 0.0724638 |
| Hypertension | 5 | 0.0724638 |
| Cerebrovascular disease | 4 | 0.057971 |
| Allergic disorders | 4 | 0.057971 |

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| | | |
|---|---|---|
| Gastrointestinal diseases | 4 | 0.057971 |
| Valvular disease | 4 | 0.057971 |
| Chronic kidney disease | 3 | 0.0434783 |
| Diabetes mellitus | 2 | 0.0289855 |
| Dyslipidaemia | 2 | 0.0289855 |
| Retinal change or optic atrophy | 5 | 0.0724638 |
| Cataract | 2 | 0.0289855 |
| Cognitive impairment or major psychosis | 3 | 0.0434783 |
| Seizures requiring therapy for ≥6 months | 2 | 0.0289855 |
| Cerebrovascular accident | 3 | 0.0434783 |
| Cranial or peripheral neuropathy | 3 | 0.0434783 |
| Transverse myelitis | 0 | 0 |
| End-stage renal disease | 3 | 0.0434783 |
| Pulmonary hypertension | 0 | 0 |
| Pulmonary fibrosis | 2 | 0.0289855 |
| Shrinking lung | 0 | 0 |
| Pleural fibrosis | 0 | 0 |
| Pulmonary infarction | 0 | 0 |
| Angina or coronary artery bypass | 0 | 0 |
| Myocardial infarction | 0 | 0 |
| Cardiomyopathy | 0 | 0 |
| Pericarditis or pericardiectomy | 0 | 0 |
| Venous thrombosis | 1 | 0.0144928 |
| Infarction or resection of bowel | 1 | 0.0144928 |
| Mesenteric insufficiency | 0 | 0 |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Chronic peritonitis | 0 | 0 | |
| | | | | | Stricture or upper gastrointestinal tract surgery | 0 | 0 | |
| | | | | | Pancreatic insufficiency | 1 | 0.0144928 | |
| | | | | | Muscle atrophy or weakness | 0 | 0 | |
| | | | | | Deforming or erosive arthritis | 0 | 0 | |
| | | | | | Osteoporosis with fracture or vertebral collapse | 2 | 0.0289855 | |
| | | | | | Avascular necrosis | 2 | 0.0289855 | |
| | | | | | Osteomyelitis | 0 | 0 | |
| | | | | | Ruptured tendon | 0 | 0 | |
| | | | | | Scarring chronic alopecia | 0 | 0 | |
| | | | | | Extensive scarring | 1 | 0.0144928 | |
| | | | | | Skin ulceration (excluding thrombosis) | 4 | 0.057971 | |
| | | | | | Premature gonadal failure | 0 | 0 | |
| | | | | | Diabetes requiring therapy | 0 | 0 | |
| | | | | | Malignancy | 0 | 0 | |
| Hersh, et al,2009, USA[103] | F: 78 | Mean 30.5±12.8 yrs | 14.0 +/-2.7 range (2–17) | Mean 16.5 yrs +/- 8.5 (range 3-47) | | 90 | | White 48.9%; Asian/Pacific Islander 16.7%; Hispanic/ Latino 15.6 %; African American 7.8%; Other 11% |
| | | | | | Pulmonary | 23.60% | 0.236 | |
| | | | | | Clotting disorders | 11.30% | 0.113 | |
| | | | | | Seizures | 23.60% | 0.236 | |
| | | | | | Myocardial infarction | 7.90% | 0.079 | |
| | | | | | Transplant | 12.4 | 0.1377778 | |
| | | | | | Dialysis | 19.1 | 0.1377778 | |
| Sousa, et al, 2016, Portugal[104] | F: 77 | Mean 25.1 ± 9.1 yrs | 13.4 ± 5.5 | | | 89 | | Caucasian 78, African 3, Others 4 |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | Hypertension | 17 | 0.1910112 | | |
| | | | | | | Diabetes | 3 | 0.0337079 | | |
| | | | | | | Hypercholesterolemia | 2 | 0.0224719 | | |
| | | | | | | Thyroid disease | 3 | 0.0337079 | | |
| | | | | | | Antiphospholipid syndrome | 10 | 0.1123596 | | |
| | | | | | | Sjögren's syndrome | 2 | 0.0224719 | | |
| Koutsonikoli, et al, 2015, Greece[105] | F: 39 | Mean 22±7.8 yrs at the end of follow up | Mean 12.5 ± 3.1 (SD) | Median 7.4 yrs and IQR (3.9–14.2) yrs | 440 person-yrs (people*avg follow up time) | | 47 | | | All Caucasians |
| | | | | | | Cataract | 4 | 0.0851064 | 0.193423598 | |
| | | | | | | Retinopathy | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Optic atrophy | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Cataract and retinopathy | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Seizures | 2 | 0.0425532 | 0.096711799 | |
| | | | | | | Cognitive impairment and seizures | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Major psychosis and seizures | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Cerebral vascular accident | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Transverse myelitis | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Single venous thrombosis | 5 | 0.106383 | 0.241779497 | |
| | | | | | | Significant tissue loss and venous thrombosis | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Avascular necrosis | 3 | 0.0638298 | 0.145067698 | |
| | | | | | | Persistent impairment of renal function or end-stage renal disease | 0 | 0 | 0 | |
| | | | | | | Pulmonary fibrosis | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Pulmonary infarction | 1 | 0.0212766 | 0.048355899 | |
| | | | | | | Premature gonadal failure | 3 | 0.0638298 | 0.145067698 | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Delayed growth | 6 | 0.1276596 | 0.290135397 |
| Gamal, et al, 2022, Egypt[106] | F: 163 | 25 yrs (median); range, 20 to 30.3 yrs | 14 (median) (range 12-15) | 12 yrs (median); (range, 6.75 to 17.25) yrs | | 186 | | |
| | | | | | Hypertension | 82 | 0.4408602 | |
| | | | | | Diabetes mellitus | 12 | 0.0645161 | |
| | | | | | Dyslipidaemia | 75 | 0.4032258 | |
| | | | | | Thyroid dysfunction | 10 | 0.0537634 | |
| | | | | | Malignancy | 1 | 0.0053763 | |
| | | | | | Osteoporosis | 27/150 | 0.18 | |
| | | | | | Avascular necrosis | 23 | 0.1236559 | |
| | | | | | End stage renal disease | 17 | 0.0913978 | |
| | | | | | Thrombosis | 22 | 0.1182796 | |
| | | | | | Raynaud's phenomenon | 51 | 0.2741935 | |
| | | | | | Pulmonary hypertension | 24 | 0.1290323 | |
| Groot, et al, 2019, Netherlands[4] | F: 101 | Median 33 yrs (range 18-65) | 14 (range 4-17) | Median 20 yrs range (1-55 yrs) | | 111 | | White 80, Non-White 31 |
| | | | | | Muscle atrophy or weakness | 1 | 0.009009 | |
| | | | | | Deforming or erosive arthritis | 10 | 0.0900901 | |
| | | | | | Osteoporosis with fracture | 6 | 0.0540541 | |
| | | | | | Avascular necrosis | 14 | 0.1261261 | |
| | | | | | Osteomyelitis | 1 | 0.009009 | |
| | | | | | Ruptured tendon | 1 | 0.009009 | |
| | | | | | Cognitive impairment or psychosis | 12 | 0.1081081 | |
| | | | | | Seizures | 10 | 0.0900901 | |
| | | | | | CVA | 7 | 0.0630631 | |
| | | | | | Cranial or Peripheral Neuropathy | 4 | 0.036036 | |
| | | | | | Transverse Myelitis | 0 | 0 | |
| | | | | | ESRD | 8 | 0.0720721 | |
| | | | | | Claudication | 0 | 0 | |

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| | | | |
|---|---|---|---|
| | | Minor tissue loss | 0 | 0 |
| | | Significant tissue loss | 4 | 0.036036 |
| | | | | |
| | | Venous thrombosis | 8 | 0.0720721 |
| | | Scarring alopecia | 8 | 0.0720721 |
| | | Scarring/ panniculum | 0 | 0 |
| | | Skin ulceration | 3 | 0.027027 |
| | | Cataracts | 8 | 0.0720721 |
| | | Retinal change | 3 | 0.027027 |
| | | Angina/bypass | 4 | 0.036036 |
| | | MI | 5 | 0.045045 |
| | | Cardiomyopathy | 1 | 0.009009 |
| | | Valvular disease | 2 | 0.018018 |
| | | Pericarditis | 0 | 0 |
| | | PAH | 1 | 0.009009 |
| | | Pulmonary fibrosis | 4 | 0.036036 |
| | | Shrinking lung | 1 | 0.009009 |
| | | Pleural Fibrosis | 0 | 0 |
| | | Pulmonary Infarction | 0 | 0 |
| | | Infarction or resection of bowel | 4 | 0.036036 |
| | | Mesenteric insufficiency | 0 | 0 |
| | | Chronic peritonitis | 0 | 0 |
| | | Stricture or upper GI surgery | 0 | 0 |
| | | Pancreatitis | 0 | 0 |
| | | Premature gonadal failure | 2 | 0.018018 |
| | | Malignancy | 2 | 0.018018 |
| | | IDDM | 1 | 0.009009 |
| Sinicato, et al, 2017, Brazil[94] | F:32 | | 35 | |
| | | | | |
| | | Low HDL-cholesterol | 5 | 0.143 |
| | | Hypertriglyceridemia | 6 | 0.171 |
| | | Insulin Resistance | 0 | 0.00 |
| | | Arterial Hypertension | 4 | 0.14 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Albumin excretion | 4 | 0.114 | | |
| **Childhood / JDM** | | | | | | | | | |
| Huber, et al, 2000, Canada[1] | F: 46 | Median 13 (range 7-26) | Median 5.8 yrs Range (1-16) | Median 7.2 yrs range (3.2 - 13.9 yrs) | | 65 | | | Caucasian 56; Asian 7, African Canadian 1, Indian 1 |
| | | | | | Calcinosis | 22 | 0.3384615 | 47.00854701 | |
| | | | | | Insulin-dependent diabetes mellitus | 3 | 0.0461538 | 6.41025641 | |
| | | | | | Connective tissue diseases | 2 (split below) | | | |
| | | | | | Scleroderma | 1 | 0.0153846 | 2.136752137 | |
| | | | | | Overlap connective tissue disease | 1 | 0.0153846 | 2.136752137 | |
| | | | | | Malignancies | 0 | 0 | | |
| Cho, et al, 2019, Korea[107] | F: 68 | Mean 12.7 ± 7.1 yrs | | 10 yrs | | 130 | | | |
| | | | | | Congestive heart failure | 1 | 0.0076923 | 0.769230769 | |
| | | | | | Cardiac arrhythmias | 1 | 0.0076923 | 0.769230769 | |
| | | | | | Valvular disease | 1 | 0.0076923 | 0.769230769 | |
| | | | | | Pulmonary circulation disorders | 2 | 0.0153846 | 1.538461538 | |
| | | | | | Peripheral vascular disorders | 3 | 0.0230769 | 2.307692308 | |
| | | | | | Hypertension, uncomplicated | 5 | 0.0384615 | 3.846153846 | |
| | | | | | Hypertension, complicated | 2 | 0.0153846 | 1.538461538 | |
| | | | | | Paralysis | 3 | 0.0230769 | 2.307692308 | |
| | | | | | Other neurological disorders | 2 | 0.0153846 | 1.538461538 | |
| | | | | | Chronic pulmonary disease | 36 | 0.2769231 | 27.69230769 | |
| | | | | | Diabetes, uncomplicated | 8 | 0.0615385 | 6.153846154 | |
| | | | | | Diabetes, complicated | 2 | 0.0153846 | 1.538461538 | |
| | | | | | Renal failure | 1 | 0.0076923 | 0.769230769 | |
| | | | | | Liver disease | 15 | 0.1153846 | 11.53846154 | |
| | | | | | Peptic ulcer disease excluding bleeding | 15 | 0.1153846 | 11.53846154 | |
| | | | | | AIDS/HIV | 1 | 0.0076923 | 0.769230769 | |

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| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Solid tumour without metastasis | 1 | 0.0076923 | 0.769230769 | |
| | | | | | | | Coagulopathy | 4 | 0.0307692 | 3.076923077 | |
| | | | | | | | Weight loss | 1 | 0.0076923 | 0.769230769 | |
| | | | | | | | Fluid and electrolyte disorders | 8 | 0.0615385 | 6.153846154 | |
| | | | | | | | Deficiency anaemia | 17 | 0.1307692 | 13.07692308 | |
| | | | | | | | Depression | 6 | 0.0461538 | 4.615384615 | |
| | | | | | | | Pulmonary disease specified | | | | |
| | | | | | | | Interstitial lung disease | 2 | 0.0153846 | 1.538461538 | |
| Concannon and Han, 2021, New Zealand[108] | F: 18 | ≤15 yrs | | Mean 8.6 yrs SD 3.7 yrs | | Mean 6.5, range (0.4–16.7) yrs | | 31 | | | Asian 5. Maori 7, pacific island 3, European 16 |
| | | | | | | | Calcinosis | 9 | 0.2903226 | 44.66501241 | |
| | | | | | | | Cutaneous vasculopathy | 5 | 0.1612903 | 24.81389578 | |
| | | | | | | | Interstitial lung disease | 6 | 0.1935484 | 29.77667494 | |
| | | | | | | | Pneumomediastinum | 2 | 0.0645161 | 9.925558313 | |
| | | | | | | | Pneumothorax | 3 | 0.0967742 | 14.88833747 | |
| | | | | | | | Surgical emphysema | 1 | 0.0322581 | 4.962779156 | |
| Okong'o, et al, 2016, South Africa[109] | F: 16 | | Median 6.75, range (2.0–9.7) yrs | Median 7.9 , range (3.4 – 9.75) year | median 4, range (0.5-84) months | Minimum of 12 months, median 55.5 months (inactive disease) 49.1 months (active disease) | | 25 | | | Indigenous African 13, mixed race 11, European ancestry 1 |
| | | | | | | | Skin | | | | |
| | | | | | | | Calcinosis | 8 | 0.32 | | |
| | | | | | | | Skin ulcers | 1 | 0.04 | | |
| | | | | | | | Oedema (Generalized, Periorbital or limb) | 0 | 0 | | |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | Alopecia | 3 | 0.12 |
| | | | | Lipodystrophy | 1 | 0.04 |
| | | | | Musculoskeletal: | | 0 |
| | | | | Arthritis | 2 | 0.08 |
| | | | | Contractures | 5 | 0.2 |
| | | | | Osteoporosis with fracture | 1 | 0.04 |
| | | | | Endocrine: | | 0 |
| | | | | Growth failure | 8 | 0.32 |
| | | | | Diabetes mellitus | 1 | 0.04 |
| | | | | Adrenal insufficiency | 0 | 0 |
| | | | | Respiratory: | | 0 |
| | | | | Interstitial lung disease (ILD) | 2 | 0.08 |
| | | | | Pulmonary haemorrhage | 1 | 0.04 |
| | | | | Gastrointestinal: | | 0 |
| | | | | Abdominal pain or bleeding | 0 | 0 |
| | | | | Ocular: | | 0 |
| | | | | Cataracts | 2 | 0.08 |
| | | | | Nervous system: | | 0 |
| | | | | Sensorineural hearing loss | 1 | 0.04 |
| | | | | Seizures, neuropathy | | 0 |
| | | | | Cardiovascular | | 0 |
| | | | | Raynaud's phenomenon | 6 | 0.24 |
| | | | | Cardiomyopathy | 0 | 0 |
| | | | | Infections: | | 0 |
| | | | | Fungal (skin) | 7 | 0.28 |
| | | | | Bacterial (staph aureus 2, gram negative 1) | 3 | 0.12 |
| | | | | TB | 2 | 0.08 |
| Ravelli, A. et al, 2010, Italy, UK[2] | F: 321 | Mean 6.9, range (0.9–17.8) yrs | Mean 7.7, range (2–25.2) yrs | | 490 | White 320 |
| | | | | Cutaneous scarring/atrophy | 195/443 | 0.4401806 |
| | | | | Calcinosis | 106/406 | 0.2610837 |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | Joint contractures | 82/448 | 0.1830357 | | |
| | | | | | Lipodystrophy | 43/444 | 0.0968468 | | |
| | | | | | Hirsutism | 44/431 | 0.1020882 | | |
| | | | | | Growth failure | 34/424 | 0.0801887 | | |
| | | | | | Osteoporosis with fractures | 26/435 | 0.0597701 | | |
| | | | | | Cataract | 9/430 | 0.0209302 | | |
| | | | | | Gastrointestinal infarction/resection | 7/444 | 0.0157658 | | |
| | | | | | Malignancy | 0/447 | 0 | | |
| Tabarki, et al, 1998, France[110] | F: 22 | Mean 6.9 yrs (range 2-14 yrs) | | Mean 4.9 yrs (range: 2.5-13 yrs) | | 36 | | | |
| | | | | | Calcinosis | 15 | 0.4166667 | 85.03401361 | |
| | | | | | Impaired function | 8 | 0.2222222 | 45.35147392 | |
| | | | | | Cataracts | 1 | 0.0277778 | 5.66893424 | |
| | | | | | Vertebral collapse | 2 | 0.0555556 | 11.33786848 | |
| | | | | | Contractures | 1 | 0.0277778 | 5.66893424 | |
| Chevalier, et al, 2021, France[111] | F: 4 | Median 6 yrs range 2-11 yrs | Median 5.5 months | Mean 14.9±8.8 yrs | | 7 | | | Caucasian 100% |
| | | | | | Polycystic ovary syndrome | 1 | 0.1428571 | 9.587727709 | |
| | | | | | Depressive disorder | 1 | 0.1428571 | 9.587727709 | |
| | | | | | mild physical dysfunction (CHAQ/HAQ >0) | 2 | 0.2857143 | 19.17545542 | |
| | | | | | Calcinosis | 2 | 0.2857143 | 19.17545542 | |
| | | | | | Lipodystrophy | 2 | 0.2857143 | 19.17545542 | |
| | | | | | Osteopenia/osteoporosis | 0 | 0 | 0 | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Cardiovascular damage | 0 | 0 | 0 |
| | | | | | Peripheral vascular damage | 0 | 0 | 0 |
| | | | | | Ocular damage | 0 | 0 | 0 |
| | | | | | Malignancy | 0 | 0 | 0 |
| | | | | | Fracture | 0 | 0 | 0 |
| | | | | | Moderate or severe disability CHAQ/HAQ>1 | 0 | 0 | 0 |
| Al-Mayouf, et al, 2017, Saudi Arabia[112] | F: 58 | Mean 6 ± 3 yrs | | Mean 5 ± 4.4 yrs | | 92 | | |
| | | | | | Permanent skin changes | 16 | 0.173913 | 34.7826087 |
| | | | | | Residual muscle weakness | 12 | 0.1304348 | 26.08695652 |
| | | | | | Calcinosis | 27 | 0.2934783 | 58.69565217 |
| | | | | | Growth failure | 37 | 0.4021739 | 80.43478261 |
| | | | | | Osteoporosis | 15 | 0.1630435 | 32.60869565 |
| | | | | | Lipodystrophy | 3 | 0.0326087 | 6.52173913 |
| | | | | | Mucinous cystic ovarian tumour | 1 | 0.0108696 | 2.173913043 |
| Sag, et al, 2021, Turkey[113] | F: 35 | Mean 8.1±4.3 yrs | Mean 8.7±4.4 | Median 4.85 yrs IQR 3.7, Mean 5.66±3.59 yrs | | 58 | | |
| | | | | | Calcinosis | 19 | 0.3275862 | 67.54354781 |
| | | | | | Pulmonary involvement | 8 | 0.137931 | 28.43938855 |
| | | | | | Skin ulcers | 4 | 0.0689655 | 14.21969428 |
| | | | | | Gastrointestinal involvement | 4 | 0.0689655 | 14.21969428 |
| | | | | | Severe muscle disease (wheel-chair dependency) | 3 | 0.0517241 | 10.66477071 |
| | | | | | Cardiac involvement | 2 | 0.0344828 | 7.109847138 |
| Ramanan, et al, 2005, Canada[114] | | | | | | | | |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | F: 20 | | Mean 8.4 yrs | Mean 34.6 months | JDM patients after June 1997 | 31 | | | Black 2; East Indian 4; Native Canadian 1; Asian 1; White 23 |
| | | | | | Calcinosis | 2 | 0.0645161 | 22.37553608 | |
| | | | | | Cataracts | 3 | 0.0967742 | 33.56330412 | |
| | F:16 | | 6.7 yrs (mean) | 4 yrs (1056 patient months) | JDM patients pre-June 1997 | 22 | | | Black 1; East Indian 1; Native Canadian 1; Asian 2; White 17 |
| | | | | | Calcinosis | 5 | 0.2272727 | 56.81818182 | |
| | | | | | Cataracts | 8 | 0.3636364 | 90.90909091 | |
| Sun, et al, 2015, Taiwan[115] | F: 26 | Mean 81.97 ± 46.63 months (25-216) | Mean 87.97± 45.90 months (28-213) | For those receiving IVIG: Mean 79 months | | 39 | | | |
| | | | | | Calcinosis | 15 | 0.3846154 | | |
| | | | | | Pulmonary involvement | 4 | 0.1025641 | Follow up time and cases were not reported coherently. Hence, no incidence rate was calculated | |
| | | | | | Malignancy | 0 | 0 | | |
| | | | | | Disability | 4 | 0.1025641 | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | GI perforation | 1 | 0.025641 | |
| | | | | | Avascular necrosis of the hip joint | 1 | 0.025641 | |
| Silverberg, et al, 2018, USA[116] | F: 915 | 0-17 | | | | 1407 | | White 543; Black 203; Hispanic 260; Asian 19; Multiracial/other 71 |
| | | | | | Obesity | 38 | 0.0270078 | |
| | | | | | Hypertension | 119 | 0.0845771 | |
| | | | | | Diabetes, uncomplicated | 18 | 0.0127932 | |
| | | | | | Late effects of cerebrovascular disease | ≤10 | ≤10/1407 | |
| | | | | | Pulmonary circulatory disorder | 15 | 0.010661 | |
| | | | | | Valvular heart disease | 17 | 0.0120824 | |
| | | | | | Peripheral and visceral atherosclerosis | ≤10 | ≤10/1407 | |
| | | | | | Any lipid abnormalities | ≤10 | ≤10/1407 | |
| | | | | | Lipodystrophy | ≤10 | ≤10/1407 | |
| | | | | | Personal history of TIA and cerebral infarction without residual deficits | ≤10 | ≤10/1407 | |
| | | | | | Arrythmia | 47 | 0.0334044 | |
| | | | | | Valvular heart disease | 17 | 0.0120824 | |
| Adulthood / JDM | | | | | | | | |
| Boros, et al, 2022, UK[117] | F:68 | Mean 20.6 yrs ±3.9 (SD) | Mean 9.2 yrs ± 4.3 (SD) | Mean 12.4 yrs ± 5.0 | | 84 | | White 69 |
| | | | | | Feel shorter than peers | 20 /80 | 0.25 | |
| | | | | | Feel weight was affected by myositis | 25/80 | 0.3125 | |
| | | | | | Puberty affected by myositis | 16/80 | 0.2 | |
| | | | | | Calcinosis | 18/82 | 0.225 | |
| | | | | | Lipodystrophy | 6/82 | 0.0731707 | |
| | | | | | Pulmonary hypertension | 2/83 | 0.0240964 | |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Mathiesen, et al, 2014, Denmark[118] | F: 37 | Mean 21.7 yrs (SD 11.1) (Range 4.2-45.9) (Non-restrictive lung); 14.2 (4.4) (10.0-19.7) (restrictive lung) | Mean 7.7 yrs (SD 4.7) (Range 1.4-17.8) (non-restrictive lung); 7.7 (4.2) (1.5-10.5) (restrictive lung) | | Mean 4.2 yrs (SD 2.9) (Range 0.610.6) (non-restrictive lung); 5.3 (2.5) (2.6- 8.5) (restrictive lung) | Mean 14.3 yrs (SD 10.7) (Range 1.1-35.1) (non-restrictive lung); 6.5 (3.0) (2.6-9.2) (restrictive lung) | | 51 | | | |
| | | | | | | | Restrictive interstitial lung disease | 4 | 0.0784314 | 12.06636501 | |
| | | | | | | | Obstructive lung function pattern | 7 | 0.1372549 | 9.598244892 | |
| Tsaltskan, et al, 2020, USA[5] | F: 39 | Median 24.0 yrs [ IQR 19.0-28.6] yrs | | Median 11.0 [IQR 8.5-15.7] | median 11.5 yrs, [ IQR 4.5-18.9] yrs | | | 49 | | | Caucasian 59.2%, African American 11.1% , Hispanic 17.7%, Other 12.0% |
| | | | | | | | Cutaneous Damage: | | | | |
| | | | | | | | Cutaneous scarring or atrophy | 28 | 0.5714286 | | |
| | | | | | | | Calcinosis | 27 | 0.5510204 | | |
| | | | | | | | Lipodystrophy | 16 | 0.3265306 | | |
| | | | | | | | Alopecia | 8 | 0.1632653 | | |
| | | | | | | | Poikiloderma | 6 | 0.122449 | | |
| | | | | | | | Skeletal Damage: | | | | |
| | | | | | | | Joint Contracture | 31 | 0.6326531 | | |
| | | | | | | | Deforming Arthropathy | 8 | 0.1632653 | | |
| | | | | | | | Osteoporosis with fracture | 4 | 0.0816327 | | |
| | | | | | | | Avascular Necrosis | 2 | 0.0408163 | | |
| | | | | | | | Endocrine Damage: | | | | |
| | | | | | | | Irregular Menses | 10/40 | 0.25 | | |
| | | | | | | | Hirsutism | 7/40 | 0.175 | | |
| | | | | | | | Diabetes | 6 | 0.122449 | | |
| | | | | | | | Amenorrhea | 4/40 | 0.1 | | |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Infertility | 1 | 0.0204082 | |
| | | | | | Pulmonary Damage: | | | |
| | | | | | Impaired lung function | 7 | 0.1428571 | |
| | | | | | Dysphonia | 4 | 0.0816327 | |
| | | | | | Pulmonary Fibrosis | 1 | 0.0204082 | |
| | | | | | Ocular Damage: | | | |
| | | | | | Cataract | 4 | 0.0816327 | |
| | | | | | Vision Loss | 3 | 0.0612245 | |
| | | | | | Gastrointestinal Damage: | | | |
| | | | | | Dysmotility | 5 | 0.1020408 | |
| | | | | | Dysphagia | 3 | 0.0612245 | |
| | | | | | Infection: | | | |
| | | | | | Chronic Infection | 5 | 0.1020408 | |
| | | | | | Multiple Infections | 4 | 0.0816327 | |
| | | | | | Cardiovascular Damage: | | | |
| | | | | | Hypertension | 4 | 0.0816327 | |
| | | | | | Peripheral Vascular Damage: | | | |
| | | | | | Tissue or pulp loss | 4 | 0.0816327 | |
| | | | | | Digit or limb loss | 1 | 0.0204082 | |
| | | | | | Malignancy | 1 | 0.0204082 | |
| Mathiesen, et al, 2012, Denmark[11] [9] | F: 38 | Mean 21.5 yrs SD 11.1 Range 4.2-45.9) | Mean 4.3 yrs SD 2.9 range (0.6-10.6) | Mean 13.9 SD 10.5 (range 2.1–35.1) yrs | | 53 | | |
| | | | | | Deforming arthropathy | 1 | 0.0188679 | 1.357404642 |
| | | | | | Pulmonary fibrosis | 3 | 0.0566038 | 4.072213927 |
| | | | | | Growth failure | 6/49 | 0.122449 | 8.809279107 |
| | | | | | Diabetes | 1 | 0.0188679 | 1.357404642 |
| | | | | | Scarring or atrophy | 21 | 0.3962264 | 28.50549749 |
| | | | | | Calcinosis | 11 | 0.2075472 | 14.93145107 |
| | | | | | Lipodystrophy | 7 | 0.1320755 | 9.501832496 |
| | | | | | Tissue or pulp loss | 1 | 0.0188679 | 1.357404642 |
| | | | | | Alopecia, poikiloderma | 0 | 0 | 0 |

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| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Joint contractures | 11 | 0.2075472 | 14.93145107 | |
| | | | | | | | Osteoporosis/osteopenia estimated by DXA | 5 | 0.0943396 | 6.787023212 | |
| | | | | | | | Osteoporotic fractures | 2 | 0.0377358 | 2.714809285 | |
| | | | | | | | Delay in secondary sexual characteristics* | 3/14 | 0.2142857 | 15.41623844 | |
| | | | | | | | Hypertriglyceridemia | 9 | 0.1698113 | 12.21664178 | |
| | | | | | | | Hirsutism, irregular menses, amenorrhoea | 0 | 0 | 0 | |
| | | | | | | | Paraesthesia | 2 | 0.0377358 | 2.714809285 | |
| | | | | | | | Arthritis | 3 | 0.0566038 | 4.072213927 | |
| | | | | | | | Impaired pulmonary function test | 4 | 0.0754717 | 5.429618569 | |
| | | | | | | | Pulmonary hypertension | 0 | 0 | 0 | |
| | | | | | | | Severe abdominal pain | 3 | 0.0566038 | 4.072213927 | |
| | | | | | | | Dysphagia, gastrointestinal dysfunction | 1 | 0.0188679 | 1.357404642 | |
| | | | | | | | Gastrointestinal surgery | 1 | 0.0188679 | 1.357404642 | |
| | | | | | | | Hypertension, cardiomyopathy, angina, myocardial infarction | 0 | 0 | 0 | |
| | | | | | | | Tissue or pulp loss, claudication | 0 | 0 | 0 | |
| | | | | | | | Cataract, visual loss | 0 | 0 | 0 | |
| | | | | | | | Chronic infection, multiple infections | 0 | 0 | 0 | |
| | | | | | | | Malignancy | 0 | 0 | 0 | |
| Sanner, et al, 2010, Norway[120] | F: 36 | Median 21.5 ( range 6.7–55.4) yrs | Mean 8.3 ±4.5 yrs | Mean 9.1 ± 4.5 yrs | minimum 24 months | Mean 16.9 ± 10.6 | | 59 | | | European 57 |
| | | | | | | | Calcinosis | 14 | 0.2372881 | 14.04071808 | |

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**Supplementary Table S4.2: Studies comparing comorbidity incidence between cases and controls of JIA/jSLE patients reported during childhood and adulthood. NR=p value not reported, NS=P value described as NS.**

| Study | Measure of effect | Control group | Comorbidity | Effect size | 95% CI | p-value |
|---|---|---|---|---|---|---|
| **Childhood/ JIA** | | | | | | |
| Schenck, et al, 2018, Germany[33] | Prevalence ratio | Diabetes registry data | Type 1 diabetes | 1.76 | (1.34-2.28) | <0.001 |
| Krause, et al, 2017, USA[37] | Rate Ratio | Non-JIA comparator subjects. Age-sex matched | Joint surgeries | 3.93 | (1.18-24.94) | NR |
| | | | Non-joint surgeries | 1.9 | (1.05-3.67) | NR |
| | | | Hospitalizations | 2.25 | (1.04-5.53) | NR |
| | | | Hospitalizations for depression | 1.41 | (0.47-4.70) | NR |
| Lee, et al, 2020, USA[38] | Adjusted Hazard ratios | Healthy children with at least 1 Wellcare visit | Type 1 diabetes | 1.81 | (1.03–3.17) | NR |
| | | | Type 2 diabetes | 1.22 | (0.79–1.90) | NR |
| | | | Composite diabetes | 1.4 | (0.94–2.09) | NR |
| | Age, sex, index date-matched Hazard ratios | Healthy children with at least 1 Wellcare visit | Type 1 diabetes | 1.89 | (1.11-3.19) | NR |
| | | | Type 2 diabetes | 1.33 | (0.88–2.02) | NR |
| | | | Composite diabetes | 1.52 | (1.05–2.21) | NR |
| | Age, sex, index date-matched Hazard ratios | Children with asthma | Type 1 diabetes | 1.46 | | NR |
| | | | Type 2 diabetes | 0.95 | (0.64–1.43) | NR |
| | | | Composite diabetes | 1.11 | (0.77–1.60) | NR |
| | Adjusted Hazard ratios | Children with asthma | Type 1 diabetes | 1.48 | (0.86–2.56) | NR |
| | | | Type 2 diabetes | 1.01 | (0.66–1.54) | NR |
| | | | Composite diabetes | 1.18 | (0.80–1.73) | NR |
| Beukelman, et al, 2012, USA[28] | Incidence rate ratio | Children with ADHD | Coccidioides | 101 | (8.1–5319) | NR |
| | | | Salmonella | 3.8 | (1.20–9.50) | NR |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | Herpes zoster | 2.1 | (1.40–3.00) | NR |
| | | | All infections | 2.4 | (1.70-3.30) | NR |
| | | | All infections except Herpes zoster | 4.2 | (2.10-7.60) | NR |
| Berthold, et al, 2022, Sweden[53] | Hazard ratios | Non-JIA comparator subjects | **Females** | | | |
| | | Sex, year of birth and residential region matched | Depression | 1.1 | (0.90–1.50) | NS |
| | | | Anxiety | 1.2 | (0.90–1.50) | NS |
| | | | **Males** | | | |
| | | | Depression | 0.8 | (0.50–1.40) | NS |
| | | | Anxiety | 0.6 | (0.40–1.10) | NS |
| Lovell, et al, 2021, USA[47] | Log risk ratio | General paediatric patients without JIA | Addison's disease | 1.65 | (1.02-2.21) | P<0.05 |
| | | | Autoimmune/Hashimoto's thyroiditis | 2.25 | (1.72 to 2.71) | P<0.05 |
| | | | Graves' disease | 0.38 | (–2.77-2.26) | NS |
| | | | Morphoea | 3.15 | (2.54-3.70) | P<0.05 |
| | | | Multiple sclerosis | 1.9 | (–1.27-3.83) | NS |
| | | | Takayasu's arteritis | 1.45 | (–1.71-3.35) | NS |
| | | | Type 1 diabetes | 1.25 | (0.75-1.70) | P<0.05 |
| | | | Vitiligo | 1.76 | (0.74-2.59) | P<0.05 |
| | | | Autoimmune disease not elsewhere classified | 2.55 | (1.78-3.21) | P<0.05 |
| | | | Chronic glomerulonephritis | 0.73 | (–2.45-2.64) | NS |
| | | | Chronic urticaria | 0.58 | (0.28-0.84) | P<0.05 |
| | | | Episcleritis/scleritis | 3.18 | (2.14-4.03) | P<0.05 |
| | | | Erythema nodosum | 1.12 | (–2.03-3.02) | NS |
| | | | Haemolytic anaemia | –0.36 | (–3.49-1.54) | NS |
| | | | Immune thrombocytopenia purpura | 0.4 | (–1.42-1.68) | NS |
| | | | Leukocytoclastic vasculitis | 4.08 | (2.19-5.49) | P<0.05 |
| | | | Pulmonary fibrosis/interstitial lung disease | 2.46 | (2.28-2.62) | P<0.05 |
| | | | Raynaud’s syndrome | 3.37 | (2.80-3.86) | P<0.05 |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | Sjögren’s syndrome/sicca syndrome | 3.27 | (2.40-5.01) | P<0.05 |
| | | | Thrombocytopenia purpura | 0.57 | (–2.61-2.46) | NS |
| | | | Vasculitis | 3.71 | (2.90-4.37) | P<0.05 |
| Simon, TA et al, 2020, USA[41] | Odd Ratio (OR) | Patients with ADHD<br>1:5 matched for age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency room visits), and calendar year of diagnosis (± 1 year) | **MarketScan** | | | |
| | | | Addison disease | 6.16 | (2.83-13.42) | NR |
| | | | Alopecia areata | 2.03 | (0.93-4.46) | NR |
| | | | Celiac disease | 3.99 | (2.66-5.98) | NR |
| | | | Chronic glomerulonephritis | Infinite | | NR |
| | | | Chronic urticaria | 3.4 | (2.74-4.22) | NR |
| | | | Crohn's disease | 16.84 | (10.91-25.99) | NR |
| | | | Erythema nodosum | 12.05 | (3.20-45.4) | NR |
| | | | Giant cell arteritis | 4.51 | (0.28-72.18 | NR |
| | | | Grave's disease | 3.23 | (1.43-7.27) | NR |
| | | | Hashimoto's thyroiditis | 3.57 | (2.48-5.14) | NR |
| | | | Haemolytic anaemia | 6.9 | (3.98-11.97) | NR |
| | | | Immune thrombocytopenia | 4.52 | (1.88-10.86 | NR |
| | | | Interstitial lung disease | 24.86 | (5.51-112.15) | NR |
| | | | Morphoea | 21.36 | (9.45-48.29) | NR |
| | | | Multiple sclerosis | 2.26 | (0.21-24.90) | NR |
| | | | Myasthenia gravis | 4.51 | (0.64-32.06) | NR |
| | | | Pernicious anaemia | 0 | | NR |
| | | | Polymyalgia rheumatica | Infinite | | NR |
| | | | Polymyositis | Infinite | | NR |
| | | | Primary biliary cirrhosis | 0 | | NR |
| | | | Psoriasis | 8.42 | (5.54-12.78) | NR |
| | | | Raynaud's syndrome | 5.7 | (3.44-9.46) | NR |
| | | | Scleritis and episcleritis | 17.19 | (6.42-46.04) | NR |
| | | | Sjogren's syndrome/sicca syndrome | Infinite | | NR |
| | | | Systemic sclerosis | 24.89 | (8.57-72.23) | NR |

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| | | | |
|---|---|---|---|
| Patients with ADHD | Type 1 diabetes | 1.7 | (1.27-2.27) | NR |
| | Ulcerative colitis | 8.34 | 4.77-14.59) | NR |
| | Uveitis | 165.1 | (108.97-250.14) | NR |
| | Vasculitis | 22.64 | (9.42-54.42) | NR |
| | Vitiligo | 2.85 | (1.39-5.88) | NR |
| | **Pharmetrics Data** | | | |
| | Addison disease | 2.51 | (1.16-5.43) | NR |
| | Alopecia areata | 2.43 | (1.274.65) | NR |
| | Celiac disease | 3.82 | (2.61-5.60) | NR |
| | Chronic glomerulonephritis | 31.57 | (3.88-256.56) | NR |
| | Chronic urticaria | 2.04 | (1.65-2.51) | NR |
| | Crohn's disease | 12.99 | (8.78-19.2) | NR |
| | Erythema nodosum | 9.02 | (2.72-29.96) | NR |
| | Giant cell arteritis | Infinite | | NR |
| | Grave's disease | 1.69 | (0.79-3.64) | NR |
| | Hashimoto's thyroiditis | 4.26 | (3.03-6.01) | NR |
| | Haemolytic anaemia | 4.52 | (2.50-8.16) | NR |
| | Immune thrombocytopenia | 4.89 | (2.23-10.72) | NR |
| | Interstitial lung disease | 12.03 | (3.19-45.33) | NR |
| | Morphoea | 18.97 | (7.15-50.33) | NR |
| | Multiple sclerosis | 3.38 | (0.76-15.11) | NR |
| | Myasthenia gravis | Infinite | | NR |
| | Pernicious anaemia | 0 | | NR |
| | Polymyalgia rheumatica | 9.02 | (0.82-99.43) | NR |
| | Polymyositis | 22.55 | (2.63-193.01) | NR |
| | Primary biliary cirrhosis | Infinite | | NR |
| | Psoriasis | 5.89 | (4.07-8.53) | NR |
| | Raynaud's syndrome | 12.16 | (7.45-19.84 | NR |
| | Scleritis and episcleritis | 13.55 | (5.76-31.88) | NR |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | Sjogren's syndrome/sicca syndrome | 72.24 | (9.58-544.36) | NR |
| | | | Systemic sclerosis | 90.34 | (12.12-673.22) | NR |
| | | | Type 1 diabetes | 1.82 | (1.37-2.41) | NR |
| | | | Ulcerative colitis | 6.49 | (3.81-11.05) | NR |
| | | | Uveitis | 133.23 | (92.47-191.95) | NR |
| | | | Vasculitis | 10.32 | (4.24-25.08) | NR |
| | | | Vitiligo | 2.61 | (1.24-5.49) | NR |
| Horneff, et al, 2016, Germany[46] | Risk ratios | German Childhood Cancer Registry report 2015 | | | | |
| | | | **Methotrexate Drug Cohort** | | | |
| | | | All malignancy | 3 | (0.75-12.0) | 0.12 |
| | | | Lymphoma & Leukaemia | 6.7 | (1.67-26.82) | 0.007 |
| | | | **Etanercept Drug Cohort** | | | |
| | | | All malignancy | 6.3 | (3.00-13.20) | <0.0001 |
| | | | Lymphoma & Leukaemia | 8 | (2.99-21.38) | <0.0001 |
| **Adulthood / JIA** | | | | | | |
| Raab, et al, 2013, Germany[59] | % difference (absolute) | Community sample in Germany | Cardiovascular diseases (as split below) | 1.60 | | 0.343 |
| | | Age and gender matched by propensity scores | Hypertension | -0.60 | | 0.759 |
| | | | Cardiac insufficiency | -0.90 | | 0.111 |
| | | | Asthma | -2.50 | | 0.098 |
| | | | Gastrointestinal diseases | -6.40 | | <0.001 |
| | | | Gastrointestinal ulcer | -1.00 | | 0.065 |
| | | | Liver and gallbladder diseases | 0.10 | | 0.125 |
| | | | Renal insufficiency | -1.00 | | 0.064 |
| | | | Diabetes mellitus | -0.70 | | 0.332 |
| | | | Depression | -4.10 | | 0.004 |
| Aggarwal, et al, 2006, India[60] | % difference (absolute) | Healthy male hospital employees | Osteopenia (Lumbar Spine) | -5.50 | | NS |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | Osteopenia (Hip) | 28.30 | | < 0.05 |
| | | | Osteopenia (Distal third of forearm) | 10.90 | | < 0.001 |
| | | | Osteoporosis (lumbar spine) | 22.40 | | NS |
| | | | Osteoporosis (hip) | 5.90 | | < 0.05 |
| | | | Osteoporosis (Distal third of forearm) | 35.70 | | < 0.001 |
| Smith, et al, 2019, USA[67] | Unadjusted Risk ratio | Women without autoimmune disease who delivered live born infants | Pre-eclampsia | 2.48 | (1.30–4.74) | P<0.05 |
| | | | Pregnancy induced hypertension without pre-eclampsia | 1.75 | (0.83–3.69) | NR |
| | | | Placenta previa and/or abruption | 1.46 | (0.46–4.68) | NR |
| | | | Preterm labour | 2.03 | (1.27–3.23) | P<0.05 |
| | | | Gestational diabetes | 1.66 | (0.87–3.16) | NR |
| | | | Fever during pregnancy | 1.19 | (0.89–1.58) | NR |
| | | | Caesarean section | 1.55 | (1.23–1.94) | P<0.05 |
| | Adjusted risk ratio | Adjusted risk ratio computed using Poisson regression | Pre-eclampsia | 1.87 | (0.71–4.96) | NR |
| | | | Pregnancy induced hypertension without preeclampsia | 0.60 | | (0.08–4.78) |
| | | | Gestational diabetes | 0.91 | (0.25–3.38) | NR |
| | | | Fever during pregnancy | 2.61 | (1.21–5.63) | P<0.05 |
| | Adjusted risk ratio | Multivariable adjustment for the prior factors | Pre-eclampsia | 2.19 | (0.82–5.88) | NR |
| | | | Gestational diabetes | 0.86 | (0.24–2.97) | NR |
| | | | Fever during pregnancy | 3.06 | (1.42–6.56) | P<0.05 |
| Simon, et al, 2020, USA[41] | Odds ratios | MarketScan patients with ADHD<br>1:5 matched age (± 2 years), sex, number of medical encounters (in-patient, out-patient, and emergency room visits), and calendar year of diagnosis (± 1 year). | Addison disease | 4.87 | (3.09-7.66) | NR |
| | | | Alopecia areata | 1.09 | (0.45-2.64) | NR |
| | | | Coeliac disease | 1.32 | (0.73-2.39) | NR |
| | | | Chronic glomerulonephritis | 2.36 | (0.59-9.43) | NR |
| | | | Chronic urticaria | 2.53 | 1.94-3.30) | NR |

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| | | | |
|---|---|---|---|
| Crohn's disease | 3.88 | (2.93-5.16) | NR |
| Erythema nodosum | 0.94 | (0.21-4.30) | NR |
| Giant cell arteritis | Infinite | | NR |
| Grave's disease | 2.15 | (1.26-3.65) | NR |
| Hashimoto's thyroiditis | 1.56 | (1.09-2.25) | NR |
| Haemolytic anaemia | 6.86 | (3.88-12.14) | NR |
| Immune thrombocytopenia | 3.97 | (2.35-6.69) | NR |
| Interstitial lung disease | 11.54 | (6.94-19.2) | NR |
| Morphoea | 3.63 | (1.59-8.28) | NR |
| Multiple sclerosis | 1.50 | (0.96-2.33) | NR |
| Myasthenia gravis | 2.36 | (0.59-9.43) | NR |
| Pernicious anaemia | 1.30 | (0.75-2.27) | NR |
| Polymyalgia rheumatica | 8.20 | (4.34-15.5) | NR |
| Polymyositis | 7.55 | (2.47-23.09) | NR |
| Primary biliary cirrhosis | 2.83 | (0.68-11.84) | NR |
| Psoriasis | 2.31 | (1.73-3.08) | NR |
| Raynaud's syndrome | 3.24 | (2.33-4.50) | NR |
| Scleritis and episcleritis | 5.25 | (2.78-9.94) | NR |
| Sjogren's syndrome/sicca syndrome | 6.82 | (4.76-9.77) | NR |
| Systemic sclerosis | 10.80 | (4.44-26.26) | NR |
| Type 1 diabetes | 2.50 | (1.98-3.14) | NR |
| Ulcerative colitis | 3.69 | (2.69-5.06) | NR |
| Uveitis | 18.46 | (13.8-24.69) | NR |
| Vasculitis | 12.13 | (7.03-20.94) | NR |
| Addison disease | 4.67 | (2.83-7.73) | NR |
| Alopecia areata | 1.66 | (0.81-3.41) | NR |
| Coeliac disease | 2.63 | (1.68-4.11) | NR |
| Chronic glomerulonephritis | 24.06 | (2.81-205.96) | NR |
| Chronic urticaria | 2.03 | (1.55-2.66) | NR |
| Crohn's disease | 3.23 | (2.36-4.42) | NR |
| Erythema nodosum | 7.70 | (2.52-23.55) | NR |

PharMetrics patients with ADHD

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| | | | | | | |
|---|---|---|---|---|---|---|
| | | | Giant cell arteritis | 3.21 | (0.54-19.19) | NR |
| | | | Grave's disease | 1.45 | (0.83-2.54) | NR |
| | | | Hashimoto's thyroiditis | 1.99 | (1.42-2.80) | NR |
| | | | Haemolytic anaemia | 5.46 | (2.73-10.94) | NR |
| | | | Immune thrombocytopenic | 4.82 | (2.51-9.27) | NR |
| | | | Interstitial lung disease | 12.89 | (6.63-25.04) | NR |
| | | | Morphoea | 4.81 | (1.69-13.73) | NR |
| | | | Multiple sclerosis | 1.52 | (0.89-2.58) | NR |
| | | | Myasthenia gravis | 3.85 | (1.03-14.34) | NR |
| | | | Pernicious anaemia | 1.24 | (0.70-2.20) | NR |
| | | | Polymyalgia rheumatica | 3.78 | (1.72-8.34) | NR |
| | | | Polymyositis | 4.01 | (1.22-13.14) | NR |
| | | | Primary biliary cirrhosis | 2.40 | (0.22-26.52) | NR |
| | | | Psoriasis | 2.93 | (2.22-3.86) | NR |
| | | | Raynaud's syndrome | 2.53 | (1.76-3.65) | NR |
| | | | Scleritis and episcleritis | 3.66 | (2.02-6.66) | NR |
| | | | Sjogren's syndrome/sicca syndrome | 6.86 | (4.70-10.02) | NR |
| | | | Systemic sclerosis | 8.83 | (3.26-23.88) | NR |
| | | | Type 1 diabetes | 2.15 | (1.66-2.78) | NR |
| | | | Ulcerative colitis | 2.41 | (1.63-3.57) | NR |
| | | | Uveitis | 29.54 | (21.51-40.58) | NR |
| | | | Vasculitis | 7.63 | (4.29-13.56) | NR |
| | | | Vitiligo | 2.16 | (1.12-4.15) | NR |
| Ma, et al, 2022, Taiwan[70] | Relative risk | Individuals without JIA | Obstructive sleep apnoea | 1.936 | (1.26-2.97) | NR |
| | Adjusted hazard ratios | Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline. | Obstructive sleep apnoea | 1.922 | (1.24–2.97) | NR |
| | Adjusted hazard ratios | Adjusted for demographic variables including age, sex, length of hospital stays, and comorbidities at baseline and allergic rhinitis | Obstructive sleep apnoea | 1.363 | (0.70-2.65) | NS |
| Zhang-Jian, et al, 2020, Taiwan[71] | | Non-JIA patients | Stillbirth | 0.92 | (0.32–2.69) | NR |

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| | | | | | | |
|---|---|---|---|---|---|---|
| | Adjusted odds ratio (for infant sex, Charlson comorbidity index, urbanization, income, occupation, maternal nationality) | 1:5 matched maternal age and birth year<br>To avoid possible confounding factors and for case control selection, only the first child from either group was included for further analysis. | Foetal distress | 0.96 | (0.65–1.41) | NR |
| | | | Foetal abnormalities, any | 0.91 | (0.64–1.29) | NR |
| | | | Pre-eclampsia | 1.42 | (0.82–2.46) | NR |
| | | | Gestational diabetes | 0.75 | (0.48–1.16) | NR |
| | | | Pregnancy-related hypertension | 1.24 | (0.79–1.96) | NR |
| | | | Antepartum haemorrhage | 1.09 | (0.81–1.47) | NR |
| | | | Severe postpartum haemorrhage | 1.20 | (0.77–1.87) | NR |
| | | | Chorioamnionitis | 1.34 | (0.49–3.66) | NR |
| | | | Caesarean delivery | 1.07 | (0.89–1.27) | NR |
| | | | Low birth weight <2500g | 1.35 | (1.02–1.79) | <0.05 |
| | | | Prematurity <37 weeks | 1.19 | (0.87–1.64) | NR |
| Krause, et al, 2017, USA[37] | Rate ratio | Non-JIA comparator<br>Age and sex matched | Joint surgeries | 8.5 | (2.27-120.10) | NR |
| | | | Non-joint surgeries | 1.92 | (0.89-4.57) | NR |
| | | | Hospitalizations | 1.79 | (0.69-5.25) | NR |
| | | | Hospitalizations for depression | 0.26 | (0.02-1.11) | NR |
| | | | Pregnancies | 0.68 | (0.38-1.20) | NS |
| **Childhood / jSLE** | | | | | | |
| Frittoli, et al, 2016, Brazil[79] | % difference (absolute) | Patients with cognitive dysfunction. Gender, age and socioeconomic status matched | Cognitive dysfunction | 4.86 | | 0.080 |
| Gad, et al, 2013, Egypt[81] | % difference (absolute) | Healthy Children. Age and sex matched | Sensorineural hearing loss | 25.00 | | NR |
| | | | Conductive hearing loss | 0 | | NR |
| Karve, et al, 2012, USA[12] | % difference (absolute) | Comparator without IBD or SLE. Age (calculated as of year 2005) and sex, and with a minimum health plan enrolment of 6 months before and 12 months after the assigned index date of the corresponding patient were matched. | Asthma | 8.50 | NA | NR |
| | | | Atopic dermatitis | 0.70 | NA | NR |
| | | | Allergic rhinitis | 2.90 | NA | NR |
| | | | Thyroiditis | 0.70 | NA | NR |
| | | | Fracture in post-index period | 2.10 | NA | NR |
| | | | Psoriasis | 0.00 | NA | NR |
| | | | Type 1 diabetes | -0.70 | NA | NR |
| | | | General diabetes | 0.00 | NA | NR |

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| | | | | | |
|---|---|---|---|---|---|
| | | | Multiple sclerosis | 1.40 | NA | NR |
| | | | Other CNS demyelinating disease | 0.00 | NA | NR |
| | | | Vitiligo | 0.00 | NA | NR |
| | | | Rheumatoid arthritis | 19.7 | NA | NR |
| Sinicato, et al, 2017, Brazil[94] | % difference (absolute) | Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched | High LDL-c levels | 33.00 | | 0.300 |
| | | | Hypercholesterolemia | 16.17 | | 0.210 |
| | | | Insulin resistance | 4.90 | | 0.177 |
| | | | Arterial hypertension | 9.80 | | 0.074 |
| | | | Albumin excretion | 12.20 | | 0.080 |
| **Adulthood/ jSLE** | | | | | | |
| Sinicato, et al, 2017, Brazil[94] | % difference (absolute) | Volunteers with no history of chronic, cardiovascular, and autoimmune disease. Age, gender, and demographic background matched | Insulin resistance | 0 | | 0.141 |
| | | | Hypertension | 11.4 | | 0.086 |
| | | | Albumin excretion | 11.4 | | 0.086 |

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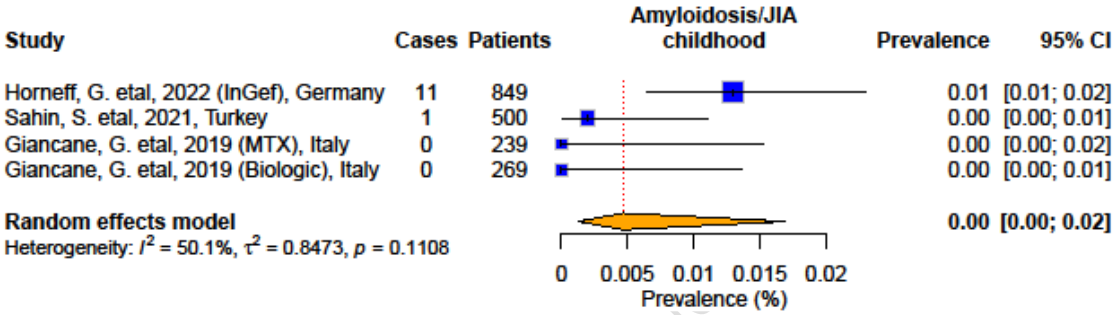
## Supplementary Figures S4.1: Meta-analysis: Pooled estimates, Forest plots and Funnel plots

### Childhood/JIA

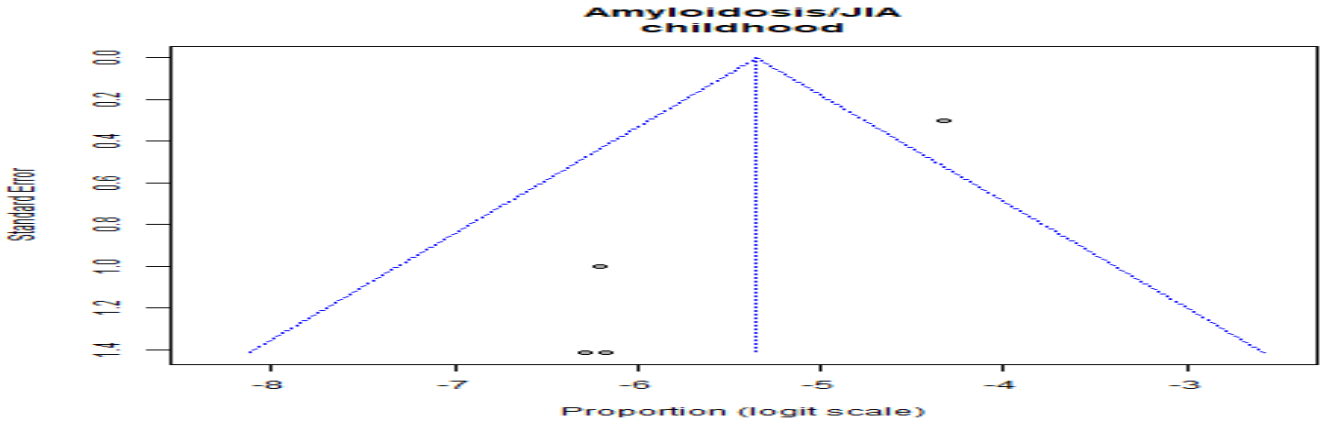
**Supplementary Figure S4.1**: Forest and Funnel Plots of studies reporting the prevalence of amyloidosis in children with Juvenile Idiopathic Arthritis

### Amyloidosis

Forest




Funnel



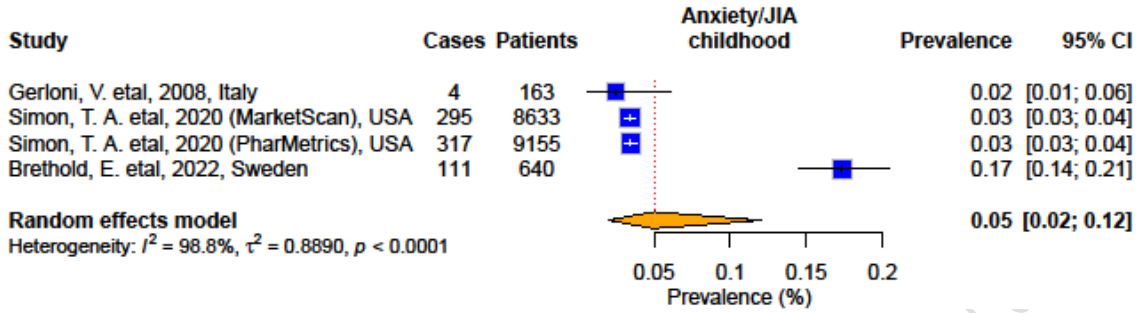

Summary statistics

```
                                    proportion           95%-CI     %W(random)
Horneff G. etal, 2022 (InGef), Germany     0.0130 [0.0065; 0.0231]     46.1
Sahin, S. etal, 2021, Turkey               0.0020 [0.0001; 0.0111]     23.4
Giancane G etal, 2019 (MTX), Italy         0.0000 [0.0000; 0.0153]     15.2
Giancane G etal, 2019 (Biologic), Italy    0.0000 [0.0000; 0.0136]     15.2
Number of studies: k = 4; Number of observations: o = 1857; Number of events: e = 12
                     proportion           95%-CI
Random effects model     0.0047 [0.0013; 0.0170]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8473 [0.0000; 11.9786]; tau = 0.9205 [0.0000; 3.4610]
 I^2 = 50.1% [0.0%; 83.5%]; H = 1.42 [1.00; 2.46]
Test of heterogeneity:
    Q d.f. p-value
 6.02    3  0.1108
```

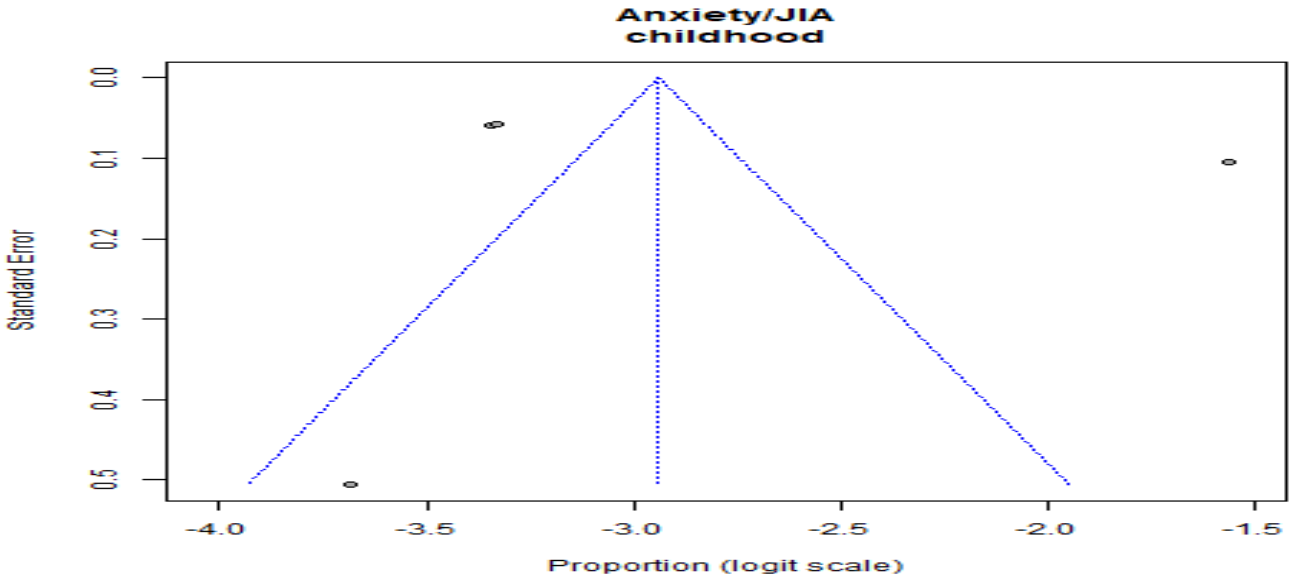
**Supplementary Figure S4.2**: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with Juvenile Idiopathic Arthritis

**Anxiety**

Forest




Funnel



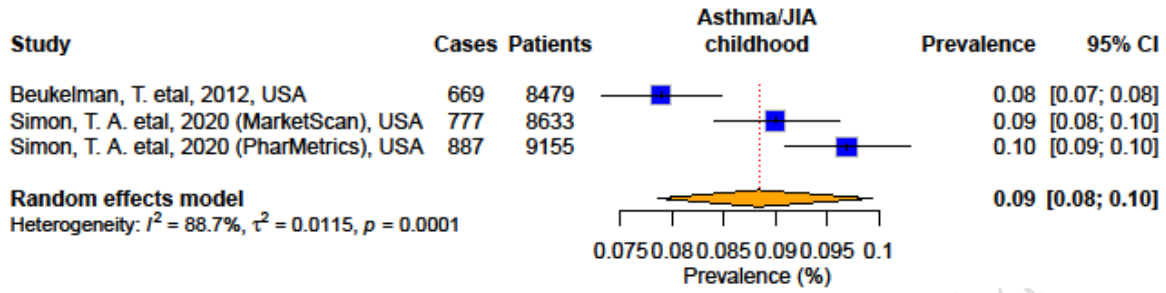

Summary statistics

```
                                        proportion           95%-CI  %W(random)
Gerloni, V. etal, 2008, Italy               0.0245 [0.0067; 0.0616]  20.7
Simon, T. A. etal, 2020 (MarketScan), USA   0.0342 [0.0304; 0.0382]  26.5
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0346 [0.0310; 0.0386]  26.5
Brethold E. etal, 2022, Sweden              0.1734 [0.1449; 0.2050]  26.3
Number of studies: k = 4; Number of observations: o = 18591; Number of events: e = 727
                    proportion           95%-CI
Random effects model     0.0502 [0.0200; 0.1206]
Quantifying heterogeneity (with 95%-CIs): tau^2 = 0.8890 [0.2565; 12.7255]; tau = 0.9429 [0.5065; 3.5673]
 I^2 = 98.8% [98.2%; 99.2%]; H = 9.15 [7.38; 11.34]
Test of heterogeneity:
      Q d.f.  p-value
 251.03    3 < 0.0001
```

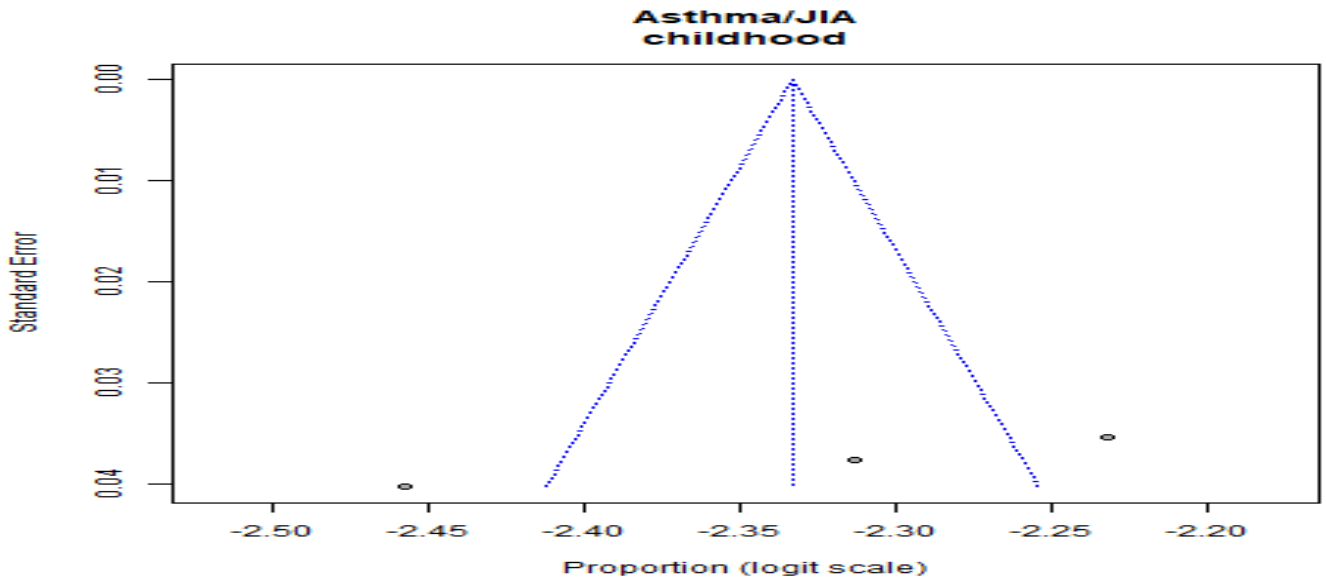
**Supplementary Figure S4.3**: Forest and Funnel Plots of studies reporting the prevalence of asthma in children with Juvenile Idiopathic Arthritis

**Asthma**

Forest




Funnel



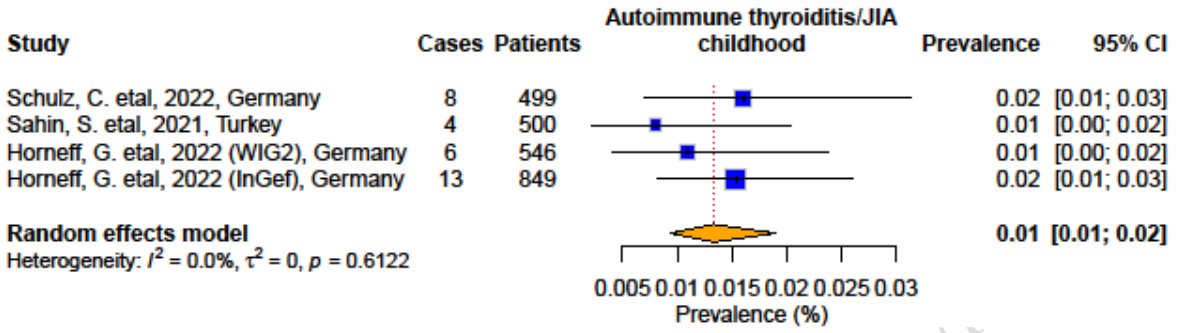

Summary statistics

```
                                        proportion           95%-CI  %W(random)
Beukelman, T. etal, 2012, USA               0.0789 [0.0732; 0.0848]  32.8
Simon, T. A. etal, 2020 (MarketScan), USA   0.0900 [0.0840; 0.0962]  33.4
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0969 [0.0909; 0.1031]  33.8
Number of studies: k = 3; Number of observations: o = 26267; Number of events: e = 2333
                     proportion           95%-CI
Random effects model     0.0884 [0.0786; 0.0993]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.0115 [0.0021; 0.5116]; tau = 0.1072 [0.0454; 0.7153]
 I^2 = 88.7% [69.0%; 95.9%]; H = 2.98 [1.80; 4.94]
Test of heterogeneity:     Q d.f. p-value
                       17.77    2  0.0001
```

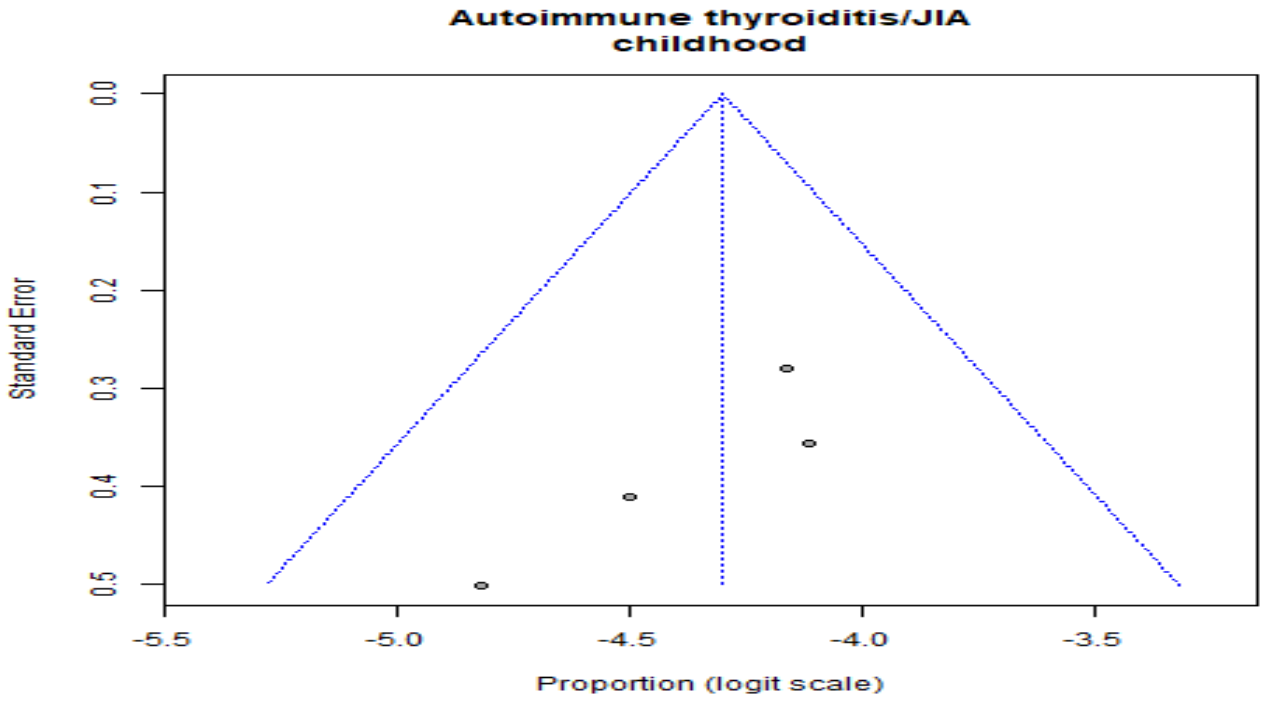
**Supplementary Figure S4.4**: Forest and Funnel Plots of studies reporting the prevalence of autoimmune thyroiditis in children with Juvenile Idiopathic Arthritis

**Autoimmune thyroiditis**

Forest




Funnel



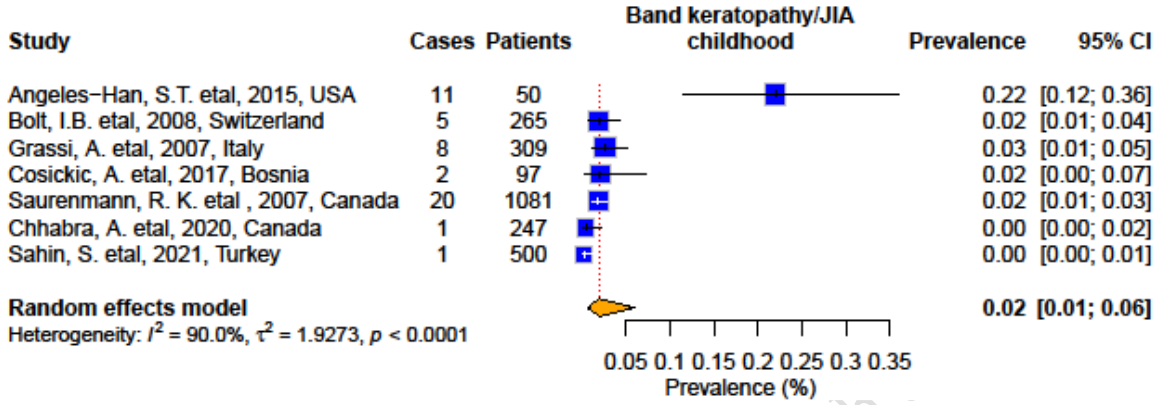

Summary statistics

```
                                    proportion           95%-CI      %W(random)
Schulz, C. etal, 2022, Germany          0.0160 [0.0069; 0.0313]      25.7
Sahin, S. etal, 2021, Turkey            0.0080 [0.0022; 0.0204]      13.0
Horneff G. etal, 2022 (WIG2), Germany   0.0110 [0.0040; 0.0238]      19.4
Horneff G. etal, 2022 (InGef), Germany  0.0153 [0.0082; 0.0260]      41.9
Number of studies: k = 4; Number of observations: o = 2394; Number of events: e = 31
                     proportion          95%-CI
Random effects model     0.0134 [0.0094; 0.0189]
Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0 [0.0000; 1.3075]; tau = 0 [0.0000; 1.1435]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.81    3  0.6122
```

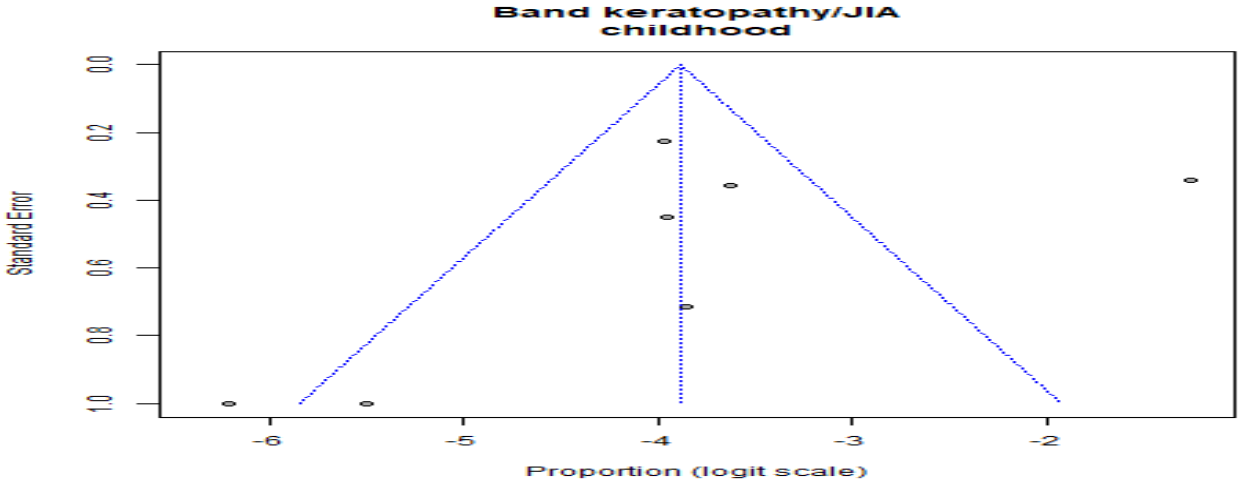
**Supplementary Figure S4.5**: Forest and Funnel Plots of studies reporting the prevalence of band keratopathy in children with Juvenile Idiopathic Arthritis

**Band keratopathy**

Forest




Funnel



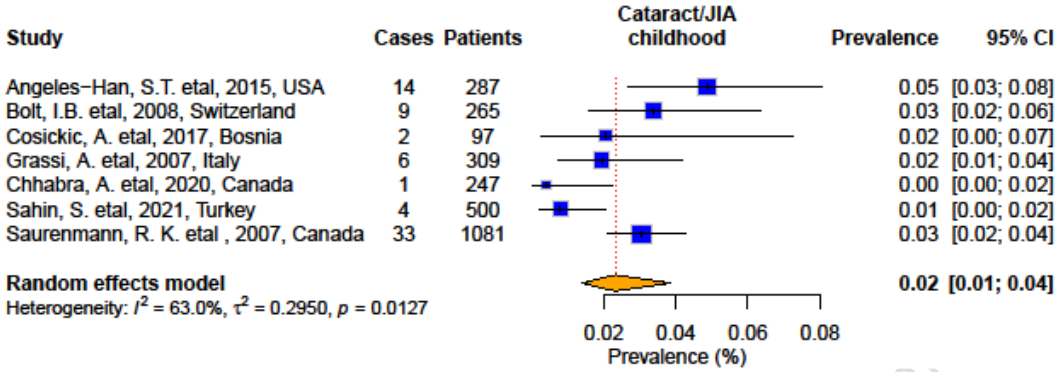

Summary statistics

```
                                    proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA        0.2200 [0.1153; 0.3596]       16.1
Bolt I.B. etal, 2008, Switzerland       0.0189 [0.0062; 0.0435]       15.4
Grassi, A. etal, 2007, Italy            0.0259 [0.0112; 0.0504]       16.0
Cosickic, A. etal, 2017, Bosnia         0.0206 [0.0025; 0.0725]       13.5
Saurenmann, R. K. etal , 2007, Canada   0.0185 [0.0113; 0.0284]       16.6
Chhabra, A. etal, 2020, Canada          0.0040 [0.0001; 0.0223]       11.2
Sahin, S. etal, 2021, Turkey            0.0020 [0.0001; 0.0111]       11.2
Number of studies: k = 7; Number of observations: o = 2549; Number of events: e = 48
                      proportion           95%-CI
Random effects model     0.0201 [0.0066; 0.0593]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9273 [0.6009; 11.4288]; tau = 1.3883 [0.7752; 3.3806]
 I^2 = 90.0% [81.9%; 94.5%]; H = 3.16 [2.35; 4.25]
Test of heterogeneity:      Q d.f.  p-value
                        | 59.88    6 < 0.0001
```

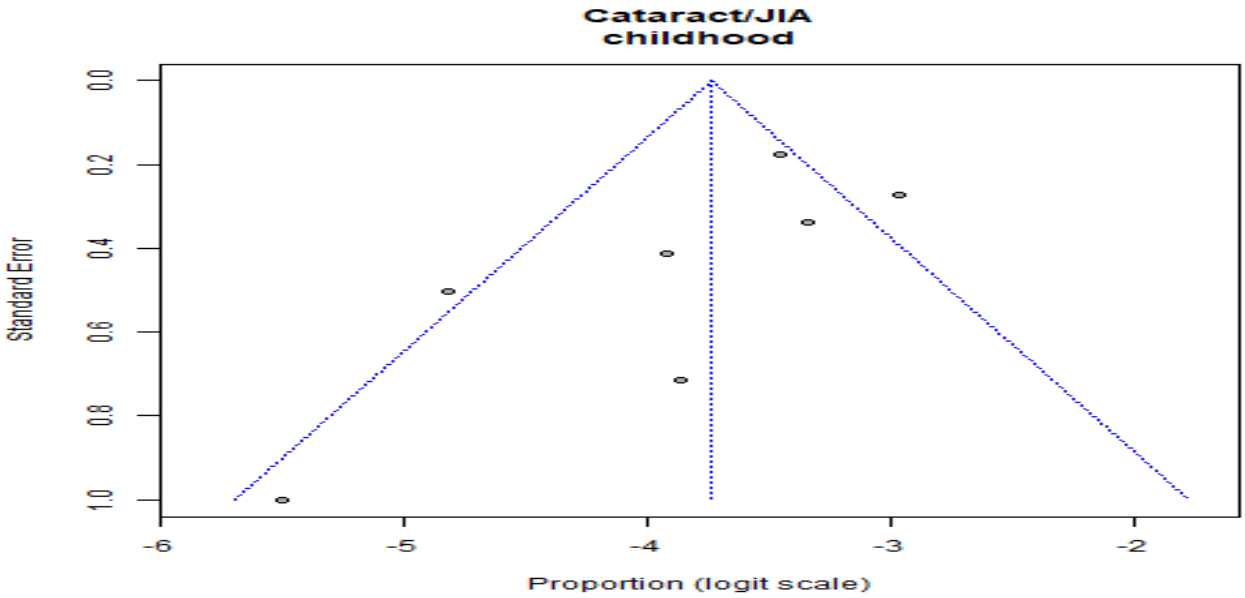
**Supplementary Figure S4.6**: Forest and Funnel Plots of studies reporting the prevalence of cataracts in children with Juvenile Idiopathic Arthritis

**Cataract**

Forest




Funnel



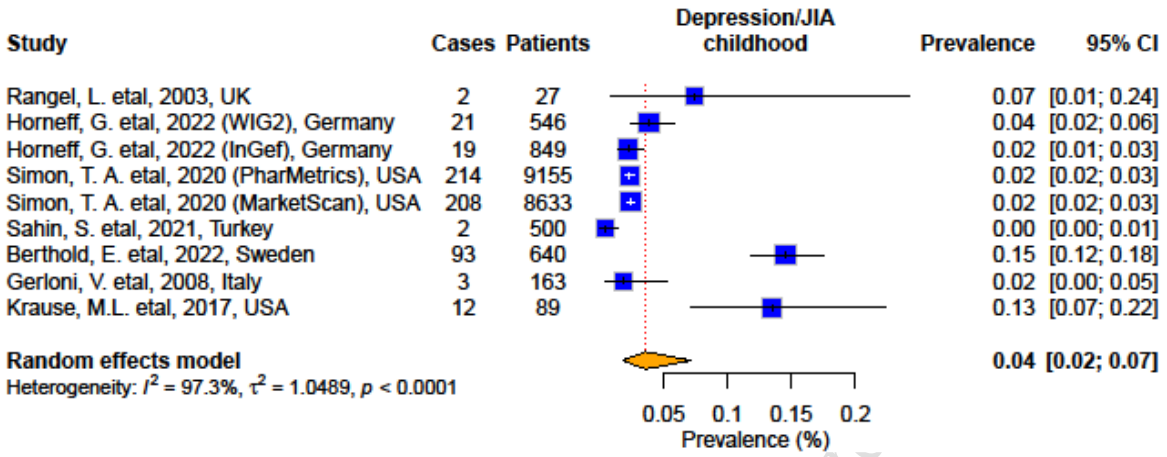

Summary statistics

```
                                    proportion           95%-CI       %W(random)
Angeles-Han S.T. etal, 2015, USA        0.0488 [0.0269; 0.0805]       19.0
Bolt I.B. etal, 2008, Switzerland       0.0340 [0.0156; 0.0635]       17.2
Cosickic, A. etal, 2017, Bosnia         0.0206 [0.0025; 0.0725]       8.7
Grassi, A. etal, 2007, Italy            0.0194 [0.0072; 0.0418]       15.2
Chhabra, A. etal, 2020, Canada          0.0040 [0.0001; 0.0223]       5.4
Sahin, S. etal, 2021, Turkey            0.0080 [0.0022; 0.0204]       12.9
Saurenmann, R. K. etal , 2007, Canada   0.0305 [0.0211; 0.0426]       21.6
Number of studies: k = 7; Number of observations: o = 2786; Number of events: e = 69
                    proportion           95%-CI
Random effects model     0.0232 [0.0139; 0.0385]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2950 [0.0217; 3.3077]; tau = 0.5432 [0.1474; 1.8187]
 I^2 = 63.0% [16.0%; 83.7%]; H = 1.64 [1.09; 2.48]
Test of heterogeneity:      Q d.f. p-value
                        16.21    6  0.0127
```

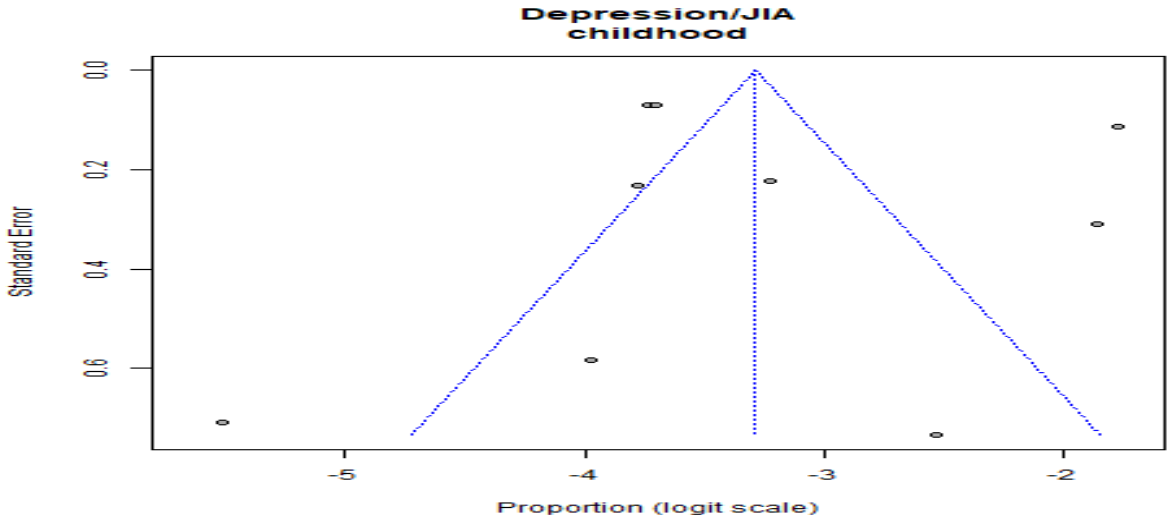
**Supplementary Figure S4.7**: Forest and Funnel Plots of studies reporting the prevalence of depression in children with Juvenile Idiopathic Arthritis

**Depression**

Forest




Funnel




Summary statistics

For reproduction

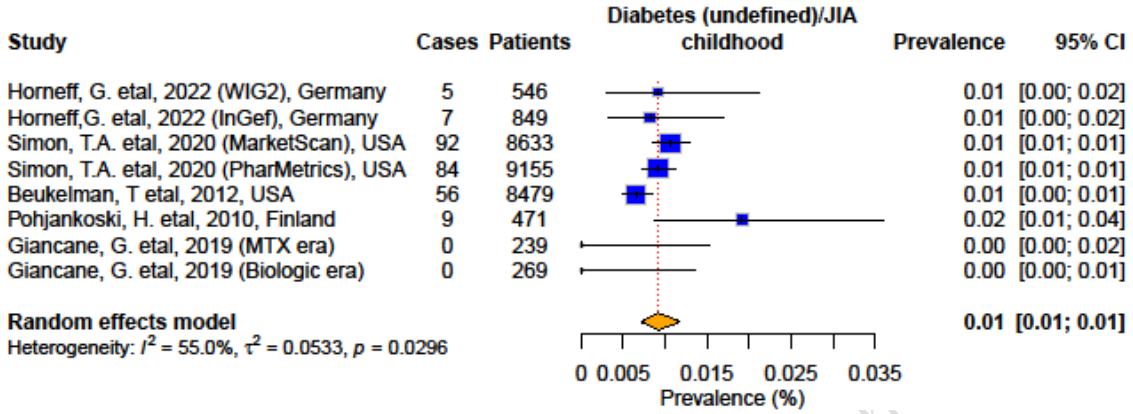
```
                                          proportion           95%-CI   %W(random)
Rangel, L. etal, 2003, UK                     0.0741 [0.0091; 0.2429]  8.4
Horneff G. etal, 2022 (WIG2), Germany         0.0385 [0.0240; 0.0582]  12.1
Horneff G. etal, 2022 (InGef), Germany        0.0224 [0.0135; 0.0347]  12.1
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0234 [0.0204; 0.0267]  12.6
Simon, T. A. etal, 2020 (MarketScan), USA     0.0241 [0.0210; 0.0276]  12.6
Sahin, S. etal, 2021                          0.0040 [0.0005; 0.0144]  8.6
Berthold, E. etal, 2022                       0.1453 [0.1189; 0.1750]  12.5
Gerloni, V. etal, 2008, Italy                 0.0184 [0.0038; 0.0528]  9.6
Krause ML et al, 2017                         0.1348 [0.0717; 0.2237]  11.6
Number of studies: k = 9; Number of observations: o = 20602; Number of events: e = 574
                     proportion           95%-CI
Random effects model     0.0361 [0.0180; 0.0711]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0489 [0.4100; 4.7484]; tau = 1.0242 [0.6403; 2.1791]
 I^2 = 97.3% [96.1%; 98.1%]; H = 6.04 [5.07; 7.18]
Test of heterogeneity:       Q d.f.  p-value
                        291.44    8 < 0.0001
```

For reproducibility only in academic research and development

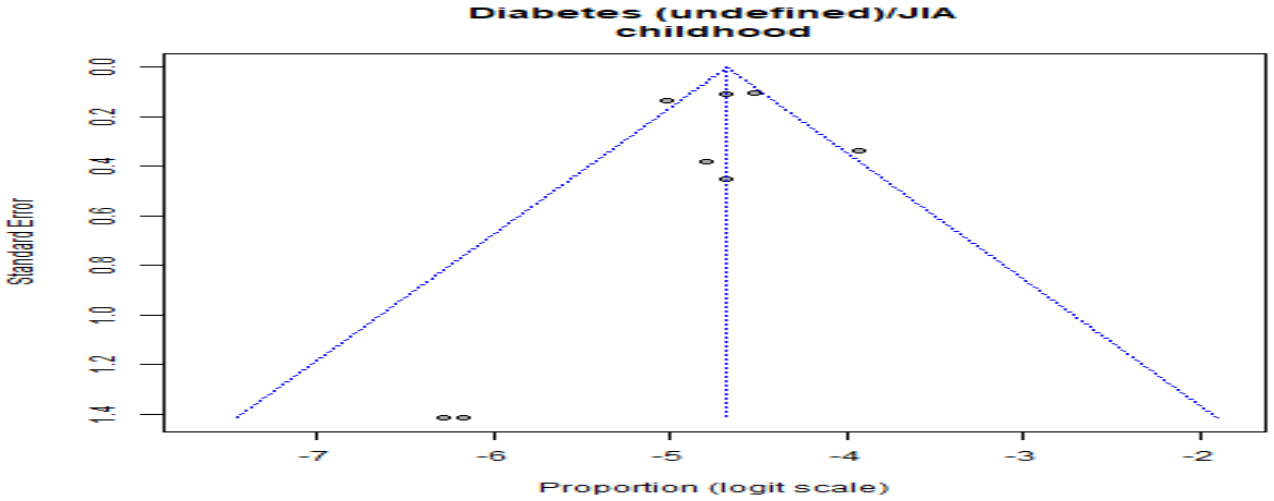
**Supplementary Figure S4.8**: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Idiopathic Arthritis

**Diabetes (undefined)**

Forest




Funnel



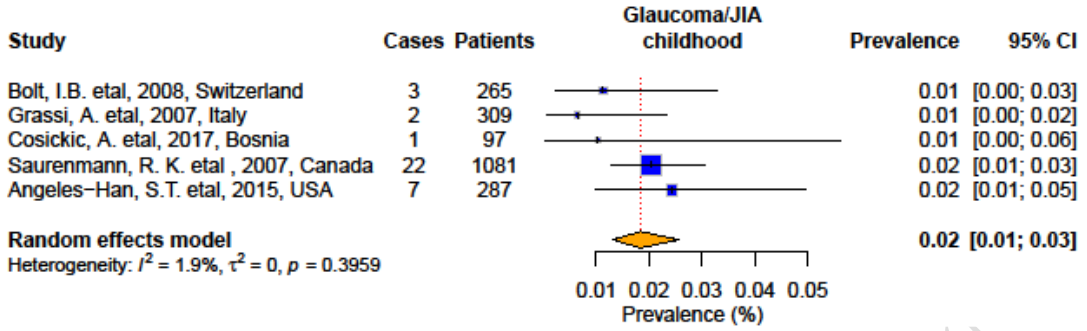

Summary statistics

```
                                        proportion           95%-CI    %W(random)
Horneff, G. etal, 2022 (WIG2), Germany      0.0092 [0.0030; 0.0212]    6.4
Horneff,G. etal, 2022 (InGef), Germany      0.0082 [0.0033; 0.0169]    8.3
Simon, TA. etal, 2020 (MarketScan), USA     0.0107 [0.0086; 0.0131]   25.6
Simon, TA. etal, 2020 (PharMetrics), USA    0.0092 [0.0073; 0.0113]   25.2
Beukelman, T etal, 2012, USA                0.0066 [0.0050; 0.0086]   23.1
Pohjankoski, H. etal, 2010, Finland         0.0191 [0.0088; 0.0360]    9.9
Giancane G etal, 2019 (MTX era)             0.0000 [0.0000; 0.0153]    0.8
Giancane G etal, 2019 (Biologic era)        0.0000 [0.0000; 0.0136]    0.8
Number of studies: k = 8; Number of observations: o = 28641; Number of events: e = 253
                    proportion           95%-CI
Random effects model     0.0092 [0.0072; 0.0118]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0533 [0.0000; 1.5500]; tau = 0.2309 [0.0000; 1.2450]
 I^2 = 55.0% [0.4%; 79.6%]; H = 1.49 [1.00; 2.22]
Test of heterogeneity:      Q d.f. p-value
                        15.55    7  0.0296
```

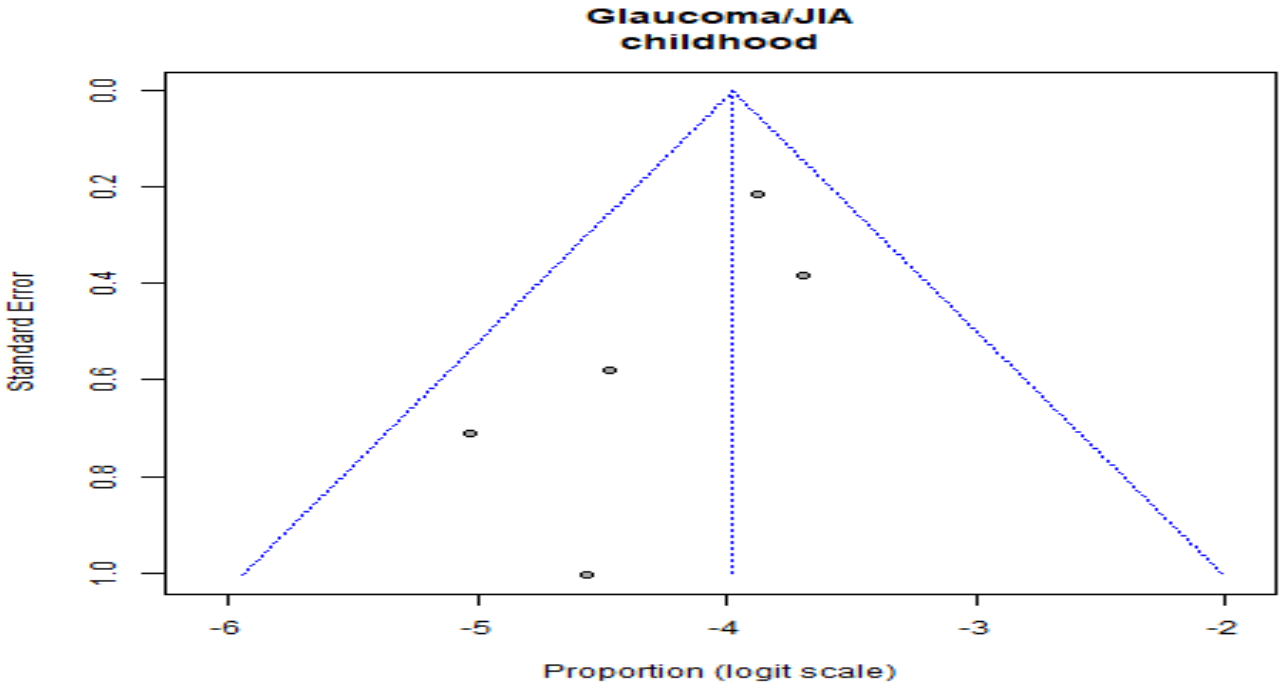
**Supplementary Figure S4.9**: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in children with Juvenile Idiopathic Arthritis

**Glaucoma**

Forest




Funnel




Summary statistics

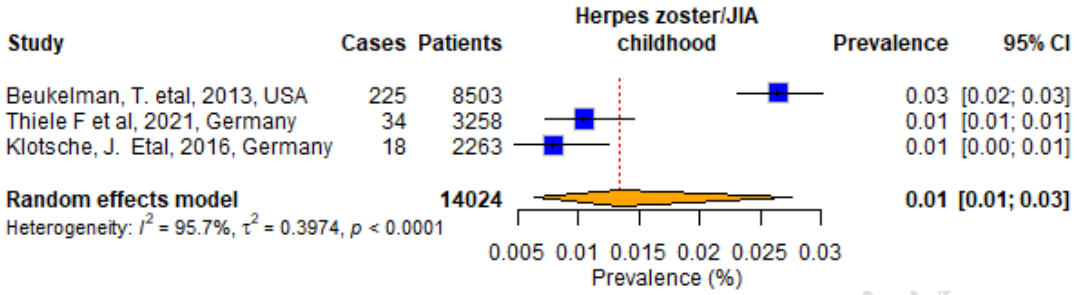
```
                                    proportion           95%-CI       %W(random)
Bolt I.B. etal, 2008, Switzerland       0.0113 [0.0023; 0.0327]       8.6
Grassi, A. etal, 2007, Italy            0.0065 [0.0008; 0.0232]       5.8
Cosickic, A. etal, 2017, Bosnia         0.0103 [0.0003; 0.0561]       2.9
Saurenmann, R. K. etal , 2007, Canada   0.0204 [0.0128; 0.0307]       62.8
Angeles-Han ST et al, 2015, USA         0.0244 [0.0099; 0.0496]       19.9
Number of studies: k = 5; Number of observations: o = 2039; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0184 [0.0133; 0.0256]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.1342]; tau = 0 [0.0000; 1.4609]
 I^2 = 1.9% [0.0%; 79.6%]; H = 1.01 [1.00; 2.21]
Test of heterogeneity:      Q d.f. p-value
                         4.08    4  0.3959
```

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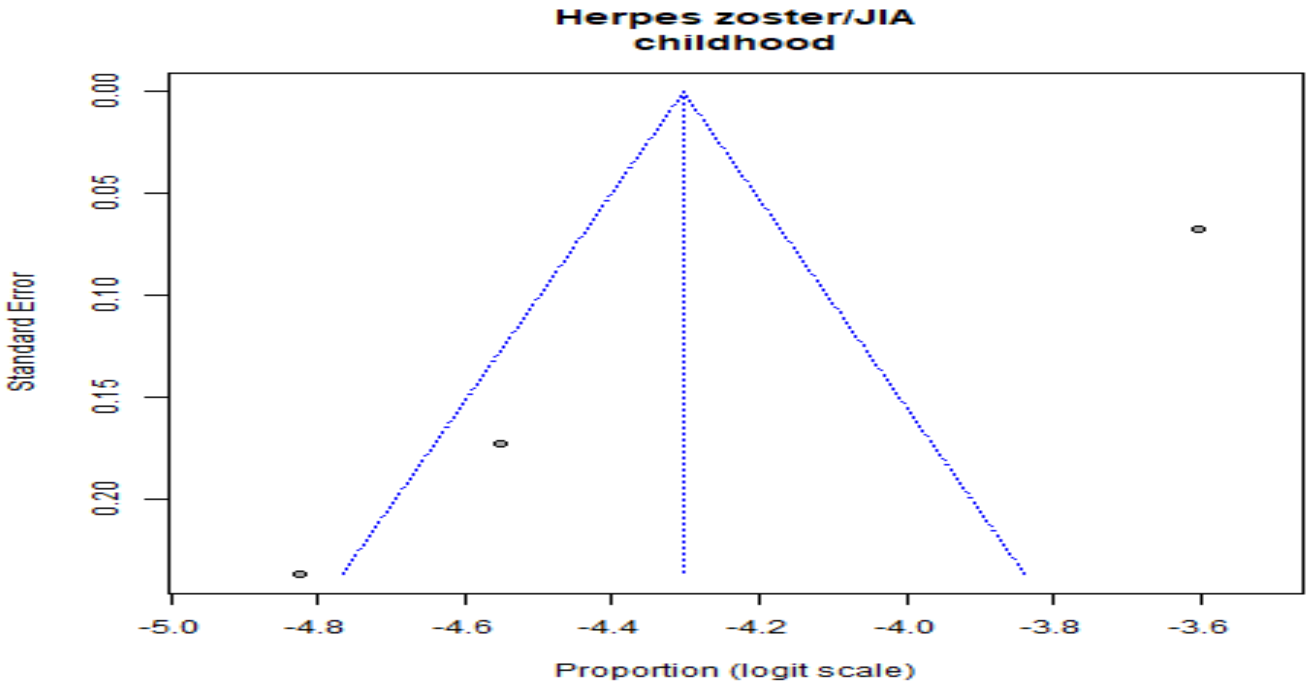
**Supplementary Figure S4.10**: Forest and Funnel Plots of studies reporting the prevalence of herpes zoster in children with Juvenile Idiopathic Arthritis

**Herpes zoster**

Forest




Funnel



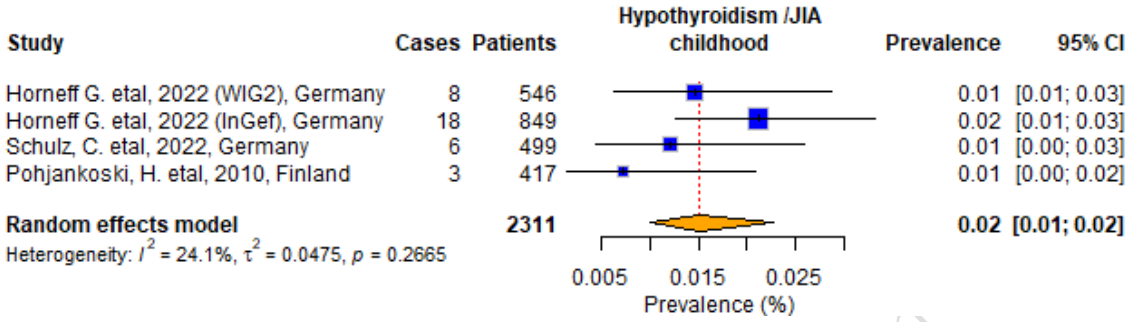

Summary statistics

```
                                  proportion           95%-CI   %W(random)
Beukelman, T. etal, 2013, USA         0.0265 [0.0232; 0.0301]         35.4
Thiele F et al, 2021, Germany         0.0104 [0.0072; 0.0146]         33.3
Klotsche, J.  Etal, 2016, Germany     0.0080 [0.0047; 0.0125]         31.4
Number of studies: k = 3; Number of observations: o = 14024; Number of events: e = 277
                       proportion           95%-CI
Random effects model     0.0133 [0.0064; 0.0275]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3974 [0.0907; 16.1850]; tau = 0.6304 [0.3012; 4.0231]
 I^2 = 95.7% [90.5%; 98.0%]; H = 4.80 [3.25; 7.10]
Test of heterogeneity:      Q d.f.  p-value
                        46.13    2 < 0.0001
```

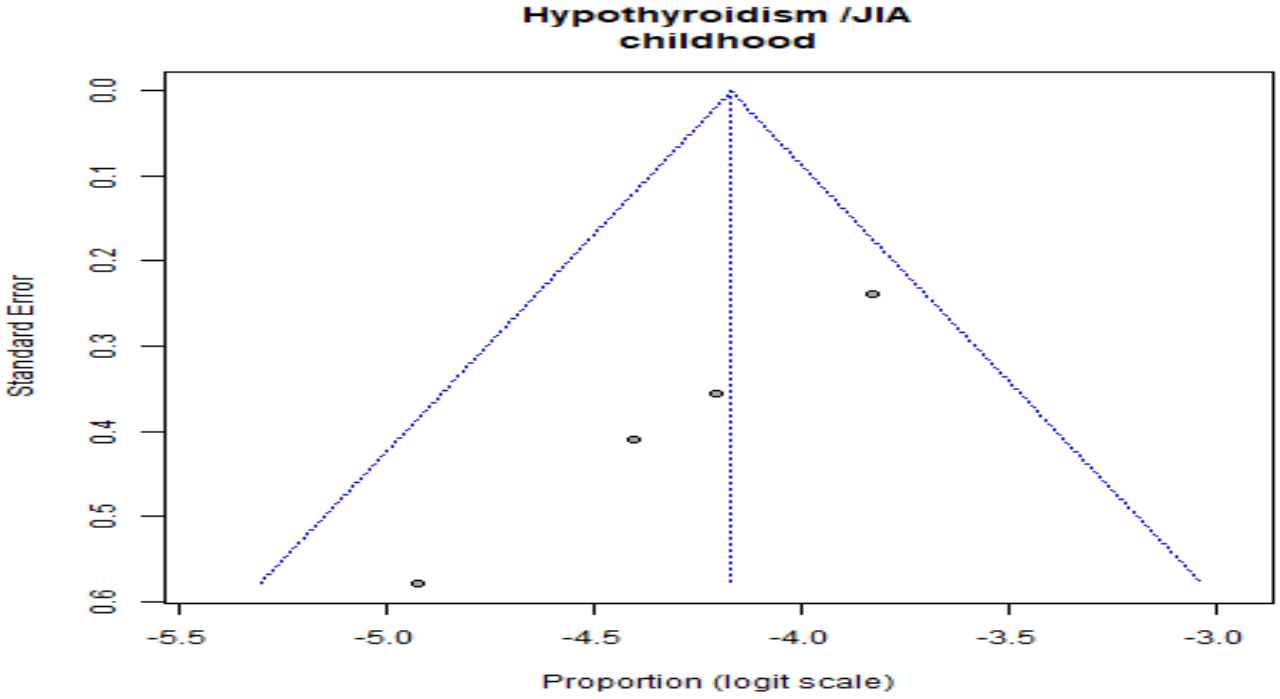
**Supplementary Figure S4.11**: Forest and Funnel Plots of studies reporting the prevalence of hypothyroidism in children with Juvenile Idiopathic Arthritis

**Hypothyroidism**

Forest




Funnel



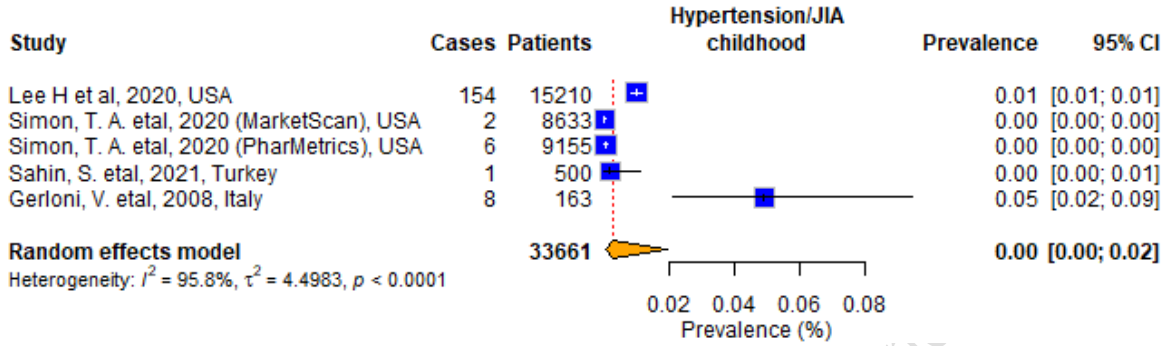

Summary statistics

```
                                    proportion           95%-CI      %W(random)
Horneff G. etal, 2022 (WIG2), Germany     0.0147 [0.0063; 0.0287]      25.4
Horneff G. etal, 2022 (InGef), Germany    0.0212 [0.0126; 0.0333]      42.5
Schulz, C. etal, 2022, Germany            0.0120 [0.0044; 0.0260]      20.5
Pohjankoski, H. etal, 2010, Finland       0.0072 [0.0015; 0.0209]      11.6
Number of studies: k = 4; Number of observations: o = 2311; Number of events: e = 35
                     proportion           95%-CI
Random effects model     0.0152 [0.0101; 0.0228]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0475 [0.0000; 2.6861]; tau = 0.2179 [0.0000; 1.6389]
 I^2 = 24.1% [0.0%; 88.4%]; H = 1.15 [1.00; 2.93]
Test of heterogeneity:     Q d.f. p-value
                        | 3.95    3  0.2665
```

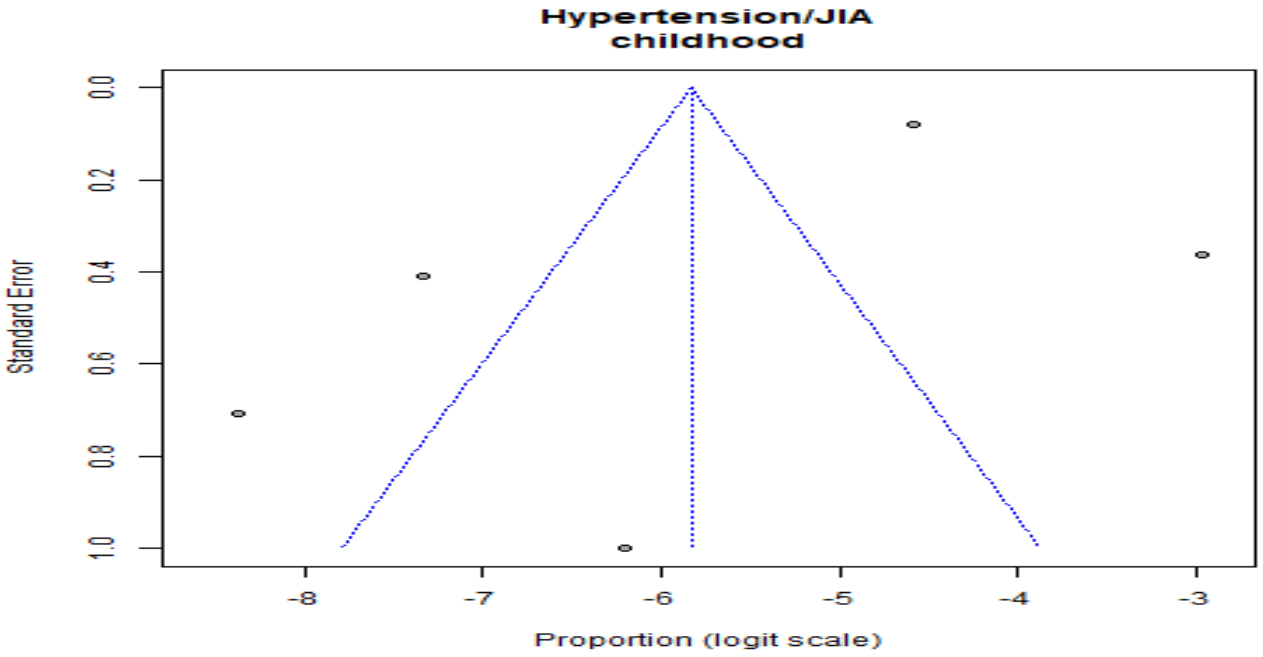
**Supplementary Figure S4.12**: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Idiopathic Arthritis

**Hypertension**

Forest




Funnel




Summary statistics

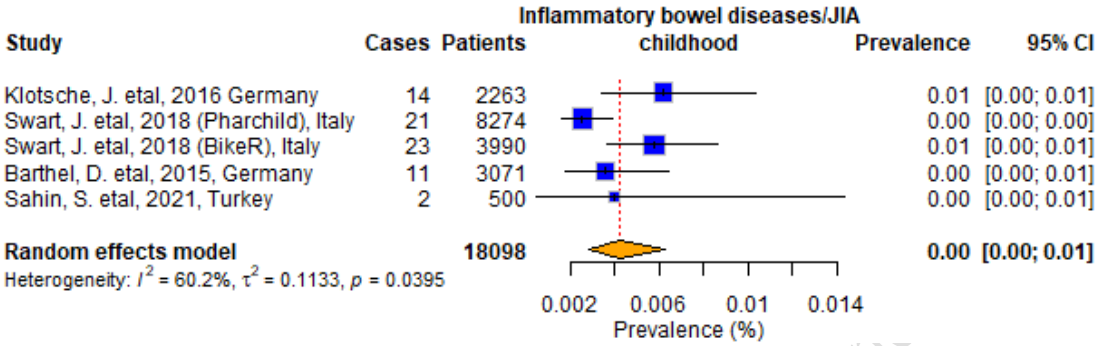
```
                                        proportion           95%-CI  %W(random)
Lee H et al, 2020, USA                      0.0101 [0.0086; 0.0118]  21.5
Simon, T. A. etal, 2020 (MarketScan), USA   0.0002 [0.0000; 0.0008]  19.3
Simon, T. A. etal, 2020 (PharMetrics), USA  0.0007 [0.0002; 0.0014]  20.7
Sahin, S. etal, 2021, Turkey                0.0020 [0.0001; 0.0111]  17.6
Gerloni, V. etal, 2008, Italy               0.0491 [0.0214; 0.0944]  20.9
Number of studies: k = 5; Number of observations: o = 33661; Number of events: e = 171
                     proportion           95%-CI
Random effects model     0.0029 [0.0004; 0.0197]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.4983 [1.4273; 38.1419]; tau = 2.1209 [1.1947; 6.1759]
 I^2 = 95.8% [92.7%; 97.6%]; H = 4.89 [3.70; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        95.46    4 < 0.0001
```

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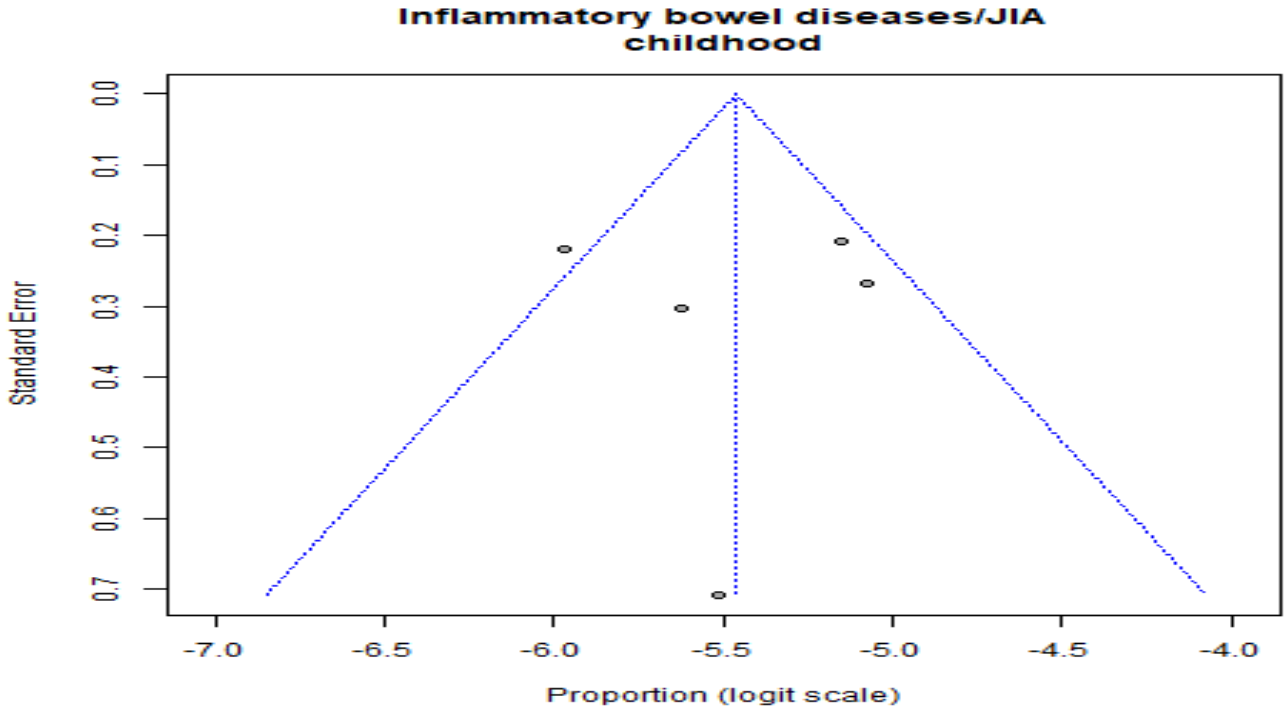
**Supplementary Figure S4.13**: Forest and Funnel Plots of studies reporting the prevalence of Inflammatory Bowel Diseases in children with Juvenile Idiopathic Arthritis

**Inflammatory bowel diseases**

Forest




Funnel



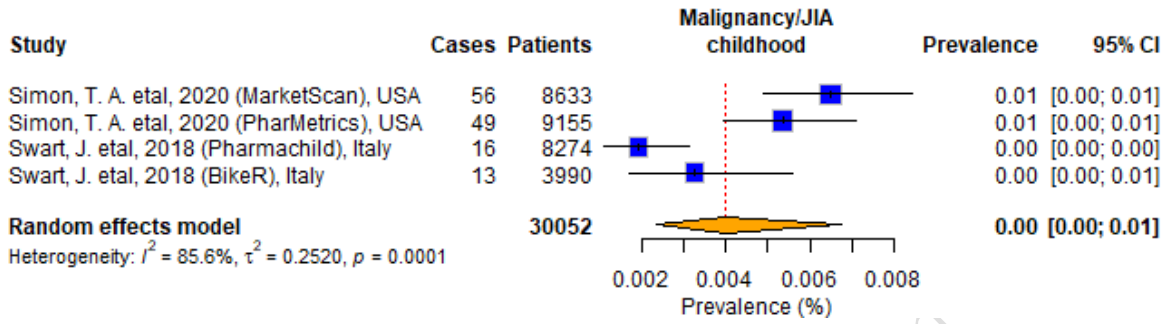

Summary statistics

```
                                        proportion           95%-CI     %W(random)
Klotsche, J. etal, 2016 Germany             0.0062 [0.0034; 0.0104]     22.0
Swart, J. etal, 2018 (Pharchild), Italy     0.0025 [0.0016; 0.0039]     25.4
Swart, J. etal, 2018 (BikeR), Italy         0.0058 [0.0037; 0.0086]     26.0
Barthel, D. etal, 2015, Germany             0.0036 [0.0018; 0.0064]     20.0
Sahin, S. etal, 2021, Turkey                0.0040 [0.0005; 0.0144]      6.6
Number of studies: k = 5; Number of observations: o = 18098; Number of events: e = 71
                    proportion           95%-CI
Random effects model     0.0042 [0.0028; 0.0063]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1133 [0.0000; 1.0495]; tau = 0.3366 [0.0000; 1.0245]
 I^2 = 60.2% [0.0%; 85.1%]; H = 1.59 [1.00; 2.59]
Test of heterogeneity:      Q d.f. p-value
                        10.06    4  0.0395
```

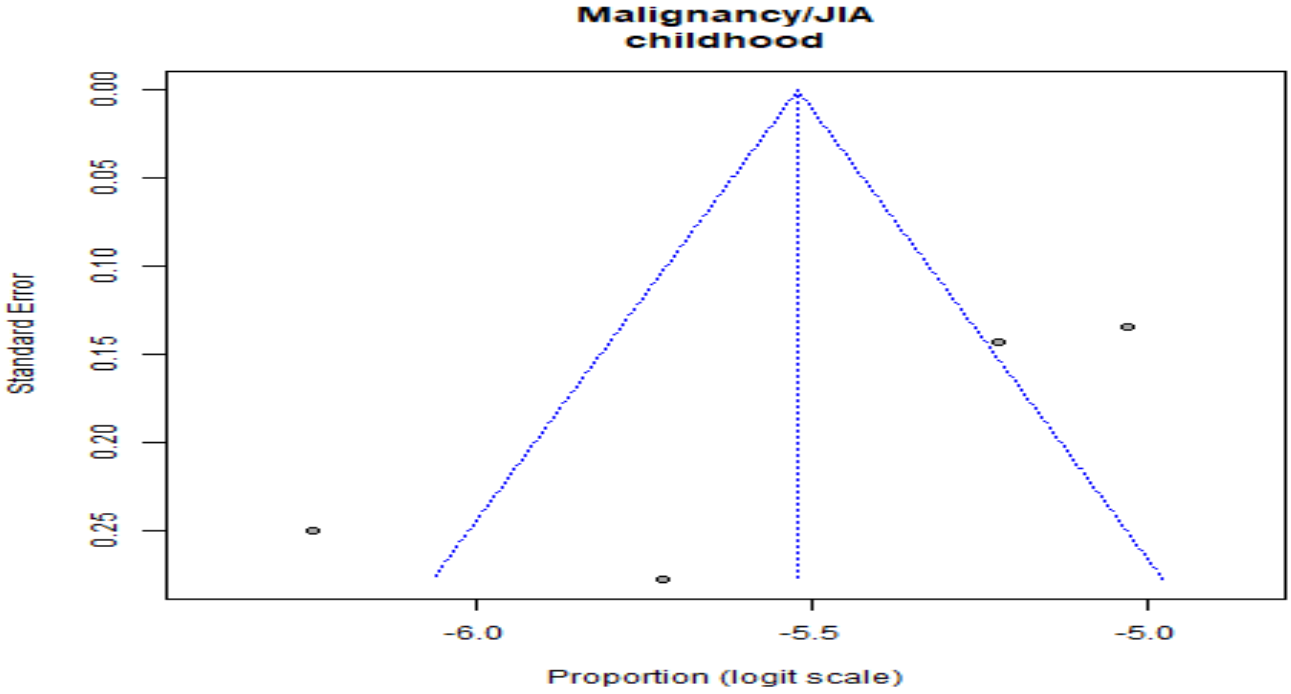
**Supplementary Figure S4.14**: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Idiopathic Arthritis

**Malignancy**

Forest




Funnel



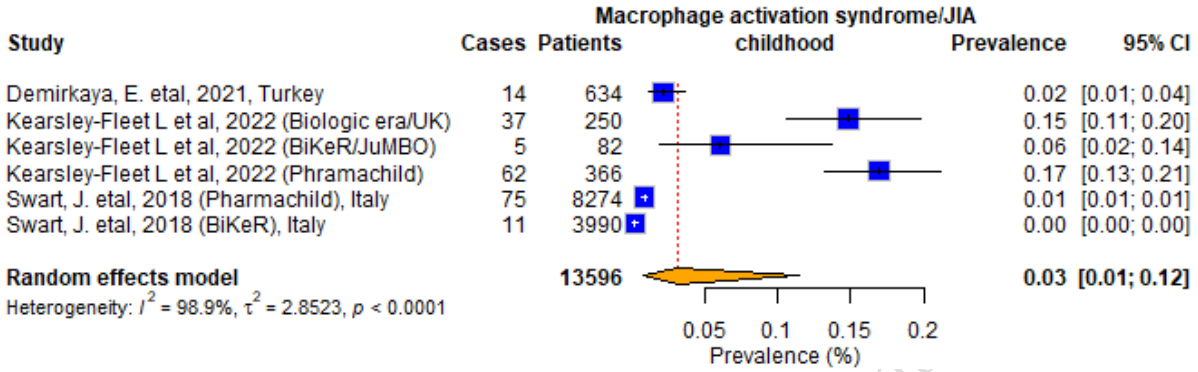

Summary statistics

```
                                          proportion           95%-CI  %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0065 [0.0049; 0.0084]  27.3
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0054 [0.0040; 0.0071]  27.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0019 [0.0011; 0.0031]  23.4
Swart, J. etal, 2018 (BikeR), Italy           0.0033 [0.0017; 0.0056]  22.4
Number of studies: k = 4; Number of observations: o = 30052; Number of events: e = 134
                     proportion           95%-CI
Random effects model     0.0040 [0.0023; 0.0068]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2520 [0.0517; 4.0772]; tau = 0.5020 [0.2273; 2.0192]
 I^2 = 85.6% [64.7%; 94.2%]; H = 2.64 [1.68; 4.14]
Test of heterogeneity:      Q d.f. p-value
                        20.88    3  0.0001
```

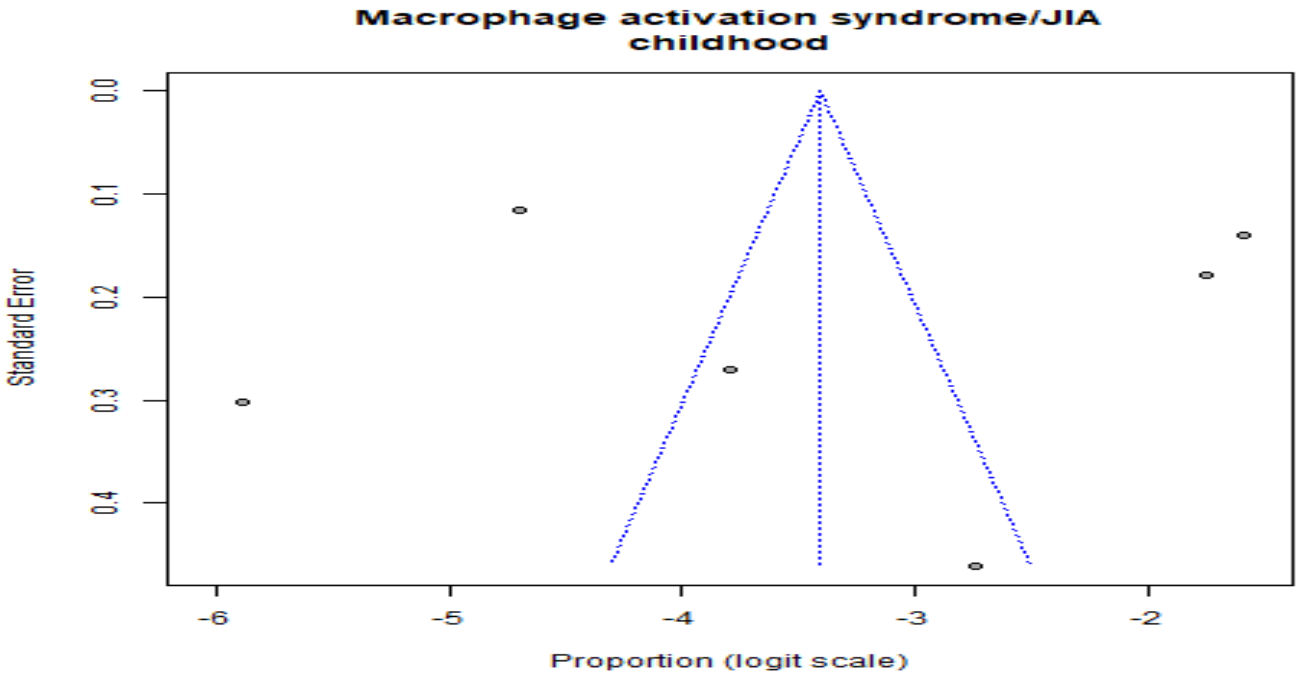
**Supplementary Figure S4.15**: Forest and Funnel Plots of studies reporting the prevalence of macrophage activation syndrome in children with Juvenile Idiopathic Arthritis

**Macrophage activation syndrome**

Forest




Funnel




Summary statistics

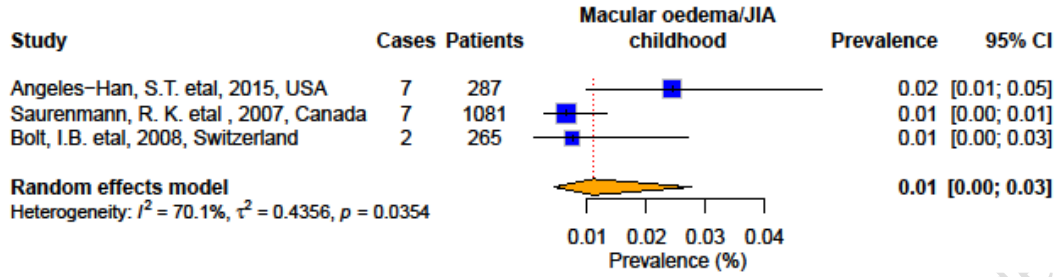
```
                                          proportion           95%-CI %W(random)
Demirkaya, E. etal, 2021, Turkey              0.0221 [0.0121; 0.0368]       16.7
Kearsley-Fleet L et al, 2022 (Biologic era/UK)  0.1480 [0.1064; 0.1982]       16.9
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)    0.0610 [0.0201; 0.1366]       15.9
Kearsley-Fleet L et al, 2022 (Phramachild)    0.1694 [0.1324; 0.2118]       17.0
Swart, J. etal, 2018 (Pharmachild), Italy     0.0091 [0.0071; 0.0113]       17.0
Swart, J. etal, 2018 (BiKeR), Italy           0.0028 [0.0014; 0.0049]       16.6
Number of studies: k = 6; Number of observations: o = 13596; Number of events: e = 204
                    proportion           95%-CI
Random effects model     0.0321 [0.0084; 0.1153]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.8523 [1.0712; 17.3534]; tau = 1.6889 [1.0350; 4.1657]
 I^2 = 98.9% [98.5%; 99.2%]; H = 9.48 [8.05; 11.16]
Test of heterogeneity:       Q d.f.  p-value
                        449.21    5 < 0.0001
```

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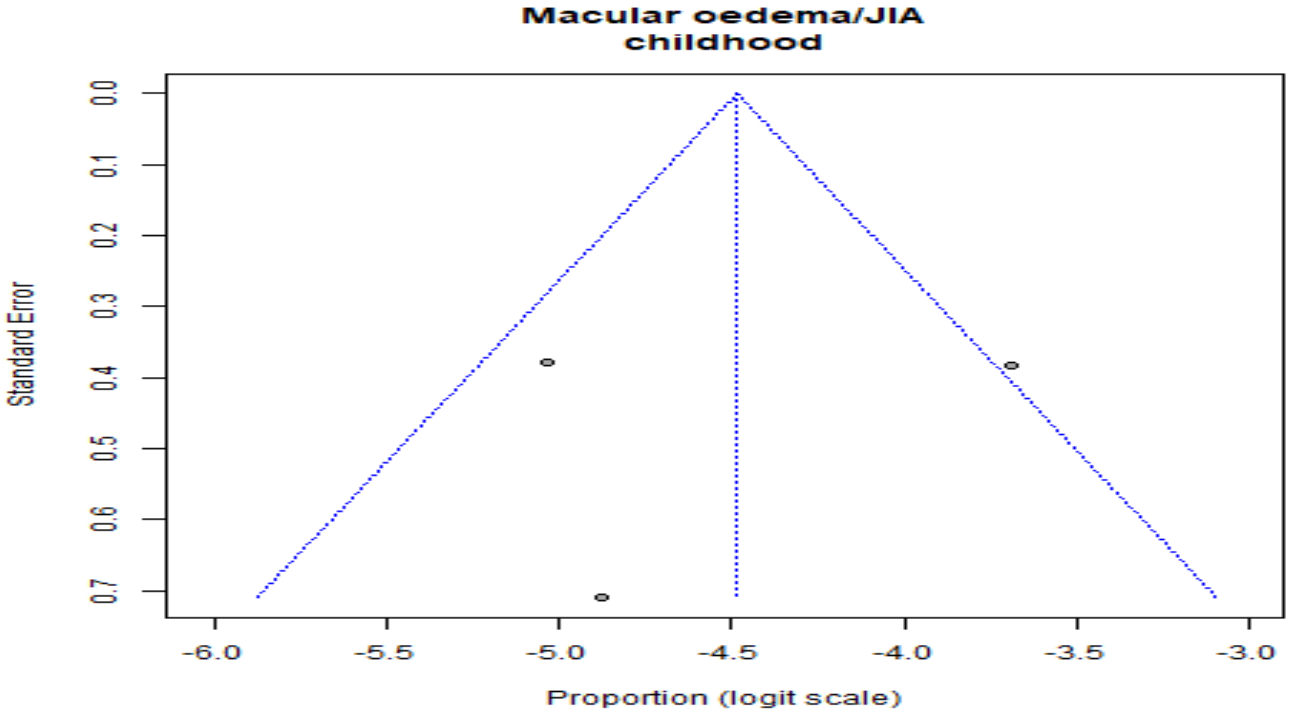
**Supplementary Figure S4.16**: Forest and Funnel Plots of studies reporting the prevalence of macular oedema in children with Juvenile Idiopathic Arthritis

**Macular Oedema**

Forest




Funnel



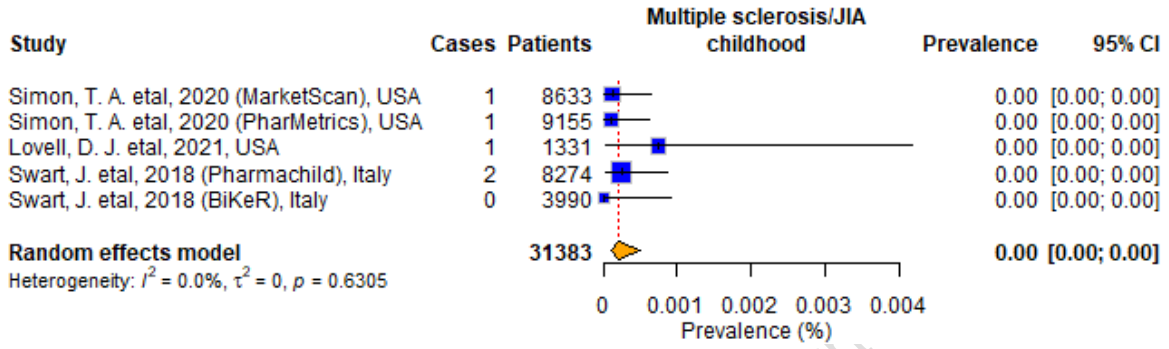

Summary statistics

```
                                  proportion           95%-CI %W(random)
Angeles-Han S.T. etal, 2015, USA      0.0244 [0.0099; 0.0496]       38.1
Saurenmann, R. K. etal , 2007, Canada 0.0065 [0.0026; 0.0133]       38.3
Bolt I.B. etal, 2008, Switzerland     0.0075 [0.0009; 0.0270]       23.6
Number of studies: k = 3; Number of observations: o = 1633; Number of events: e = 16
                    proportion           95%-CI
Random effects model     0.0112 [0.0045; 0.0276]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4356 [0.0000; 21.1933]; tau = 0.6600 [0.0000; 4.6036]
 I^2 = 70.1% [0.0%; 91.2%]; H = 1.83 [1.00; 3.38]
Test of heterogeneity:     Q d.f. p-value
                        6.68    2  0.0354
```

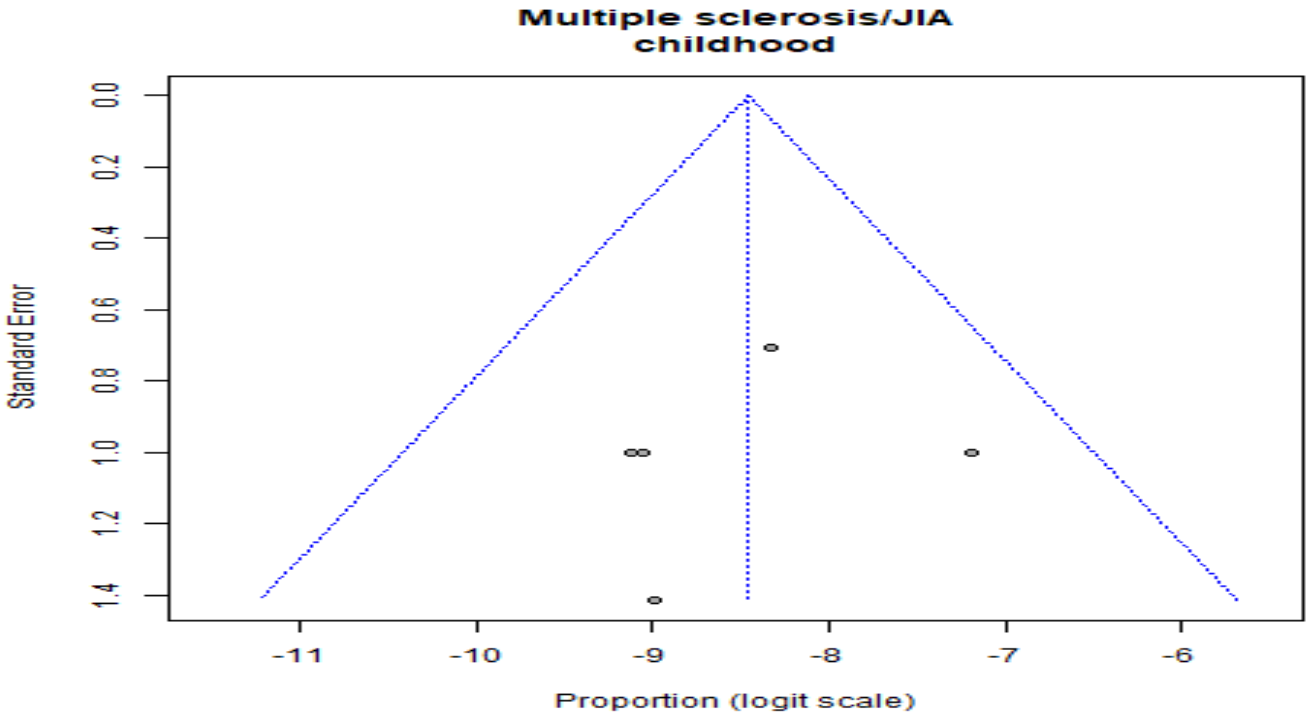
**Supplementary Figure S4.17**: Forest and Funnel Plots of studies reporting the prevalence of Multiple Sclerosis in children with Juvenile Idiopathic Arthritis

**Multiple sclerosis**

Forest




Funnel




Summary statistics

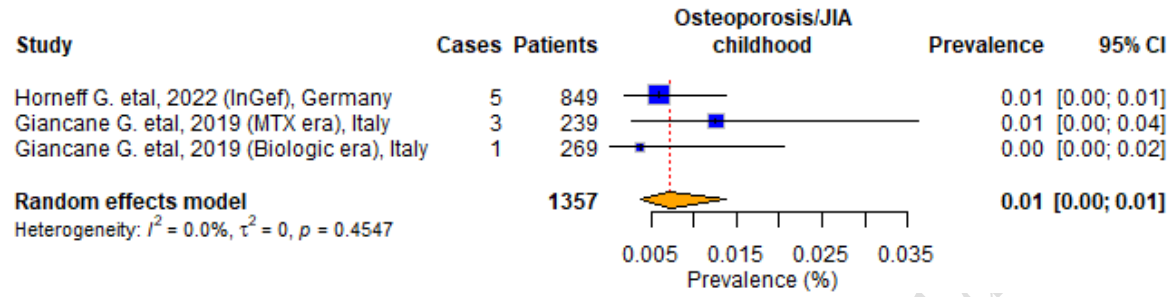
```
                                          proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0001 [0.0000; 0.0006]       18.2
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0001 [0.0000; 0.0006]       18.2
Lovell, D. J. etal, 2021, USA                 0.0008 [0.0000; 0.0042]       18.2
Swart, J. etal, 2018 (Pharmachild), Italy     0.0002 [0.0000; 0.0009]       36.4
Swart, J. etal, 2018 (BiKeR), Italy           0.0000 [0.0000; 0.0009]        9.1
Number of studies: k = 5; Number of observations: o = 31383; Number of events: e = 5
                     proportion           95%-CI
Random effects model     0.0002 [0.0001; 0.0005]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 4.4533]; tau = 0 [0.0000; 2.1103]
 I^2 = 0.0% [0.0%; 79.2%]; H = 1.00 [1.00; 2.19]
Test of heterogeneity:     Q d.f. p-value
                        2.58    4  0.6305
```

=For reproducibility only in academic research and development=

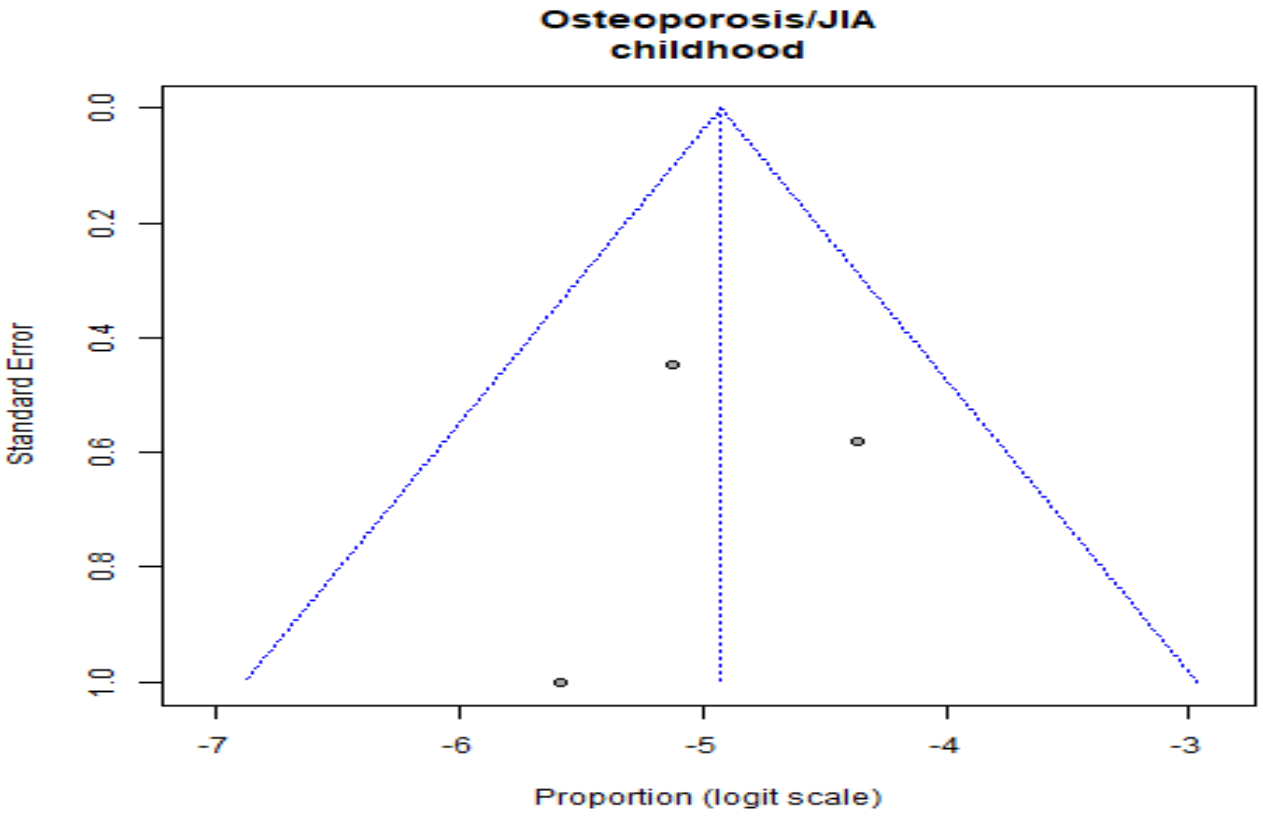
**Supplementary Figure S4.18**: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with Juvenile Idiopathic Arthritis

**Osteoporosis**

Forest




Funnel



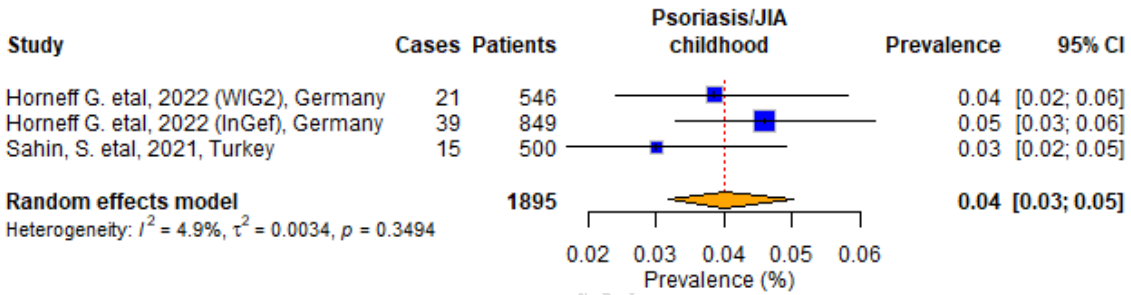

Summary statistics

```
                                          proportion           95%-CI %W(random)
Horneff G. etal, 2022 (InGef), Germany           0.0059 [0.0019; 0.0137]       55.7
Giancane G. etal, 2019 (MTX era), Italy          0.0126 [0.0026; 0.0362]       33.2
Giancane G. etal, 2019 (Biologic era), Italy     0.0037 [0.0001; 0.0205]       11.2
Number of studies: k = 3; Number of observations: o = 1357; Number of events: e = 9
                     proportion           95%-CI
Random effects model     0.0072 [0.0037; 0.0138]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 14.5276]; tau = 0 [0.0000; 3.8115]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        | 1.58    2  0.4547
```

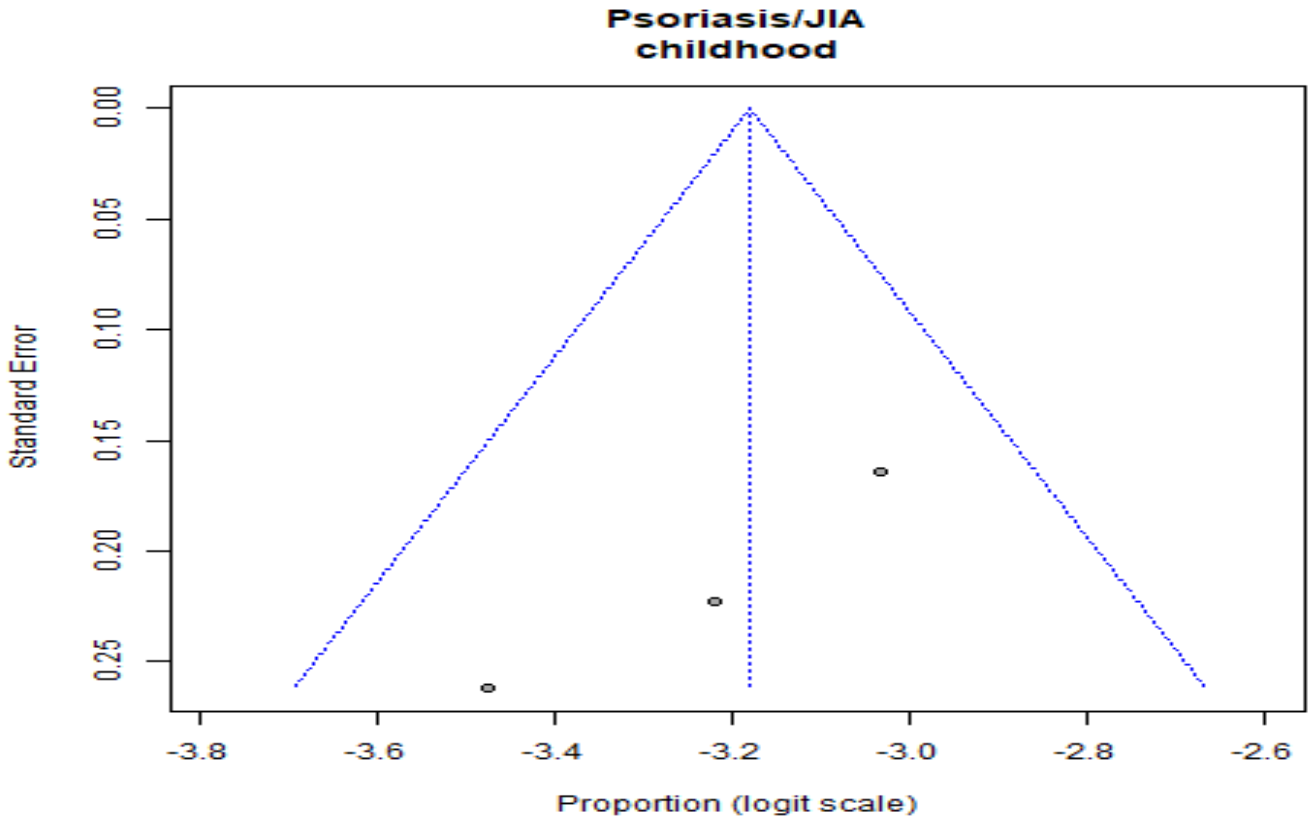
**Supplementary Figure S4.19**: Forest and Funnel Plots of studies reporting the prevalence of psoriasis in children with Juvenile Idiopathic Arthritis

**Psoriasis**

Forest




Funnel



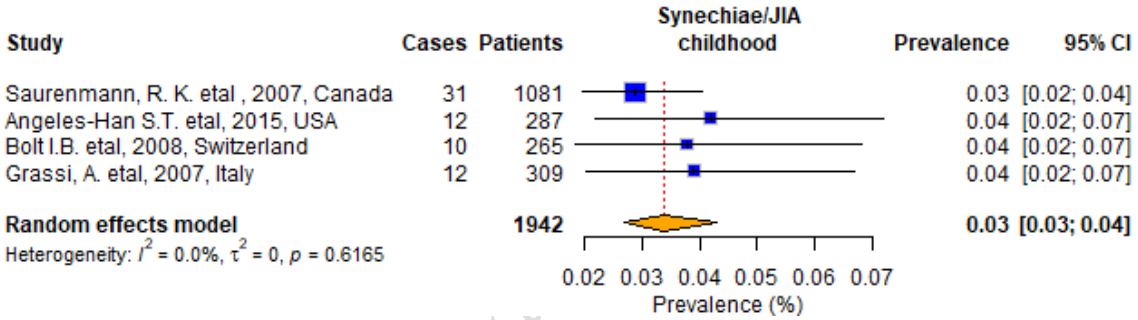
Summary statistics

```
                                        proportion           95%-CI %W(random)
Horneff G. etal, 2022 (WIG2), Germany       0.0385 [0.0240; 0.0582]       28.7
Horneff G. etal, 2022 (InGef), Germany      0.0459 [0.0329; 0.0623]       50.2
Sahin, S. etal, 2021, Turkey                0.0300 [0.0169; 0.0490]       21.1
Number of studies: k = 3; Number of observations: o = 1895; Number of events: e = 75
                     proportion           95%-CI
Random effects model     0.0399 [0.0316; 0.0503]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0034 [0.0000; 1.9016]; tau = 0.0579 [0.0000; 1.3790]
 I^2 = 4.9% [0.0%; 90.1%]; H = 1.03 [1.00; 3.18]
Test of heterogeneity:     Q d.f. p-value
                        2.10    2  0.3494
```

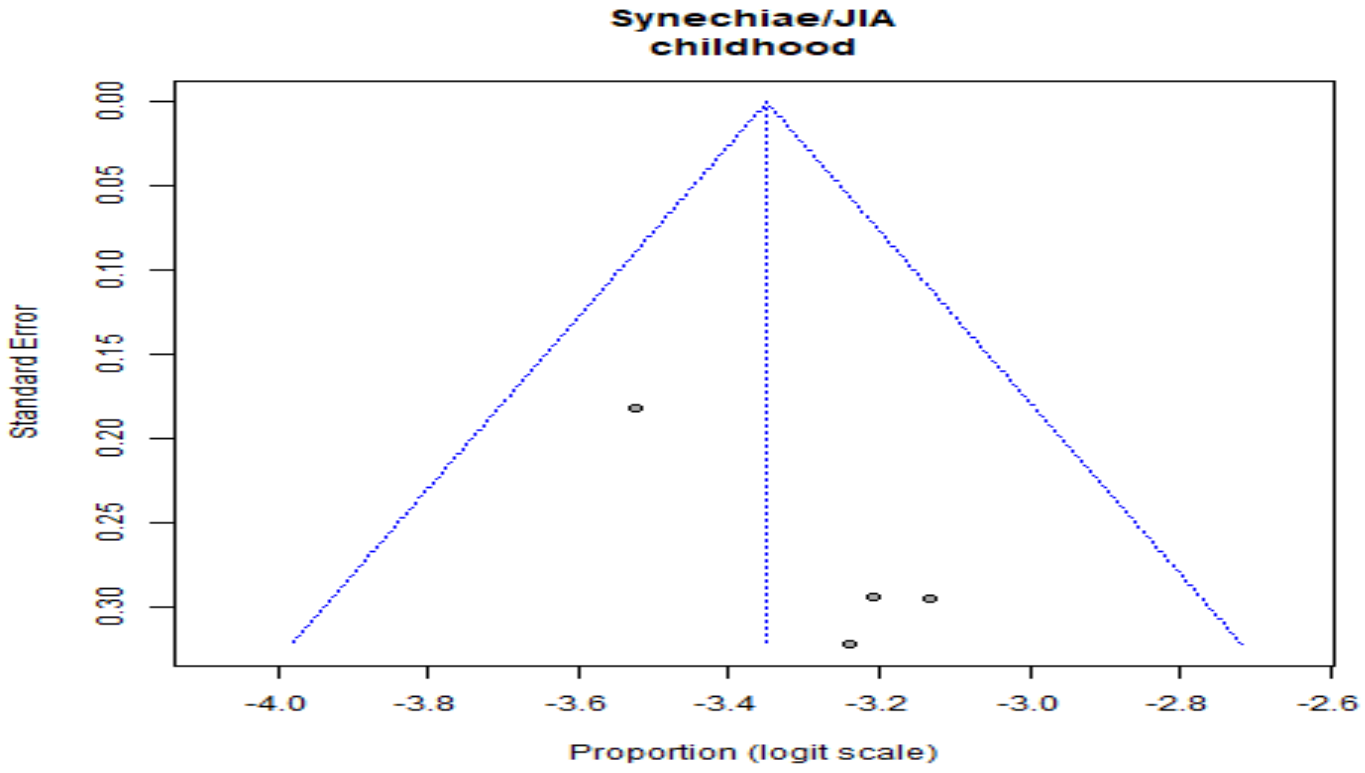
**Supplementary Figure S4.20**: Forest and Funnel Plots of studies reporting the prevalence of synechiae in children with Juvenile Idiopathic Arthritis

**Synechiae**

Forest




Funnel



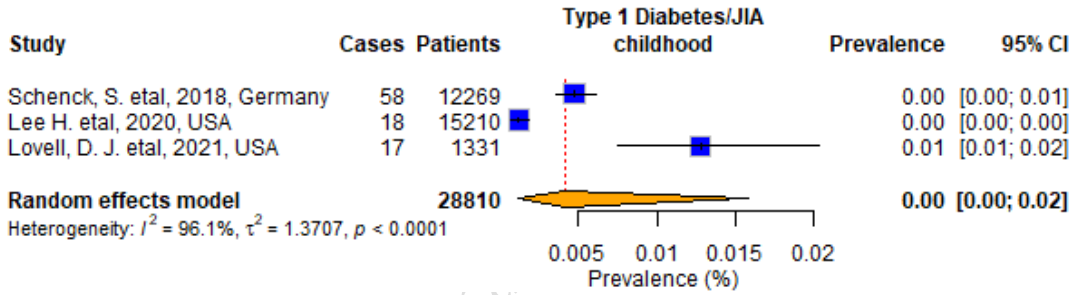
Summary statistics

```
                                    proportion           95%-CI %W(random)
Saurenmann, R. K. etal , 2007, Canada     0.0287 [0.0196; 0.0405]       48.0
Angeles-Han S.T. etal, 2015, USA          0.0418 [0.0218; 0.0719]       18.3
Bolt I.B. etal, 2008, Switzerland         0.0377 [0.0182; 0.0683]       15.3
Grassi, A. etal, 2007, Italy              0.0388 [0.0202; 0.0669]       18.4
Number of studies: k = 4; Number of observations: o = 1942; Number of events: e = 65
                     proportion           95%-CI
Random effects model     0.0339 [0.0267; 0.0430]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.3558]; tau = 0 [0.0000; 0.5965]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.79    3  0.6165
```

**Supplementary Figure S4.21**: Forest and Funnel Plots of studies reporting the prevalence of Type 1 diabetes in children with Juvenile Idiopathic Arthritis

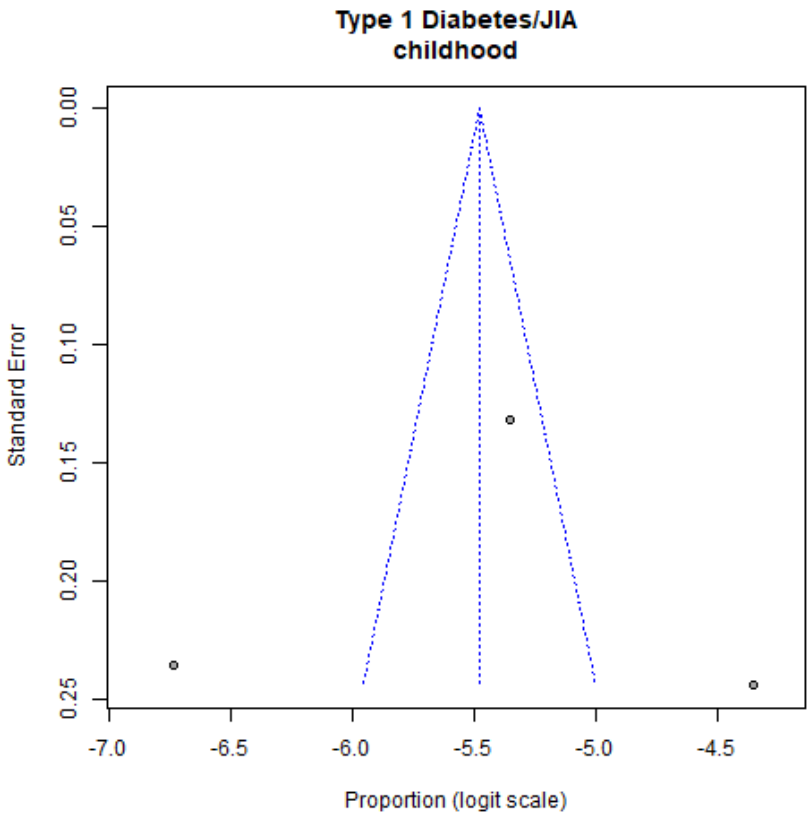
**Type 1 Diabetes**

Forest




Funnel

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Summary statistics

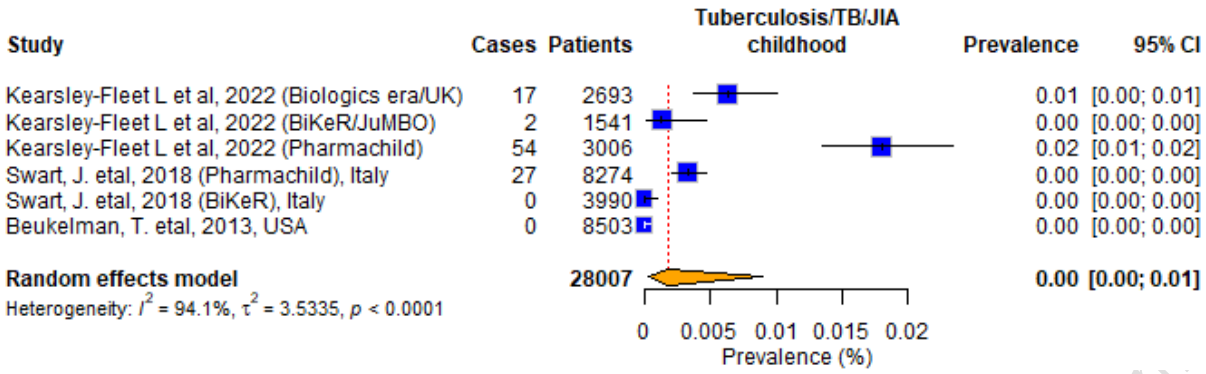
```
                                proportion           95%-CI %W(random)
Schenck, S. etal, 2018, Germany     0.0047 [0.0036; 0.0061]       34.0
Lee H. etal, 2020, USA              0.0012 [0.0007; 0.0019]       33.1
Lovell, D. J. etal, 2021, USA       0.0128 [0.0075; 0.0204]       33.0
Number of studies: k = 3; Number of observations: o = 28810; Number of events: e = 93
                     proportion           95%-CI
Random effects model     0.0042 [0.0011; 0.0158]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3707 [0.3335; 56.8637]; tau = 1.1708 [0.5775; 7.5408]
 I^2 = 96.1% [91.6%; 98.2%]; H = 5.04 [3.45; 7.37]
Test of heterogeneity:      Q d.f.  p-value
                         50.80    2 < 0.0001
```

**Supplementary Figure S4.22**: Forest and Funnel Plots of studies reporting the prevalence of tuberculosis in children with Juvenile Idiopathic Arthritis

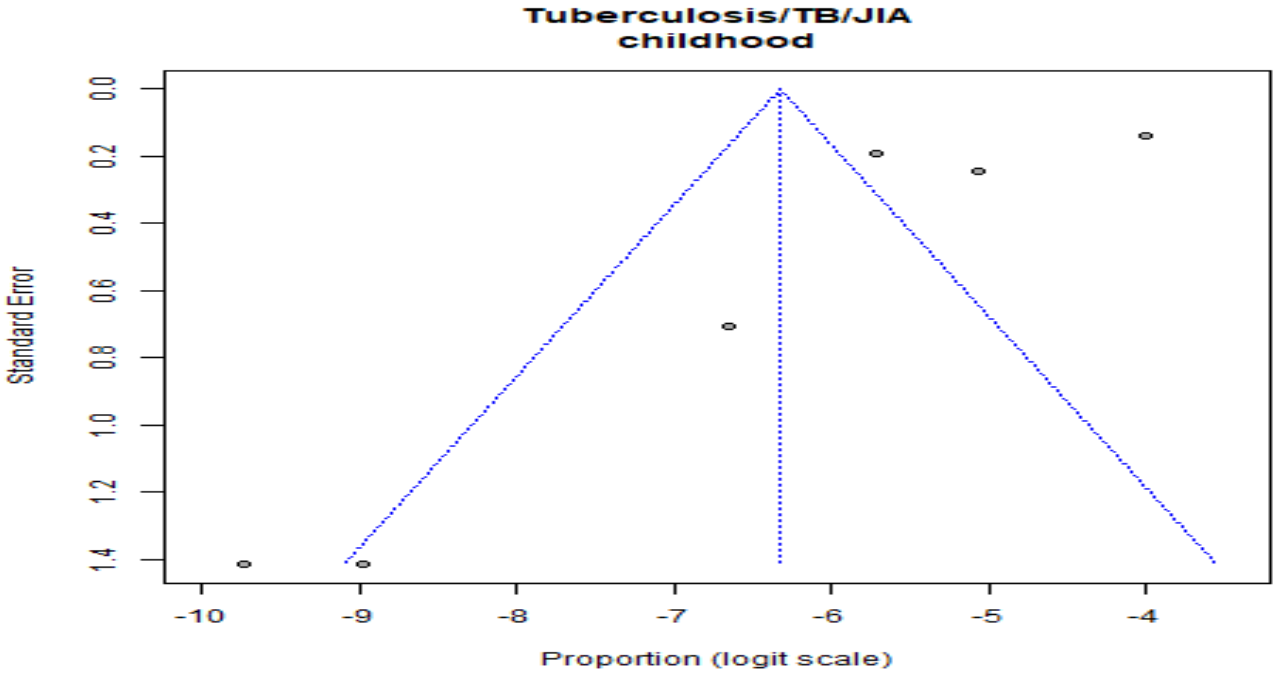
**Tuberculosis/TB**

Forest

=For reproducibility, academic research and development=




Funnel



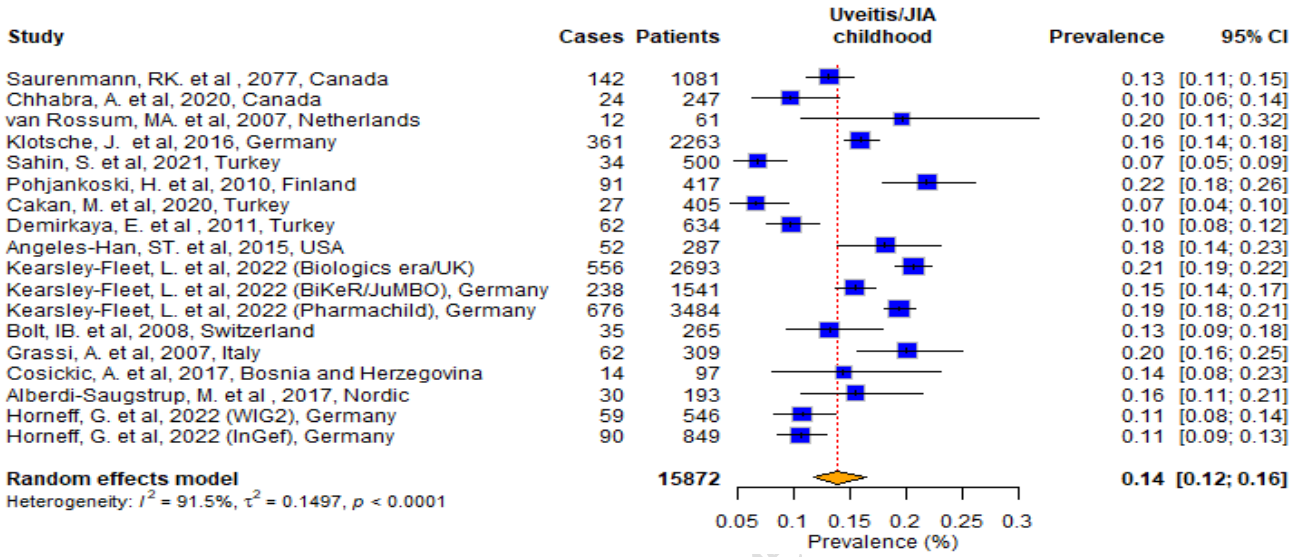

Summary statistics

```
                                          proportion           95%-CI %W(random)
Kearsley-Fleet L et al, 2022 (Biologics era/UK)     0.0063 [0.0037; 0.0101]       19.2
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)          0.0013 [0.0002; 0.0047]       17.1
Kearsley-Fleet L et al, 2022 (Pharmachild)          0.0180 [0.0135; 0.0234]       19.4
Swart, J. etal, 2018 (Pharmachild), Italy           0.0033 [0.0022; 0.0047]       19.3
Swart, J. etal, 2018 (BiKeR), Italy                 0.0000 [0.0000; 0.0009]       12.5
Beukelman, T. etal, 2013, USA                       0.0000 [0.0000; 0.0004]       12.5
Number of studies: k = 6; Number of observations: o = 28007; Number of events: e = 100
                     proportion           95%-CI
Random effects model     0.0018 [0.0004; 0.0090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.5335 [0.9812; 29.4146]; tau = 1.8797 [0.9905; 5.4235]
 I^2 = 94.1% [89.8%; 96.6%]; H = 4.13 [3.13; 5.45]
Test of heterogeneity:      Q d.f.  p-value
                        85.36    5 < 0.0001
```

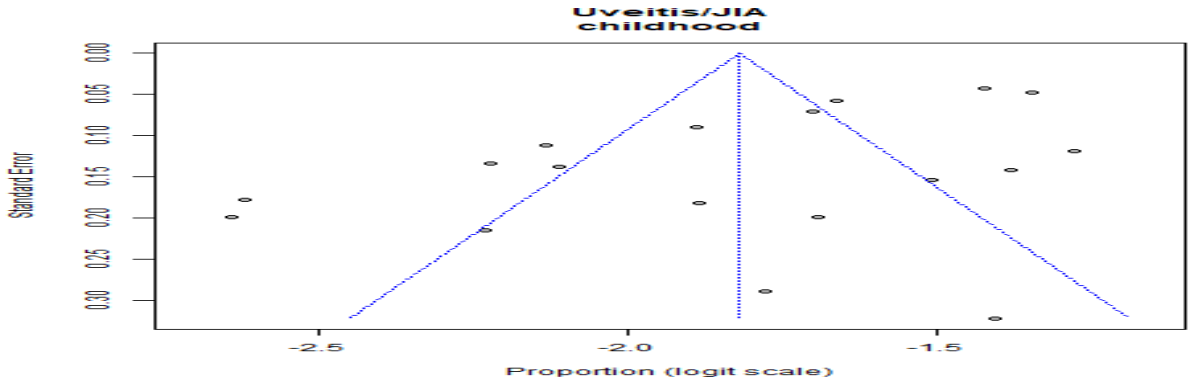
**Supplementary Figure S4.23**: Forest and Funnel Plots of studies reporting the prevalence of uveitis in children with Juvenile Idiopathic Arthritis

**Uveitis**

Forest




Funnel



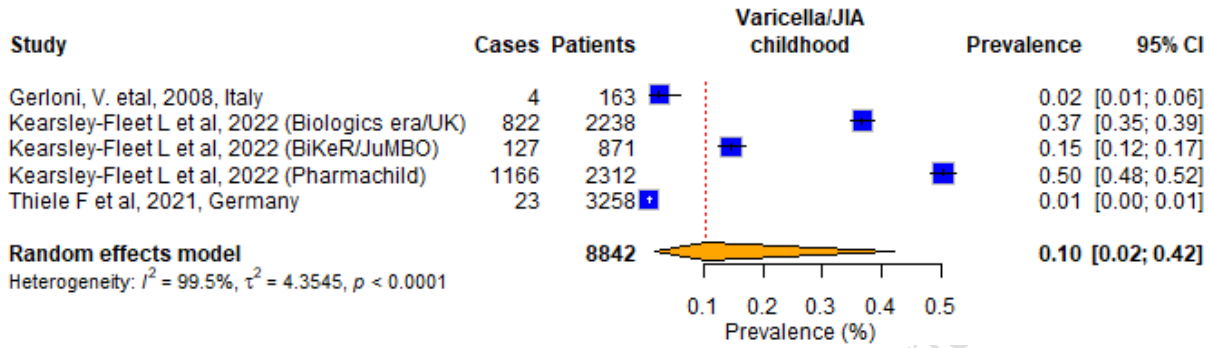

Summary statistics

```
                                                    proportion           95%-CI %W(random)
Saurenmann, RK. et al , 2077, Canada                    0.1314 [0.1118; 0.1530]        6.1
Chhabra, A. et al, 2020, Canada                         0.0972 [0.0633; 0.1411]        4.9
van Rossum, MA. et al, 2007, Netherlands                0.1967 [0.1060; 0.3184]        3.8
Klotsche, J.  et al, 2016, Germany                      0.1595 [0.1447; 0.1753]        6.3
Sahin, S. et al, 2021, Turkey                           0.0680 [0.0475; 0.0937]        5.3
Pohjankoski, H. et al, 2010, Finland                    0.2182 [0.1795; 0.2610]        5.9
Cakan, M. et al, 2020, Turkey                           0.0667 [0.0444; 0.0955]        5.1
Demirkaya, E. et al , 2011, Turkey                      0.0978 [0.0758; 0.1236]        5.8
Angeles-Han, ST. et al, 2015, USA                       0.1812 [0.1384; 0.2307]        5.6
Kearsley-Fleet, L. et al, 2022 (Biologics era/UK)       0.2065 [0.1913; 0.2222]        6.4
Kearsley-Fleet, L. et al, 2022 (BiKeR/JuMBO), Germany   0.1544 [0.1367; 0.1735]        6.3
Kearsley-Fleet, L. et al, 2022 (Pharmachild), Germany   0.1940 [0.1810; 0.2076]        6.4
Bolt, IB. et al, 2008, Switzerland                      0.1321 [0.0937; 0.1789]        5.3
Grassi, A. et al, 2007, Italy                           0.2006 [0.1574; 0.2497]        5.7
Cosickic, A. et al, 2017, Bosnia and Herzegovina        0.1443 [0.0812; 0.2303]        4.2
Alberdi-Saugstrup, M. et al , 2017, Nordic              0.1554 [0.1074; 0.2144]        5.1
Horneff, G. et al, 2022 (WIG2), Germany                 0.1081 [0.0833; 0.1372]        5.7
Horneff, G. et al, 2022 (InGef), Germany                0.1060 [0.0861; 0.1287]        6.0
Number of studies: k = 18; Number of observations: o = 15872; Number of events: e = 2565
                        proportion           95%-CI
Random effects model        0.1395 [0.1179; 0.1643]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1497 [0.0741; 0.3706]; tau = 0.3870 [0.2721; 0.6088]
 I^2 = 91.5% [88.0%; 93.9%]; H = 3.42 [2.89; 4.05]
Test of heterogeneity:       Q d.f.  p-value
                      | 198.93    17 < 0.0001
```

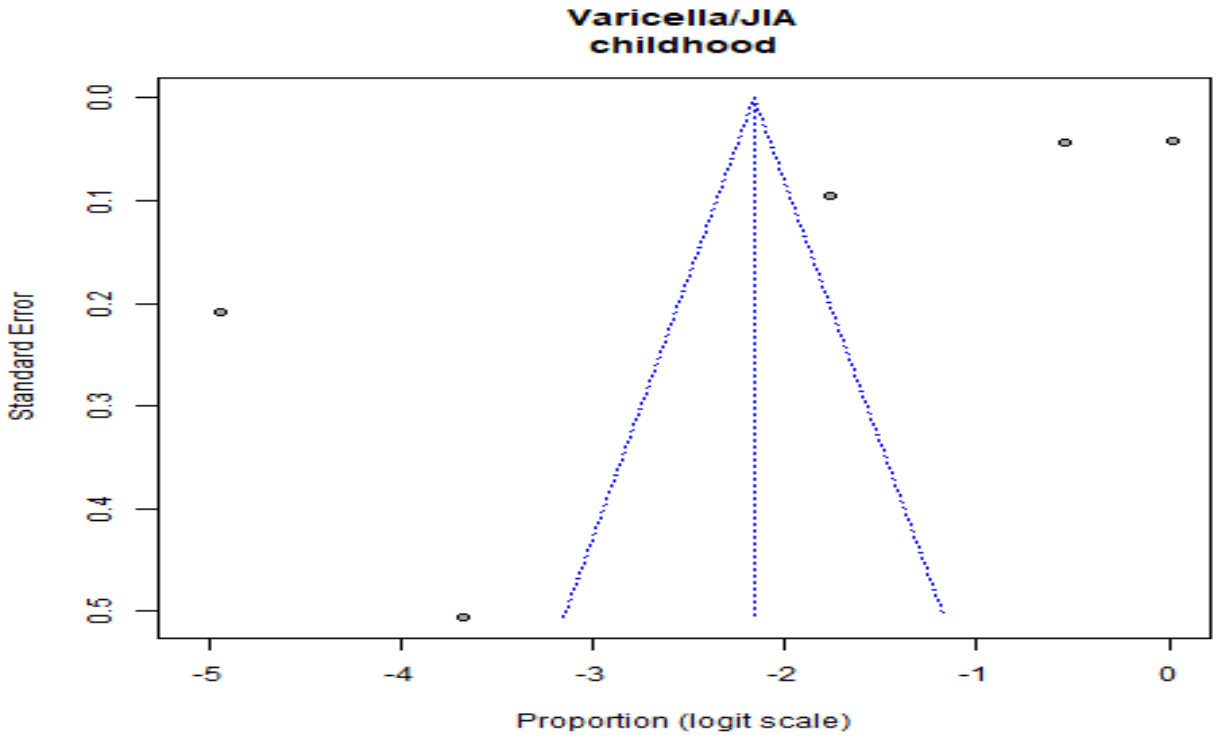
**Supplementary Figure S4.24**: Forest and Funnel Plots of studies reporting the prevalence of varicella in children with Juvenile Idiopathic Arthritis

**Varicella**

Forest




Funnel




Summary statistics

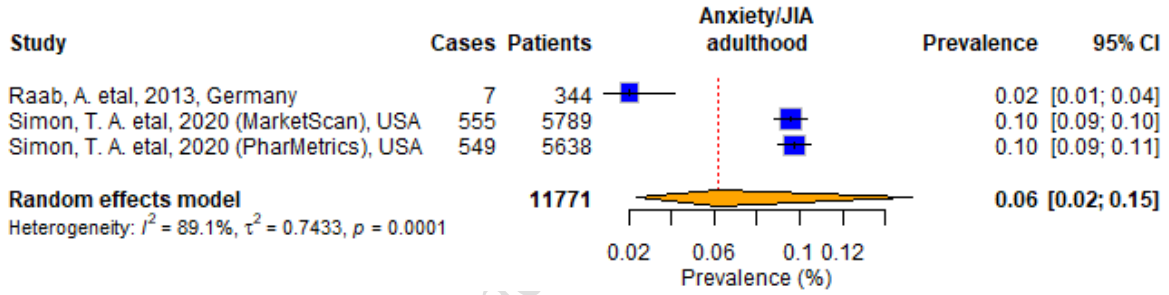
```
                                                proportion           95%-CI %W(random)
Gerloni, V. etal, 2008, Italy                       0.0245 [0.0067; 0.0616]       19.2
Kearsley-Fleet L et al, 2022 (Biologics era/UK)     0.3673 [0.3473; 0.3877]       20.3
Kearsley-Fleet L et al, 2022 (BiKeR/JuMBO)          0.1458 [0.1230; 0.1710]       20.2
Kearsley-Fleet L et al, 2022 (Pharmachild)          0.5043 [0.4837; 0.5249]       20.3
Thiele F et al, 2021, Germany                       0.0071 [0.0045; 0.0106]       20.1
Number of studies: k = 5; Number of observations: o = 8842; Number of events: e = 2142
                     proportion           95%-CI
Random effects model     0.1031 [0.0179; 0.4204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 4.3545 [1.5277; 36.2489]; tau = 2.0868 [1.2360; 6.0207]
 I^2 = 99.5% [99.4%; 99.6%]; H = 14.32 [12.48; 16.44]
Test of heterogeneity:       Q d.f.  p-value
                        820.56    4 < 0.0001
```

**Adulthood/JIA**

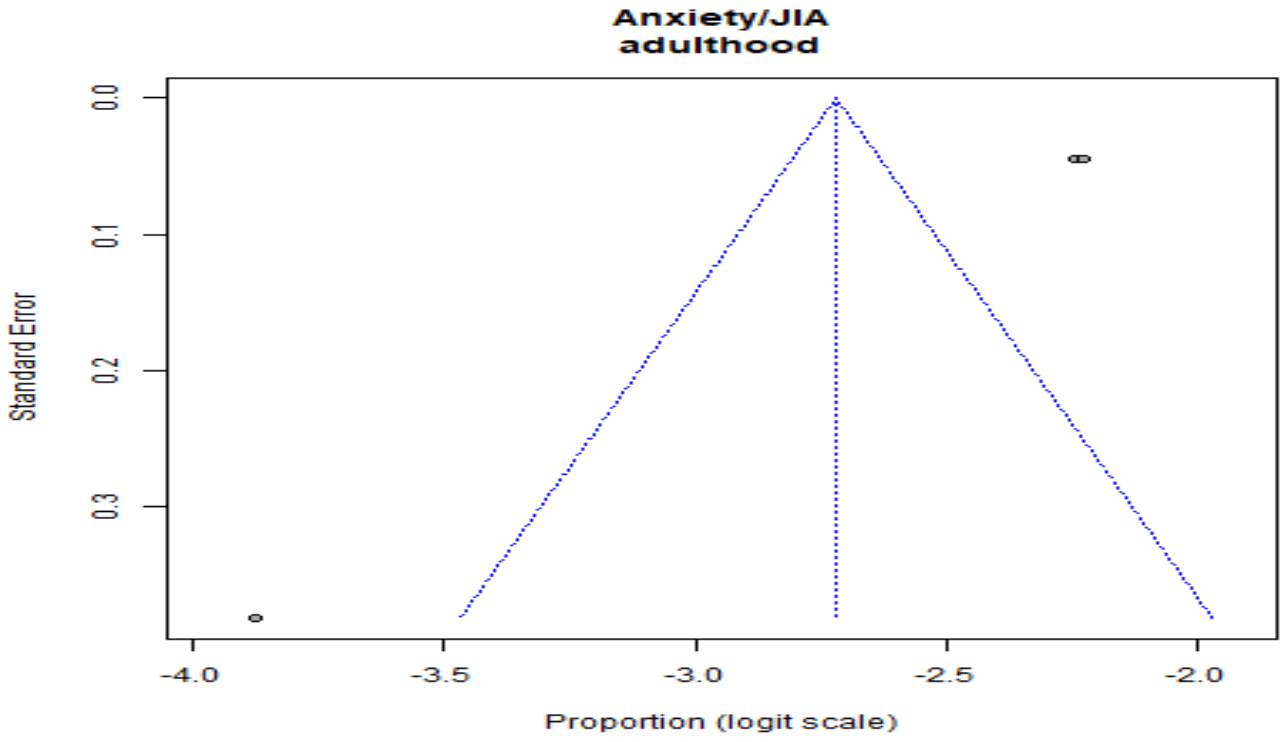
**Supplementary Figure S4.25**: Forest and Funnel Plots of studies reporting the prevalence of anxiety in adults with Juvenile Idiopathic Arthritis

**Anxiety**

Forest




Funnel



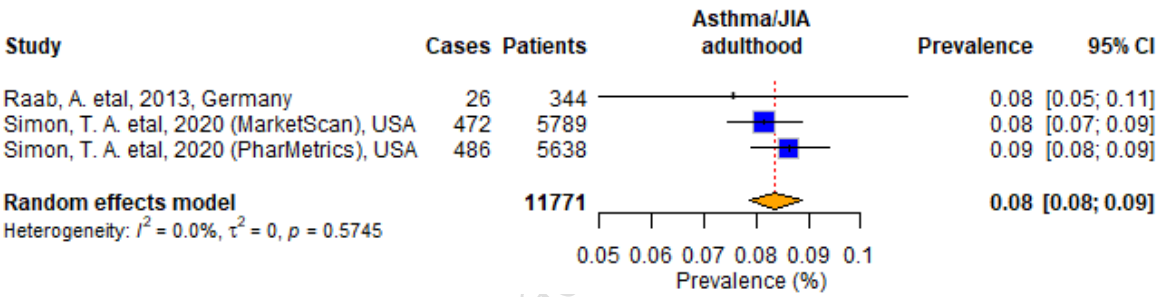
Summary statistics

```
                                          proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                  0.0203 [0.0082; 0.0415]       29.5
Simon, T. A. etal, 2020 (MarketScan), USA     0.0959 [0.0884; 0.1037]       35.2
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0974 [0.0898; 0.1054]       35.2
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 1111
                    proportion           95%-CI
Random effects model     0.0618 [0.0236; 0.1525]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7433 [0.1448; 35.2654]; tau = 0.8622 [0.3806; 5.9385]
 I^2 = 89.1% [70.3%; 96.0%]; H = 3.03 [1.83; 5.01]
Test of heterogeneity:      Q d.f. p-value
                        18.36    2  0.0001
```

**Supplementary Figure S4.26**: Forest and Funnel Plots of studies reporting the prevalence of asthma in adults with Juvenile Idiopathic Arthritis

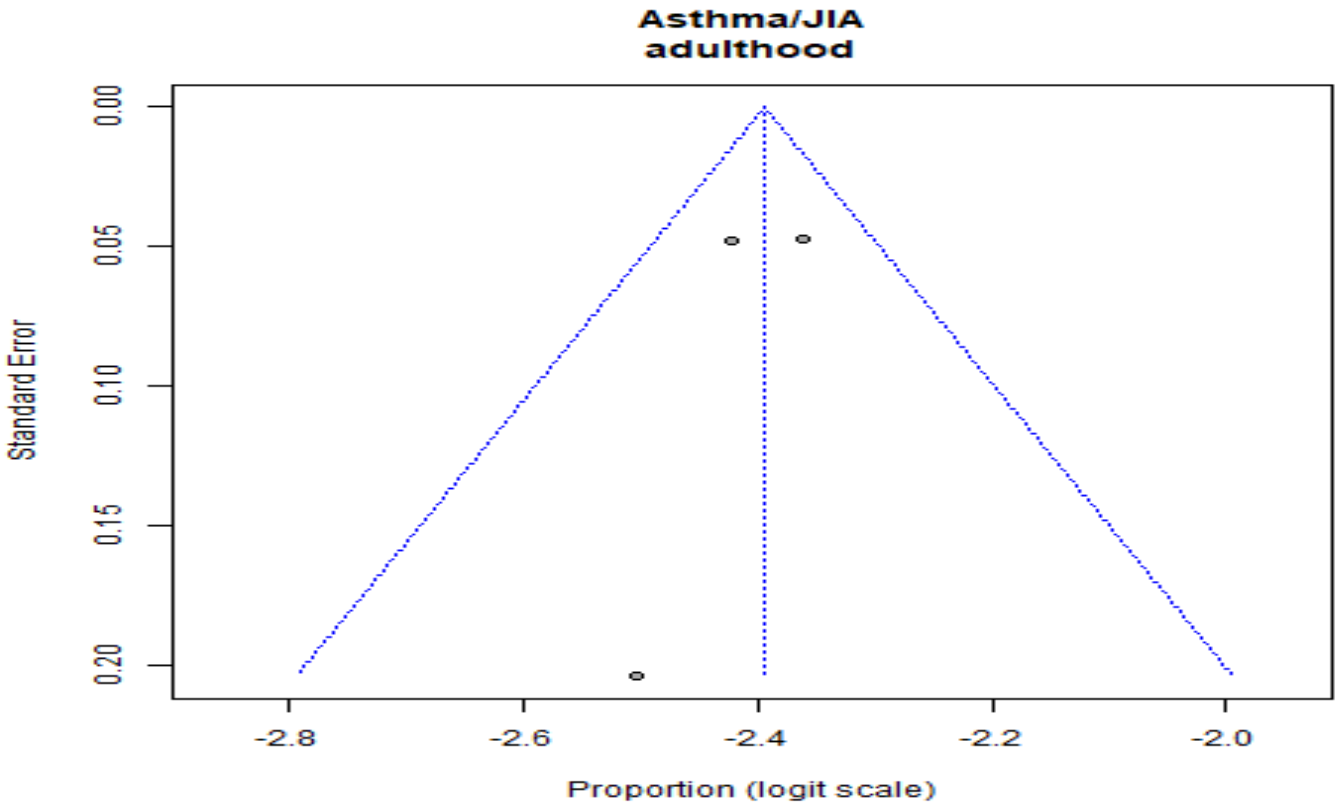
**Asthma**

Forest




Funnel

For reproducibility only in aca nd developme



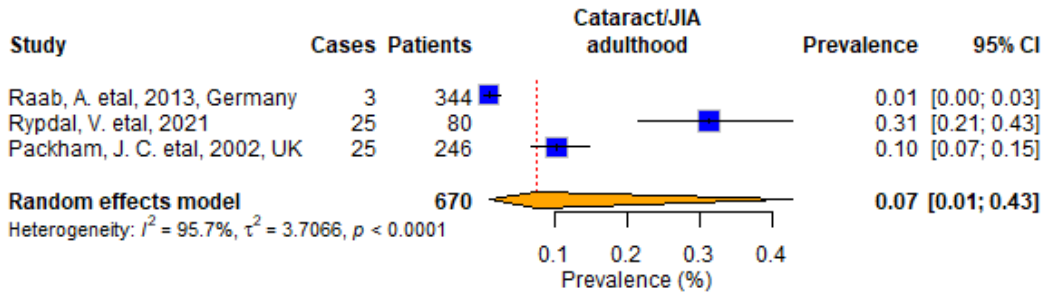

Summary statistics

```
                                          proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                  0.0756 [0.0500; 0.1088]        2.7
Simon, T. A. etal, 2020 (MarketScan), USA     0.0815 [0.0746; 0.0889]       48.1
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0862 [0.0790; 0.0938]       49.3
Number of studies: k = 3; Number of observations: o = 11771; Number of events: e = 984
                     proportion           95%-CI
Random effects model     0.0836 [0.0788; 0.0888]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 0.1803]; tau = 0 [0.0000; 0.4246]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        | 1.11    2  0.5745
```

**Supplementary Figure S4.27**: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with Juvenile Idiopathic Arthritis

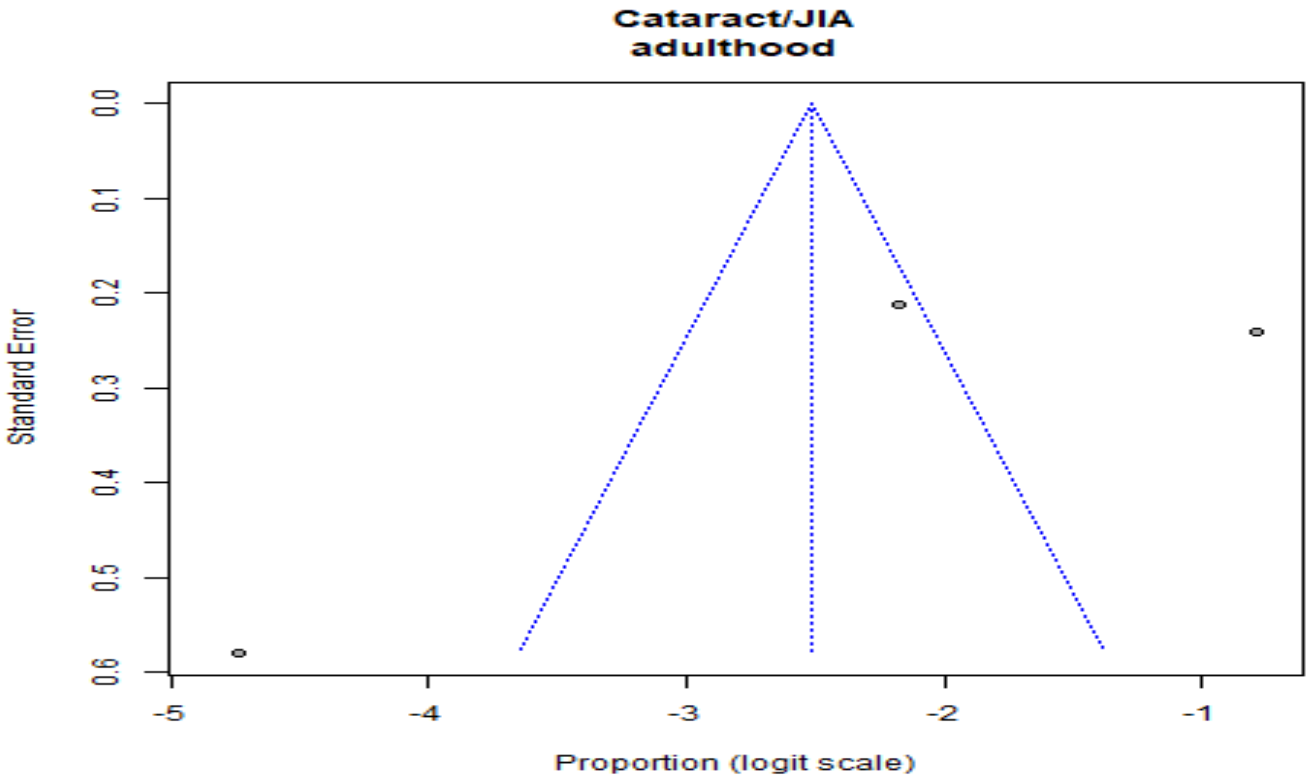
**Cataract**

Forest




reproduction

Funnel




Summary statistics

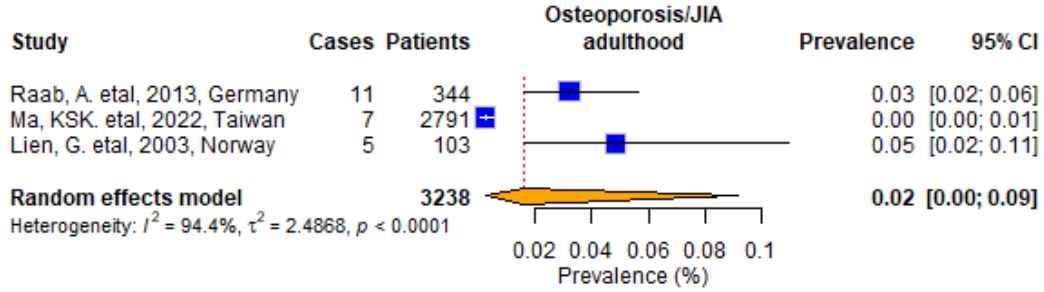
```
                              proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany      0.0087 [0.0018; 0.0253]       31.7
Rypdal, V. etal, 2021             0.3125 [0.2135; 0.4259]       34.1
Packham, J. C. etal, 2002, UK     0.1016 [0.0669; 0.1464]       34.2
Number of studies: k = 3
Number of observations: o = 670
Number of events: e = 53
                     proportion           95%-CI
Random effects model     0.0748 [0.0087; 0.4266]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 3.7066 [0.8680; >100.0000]; tau = 1.9252 [0.9316; >10.0000]
 I^2 = 95.7% [90.6%; 98.0%]; H = 4.82 [3.26; 7.11]
Test of heterogeneity:      Q d.f.  p-value
                        46.40    2 < 0.0001
```

**Supplementary Figure S4.28**: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in adults with Juvenile Idiopathic Arthritis

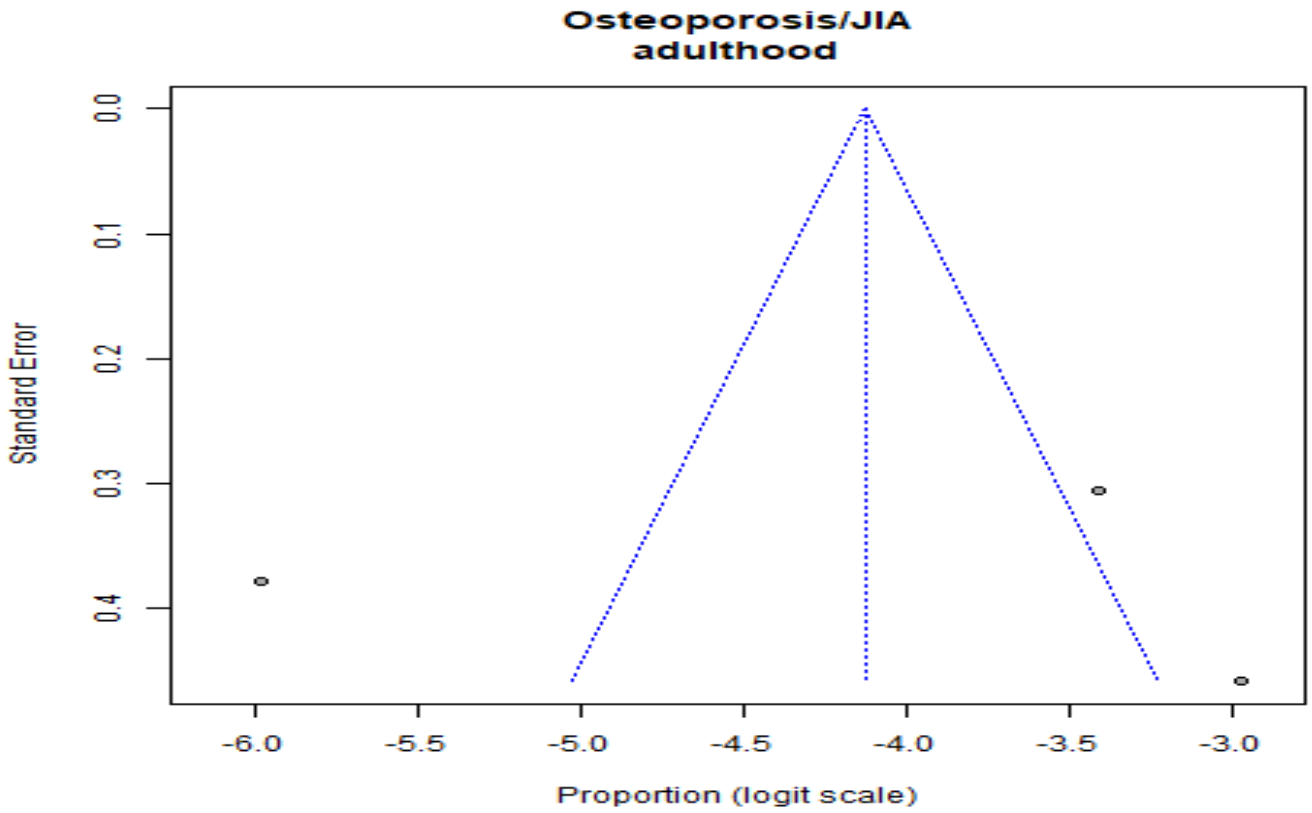
**Osteoporosis**

Forest

For reproduction academic




Funnel



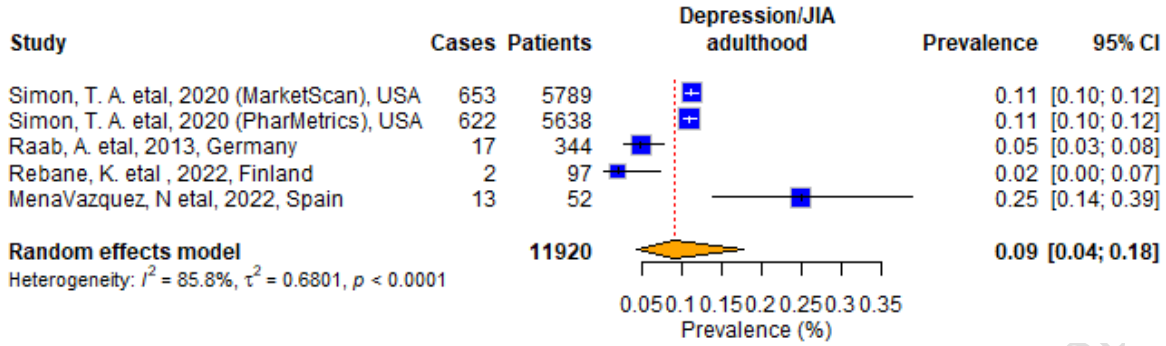

Summary statistics

```
                              proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany     0.0320 [0.0161; 0.0565]       34.0
Ma, KSK. etal, 2022, Taiwan      0.0025 [0.0010; 0.0052]       33.4
Lien, G. etal, 2003, Norway      0.0485 [0.0159; 0.1097]       32.6
Number of studies: k = 3; Number of observations: o = 3238; Number of events: e = 23
                    proportion           95%-CI
Random effects model     0.0158 [0.0026; 0.0918]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 2.4868 [0.5640; >100.0000]; tau = 1.5770 [0.7510; >10.0000]
 I^2 = 94.4% [87.1%; 97.6%]; H = 4.24 [2.78; 6.45]
Test of heterogeneity:      Q d.f.  p-value
                        35.90    2 < 0.0001
```

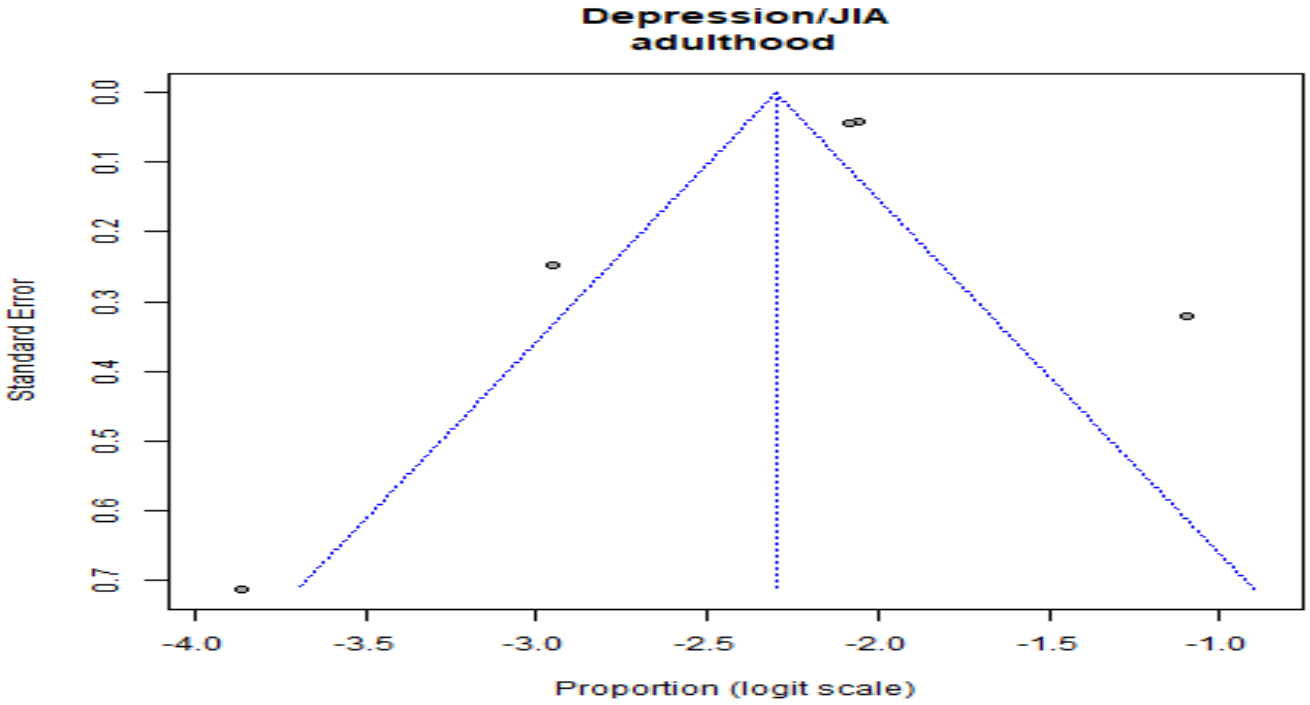
**Supplementary Figure S4.29**: Forest and Funnel Plots of studies reporting the prevalence of depression in adults with Juvenile Idiopathic Arthritis

**Depression**

Forest




Funnel



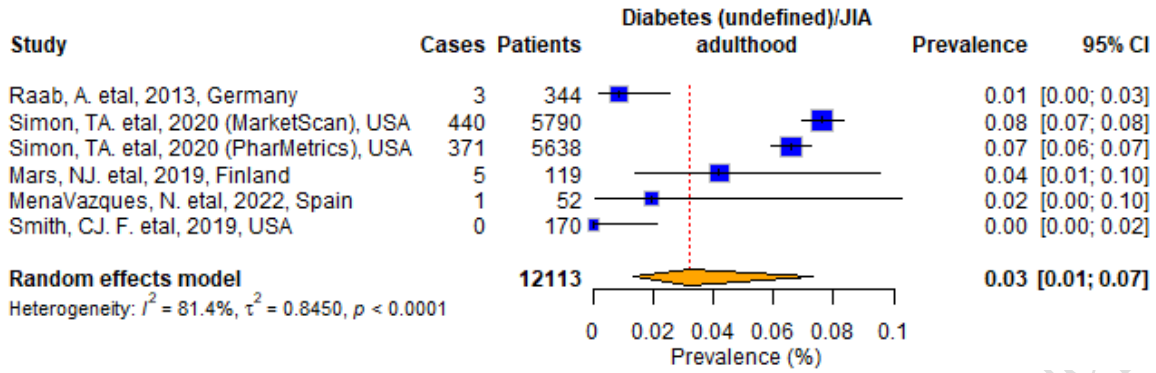

Summary statistics

```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.1128 [0.1048; 0.1212]       22.9
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1103 [0.1023; 0.1188]       22.9
Raab, A. etal, 2013, Germany                  0.0494 [0.0290; 0.0779]       21.1
Rebane, K. etal , 2022, Finland               0.0206 [0.0025; 0.0725]       13.1
MenaVazquez, N etal, 2022, Spain              0.2500 [0.1403; 0.3895]       20.0
Number of studies: k = 5; Number of observations: o = 11920; Number of events: e = 1307
                    proportion           95%-CI
Random effects model     0.0911 [0.0442; 0.1787]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6801 [0.1703; 8.6878]; tau = 0.8247 [0.4127; 2.9475]
 I^2 = 85.8% [68.9%; 93.5%]; H = 2.66 [1.79; 3.94]
Test of heterogeneity:      Q d.f.  p-value
                        28.23    4 < 0.0001
```

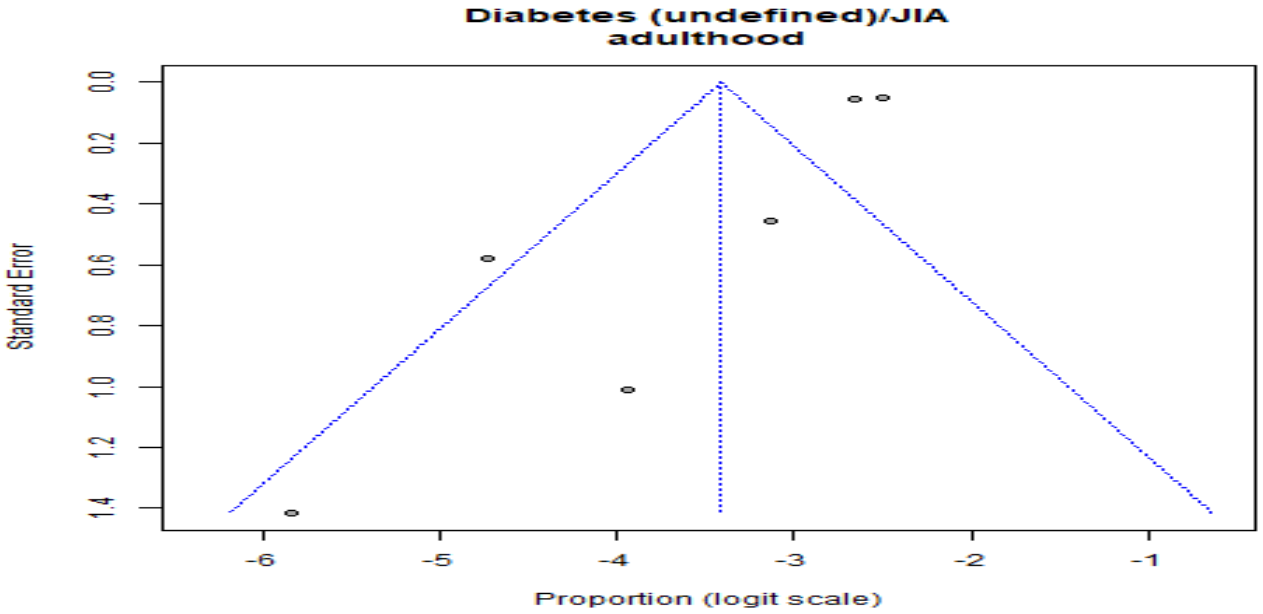
**Supplementary Figure S4.30**: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with Juvenile Idiopathic Arthritis

**Diabetes (undefined)**

Forest




Funnel



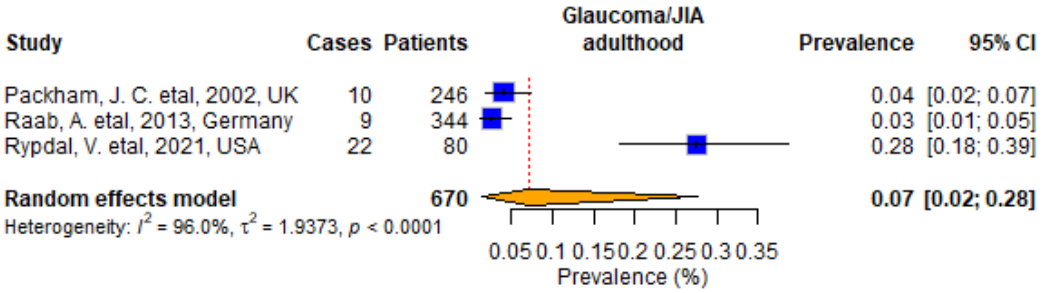

Summary statistics

```
                                        proportion           95%-CI %W(random)
Raab, A. etal, 2013, Germany                0.0087 [0.0018; 0.0253]       16.8
Simon, TA. etal, 2020 (MarketScan), USA     0.0760 [0.0693; 0.0831]       23.4
Simon, TA. etal, 2020 (PharMetrics), USA    0.0658 [0.0595; 0.0726]       23.4
Mars, NJ. etal, 2019, Finland               0.0420 [0.0138; 0.0953]       18.8
MenaVazques, N. etal, 2022, Spain           0.0192 [0.0005; 0.1026]       10.6
Smith, CJ. F. etal, 2019, USA               0.0000 [0.0000; 0.0215]        7.0
Number of studies: k = 6; Number of observations: o = 12113; Number of events: e = 820
                     proportion           95%-CI
Random effects model     0.0319 [0.0136; 0.0731]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.8450 [0.1589; 9.2267]; tau = 0.9192 [0.3986; 3.0375]
 I^2 = 81.4% [60.1%; 91.3%]; H = 2.32 [1.58; 3.39]
Test of heterogeneity:      Q d.f.  p-value
                        26.84    5 < 0.0001
```

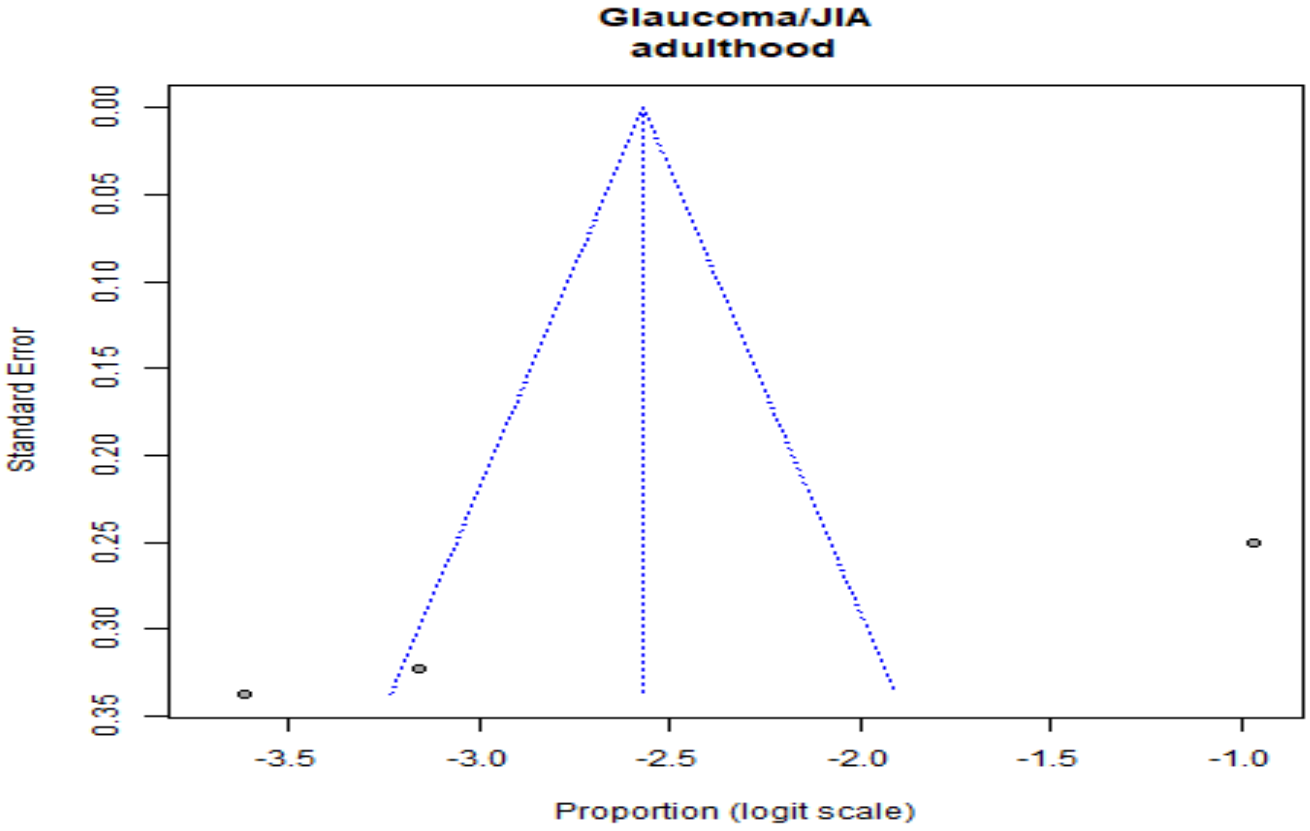
**Supplementary Figure S4.31**: Forest and Funnel Plots of studies reporting the prevalence of glaucoma in adults with Juvenile Idiopathic Arthritis

**Glaucoma**

Forest




Funnel



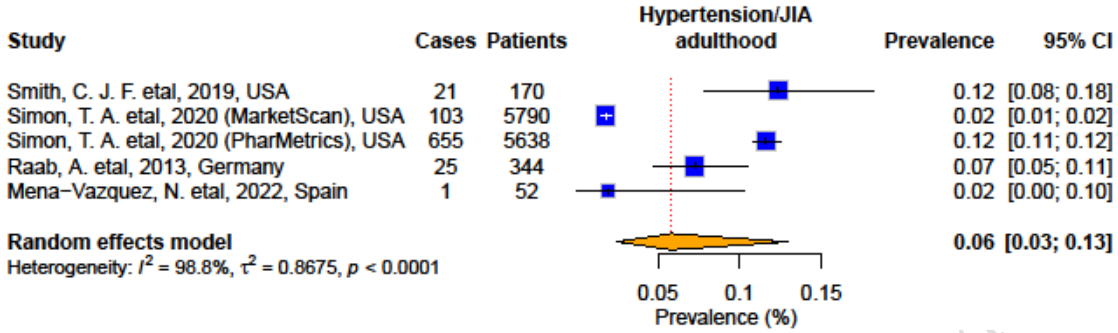

Summary statistics

```
                          proportion           95%-CI %W(random)
Packham, J. C. etal, 2002, UK     0.0407 [0.0197; 0.0735]       33.2
Raab, A. etal, 2013, Germany      0.0262 [0.0120; 0.0491]       33.0
Rypdal, V. etal, 2021, USA        0.2750 [0.1810; 0.3862]       33.8
Number of studies: k = 3; Number of observations: o = 670; Number of events: e = 41
                     proportion           95%-CI
Random effects model     0.0711 [0.0150; 0.2774]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.9373 [0.4633; 79.0548]; tau = 1.3919 [0.6807; 8.8913]
 I^2 = 96.0% [91.5%; 98.1%]; H = 5.02 [3.43; 7.35]
Test of heterogeneity:      Q d.f.  p-value
                        50.50    2 < 0.0001
```

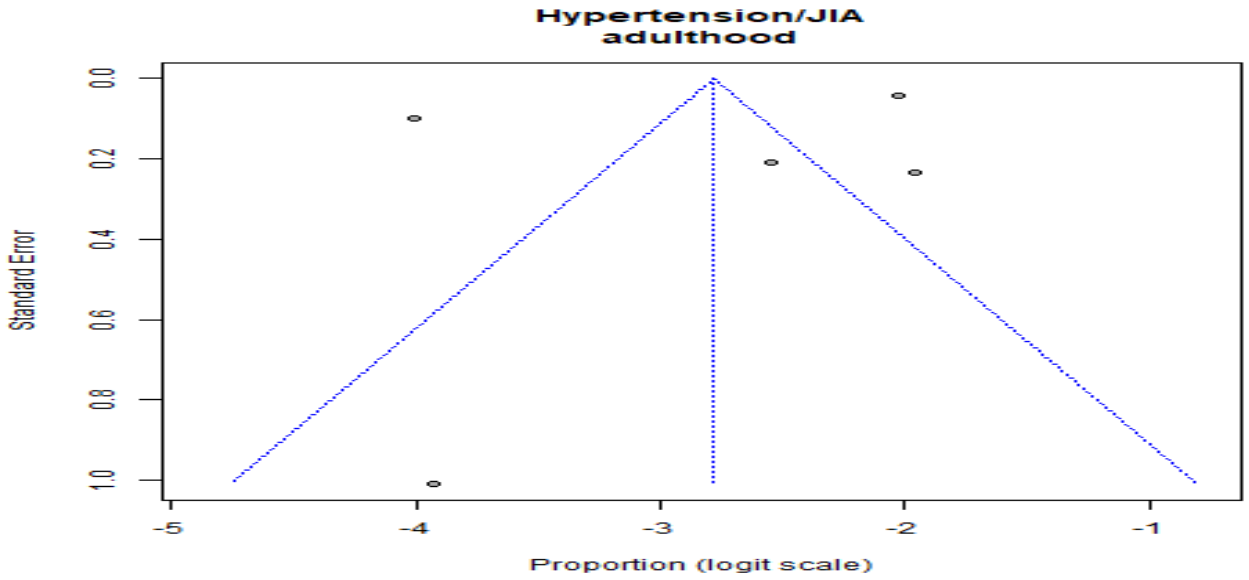
**Supplementary Figure S4.32**: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with Juvenile Idiopathic Arthritis

**Hypertension**

Forest




Funnel




Summary statistics

For reproduction

```
                                          proportion   95% CI                  %W(random)
Smith, C. J. F. etal, 2019, USA               0.1235   [0.0781; 0.1826]        21.7
Simon, T. A. etal, 2020 (MarketScan), USA     0.0178   [0.0145; 0.0215]        22.8
Simon, T. A. etal, 2020 (PharMetrics), USA    0.1162   [0.1079; 0.1248]        23.0
Raab, A. etal, 2013, Germany                  0.0727   [0.0476; 0.1054]        21.9
Mena-Vazquez, N. etal, 2022, Spain            0.0192   [0.0005; 0.1026]        10.6

Number of studies: k = 5
Number of observations: o = 11994
Number of events: e = 805

                     proportion         95% CI
Random effects model     0.0584         [0.0252; 0.1296]

Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.8675 [0.2585; 8.1278]; tau = 0.9314 [0.5084; 2.8509]
 I^2 = 98.8% [98.3%; 99.2%]; H = 9.28 [7.71; 11.17]

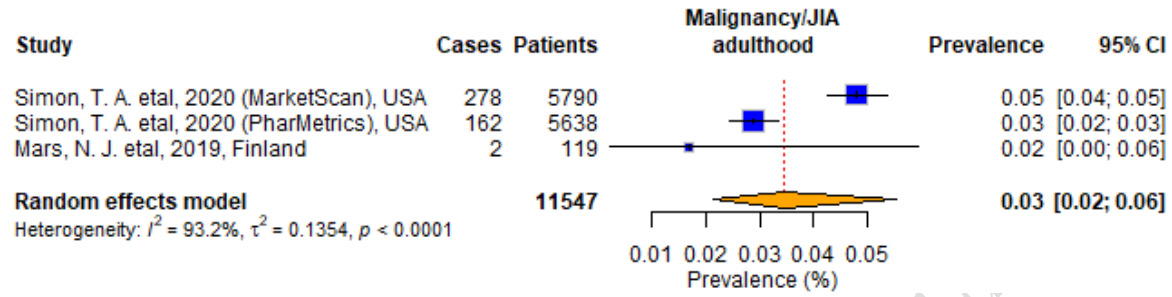
Test of heterogeneity:
      Q d.f.  p-value
 344.45    4 < 0.0001
```

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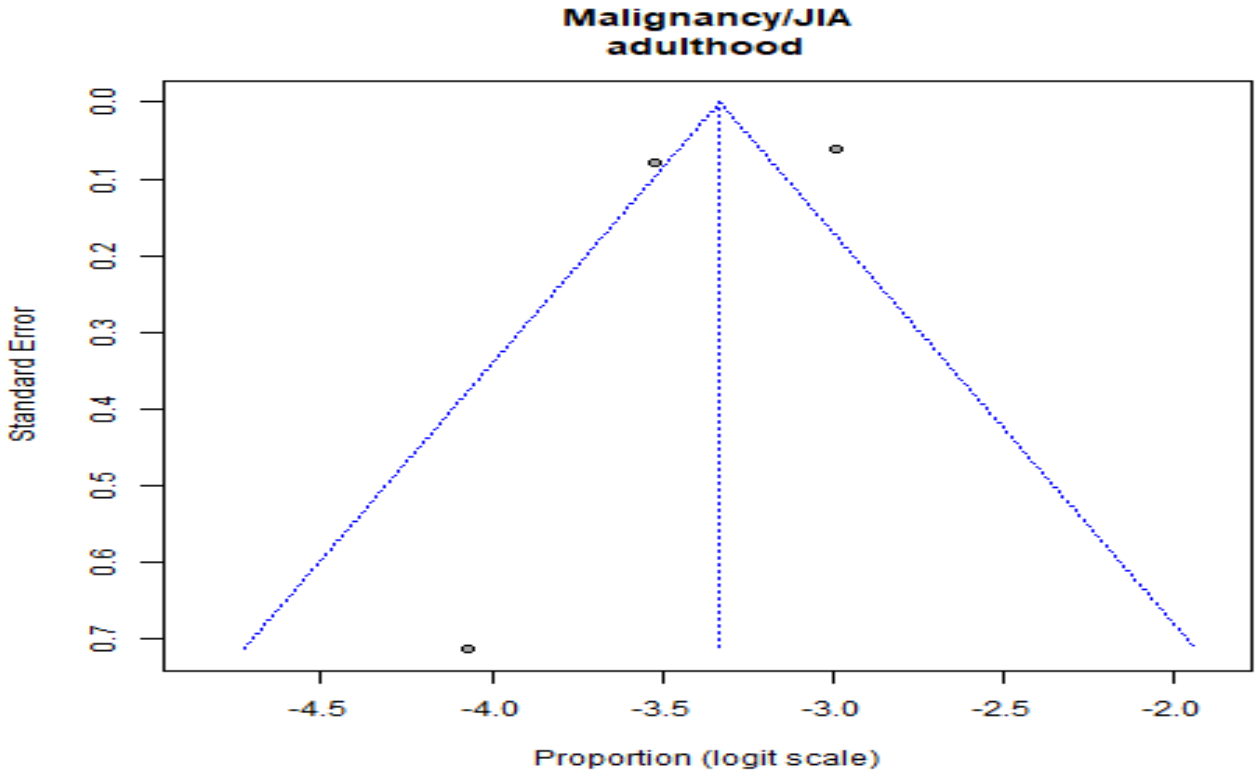
**Supplementary Figure S4.33**: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with Juvenile Idiopathic Arthritis

**Malignancy**

Forest




Funnel




Summary statistics

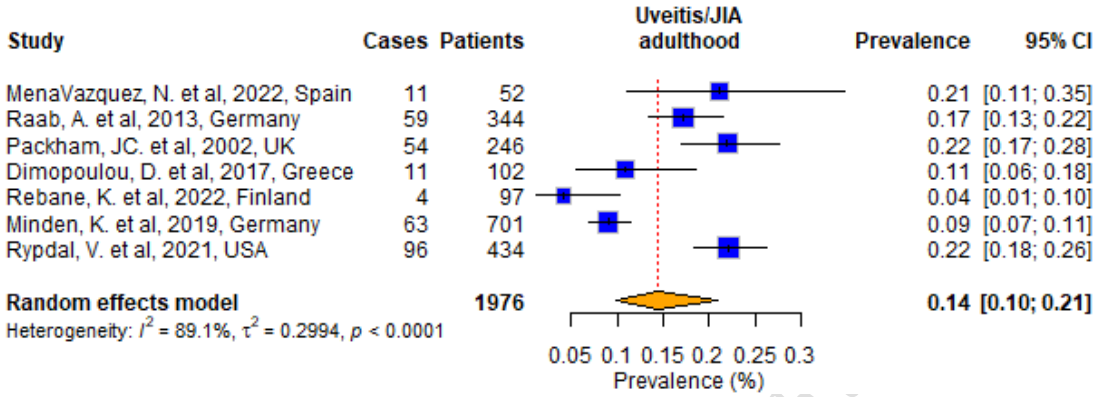
```
                                        proportion           95%-CI %W(random)
Simon, T. A. etal, 2020 (MarketScan), USA     0.0480 [0.0426; 0.0538]       45.5
Simon, T. A. etal, 2020 (PharMetrics), USA    0.0287 [0.0245; 0.0334]       44.7
Mars, N. J. etal, 2019, Finland               0.0168 [0.0020; 0.0594]        9.8
Number of studies: k = 3; Number of observations: o = 11547; Number of events: e = 442
                    proportion           95%-CI
Random effects model     0.0345 [0.0214; 0.0553]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1354 [0.0182; 11.3036]; tau = 0.3679 [0.1350; 3.3621]
 I^2 = 93.2% [83.6%; 97.2%]; H = 3.85 [2.47; 5.99]
Test of heterogeneity:      Q d.f.  p-value
                        29.61    2 < 0.0001
```

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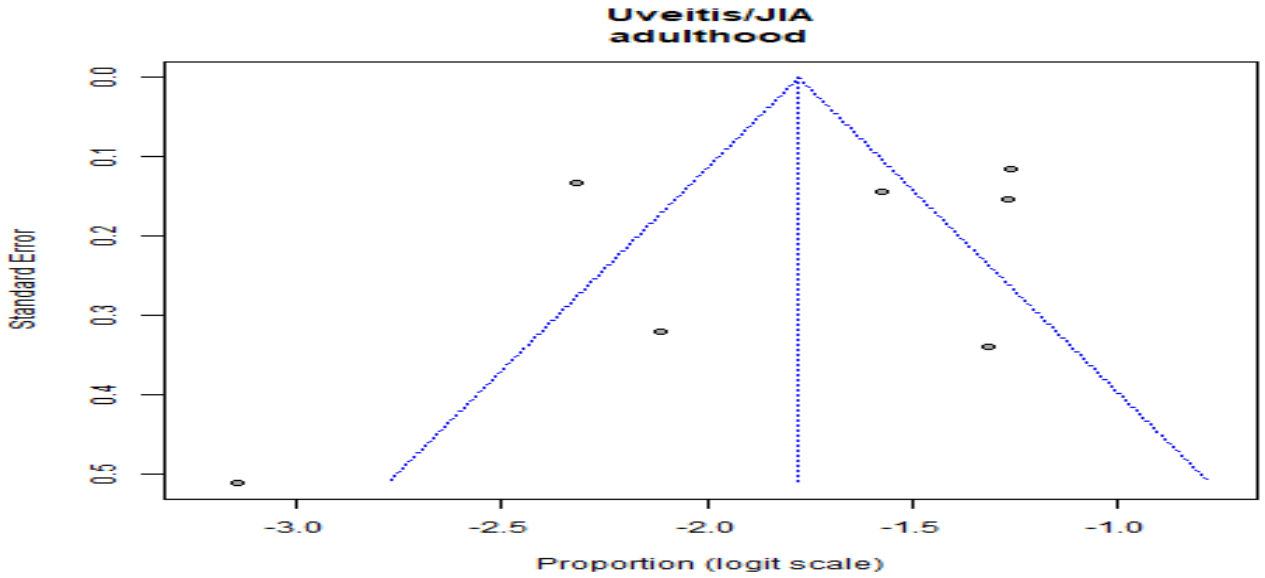
**Supplementary Figure S4.34**: Forest and Funnel Plots of studies reporting the prevalence of uveitis in adults with Juvenile Idiopathic Arthritis

**Uveitis**

Forest




Funnel




Summary statistics

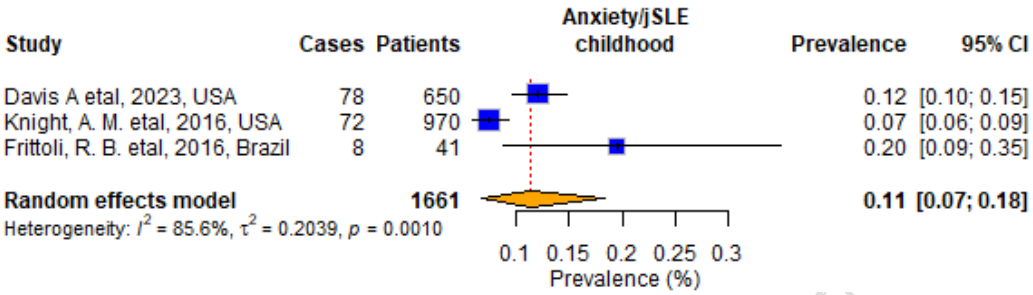
```
                                proportion           95%-CI %W(random)
MenaVazquez, N. et al, 2022, Spain     0.2115 [0.1106; 0.3470]       12.5
Raab, A. et al, 2013, Germany          0.1715 [0.1332; 0.2156]       16.2
Packham, JC. et al, 2002, UK           0.2195 [0.1694; 0.2765]       16.1
Dimopoulou, D. et al, 2017, Greece     0.1078 [0.0551; 0.1848]       12.9
Rebane, K. et al, 2022, Finland        0.0412 [0.0113; 0.1022]        9.3
Minden, K. et al, 2019, Germany        0.0899 [0.0698; 0.1135]       16.4
Rypdal, V. et al, 2021, USA            0.2212 [0.1830; 0.2632]       16.6
Number of studies: k = 7; Number of observations: o = 1976; Number of events: e = 298
                    proportion           95%-CI
Random effects model     0.1446 [0.0976; 0.2090]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2994 [0.0932; 2.2766]; tau = 0.5472 [0.3053; 1.5089]
 I^2 = 89.1% [79.9%; 94.0%]; H = 3.02 [2.23; 4.09]
Test of heterogeneity:      Q d.f.  p-value
                        54.84    6 < 0.0001
```

**Childhood/jSLE**

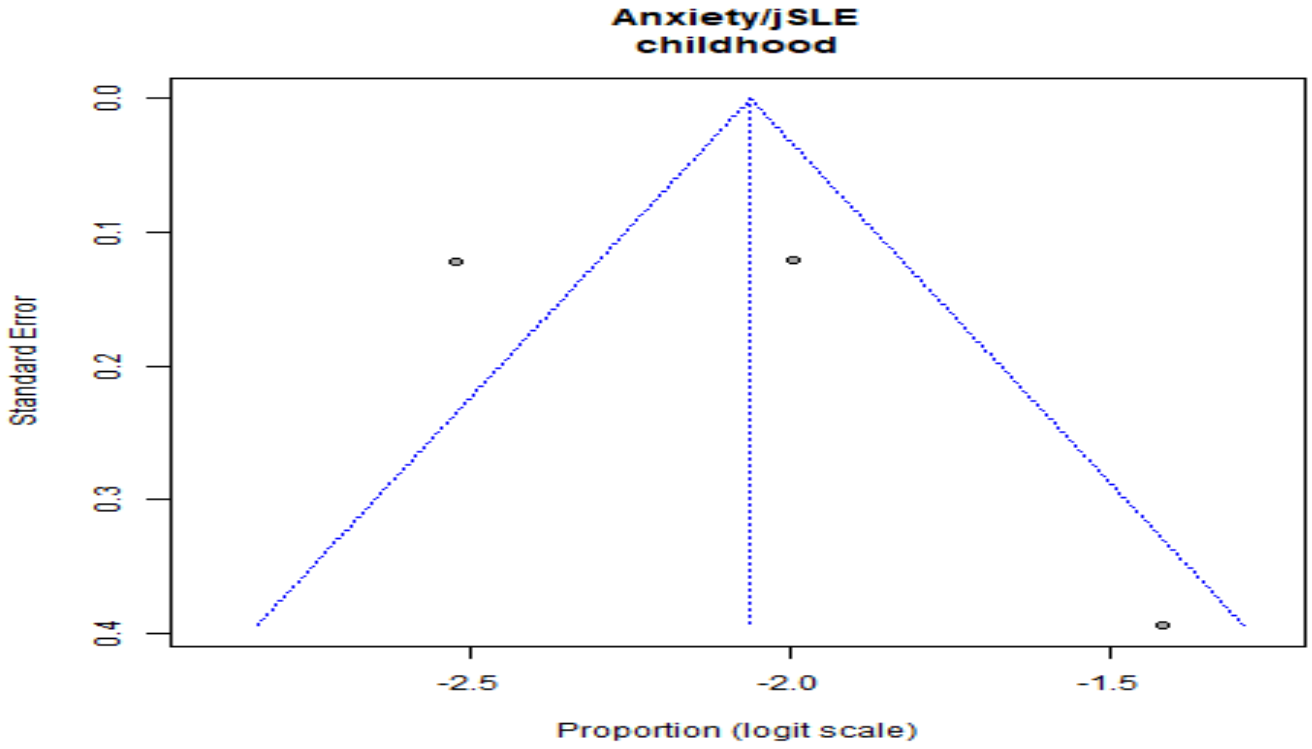
**Supplementary Figure S4.35**: Forest and Funnel Plots of studies reporting the prevalence of anxiety in children with juvenile-onset Systemic Lupus Erythematosus

**Anxiety**

Forest




Funnel



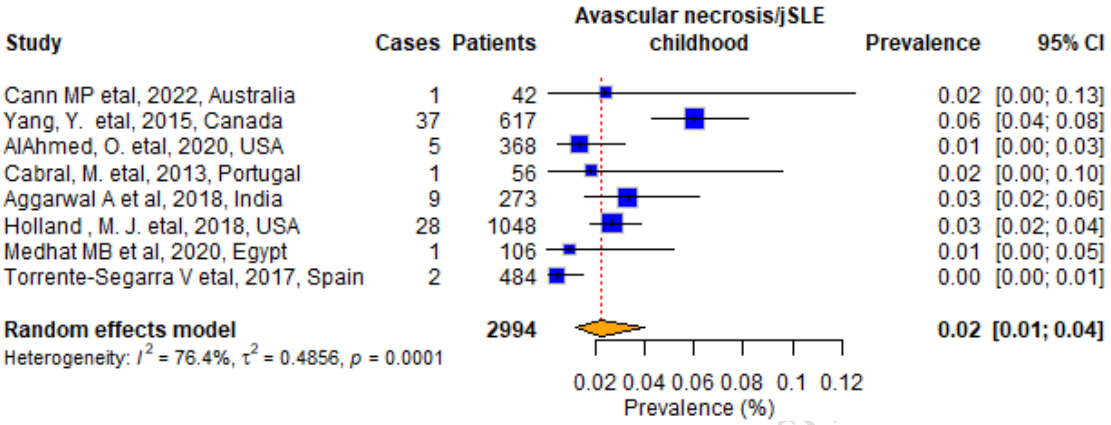

Summary statistics

```
                                proportion           95%-CI %W(random)
Davis A etal, 2023, USA             0.1200 [0.0960; 0.1475]       38.4
Knight, A. M. etal, 2016, USA       0.0742 [0.0585; 0.0926]       38.3
Frittoli, R. B. etal, 2016, Brazil  0.1951 [0.0882; 0.3487]       23.3
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 158
                    proportion           95%-CI
Random effects model     0.1129 [0.0673; 0.1833]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2039 [0.0224; 12.0079]; tau = 0.4515 [0.1498; 3.4652]
 I^2 = 85.6% [57.7%; 95.1%]; H = 2.63 [1.54; 4.50]
Test of heterogeneity:      Q d.f. p-value
                        13.84    2  0.0010
```

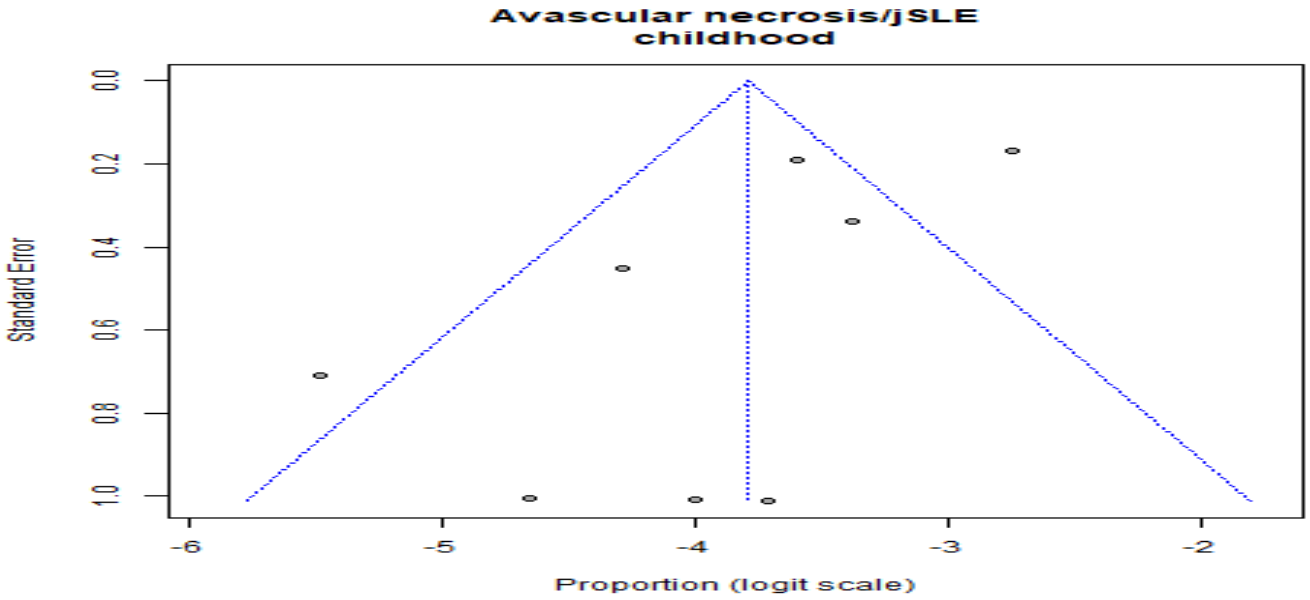
**Supplementary Figure S4.36**: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in children with juvenile-onset Systemic Lupus Erythematosus

**Avascular necrosis**

Forest




Funnel



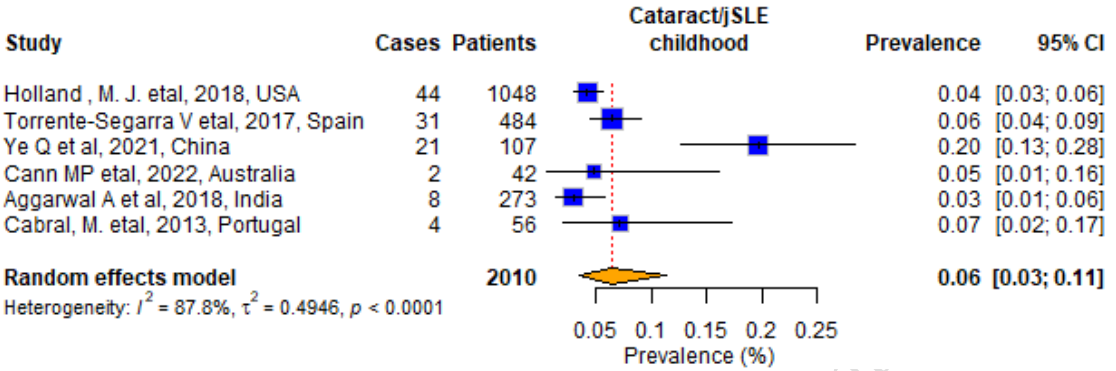

Summary statistics

```
                                    proportion           95%-CI %W(random)
Cann MP etal, 2022, Australia           0.0238 [0.0006; 0.1257]        6.6
Yang, Y.  etal, 2015, Canada            0.0600 [0.0426; 0.0817]       19.5
AlAhmed, O. etal, 2020, USA             0.0136 [0.0044; 0.0314]       14.5
Cabral, M. etal, 2013, Portugal         0.0179 [0.0005; 0.0955]        6.7
Aggarwal A et al, 2018, India           0.0330 [0.0152; 0.0617]       16.7
Holland , M. J. etal, 2018, USA         0.0267 [0.0178; 0.0384]       19.2
Medhat MB et al, 2020, Egypt            0.0094 [0.0002; 0.0514]        6.7
Torrente-Segarra V etal, 2017, Spain    0.0041 [0.0005; 0.0148]       10.1
Number of studies: k = 8; Number of observations: o = 2994; Number of events: e = 84
                     proportion           95%-CI
Random effects model     0.0220 [0.0120; 0.0402]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4856 [0.0757; 2.5590]; tau = 0.6968 [0.2752; 1.5997]
 I^2 = 76.4% [52.8%; 88.2%]; H = 2.06 [1.46; 2.91]
Test of heterogeneity:      Q d.f. p-value
                        29.62    7  0.0001
```

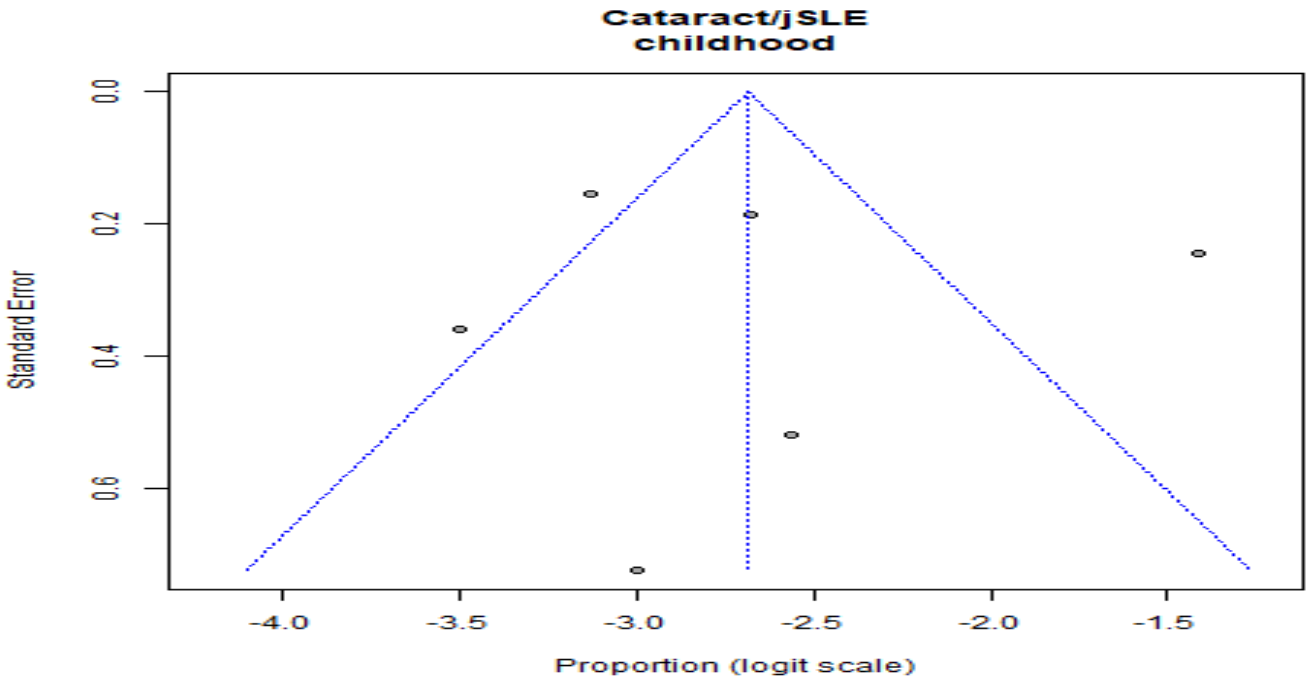
**Supplementary Figure S4.37**: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with juvenile-onset Systemic Lupus Erythematosus

**Cataract**

Forest




Funnel



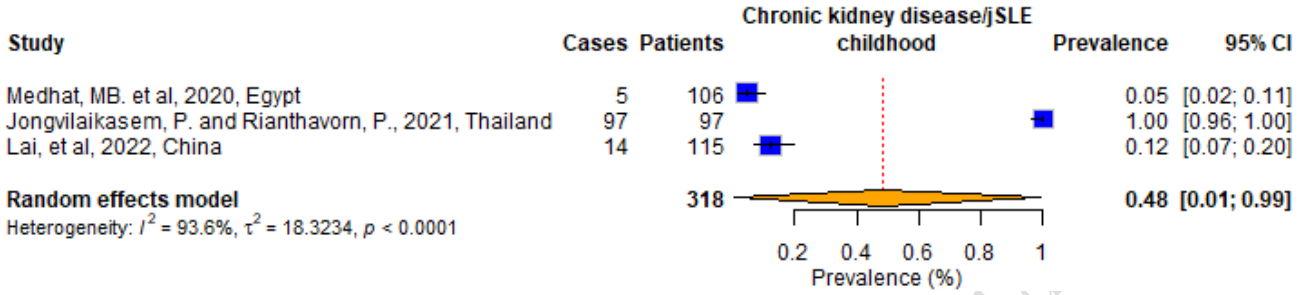

Summary statistics

```
                                  proportion           95%-CI %W(random)
Holland , M. J. etal, 2018, USA       0.0420 [0.0307; 0.0560]       20.3
Torrente-Segarra V etal, 2017, Spain  0.0640 [0.0439; 0.0897]       19.9
Ye Q et al, 2021, China               0.1963 [0.1258; 0.2842]       19.0
Cann MP etal, 2022, Australia         0.0476 [0.0058; 0.1616]       10.3
Aggarwal A et al, 2018, India         0.0293 [0.0127; 0.0569]       16.9
Cabral, M. etal, 2013, Portugal       0.0714 [0.0198; 0.1729]       13.8
Number of studies: k = 6; Number of observations: o = 2010; Number of events: e = 110
                     proportion           95%-CI
Random effects model     0.0639 [0.0349; 0.1141]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4946 [0.1265; 3.0315]; tau = 0.7032 [0.3557; 1.7411]
 I^2 = 87.8% [75.9%; 93.8%]; H = 2.86 [2.04; 4.02]
Test of heterogeneity:      Q d.f.  p-value
                        40.98    5 < 0.0001
```

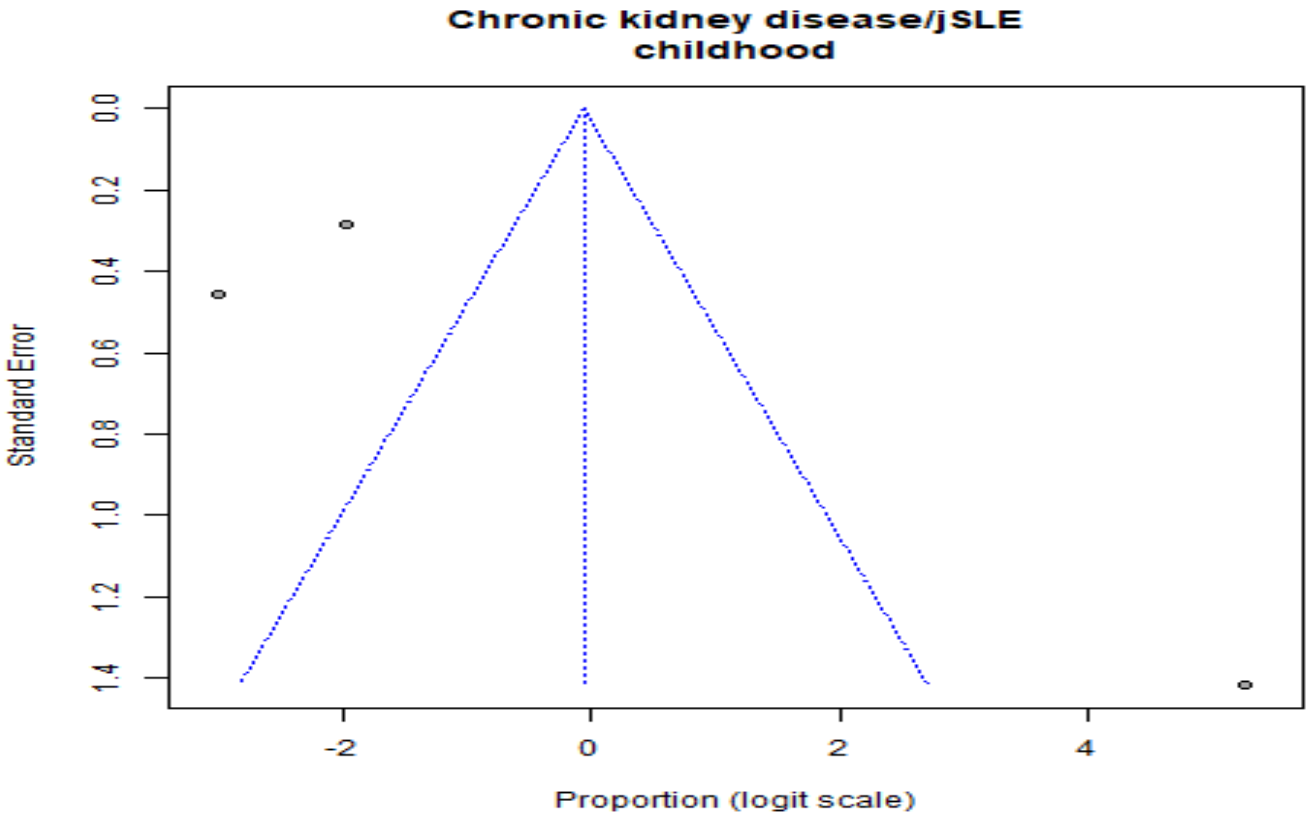
**Supplementary Figure S4.38**: Forest and Funnel Plots of studies reporting the prevalence of chronic kidney disease in children with juvenile-onset Systemic Lupus Erythematosus

**Chronic kidney disease**

Forest




Funnel




Summary statistics

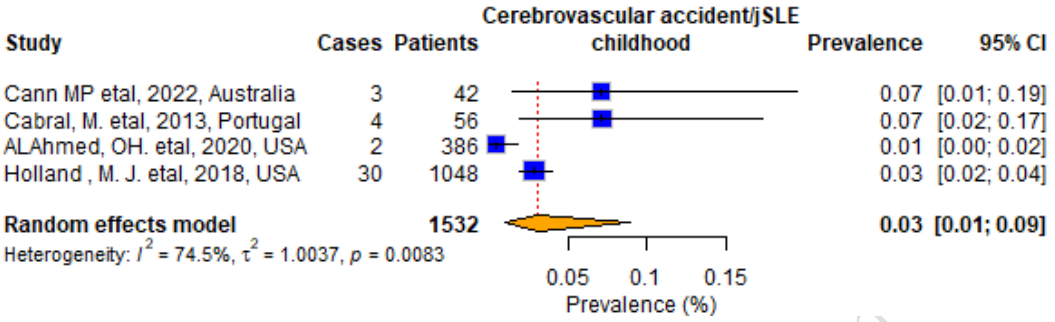
```
                                                  proportion       95%-CI          %W(random)
Medhat, MB. et al, 2020, Egypt                        0.0472       [0.0155; 0.1067]        34.3
Jongvilaikasem, P. and Rianthavorn, P., 2021, Thailand     1.0000       [0.9627; 1.0000]        31.2
Lai, et al, 2022, China                               0.1217       [0.0682; 0.1958]        34.5
Number of studies: k = 3; Number of observations: o = 318; Number of events: e = 116
                     proportion           95%-CI
Random effects model     0.4838 [0.0067; 0.9924]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 18.3234 [4.1403; >183.2337]; tau = 4.2806 [2.0348; >13.5364]
 I^2 = 93.6% [84.6%; 97.3%]; H = 3.94 [2.55; 6.10]
Test of heterogeneity:      Q d.f.  p-value
                        31.09    2 < 0.0001
```

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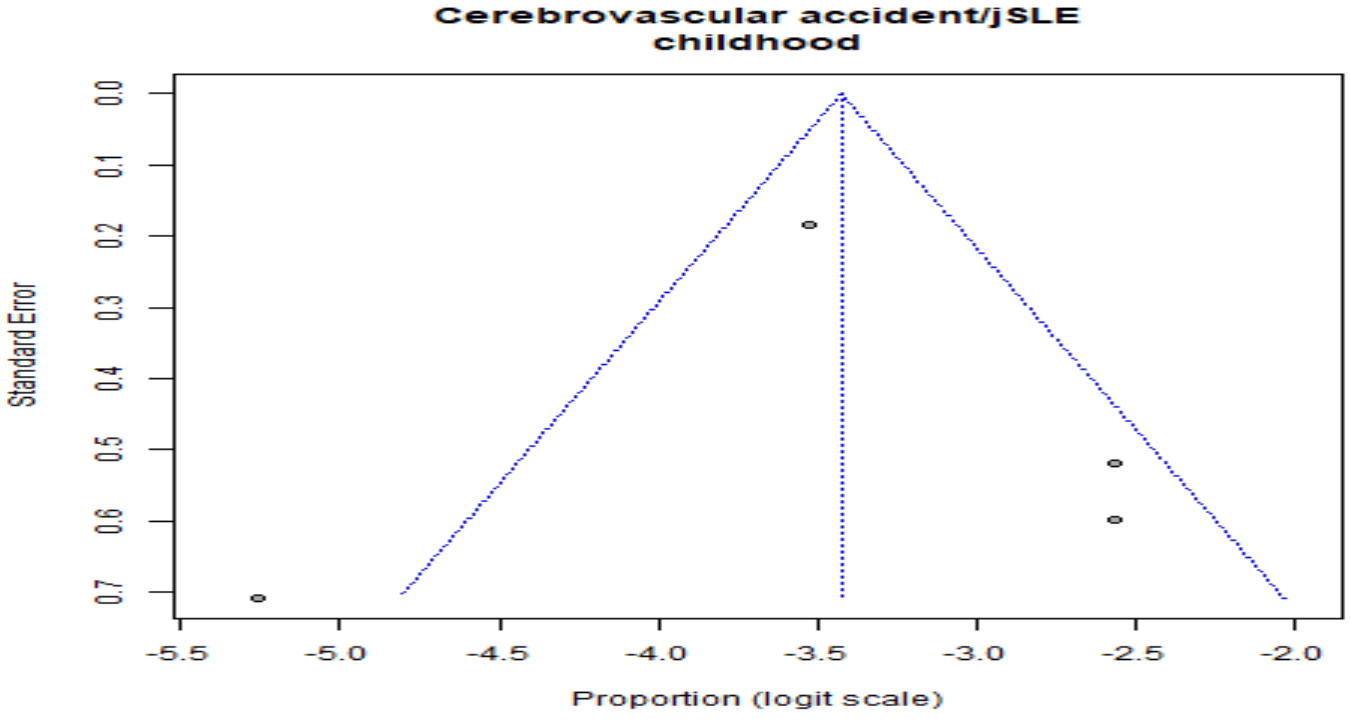
**Supplementary Figure S4.39**: Forest and Funnel Plots of studies reporting the prevalence of cerebrovascular accident in children with juvenile-onset Systemic Lupus Erythematosus

**Cerebrovascular accident**

Forest




Funnel



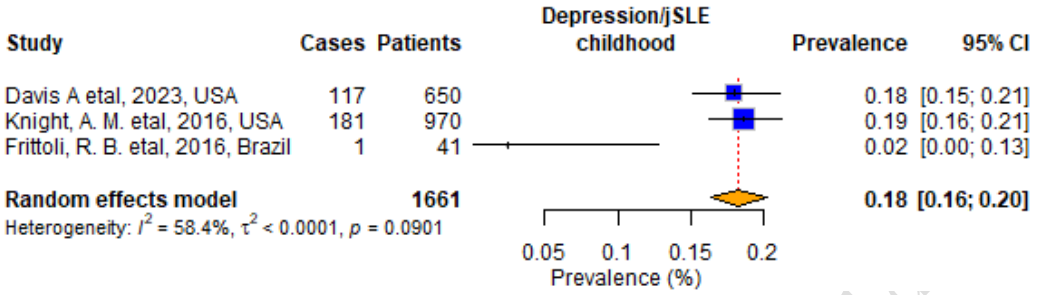

Summary statistics

```
                                proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia       0.0714      [0.0150; 0.1948]        23.3
Cabral, M. etal, 2013, Portugal     0.0714      [0.0198; 0.1729]        25.0
ALAhmed, OH. etal, 2020, USA        0.0052      [0.0006; 0.0186]        21.1
Holland , M. J. etal, 2018, USA     0.0286      [0.0194; 0.0406]        30.6
Number of studies: k = 4; Number of observations: o = 1532; Number of events: e = 39
                     proportion           95%-CI
Random effects model     0.0315 [0.0106; 0.0893]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0037 [0.0947; 21.9732]; tau = 1.0018 [0.3078; 4.6876]
 I^2 = 74.5% [28.9%; 90.8%]; H = 1.98 [1.19; 3.30]
Test of heterogeneity:      Q d.f. p-value
                        11.75    3  0.0083
```

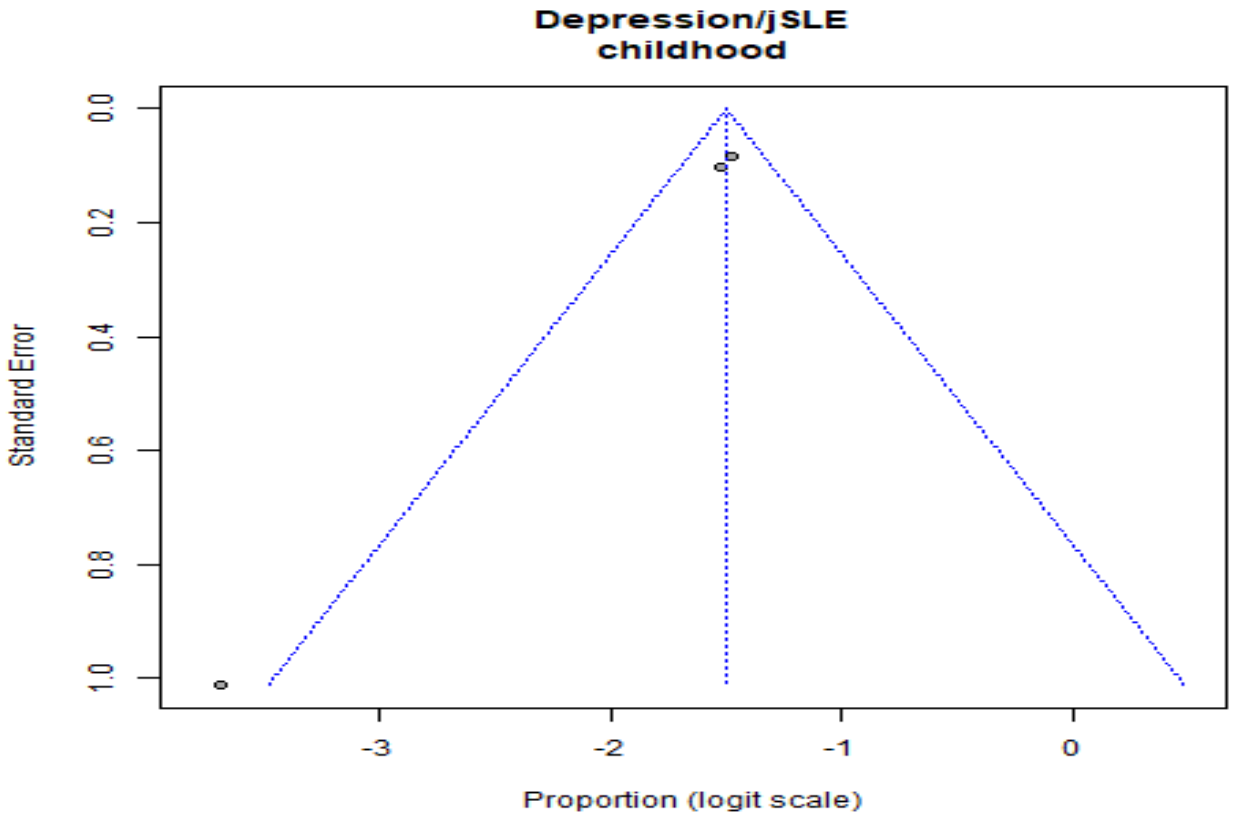
**Supplementary Figure S4.40**: Forest and Funnel Plots of studies reporting the prevalence of depression in children with juvenile-onset Systemic Lupus Erythematosus

**Depression**

Forest




Funnel




Summary statistics

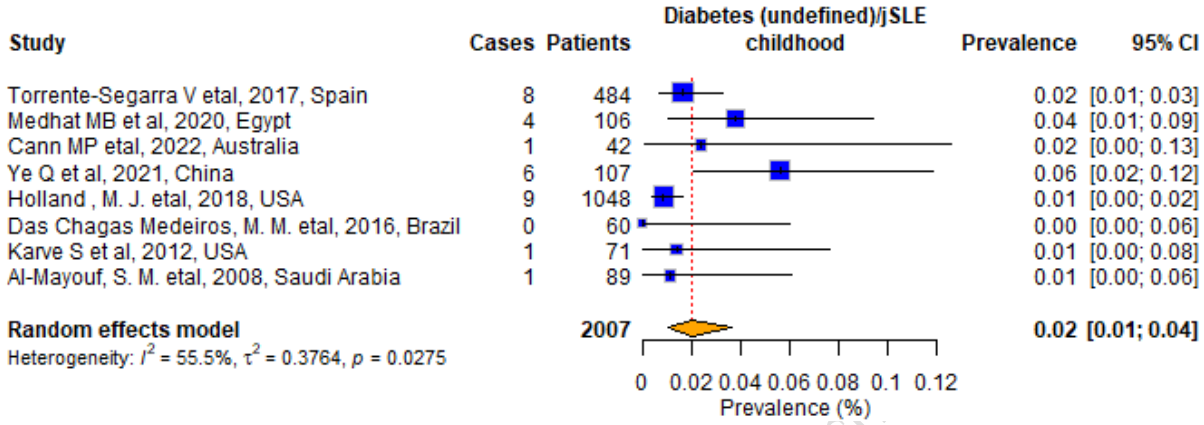
```
                                  proportion   95%-CI          %W(random)
Davis A etal, 2023, USA               0.1800   [0.1512; 0.2117]        39.3
Knight, A. M. etal, 2016, USA         0.1866   [0.1625; 0.2126]        60.3
Frittoli, R. B. etal, 2016, Brazil    0.0244   [0.0006; 0.1286]         0.4
Number of studies: k = 3; Number of observations: o = 1661; Number of events: e = 299
                     proportion           95%-CI
Random effects model     0.1827 [0.1647; 0.2021]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 62.7424]; tau = 0.0018 [0.0000; 7.9210]
 I^2 = 58.4% [0.0%; 88.2%]; H = 1.55 [1.00; 2.91]
Test of heterogeneity:     Q d.f. p-value
                        4.81    2  0.0901
```

=For reproducibility only in academic research and development=

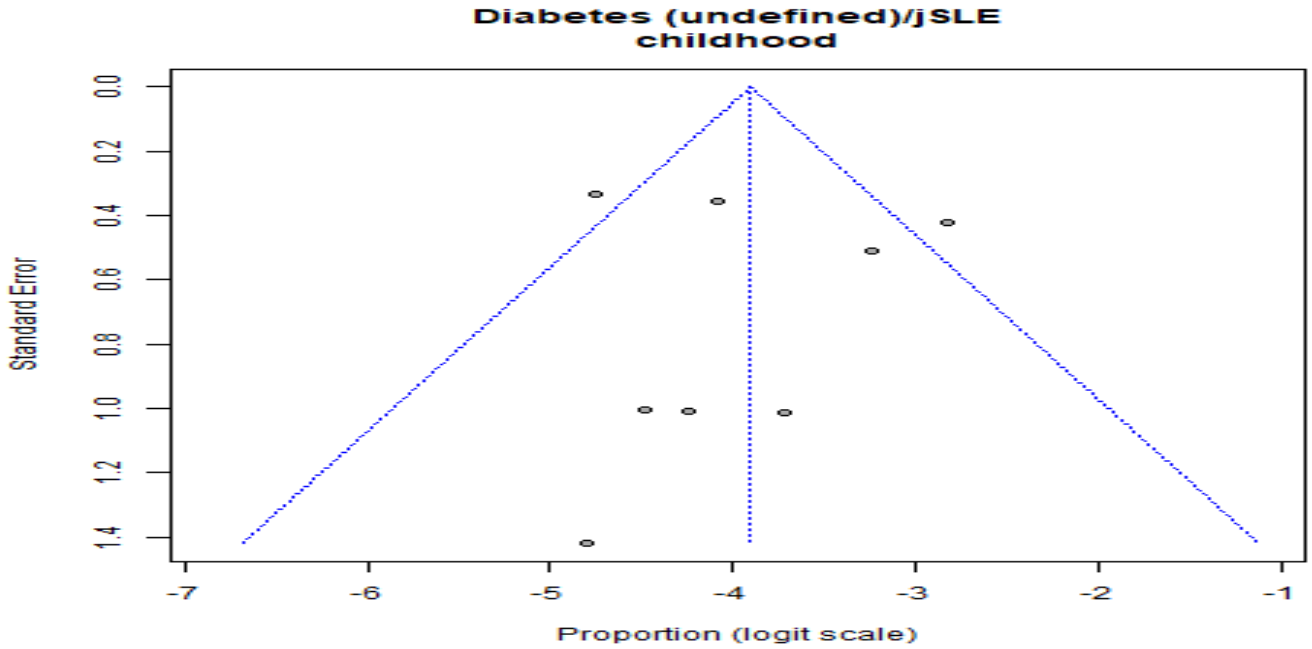
**Supplementary Figure S4.41**: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with juvenile-onset Systemic Lupus Erythematosus

**Diabetes (undefined)**

Forest




Funnel



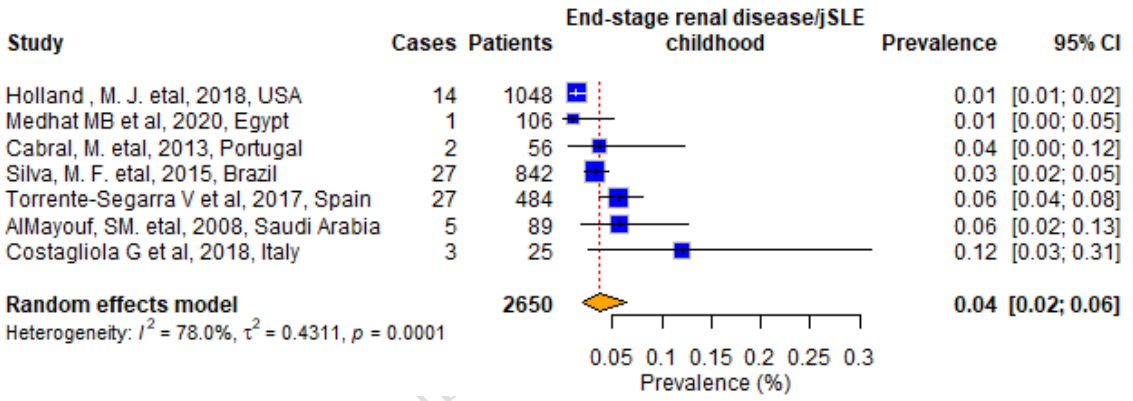

Summary statistics

```
                                          proportion         95%-CI          %W(random)
Torrente-Segarra V etal, 2017, Spain          0.0165        [0.0072; 0.0323]        19.9
Medhat MB et al, 2020, Egypt                  0.0377        [0.0104; 0.0938]        15.7
Cann MP etal, 2022, Australia                 0.0238        [0.0006; 0.1257]         7.1
Ye Q et al, 2021, China                       0.0561        [0.0209; 0.1181]        18.1
Holland , M. J. etal, 2018, USA               0.0086        [0.0039; 0.0162]        20.5
Das Chagas Medeiros, M. M. etal, 2016, Brazil 0.0000        [0.0000; 0.0596]         4.2
Karve S et al, 2012, USA                      0.0141        [0.0004; 0.0760]         7.2
Al-Mayouf, S. M. etal, 2008, Saudi Arabia     0.0112        [0.0003; 0.0610]         7.2
Number of studies: k = 8; Number of observations: o = 2007; Number of events: e = 30
                     proportion           95%-CI
Random effects model     0.0198 [0.0107; 0.0362]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3764 [0.0000; 1.6458]; tau = 0.6135 [0.0000; 1.2829]
 I^2 = 55.5% [1.9%; 79.9%]; H = 1.50 [1.01; 2.23]
Test of heterogeneity:       Q d.f. p-value
                         15.75    7  0.0275
```

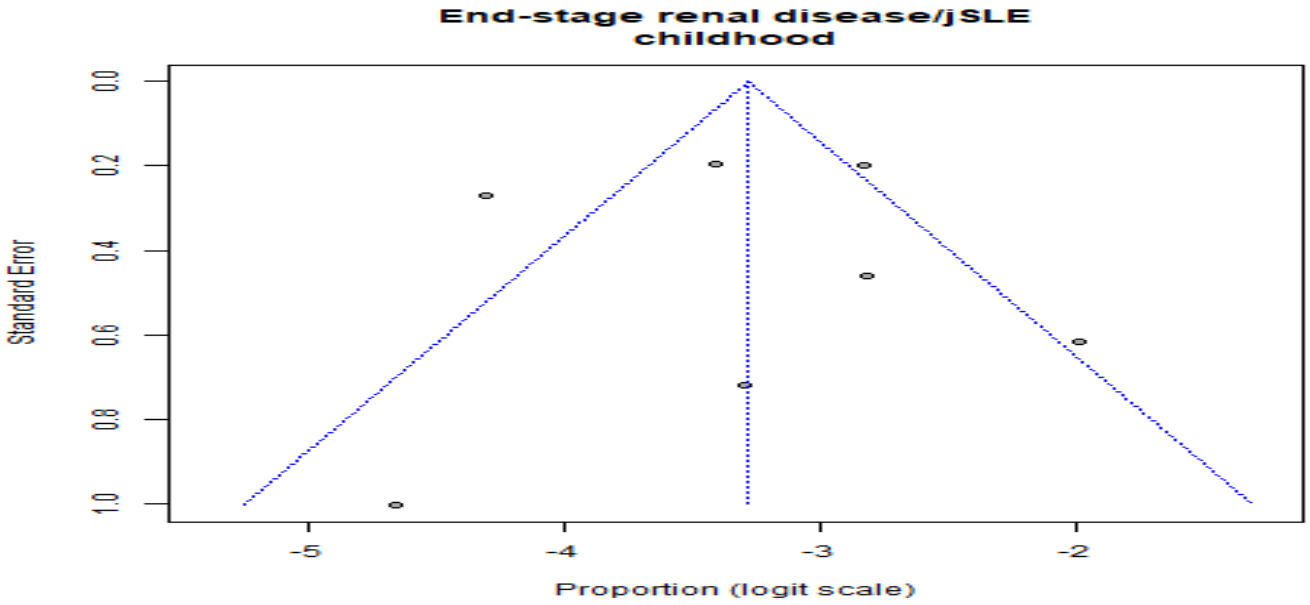
**Supplementary Figure S4.42**: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in children with juvenile-onset Systemic Lupus Erythematosus

**End-stage renal disease**

Forest




Funnel




Summary statistics

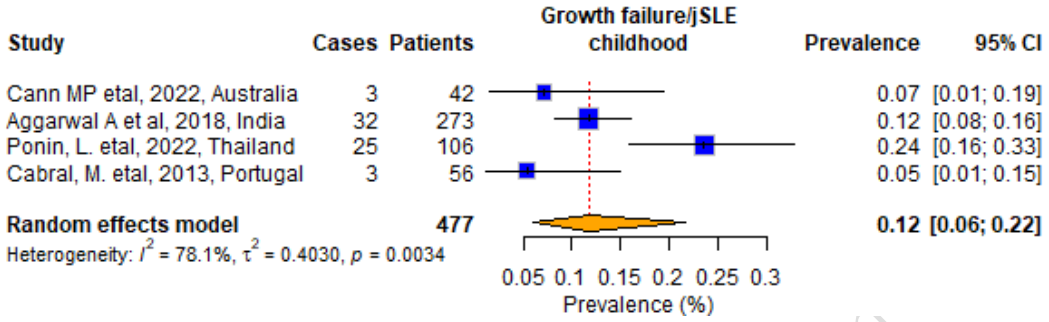
```
                                     proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA          0.0134       [0.0073; 0.0223]        18.4
Medhat MB et al, 2020, Egypt             0.0094       [0.0002; 0.0514]        6.4
Cabral, M. etal, 2013, Portugal          0.0357       [0.0044; 0.1231]        9.8
Silva, M. F. etal, 2015, Brazil          0.0321       [0.0212; 0.0463]        19.8
Torrente-Segarra V et al, 2017, Spain    0.0558       [0.0371; 0.0801]        19.7
AlMayouf, SM. etal, 2008, Saudi Arabia   0.0562       [0.0185; 0.1263]        14.4
Costagliola G et al, 2018, Italy         0.1200       [0.0255; 0.3122]        11.5
Number of studies: k = 7; Number of observations: o = 2650; Number of events: e = 79
                    proportion           95%-CI
Random effects model     0.0362 [0.0203; 0.0639]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4311 [0.0842; 3.5598]; tau = 0.6566 [0.2901; 1.8867]
 I^2 = 78.0% [54.3%; 89.4%]; H = 2.13 [1.48; 3.07]
Test of heterogeneity:      Q d.f. p-value
                        27.24    6  0.0001
```

=For reproducibility only in academic research and develop

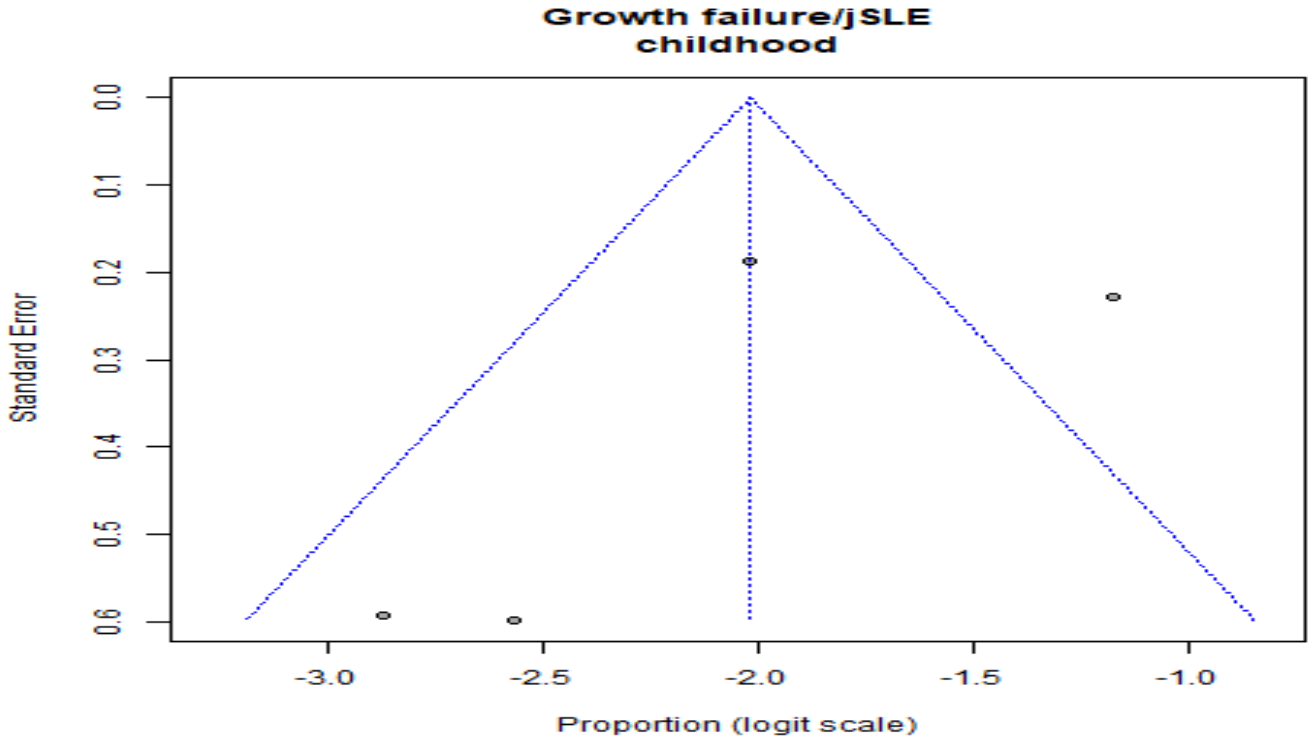
**Supplementary Figure S4.43**: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with juvenile-onset Systemic Lupus Erythematosus

**Growth failure**

Forest




Funnel



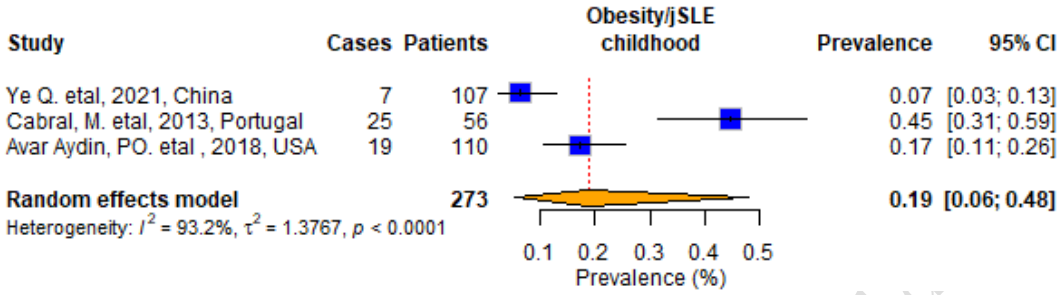

Summary statistics

```
                                proportion      95%-CI          %W(random)
Cann MP etal, 2022, Australia       0.0714      [0.0150; 0.1948]        18.4
Aggarwal A et al, 2018, India       0.1172      [0.0816; 0.1614]        32.1
Ponin, L. etal, 2022, Thailand      0.2358      [0.1588; 0.3282]        30.9
Cabral, M. etal, 2013, Portugal     0.0536      [0.0112; 0.1487]        18.6
Number of studies: k = 4; Number of observations: o = 477; Number of events: e = 63
                     proportion           95%-CI
Random effects model     0.1173 [0.0599; 0.2170]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4030 [0.0383; 7.5155]; tau = 0.6349 [0.1957; 2.7414]
 I^2 = 78.1% [40.8%; 91.9%]; H = 2.14 [1.30; 3.51]
Test of heterogeneity:      Q d.f. p-value
                        13.69    3  0.0034
```

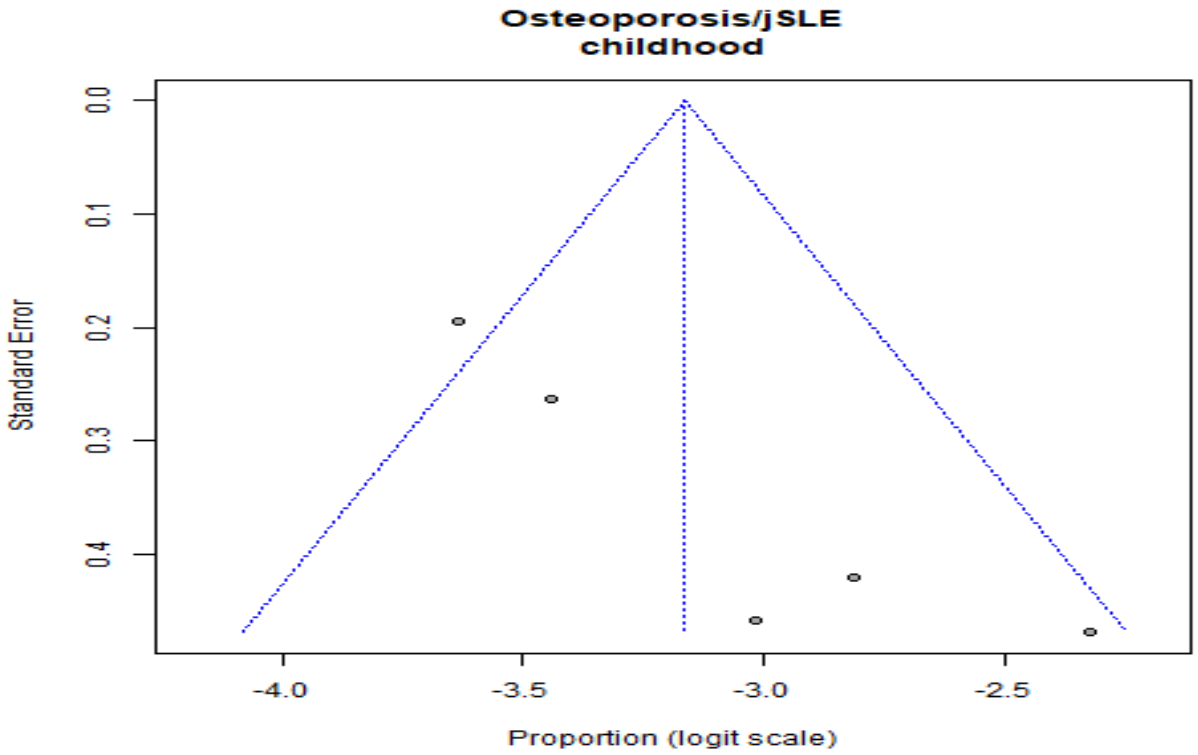
**Supplementary Figure S4.44**: Forest and Funnel Plots of studies reporting the prevalence of obesity in children with juvenile-onset Systemic Lupus Erythematosus

**Obesity**

Forest




Funnel



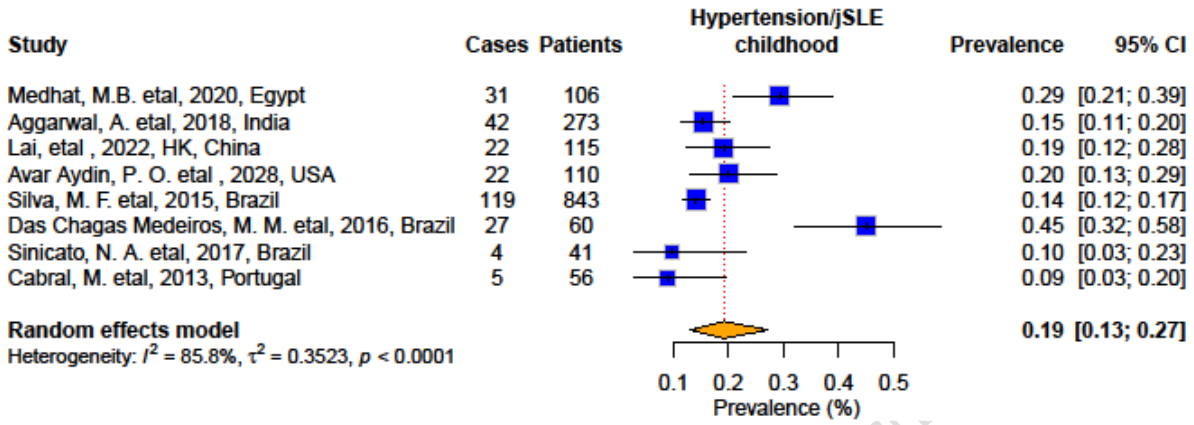

Summary statistics

```
                                proportion     95%-CI          %W(random)
Ye Q. etal, 2021, China             0.0654     [0.0267; 0.1302]        32.1
Cabral, M. etal, 2013, Portugal     0.4464     [0.3134; 0.5853]        33.9
Avar Aydin, PO. etal , 2018, USA    0.1727     [0.1073; 0.2565]        34.1
Number of studies: k = 3; Number of observations: o = 273; Number of events: e = 51
                     proportion           95%-CI
Random effects model     0.1886 [0.0556; 0.4784]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.3767 [0.2984; 59.0990]; tau = 1.1733 [0.5463; 7.6876]
 I^2 = 93.2% [83.5%; 97.2%]; H = 3.83 [2.46; 5.97]
Test of heterogeneity:      Q d.f.  p-value
                        29.39    2 < 0.0001
```

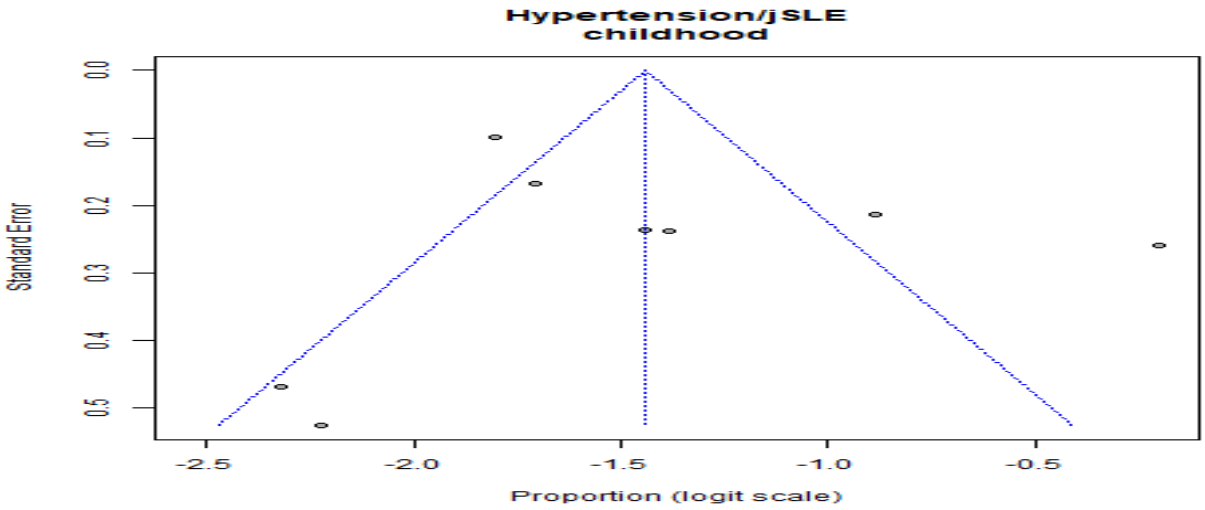
**Supplementary Figure S4.45**: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with juvenile-onset Systemic Lupus Erythematosus

**Hypertension**

Forest




Funnel




Summary statistics

For reproduction

```
                                          proportion        95% CI                %W(random)
Medhat, M.B. etal, 2020, Egypt                0.2925        [0.2081; 0.3887]        13.6
Aggarwal, A. etal, 2018, India                0.1538        [0.1132; 0.2022]        14.2
Lai, etal , 2022, HK, China                   0.1913        [0.1239; 0.2752]        13.2
Avar Aydin, P. O. etal , 2028, USA            0.2000        [0.1298; 0.2870]        13.2
Silva, M. F. etal, 2015, Brazil               0.1412        [0.1184; 0.1665]        14.9
Das Chagas Medeiros, M. M. etal, 2016, Brazil 0.4500        [0.3212; 0.5839]        12.9
Sinicato, N. A. etal, 2017, Brazil            0.0976        [0.0272; 0.2313]        8.6
Cabral, M. etal, 2013, Portugal               0.0893        [0.0296; 0.1962]        9.4

Number of studies: k = 8
Number of observations: o = 1604
Number of events: e = 272

                    proportion 95% CI
Random effects model     0.1914 [0.1305; 0.2718]

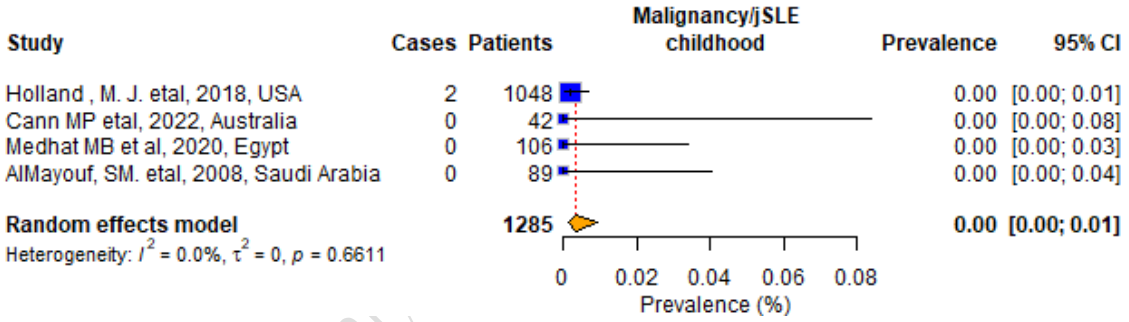
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.3523 [0.1124; 1.9017]; tau = 0.5936 [0.3352; 1.3790]
 I^2 = 85.8% [74.0%; 92.3%]; H = 2.65 [1.96; 3.59]

Test of heterogeneity:
     Q d.f.  p-value
 49.34    7 < 0.0001
```

**Supplementary Figure S4.46**: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with juvenile-onset Systemic Lupus Erythematosus

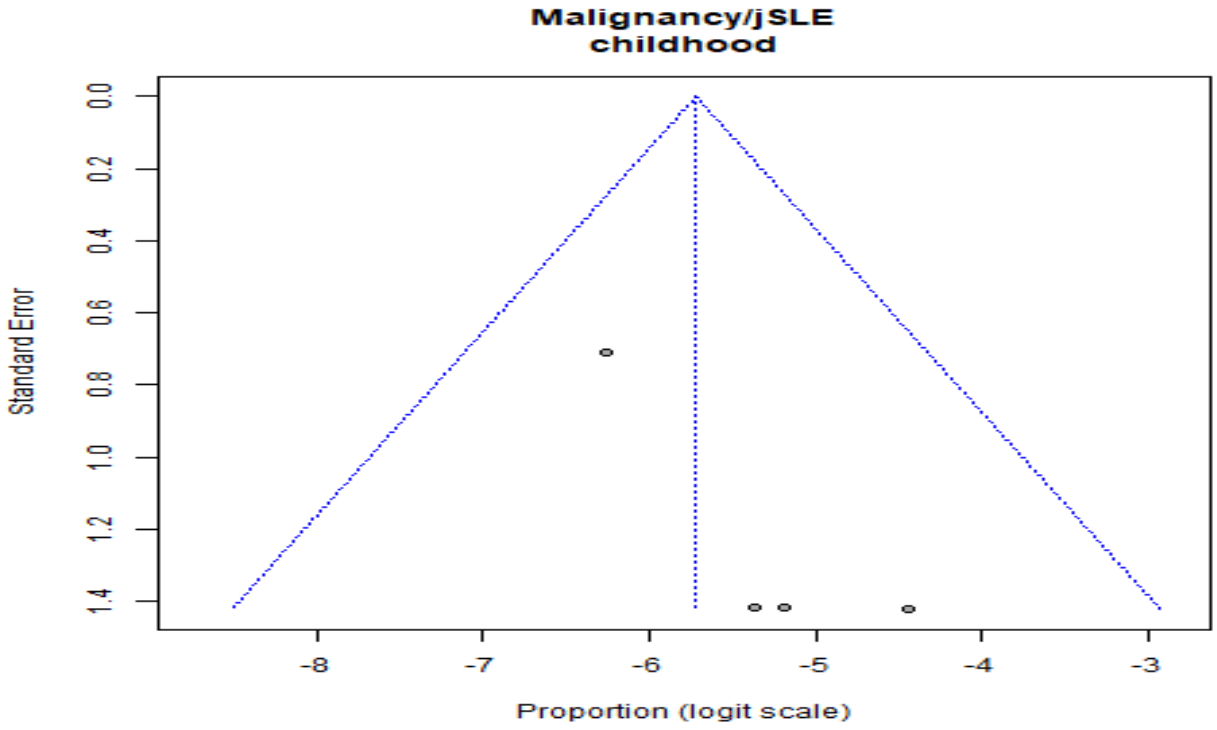
**Malignancy**

Forest




Funnel

=For reproducibility only, search and development=




Summary statistics

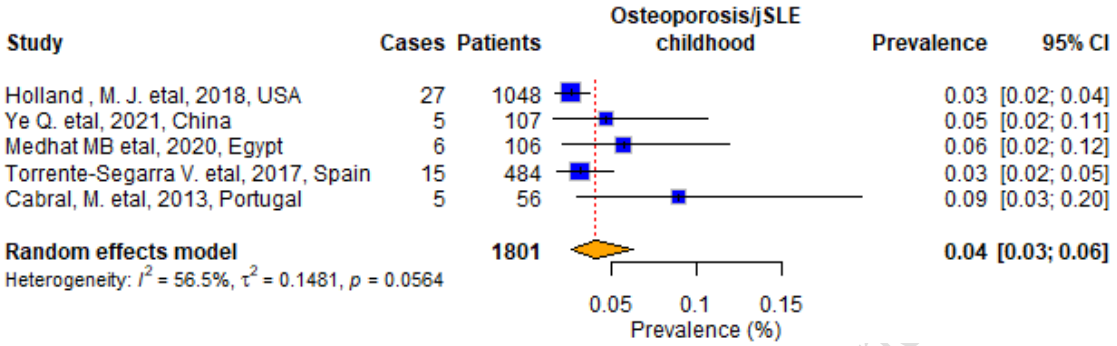
```
                                        proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0019       [0.0002; 0.0069]        57.3
Cann MP etal, 2022, Australia               0.0000       [0.0000; 0.0841]        14.2
Medhat MB et al, 2020, Egypt                0.0000       [0.0000; 0.0342]        14.3
AlMayouf, SM. etal, 2008, Saudi Arabia      0.0000       [0.0000; 0.0406]        14.3
Number of studies: k = 4; Number of observations: o = 1285; Number of events: e = 2
                      proportion           95%-CI
Random effects model     0.0033 [0.0011; 0.0093]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 6.5718]; tau = 0 [0.0000; 2.5635]
 I^2 = 0.0% [0.0%; 84.7%]; H = 1.00 [1.00; 2.56]
Test of heterogeneity:     Q d.f. p-value
                        1.59    3  0.6611
```

For reproducibility

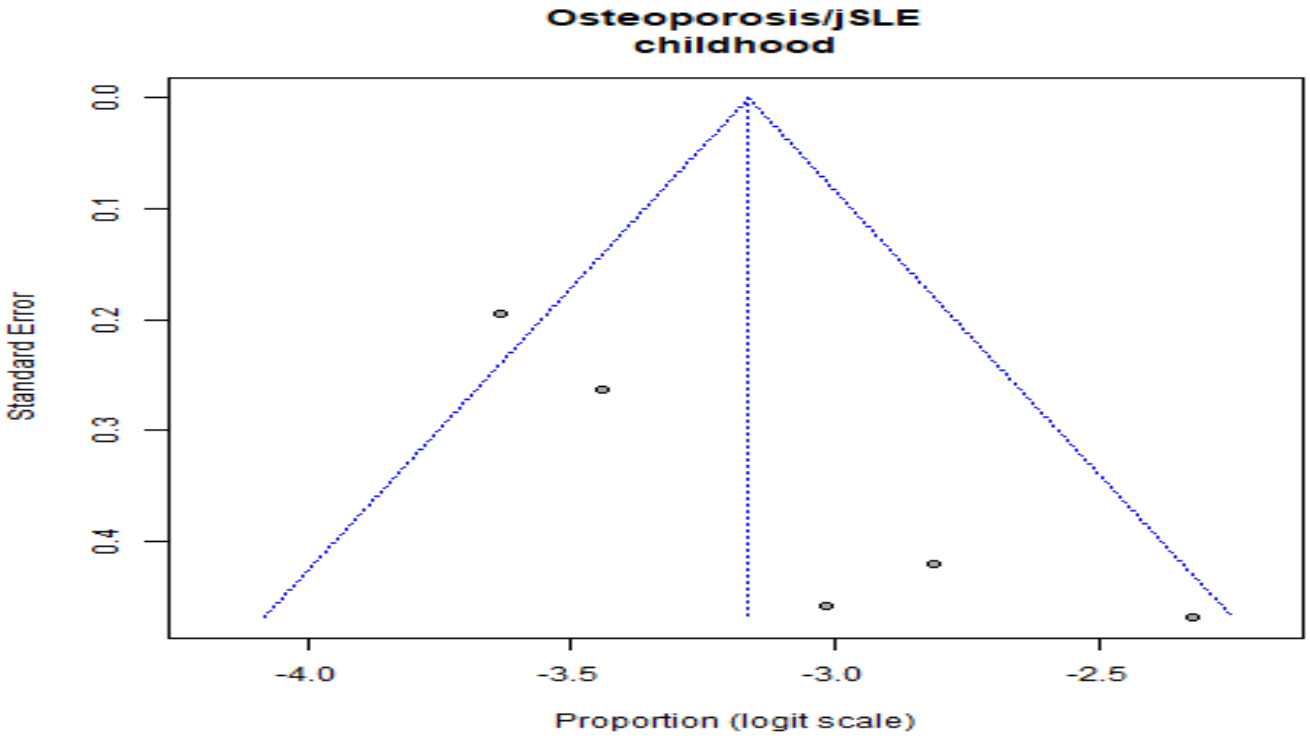
**Supplementary Figure S4.47**: Forest and Funnel Plots of studies reporting the prevalence of osteoporosis in children with juvenile-onset Systemic Lupus Erythematosus

**Osteoporosis**

Forest




Funnel




Summary statistics

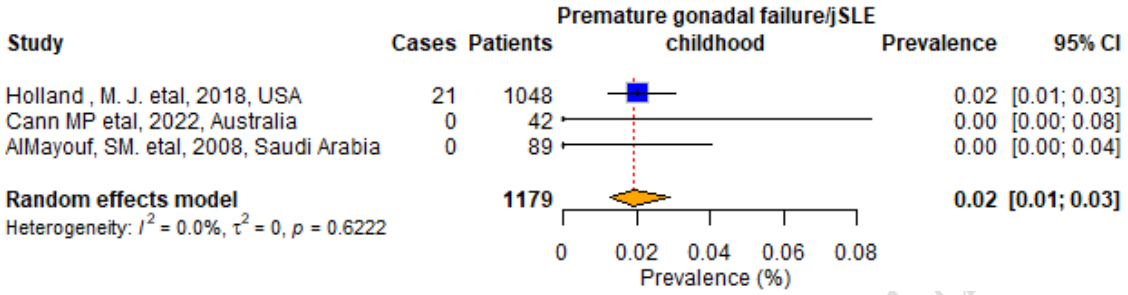
```
                                    proportion        95%-CI          %W(random)
Holland , M. J. etal, 2018, USA           0.0258       [0.0170; 0.0373]        28.9
Ye Q. etal, 2021, China                   0.0467       [0.0153; 0.1057]        15.0
Medhat MB etal, 2020, Egypt               0.0566       [0.0211; 0.1191]        16.6
Torrente-Segarra V. etal, 2017, Spain     0.0310       [0.0174; 0.0506]        24.8
Cabral, M. etal, 2013, Portugal           0.0893       [0.0296; 0.1962]        14.6
Number of studies: k = 5; Number of observations: o = 1801; Number of events: e = 58
                     proportion           95%-CI
Random effects model     0.0405 [0.0261; 0.0624]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.1481 [0.0000; 2.0934]; tau = 0.3848 [0.0000; 1.4469]
 I^2 = 56.5% [0.0%; 83.9%]; H = 1.52 [1.00; 2.49]
Test of heterogeneity:     Q d.f. p-value
                        9.19    4  0.0564
```

For reproducibility only in academic research and developme

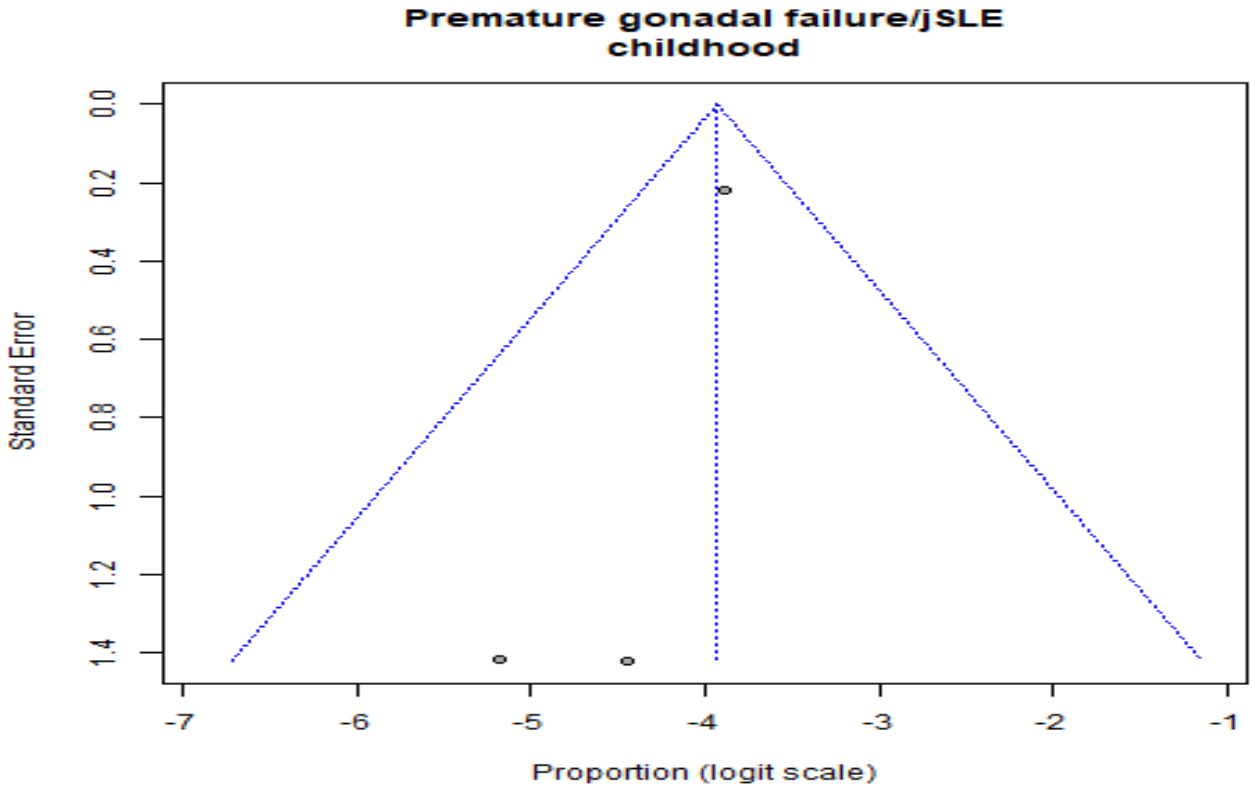
**Supplementary Figure S4.48**: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in children with juvenile-onset Systemic Lupus Erythematosus

**Premature gonadal failure**

Forest




Funnel




Summary statistics

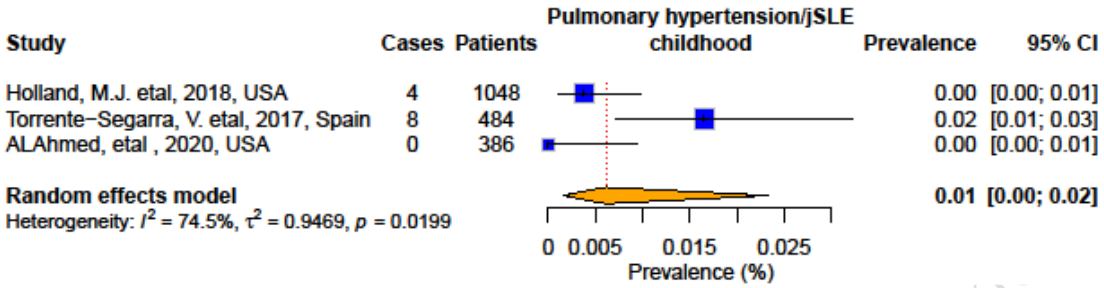
```
                                     proportion       95%-CI          %W(random)
Holland , M. J. etal, 2018, USA          0.0200       [0.0124; 0.0305]        95.4
Cann MP etal, 2022, Australia            0.0000       [0.0000; 0.0841]        2.3
AlMayouf, SM. etal, 2008, Saudi Arabia   0.0000       [0.0000; 0.0406]        2.3
Number of studies: k = 3; Number of observations: o = 1179; Number of events: e = 21
                     proportion           95%-CI
Random effects model     0.0192 [0.0127; 0.0290]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.6365]; tau = 0 [0.0000; 3.9543]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.95    2  0.6222
```

=For reproducibility only in academic research and developmen

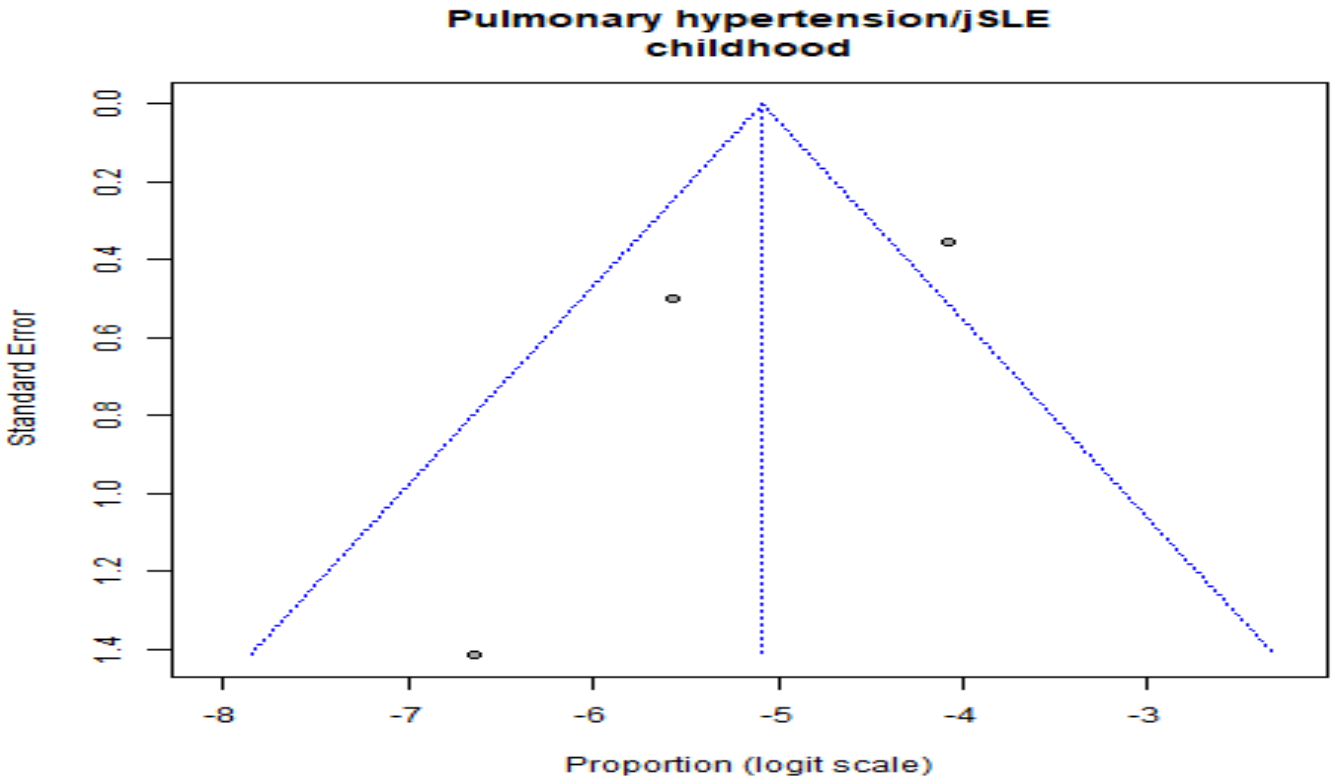
**Supplementary Figure S4.49**: Forest and Funnel Plots of studies reporting the prevalence of pulmonary hypertension in children with juvenile-onset Systemic Lupus Erythematosus

**Pulmonary hypertension**

Forest




Funnel




Summary statistics

```
                                        proportion             95%-CI       %W(random)
Holland, MJ. etal, 2018, USA                0.0038 [0.0010; 0.0097]       39.7
Torrente-Segarra, V. etal, 2017, Spain      0.0165 [0.0072; 0.0323]       44.2
AlAhmed, etal , 2020, USA                   0.0000 [0.0000; 0.0095]       16.1

Number of studies: k = 3
Number of observations: o = 1918
Number of events: e = 12

                      proportion           95%-CI
Random effects model      0.0061 [0.0016; 0.0233]

Quantifying heterogeneity (with 95%-CIs):
 tau^2 = 0.9469 [0.0140; 64.4784]; tau = 0.9731 [0.1183; 8.0298]
 I^2 = 74.5% [15.0%; 92.3%]; H = 1.98 [1.08; 3.61]

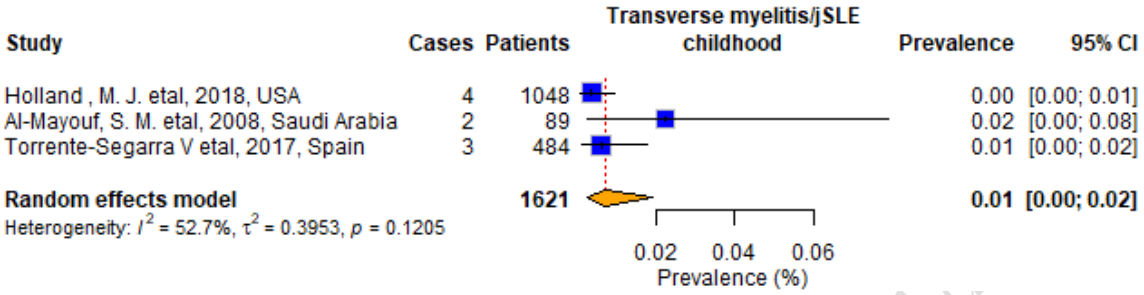
Test of heterogeneity:
    Q d.f. p-value
 7.83    2  0.0199
```

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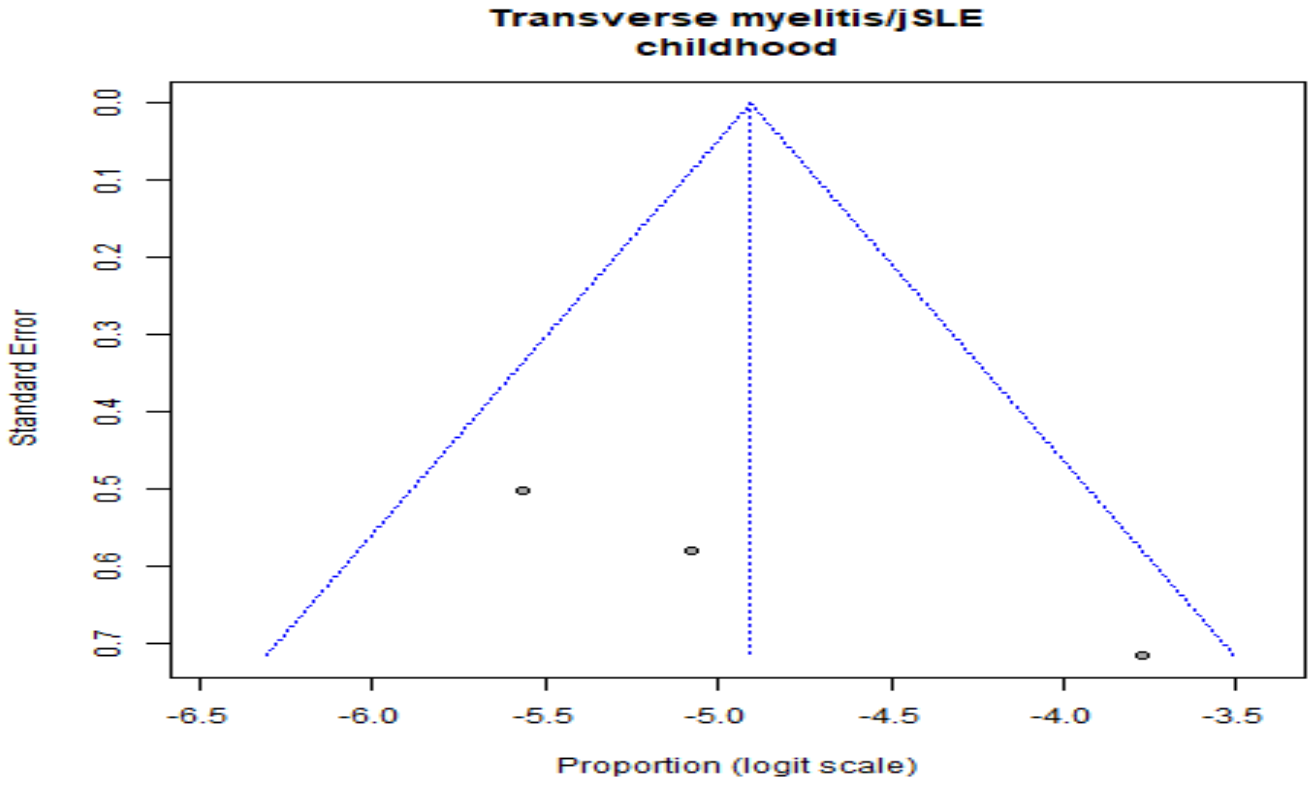
**Supplementary Figure S4.50**: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in children with juvenile-onset Systemic Lupus Erythematosus

**Transverse myelitis**

Forest




Funnel



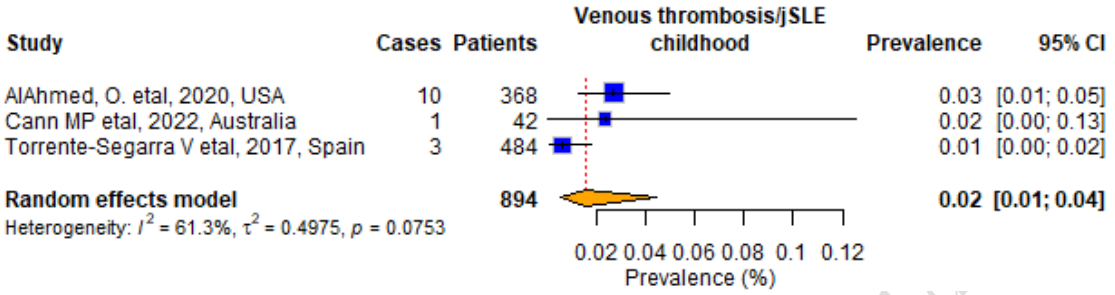

Summary statistics

```
                                        proportion    95%-CI          %W(random)
Holland , M. J. etal, 2018, USA             0.0038    [0.0010; 0.0097]        38.5
Al-Mayouf, S. M. etal, 2008, Saudi Arabia   0.0225    [0.0027; 0.0788]        27.4
Torrente-Segarra V etal, 2017, Spain        0.0062    [0.0013; 0.0180]        34.1
Number of studies: k = 3; Number of observations: o = 1621; Number of events: e = 9
                     proportion           95%-CI
Random effects model     0.0073 [0.0028; 0.0193]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3953 [0.0000; 33.4831]; tau = 0.6287 [0.0000; 5.7865]
 I^2 = 52.7% [0.0%; 86.4%]; H = 1.45 [1.00; 2.71]
Test of heterogeneity:     Q d.f. p-value
                        4.23    2  0.1205
```

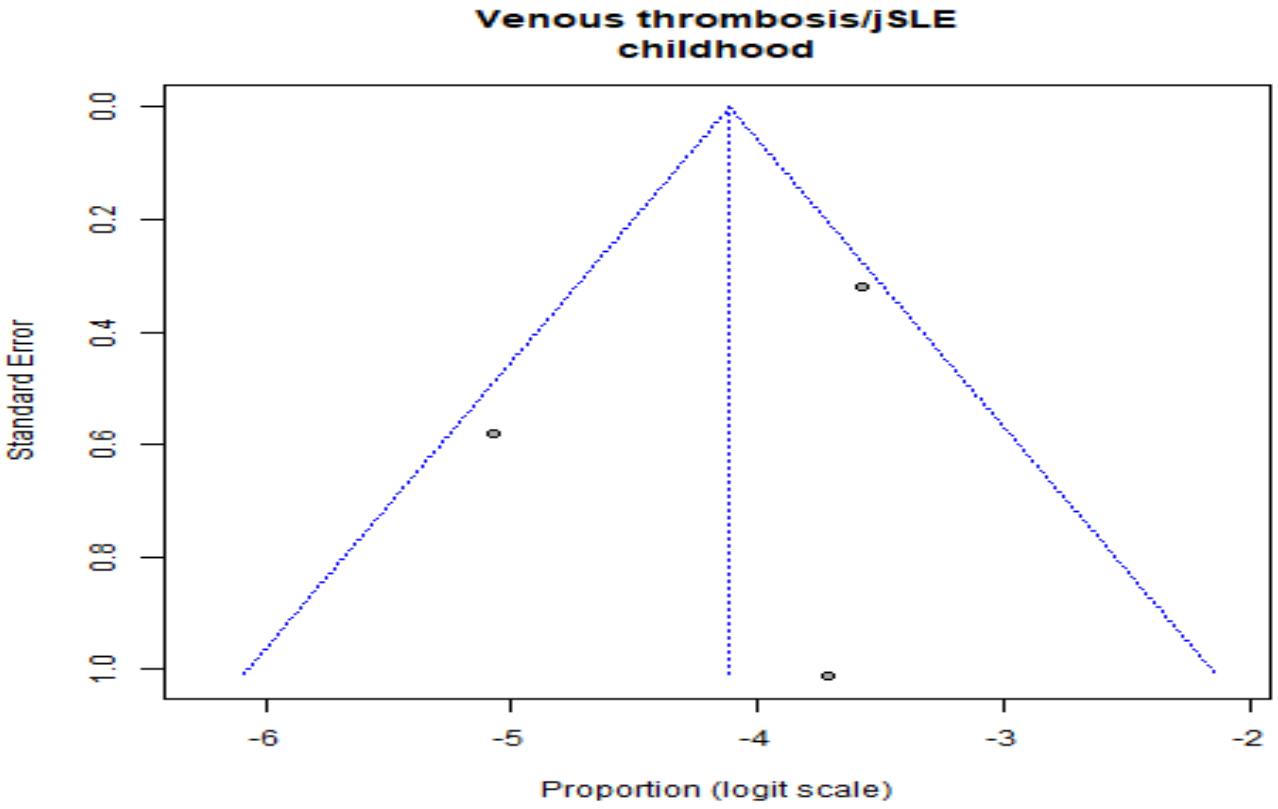
**Supplementary Figure S4.51**: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in children with juvenile-onset Systemic Lupus Erythematosus

**Venous thrombosis**

Forest




Funnel




Summary statistics

```
                                    proportion         95%-CI  %W(random)
AlAhmed, O. etal, 2020, USA             0.0272 [0.0131; 0.0494]        47.3
Cann MP etal, 2022, Australia           0.0238 [0.0006; 0.1257]        18.6
Torrente-Segarra V etal, 2017, Spain    0.0062 [0.0013; 0.0180]        34.1
Number of studies: k = 3; Number of observations: o = 894; Number of events: e = 14
                      proportion           95%-CI
Random effects model     0.0161 [0.0057; 0.0444]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4975 [0.0000; 26.7942]; tau = 0.7053 [0.0000; 5.1763]
 I^2 = 61.3% [0.0%; 89.0%]; H = 1.61 [1.00; 3.01]
Test of heterogeneity:     Q d.f. p-value
                        5.17    2  0.0753
```

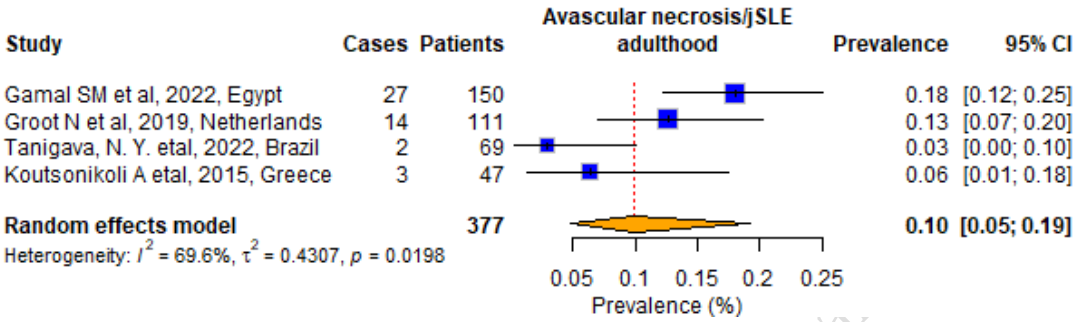
For reproducibility only in academic research and development

**Adulthood/jSLE**

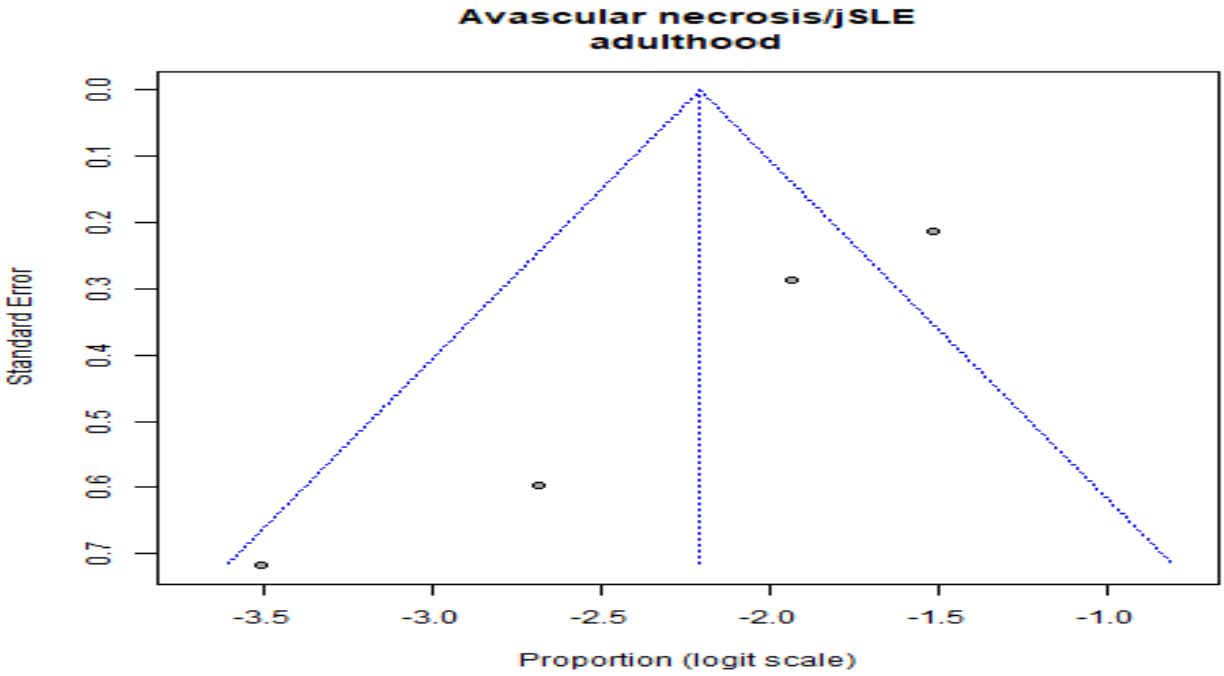
**Supplementary Figure S4.52**: Forest and Funnel Plots of studies reporting the prevalence of avascular necrosis in adults with juvenile-onset Systemic Lupus Erythematosus

**Avascular necrosis**

Forest




Funnel



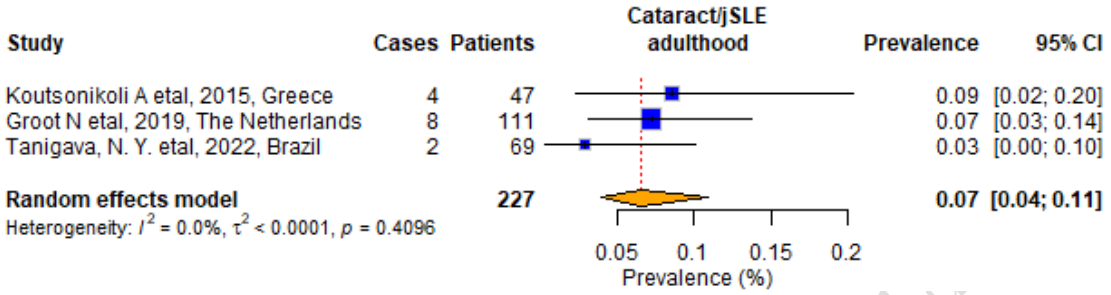

Summary statistics

```
                                proportion   95%-CI          %W(random)
Gamal SM et al, 2022, Egypt         0.1800   [0.1221; 0.2510]       32.9
Groot N et al, 2019, Netherlands    0.1261   [0.0707; 0.2026]       30.6
Tanigava, N. Y. etal, 2022, Brazil  0.0290   [0.0035; 0.1008]       16.6
Koutsonikoli A etal, 2015, Greece   0.0638   [0.0134; 0.1754]       19.9
Number of studies: k = 4; Number of observations: o = 377; Number of events: e = 46
                    proportion           95%-CI
Random effects model     0.0990 [0.0482; 0.1928]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4307 [0.0084; 10.4154]; tau = 0.6563 [0.0918; 3.2273]
 I^2 = 69.6% [12.4%; 89.4%]; H = 1.81 [1.07; 3.08]
Test of heterogeneity:     Q d.f. p-value
                        9.86    3  0.0198
```

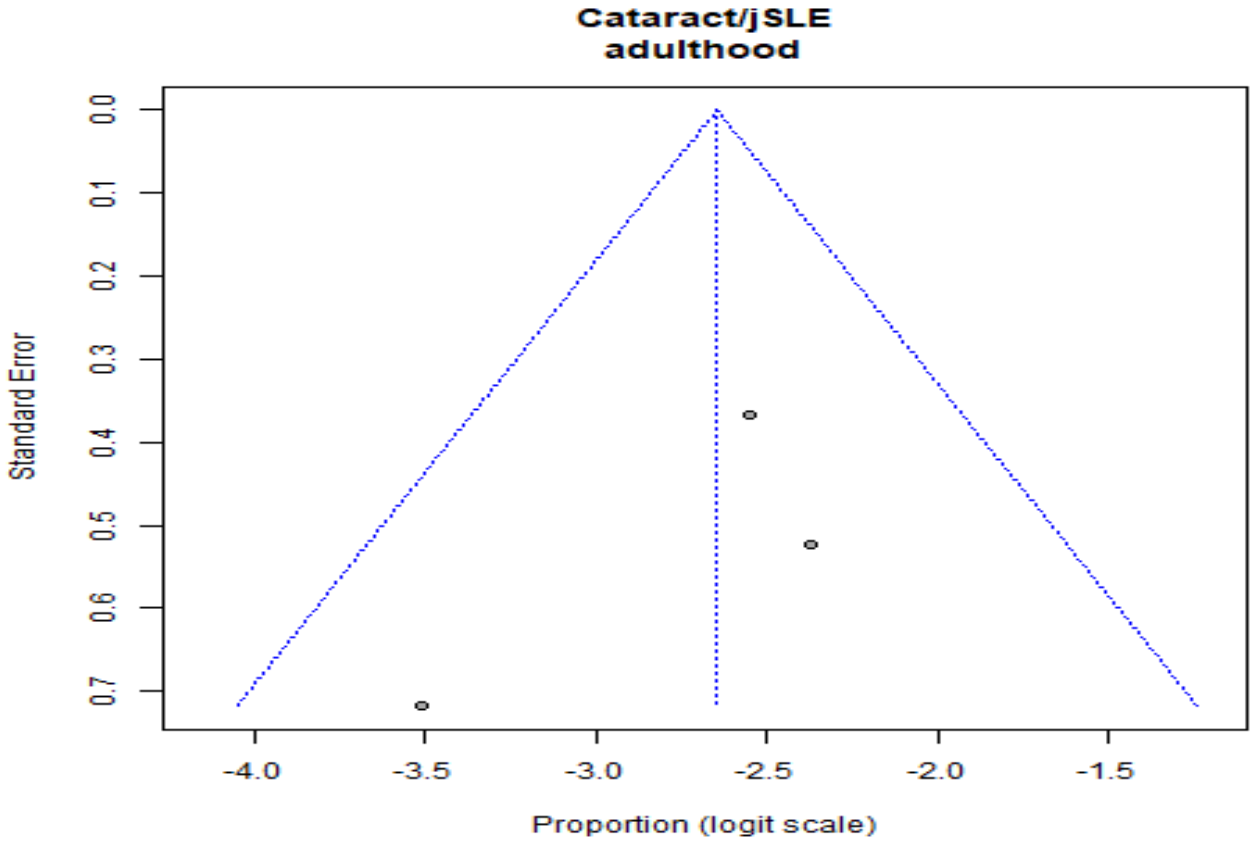
**Supplementary Figure S4.53**: Forest and Funnel Plots of studies reporting the prevalence of cataract in adults with juvenile-onset Systemic Lupus Erythematosus

**Cataract**

Forest




Funnel




Summary statistics

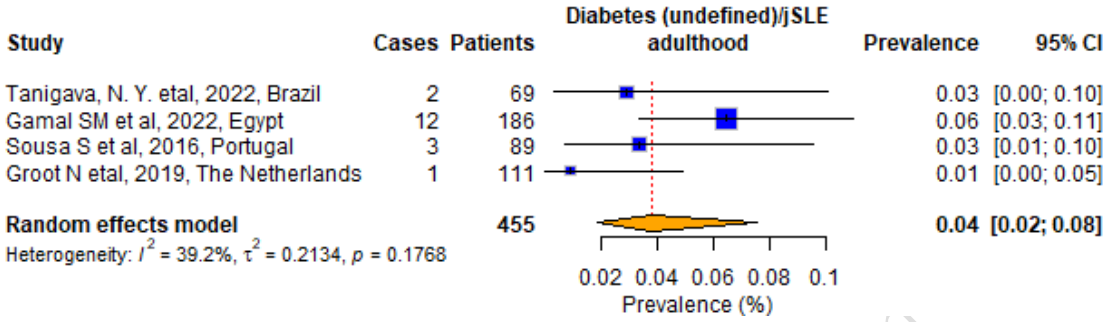
```
                                    proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece       0.0851  [0.0237; 0.2038]        28.1
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        57.0
Tanigava, N. Y. etal, 2022, Brazil      0.0290  [0.0035; 0.1008]        14.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                     proportion           95%-CI
Random effects model     0.0662 [0.0395; 0.1087]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 14.3203]; tau = 0.0004 [0.0000; 3.7842]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.78    2  0.4096
```

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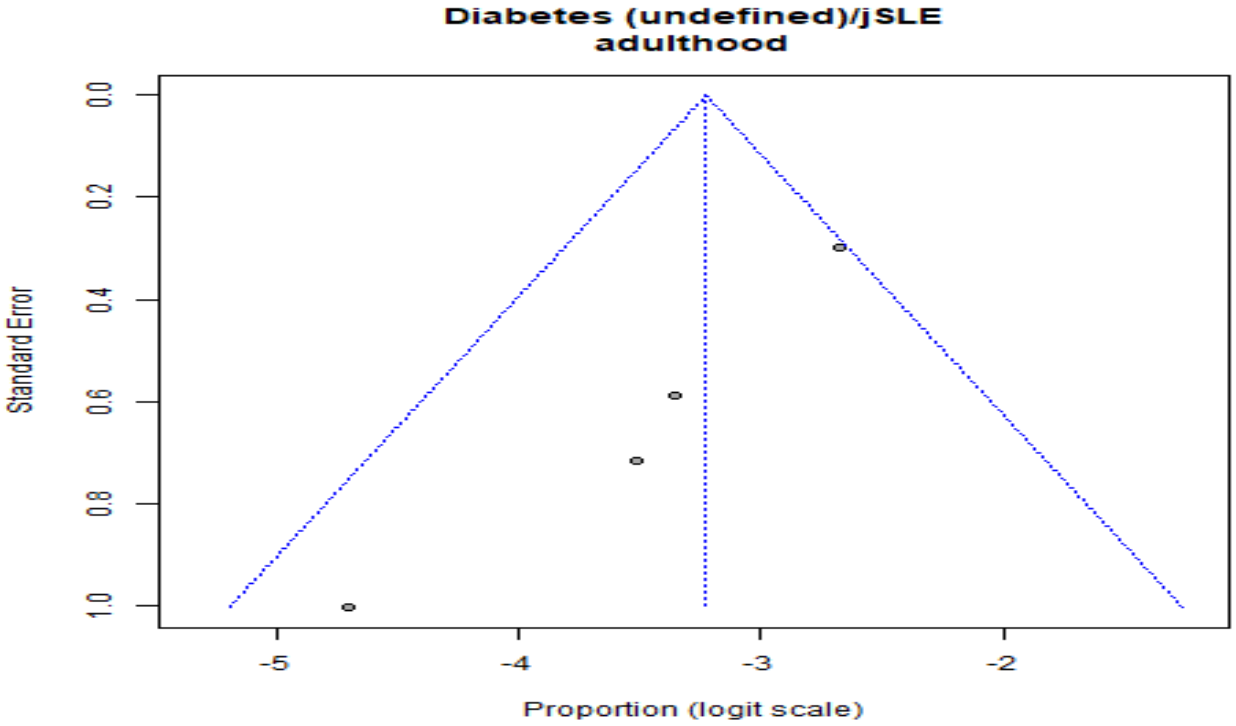
**Supplementary Figure S4.54**: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in adults with juvenile-onset Systemic Lupus Erythematosus

**Diabetes (undefined)**

Forest




Funnel



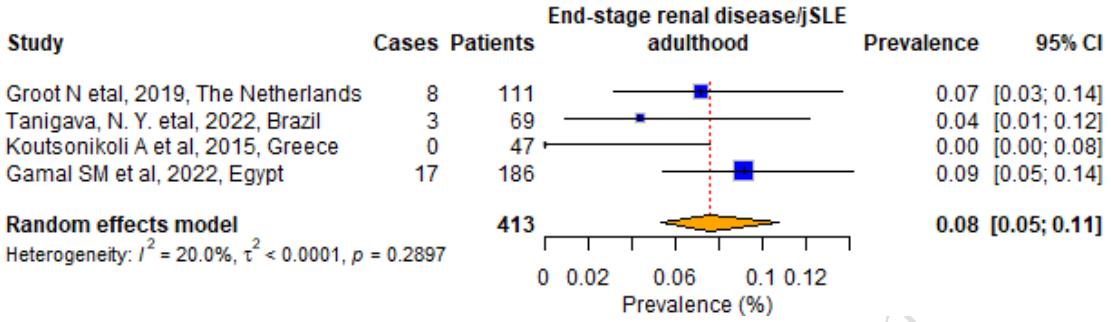

Summary statistics

```
                                proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0290  [0.0035; 0.1008]       18.8
Gamal SM et al, 2022, Egypt            0.0645  [0.0338; 0.1100]       45.4
Sousa S et al, 2016, Portugal          0.0337  [0.0070; 0.0954]       24.6
Groot N etal, 2019, The Netherlands    0.0090  [0.0002; 0.0492]       11.2
Number of studies: k = 4; Number of observations: o = 455; Number of events: e = 18
                    proportion          95%-CI
Random effects model     0.0382 [0.0188; 0.0758]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.2134 [0.0000; 9.2209]; tau = 0.4620 [0.0000; 3.0366]
 I^2 = 39.2% [0.0%; 79.3%]; H = 1.28 [1.00; 2.20]
Test of heterogeneity:     Q d.f. p-value
                        4.93    3  0.1768
```

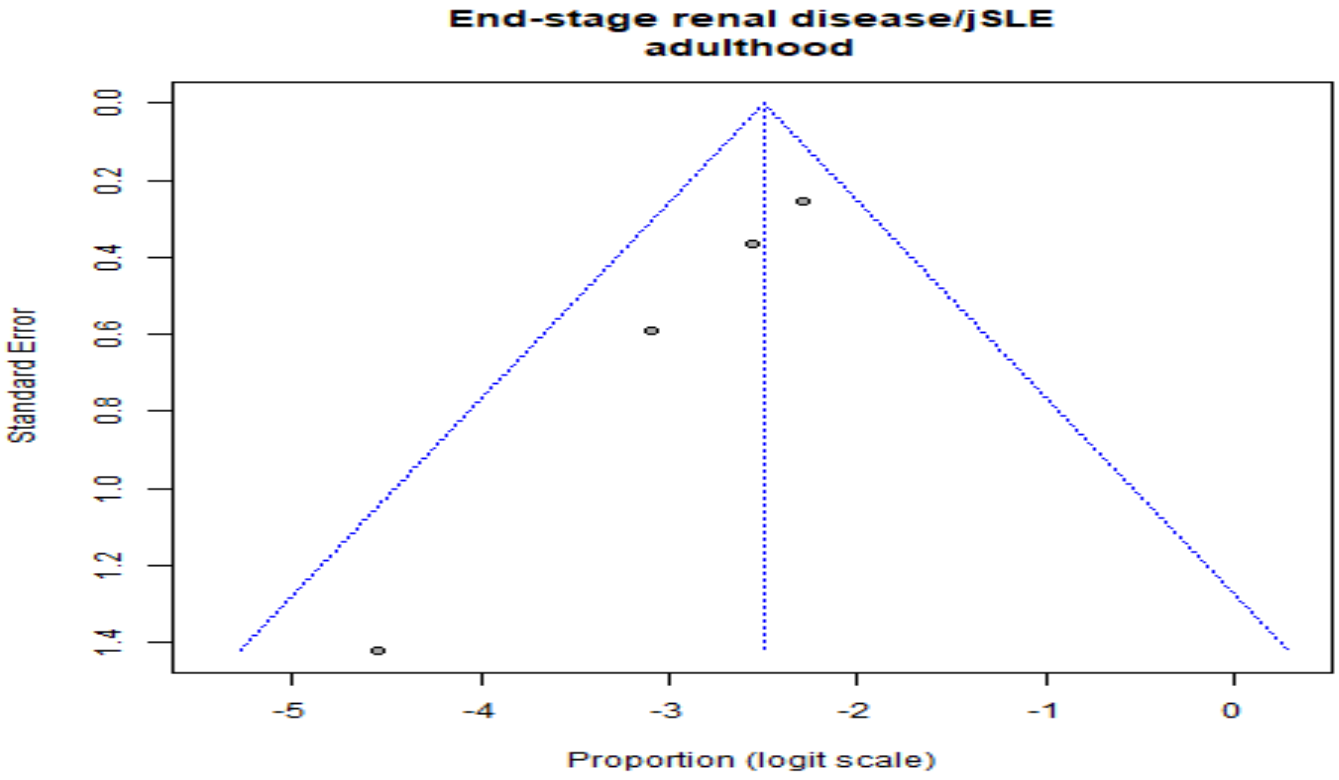
**Supplementary Figure S4.55**: Forest and Funnel Plots of studies reporting the prevalence of end-stage renal disease in adults with juvenile-onset Systemic Lupus Erythematosus

**End-stage renal disease**

Forest




Funnel




Summary statistics

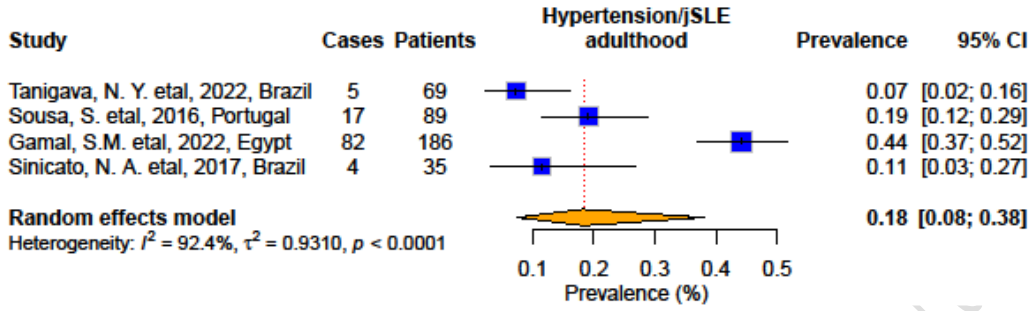
```
                                   proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        28.3
Tanigava, N. Y. etal, 2022, Brazil      0.0435  [0.0091; 0.1218]        10.9
Koutsonikoli A et al, 2015, Greece      0.0000  [0.0000; 0.0755]        1.9
Gamal SM et al, 2022, Egypt             0.0914  [0.0541; 0.1423]        58.9
Number of studies: k = 4; Number of observations: o = 413; Number of events: e = 28
                     proportion           95%-CI
Random effects model     0.0759 [0.0531; 0.1075]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 12.7842]; tau = 0.0006 [0.0000; 3.5755]
 I^2 = 20.0% [0.0%; 87.8%]; H = 1.12 [1.00; 2.86]
Test of heterogeneity:     Q d.f. p-value
                        3.75    3  0.2897
```

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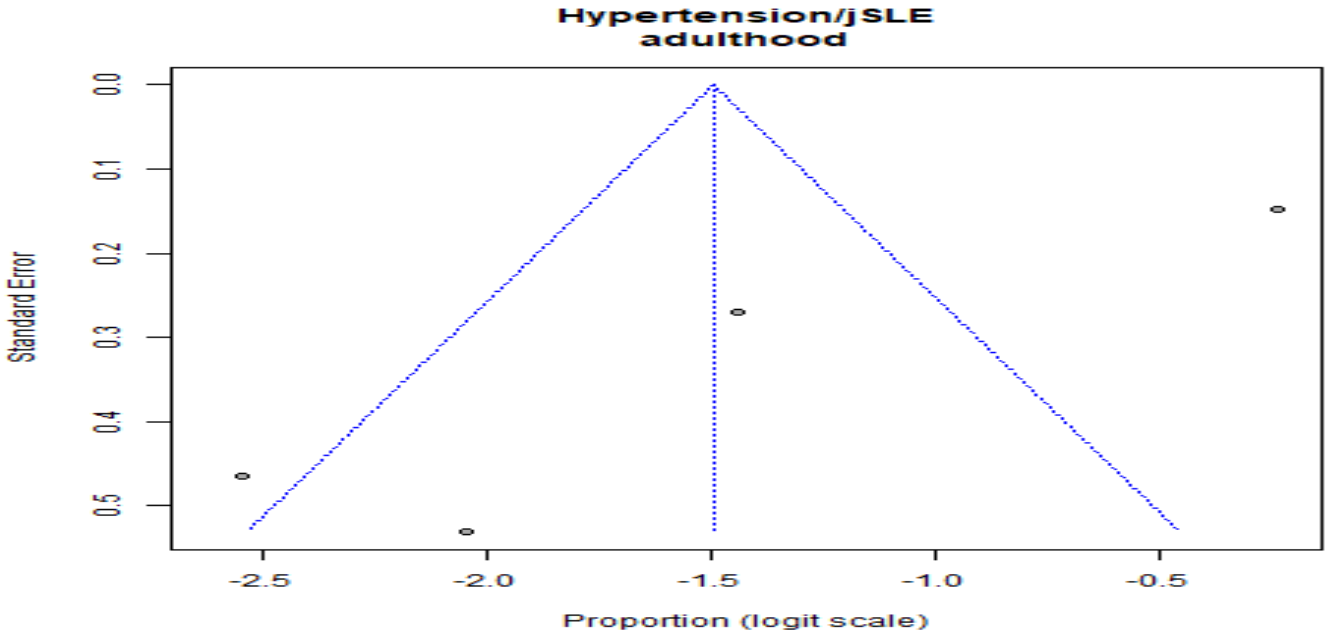
**Supplementary Figure S4.56**: Forest and Funnel Plots of studies reporting the prevalence of hypertension in adults with juvenile-onset Systemic Lupus Erythematosus

**Hypertension**

Forest




Funnel




Summary statistics

```
                                  proportion   95% CI                 %W(random)
Tanigava, N. Y. etal, 2022, Brazil     0.0725   [0.0239; 0.1611]        23.3
Sousa, S. etal, 2016, Portugal         0.1910   [0.1154; 0.2881]        26.6
Gamal, S.M. etal, 2022, Egypt          0.4409   [0.3683; 0.5154]        28.0
Sinicato, N. A. etal, 2017, Brazil     0.1143   [0.0320; 0.2674]        22.0

Number of studies: k = 4
Number of observations: o = 379
Number of events: e = 108

                    proportion 95% CI
Random effects model     0.1830 [0.0752; 0.3816]

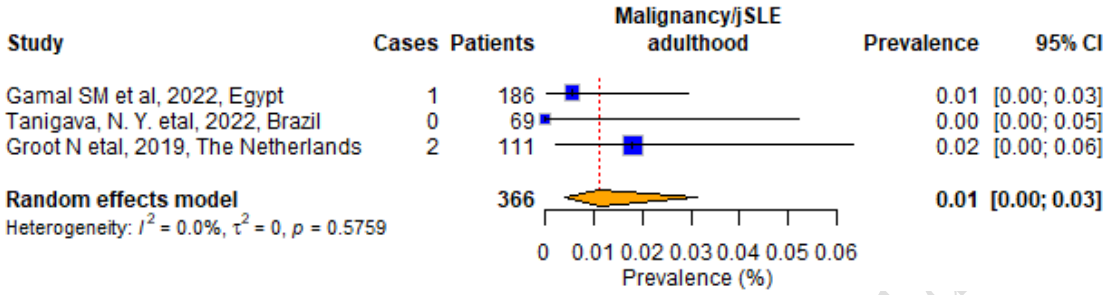
Quantifying heterogeneity (with 95% CIs):
 tau^2 = 0.9310 [0.2222; 13.6986]; tau = 0.9649 [0.4714; 3.7012]
 I^2 = 92.4% [83.8%; 96.5%]; H = 3.63 [2.49; 5.31]

Test of heterogeneity:
     Q d.f.  p-value
 39.62    3 < 0.0001
```

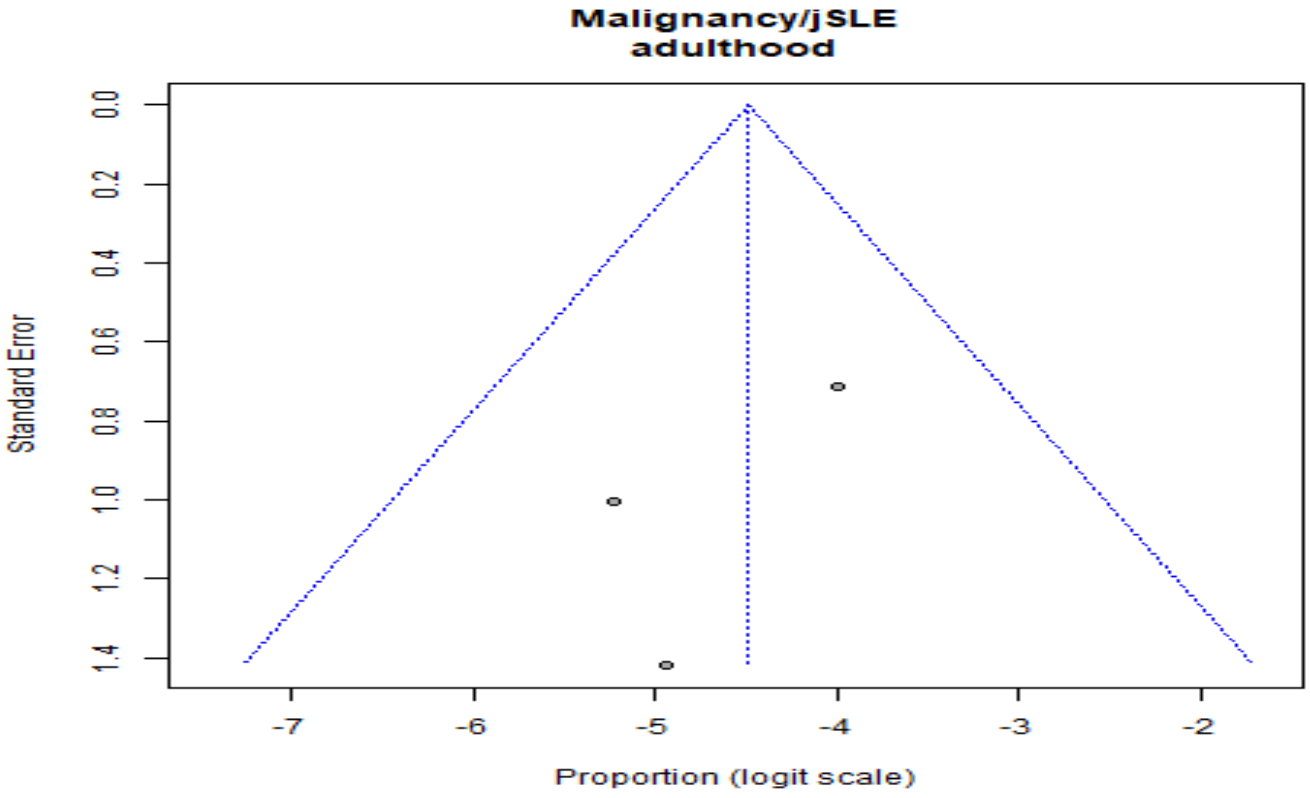
**Supplementary Figure S4.57**: Forest and Funnel Plots of studies reporting the prevalence of malignancy in adults with juvenile-onset Systemic Lupus Erythematosus

**Malignancy**

Forest




Funnel




Summary statistics

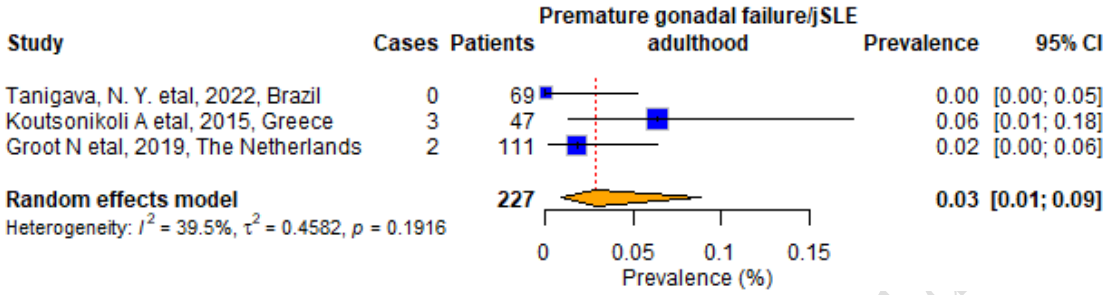
```
                                   proportion  95%-CI          %W(random)
Gamal SM et al, 2022, Egypt            0.0054  [0.0001; 0.0296]        28.8
Tanigava, N. Y. etal, 2022, Brazil     0.0000  [0.0000; 0.0521]        14.4
Groot N etal, 2019, The Netherlands    0.0180  [0.0022; 0.0636]        56.8
Number of studies: k = 3; Number of observations: o = 366; Number of events: e = 3
                     proportion           95%-CI
Random effects model     0.0112 [0.0039; 0.0314]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 15.3931]; tau = 0 [0.0000; 3.9234]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        1.10    2  0.5759
```

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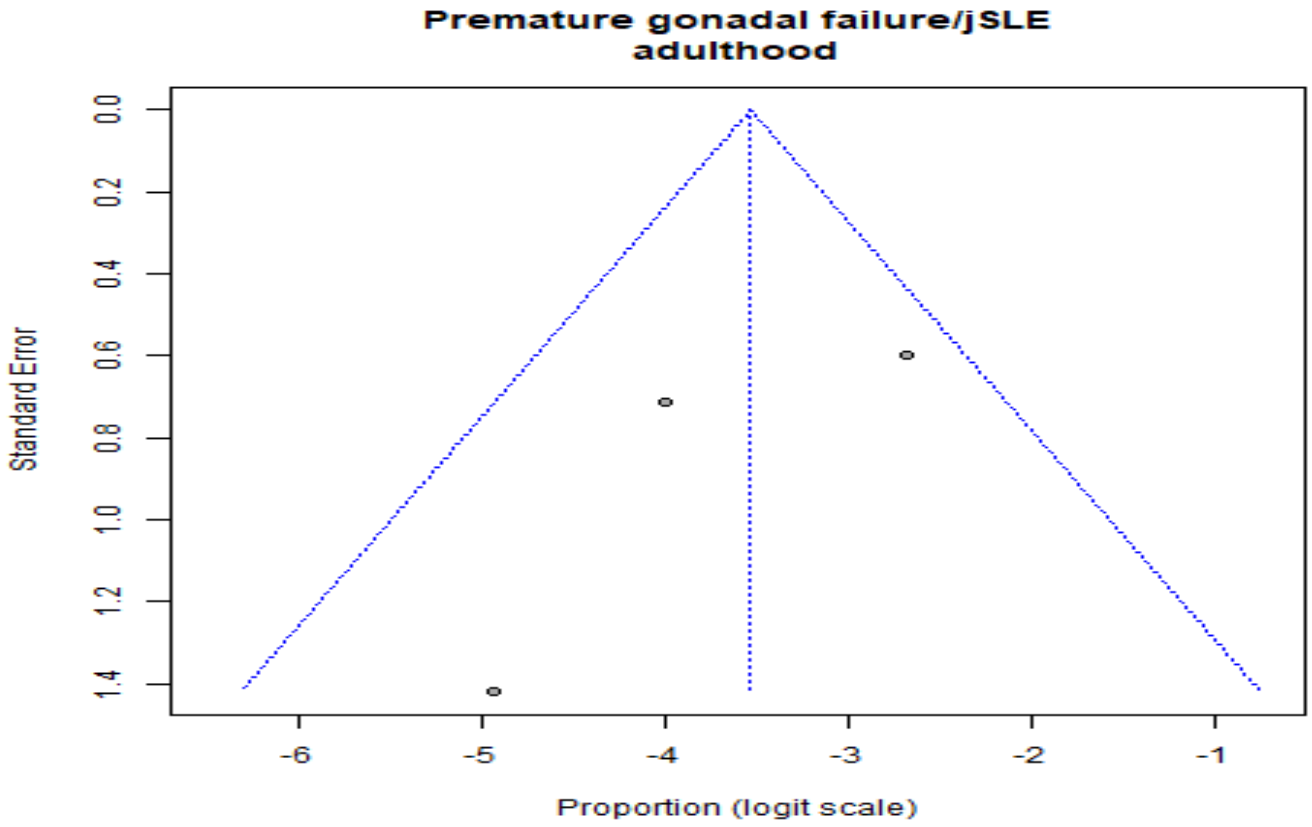
**Supplementary Figure S4.58**: Forest and Funnel Plots of studies reporting the prevalence of premature gonadal failure in adults with juvenile-onset Systemic Lupus Erythematosus

**Premature gonadal failure**

Forest




Funnel




Summary statistics

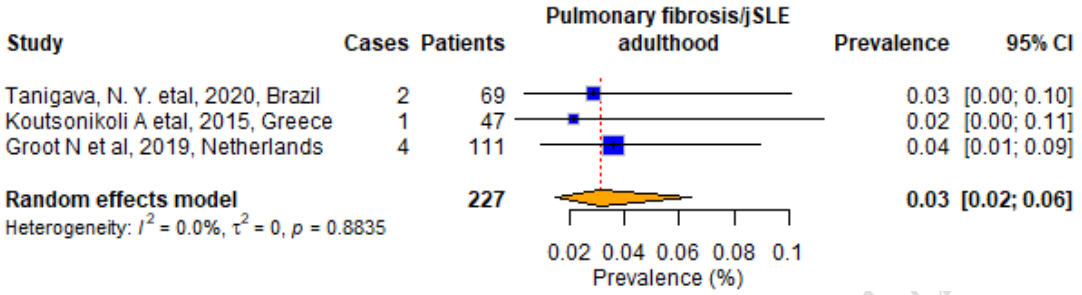
```
                                    proportion  95%-CI          %W(random)
Tanigava, N. Y. etal, 2022, Brazil      0.0000  [0.0000; 0.0521]        15.2
Koutsonikoli A etal, 2015, Greece       0.0638  [0.0134; 0.1754]        46.1
Groot N etal, 2019, The Netherlands     0.0180  [0.0022; 0.0636]        38.8
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 5
                     proportion           95%-CI
Random effects model     0.0283 [0.0087; 0.0882]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.4582 [0.0000; 49.3226]; tau = 0.6769 [0.0000; 7.0230]
 I^2 = 39.5% [0.0%; 81.3%]; H = 1.29 [1.00; 2.31]
Test of heterogeneity:     Q d.f. p-value
                        3.30    2  0.1916
```

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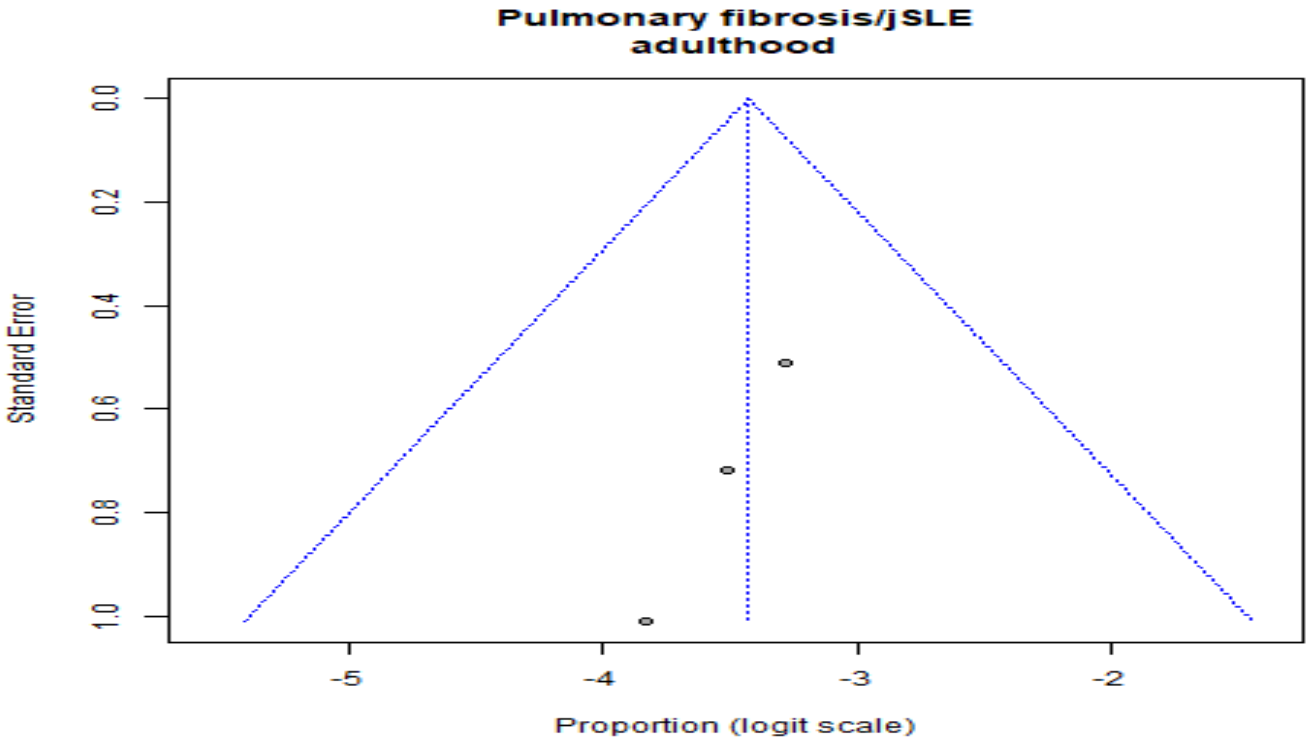
**Supplementary Figure S4.59**: Forest and Funnel Plots of studies reporting the prevalence of pulmonary fibrosis in adults with juvenile-onset Systemic Lupus Erythematosus

**Pulmonary fibrosis**

Forest




Funnel



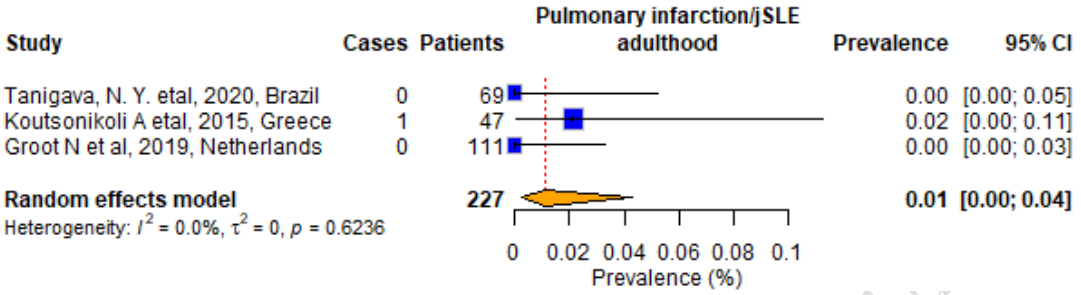

Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0290   [0.0035; 0.1008]       28.7
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       14.4
Groot N et al, 2019, Netherlands       0.0360   [0.0099; 0.0897]       56.9
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 7
                    proportion           95%-CI
Random effects model     0.0314 [0.0150; 0.0644]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 2.2629]; tau = 0 [0.0000; 1.5043]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:    Q d.f. p-value
                       0.25    2  0.8835
```

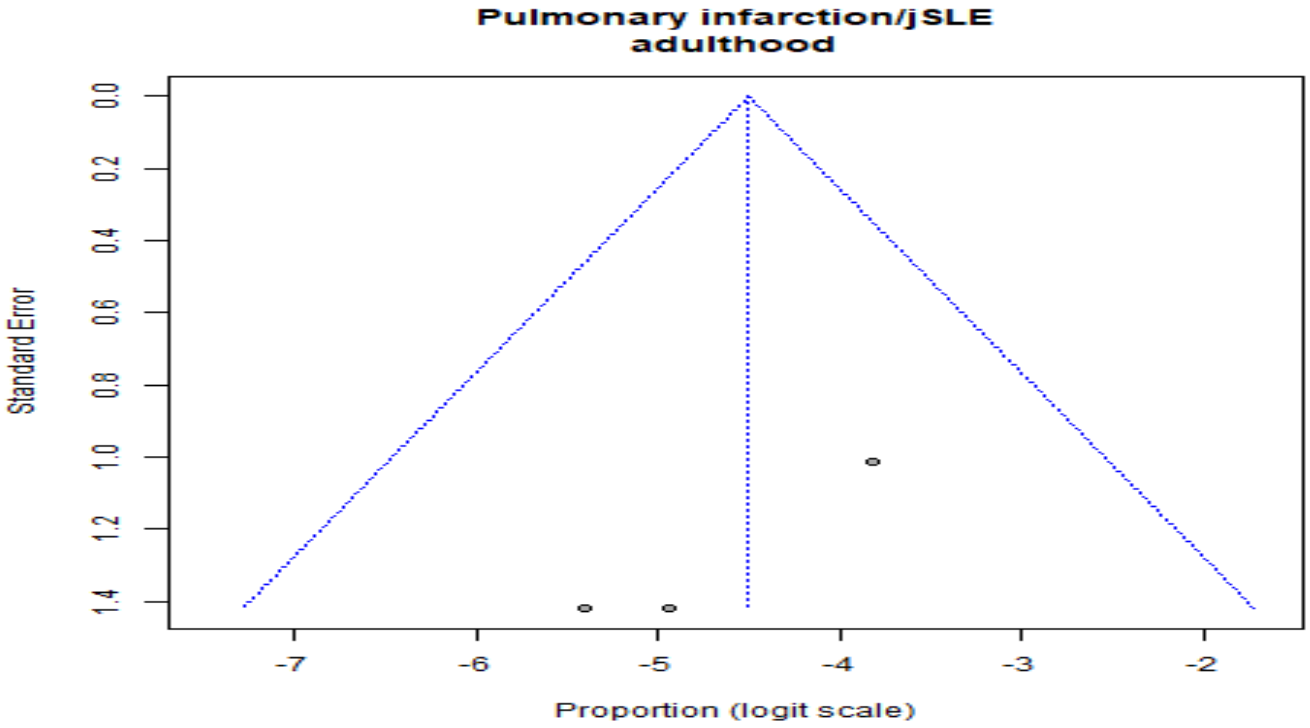
**Supplementary Figure S4.60**: Forest and Funnel Plots of studies reporting the prevalence of pulmonary infarction in adults with juvenile-onset Systemic Lupus Erythematosus

**Pulmonary infarction**

Forest




Funnel



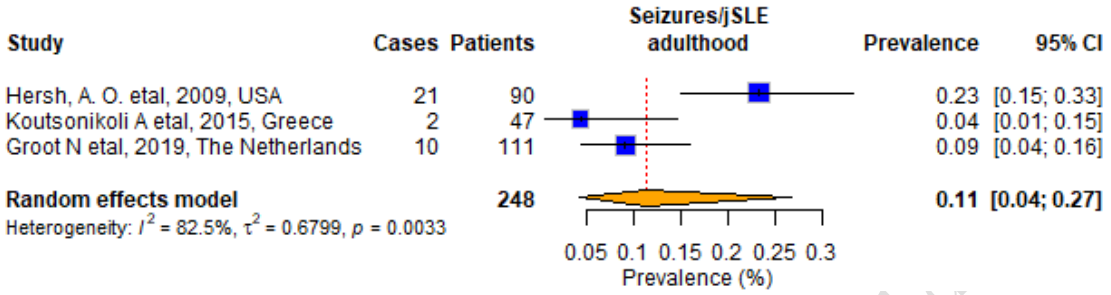

Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tanigava, N. Y. etal, 2020, Brazil     0.0000   [0.0000; 0.0521]       25.2
Koutsonikoli A etal, 2015, Greece      0.0213   [0.0005; 0.1129]       49.6
Groot N et al, 2019, Netherlands       0.0000   [0.0000; 0.0327]       25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                     proportion           95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

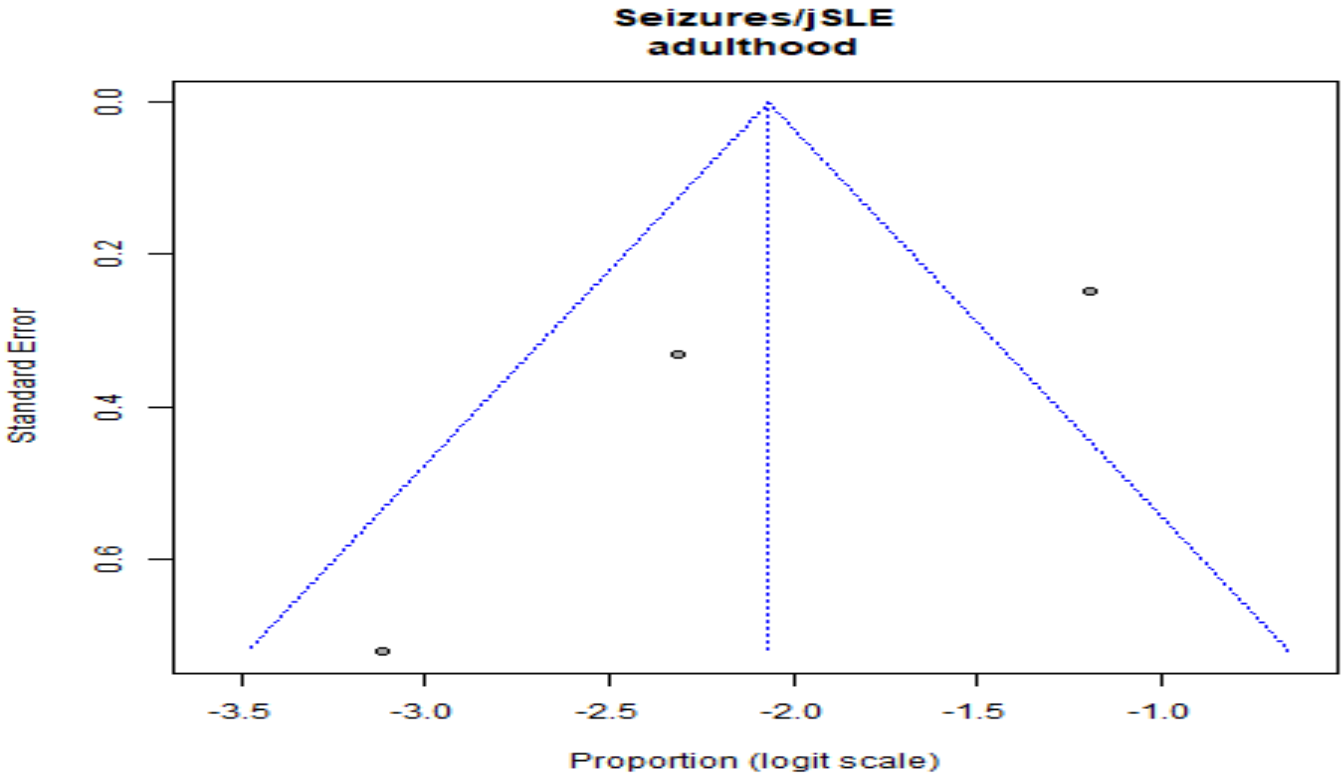
**Supplementary Figure S4.61**: Forest and Funnel Plots of studies reporting the prevalence of seizures in adults with juvenile-onset Systemic Lupus Erythematosus

**Seizures**

Forest




Funnel



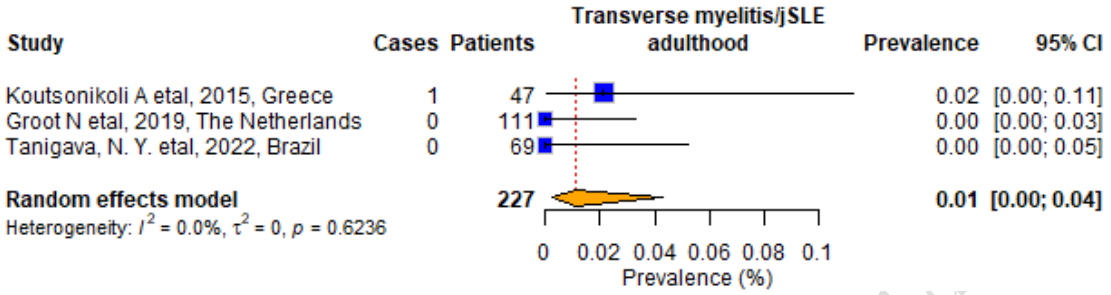

Summary statistics

```
                                proportion  95%-CI          %W(random)
Hersh, A. O. etal, 2009, USA        0.2333  [0.1506; 0.3343]       39.1
Koutsonikoli A etal, 2015, Greece   0.0426  [0.0052; 0.1454]       24.1
Groot N etal, 2019, The Netherlands 0.0901  [0.0441; 0.1594]       36.7
Number of studies: k = 3; Number of observations: o = 248; Number of events: e = 33
                     proportion           95%-CI
Random effects model     0.1124 [0.0422; 0.2668]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.6799 [0.0646; 36.6266]; tau = 0.8246 [0.2542; 6.0520]
 I^2 = 82.5% [46.4%; 94.3%]; H = 2.39 [1.37; 4.19]
Test of heterogeneity:      Q d.f. p-value
                        11.43    2  0.0033
```

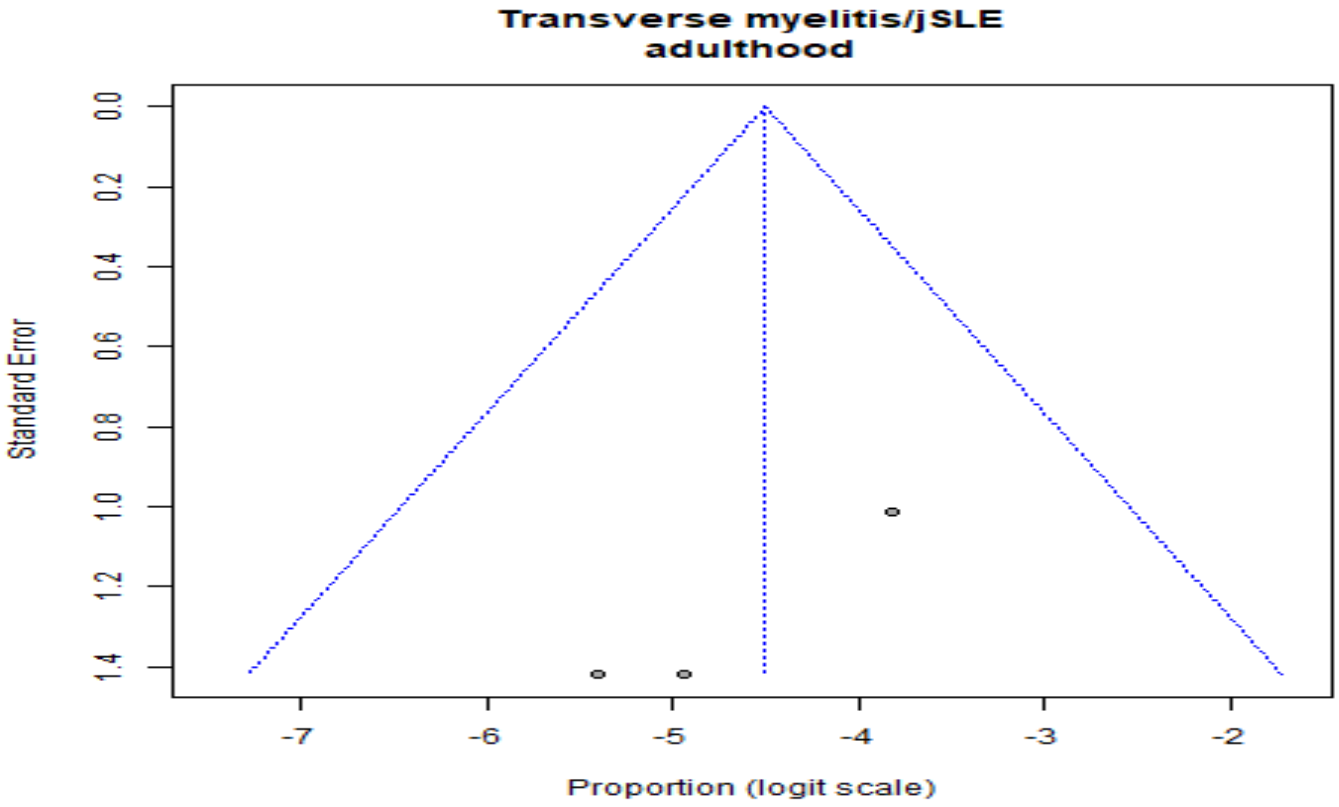
**Supplementary Figure S4.62**: Forest and Funnel Plots of studies reporting the prevalence of transverse myelitis in adults with juvenile-onset Systemic Lupus Erythematosus

**Transverse myelitis**

Forest




Funnel




Summary statistics

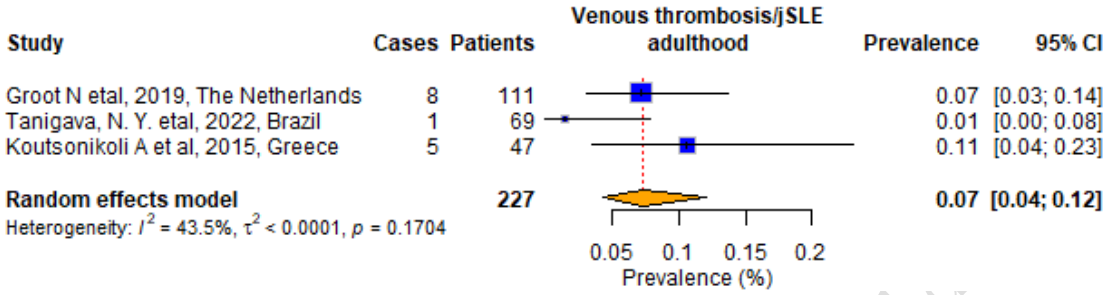
```
                                    proportion  95%-CI          %W(random)
Koutsonikoli A etal, 2015, Greece       0.0213  [0.0005; 0.1129]        49.6
Groot N etal, 2019, The Netherlands     0.0000  [0.0000; 0.0327]        25.2
Tanigava, N. Y. etal, 2022, Brazil      0.0000  [0.0000; 0.0521]        25.2
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 1
                     proportion           95%-CI
Random effects model     0.0109 [0.0027; 0.0427]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0 [0.0000; 24.5185]; tau = 0 [0.0000; 4.9516]
 I^2 = 0.0% [0.0%; 89.6%]; H = 1.00 [1.00; 3.10]
Test of heterogeneity:     Q d.f. p-value
                        0.94    2  0.6236
```

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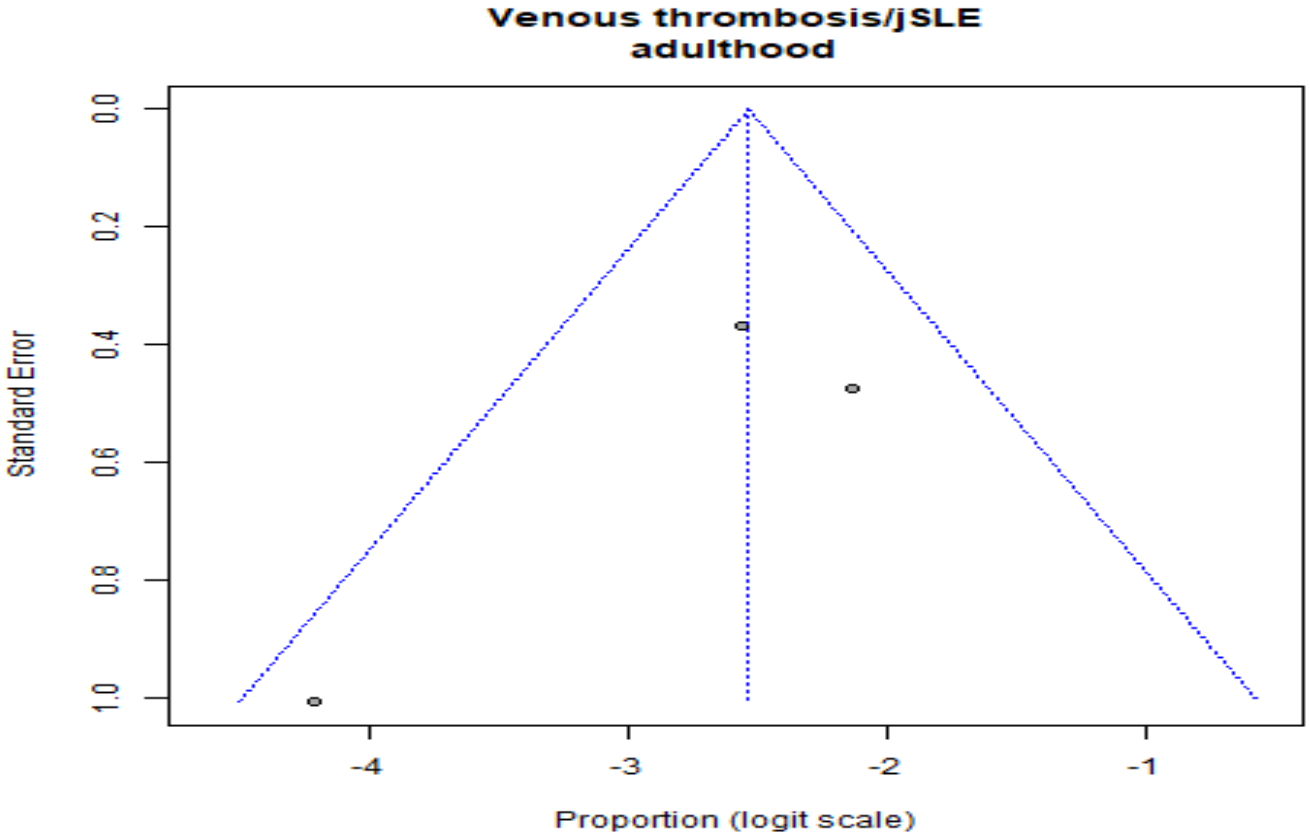
**Supplementary Figure S4.63**: Forest and Funnel Plots of studies reporting the prevalence of venous thrombosis in adults with juvenile-onset Systemic Lupus Erythematosus

**Venous thrombosis**

Forest




Funnel




Summary statistics

```
                                    proportion  95%-CI          %W(random)
Groot N etal, 2019, The Netherlands     0.0721  [0.0316; 0.1371]        57.6
Tanigava, N. Y. etal, 2022, Brazil      0.0145  [0.0004; 0.0781]         7.7
Koutsonikoli A et al, 2015, Greece      0.1064  [0.0355; 0.2310]        34.7
Number of studies: k = 3; Number of observations: o = 227; Number of events: e = 14
                     proportion           95%-CI
Random effects model     0.0735 [0.0439; 0.1204]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 47.4982]; tau = 0.0006 [0.0000; 6.8919]
 I^2 = 43.5% [0.0%; 83.1%]; H = 1.33 [1.00; 2.43]
Test of heterogeneity:     Q d.f. p-value
                        3.54    2  0.1704
```

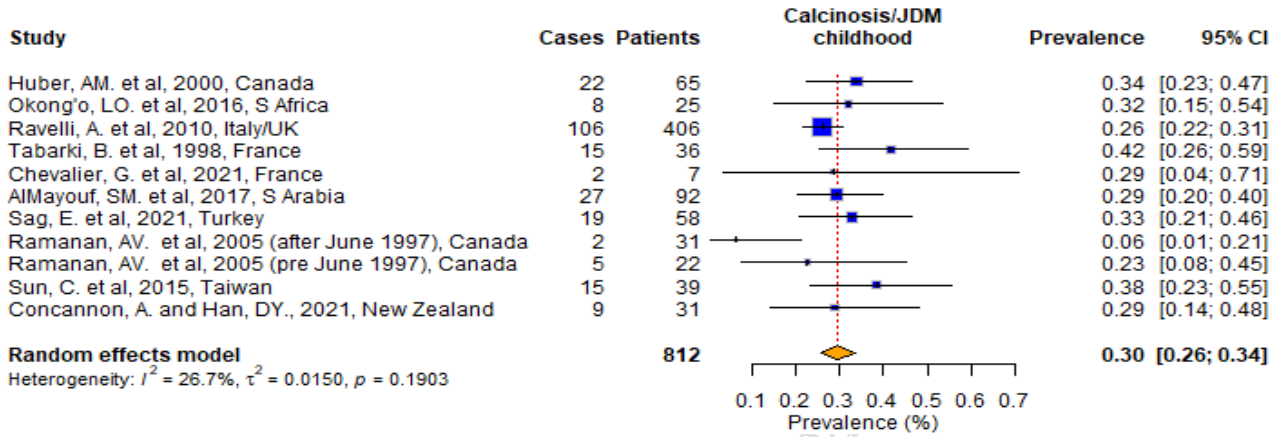
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## Childhood/JDM

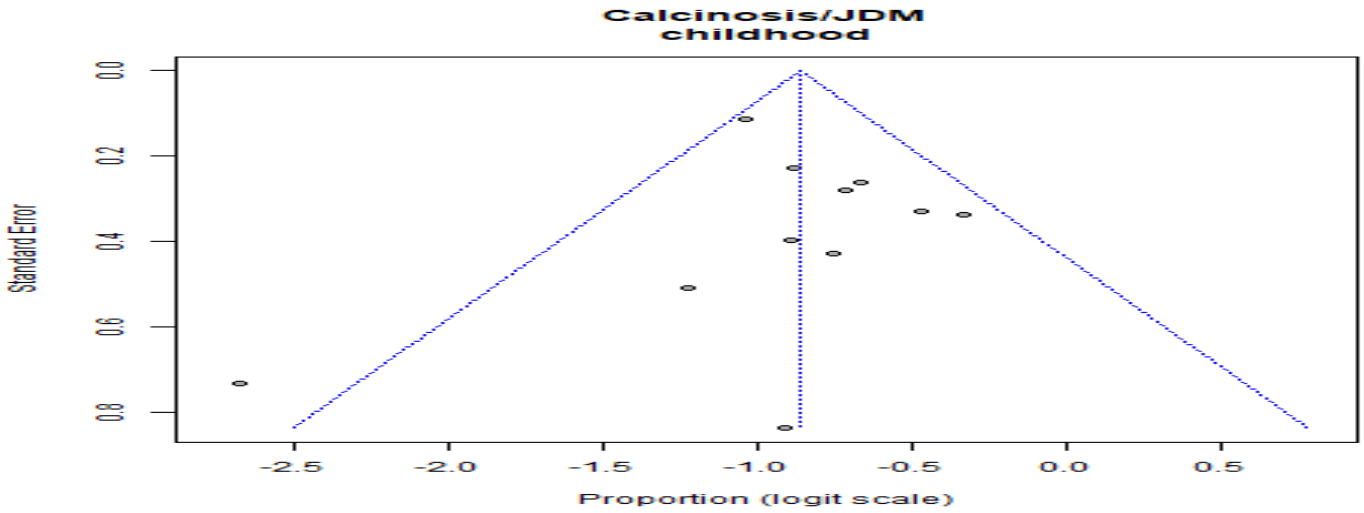
**Supplementary Figure S4.64**: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in children with Juvenile Dermatomyositis

### Calcinosis

Forest




Funnel




Summary statistics

For reproduction

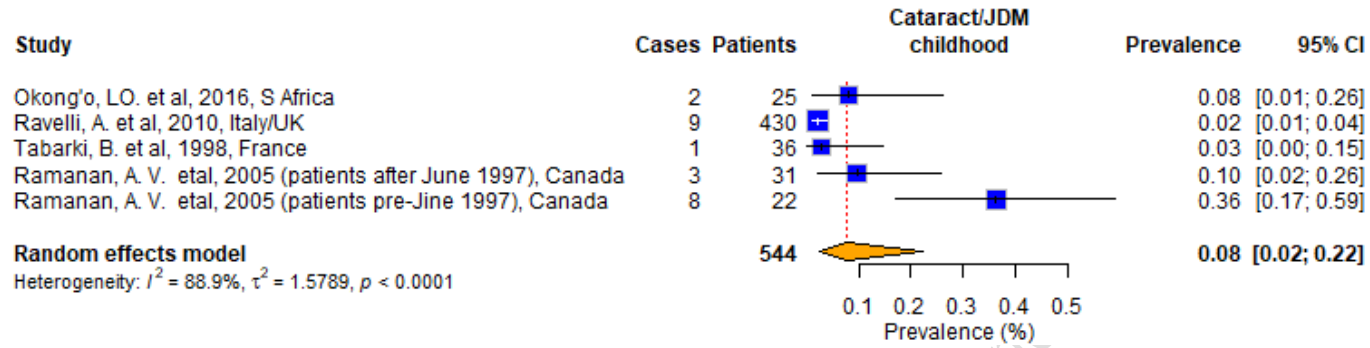
```
                                                    proportion  95%-CI          %W(random)
Huber, AM. et al, 2000, Canada                          0.3385  [0.2257; 0.4665]        11.2
Okong'o, LO. et al, 2016, S Africa                      0.3200  [0.1495; 0.5350]         4.7
Ravelli, A. et al, 2010, Italy/UK                       0.2611  [0.2190; 0.3067]        33.6
Tabarki, B. et al, 1998, France                         0.4167  [0.2551; 0.5924]         7.2
Chevalier, G. et al, 2021, France                       0.2857  [0.0367; 0.7096]         1.3
AlMayouf, SM. et al, 2017, S Arabia                     0.2935  [0.2031; 0.3976]        13.9
Sag, E. et al, 2021, Turkey                             0.3276  [0.2101; 0.4634]        10.0
Ramanan, AV.  et al, 2005 (after June 1997), Canada     0.0645  [0.0079; 0.2142]         1.7
Ramanan, AV.  et al, 2005 (pre June 1997), Canada       0.2273  [0.0782; 0.4537]         3.4
Sun, C. et al, 2015, Taiwan                             0.3846  [0.2336; 0.5538]         7.6
Concannon, A. and Han, DY., 2021, New Zealand           0.2903  [0.1422; 0.4804]         5.4
Number of studies: k = 11; Number of observations: o = 812; Number of events: e = 230
                     proportion           95%-CI
Random effects model     0.2970 [0.2591; 0.3381]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.0150 [0.0000; 0.7653]; tau = 0.1226 [0.0000; 0.8748]
 I^2 = 26.7% [0.0%; 63.7%]; H = 1.17 [1.00; 1.66]
Test of heterogeneity:       Q d.f. p-value
                         13.63   10  0.1903
```

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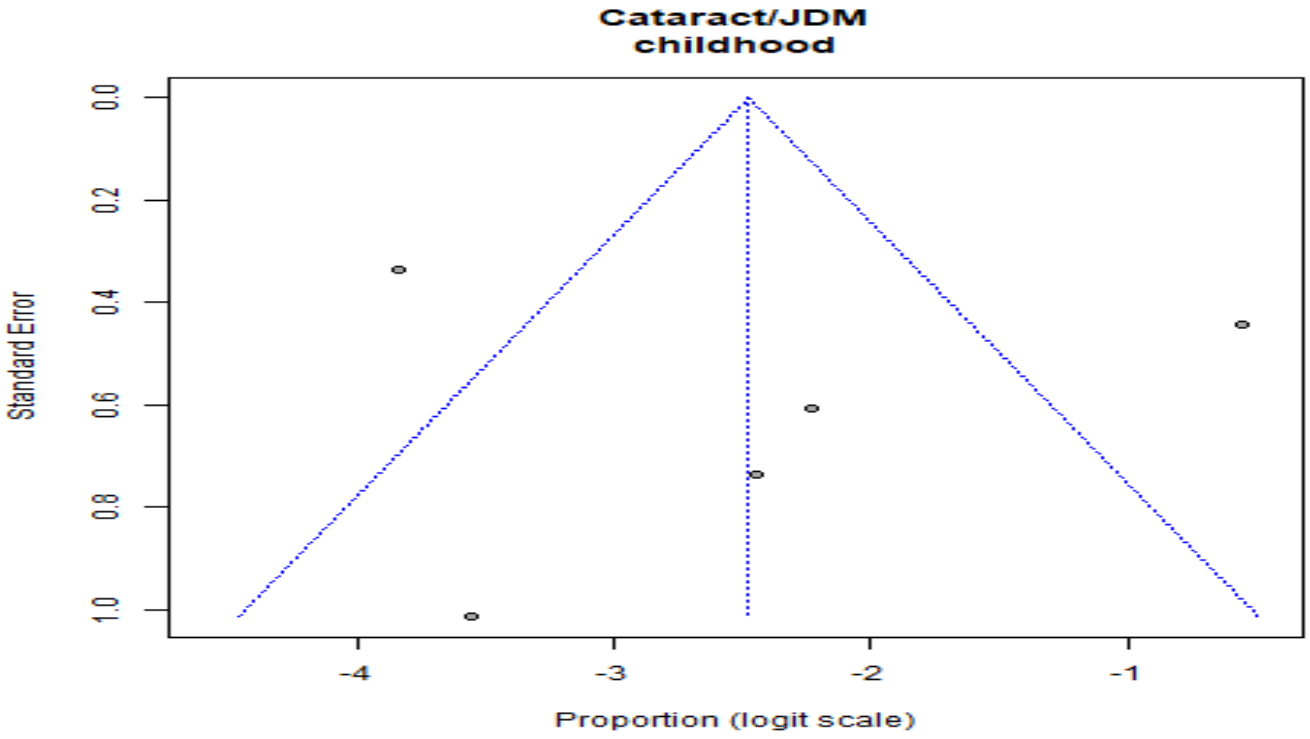
**Supplementary Figure S4.65**: Forest and Funnel Plots of studies reporting the prevalence of cataract in children with Juvenile Dermatomyositis

**Cataract**

Forest




Funnel




Summary statistics

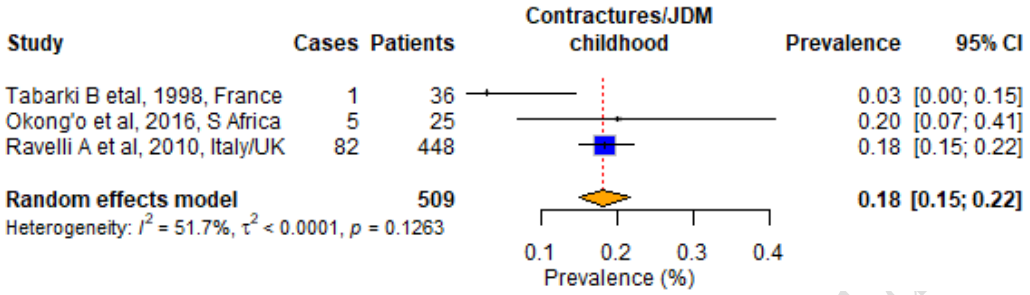
```
                                                        proportion        95%-CI          %W(random)
Okong'o, LO. et al, 2016, S Africa                          0.0800        [0.0098; 0.2603]        18.7
Ravelli, A. et al, 2010, Italy/UK                           0.0209        [0.0096; 0.0394]        23.4
Tabarki, B. et al, 1998, France                             0.0278        [0.0007; 0.1453]        15.2
Ramanan, A. V.  etal, 2005 (patients after June 1997), Canada     0.0968        [0.0204; 0.2575]        20.4
Ramanan, A. V.  etal, 2005 (patients pre-June 1997), Canada       0.3636        [0.1720; 0.5934]        22.3
Number of studies: k = 5; Number of observations: o = 544; Number of events: e = 23
                     proportion           95%-CI
Random effects model     0.0775 [0.0239; 0.2239]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5789 [0.3765; 13.6521]; tau = 1.2565 [0.6136; 3.6949]
 I^2 = 88.9% [76.9%; 94.7%]; H = 3.01 [2.08; 4.35]
Test of heterogeneity:      Q d.f.  p-value
                        36.16    4 < 0.0001
```

=For reproducibility only in academic research and developmen

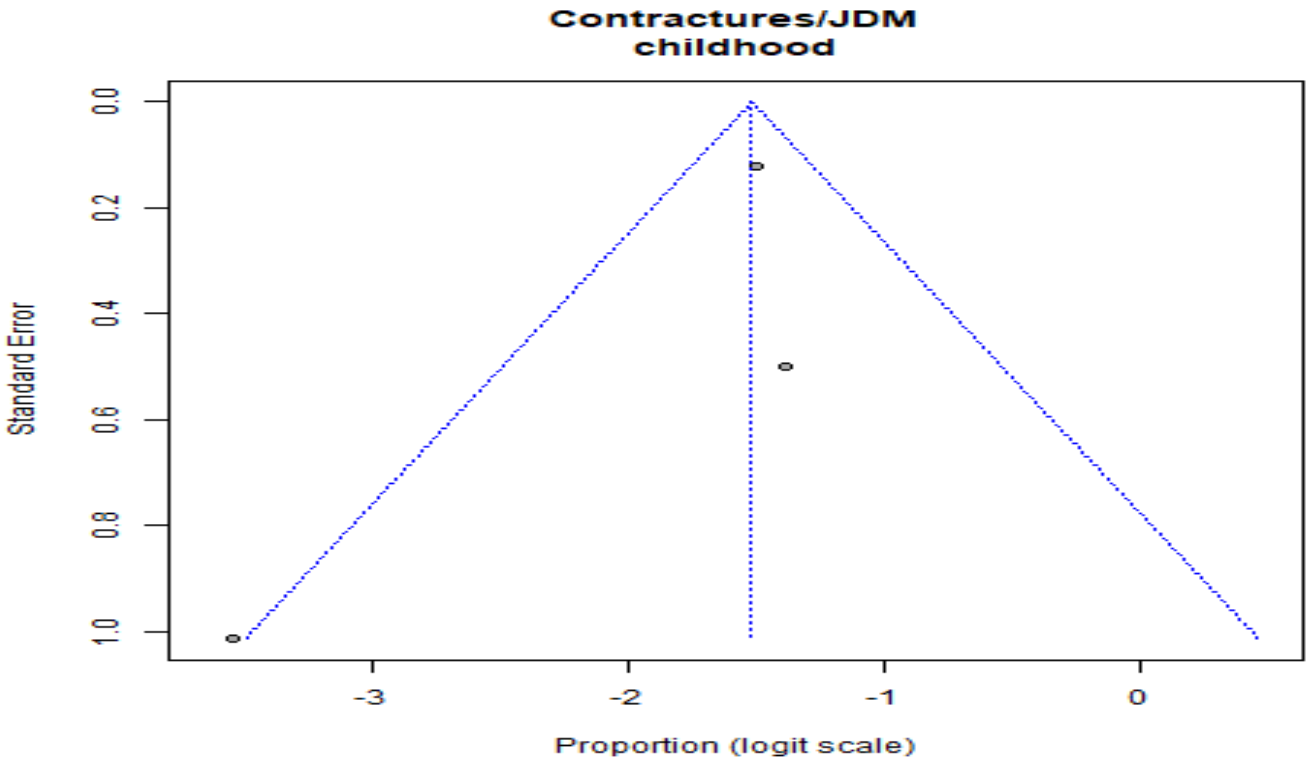
**Supplementary Figure S4.66**: Forest and Funnel Plots of studies reporting the prevalence of contractures in children with Juvenile Dermatomyositis

**Contractures**

Forest




Funnel



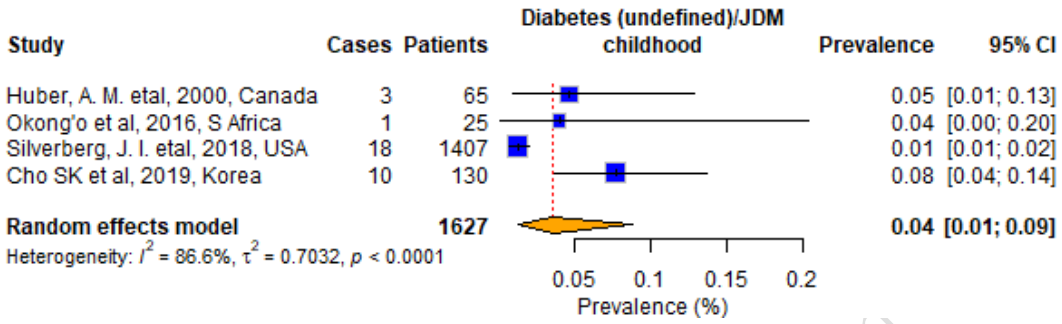

Summary statistics

```
                                proportion      95%-CI          %W(random)
Tabarki B etal, 1998, France        0.0278      [0.0007; 0.1453]        1.4
Okong'o et al, 2016, S Africa       0.2000      [0.0683; 0.4070]        5.6
Ravelli A et al, 2010, Italy/UK     0.1830      [0.1483; 0.2220]       93.1
Number of studies: k = 3; Number of observations: o = 509; Number of events: e = 88
                     proportion           95%-CI
Random effects model     0.1798 [0.1482; 0.2164]
Quantifying heterogeneity (with 95%-CIs):  tau^2 < 0.0001 [0.0000; 58.2361]; tau = 0.0020 [0.0000; 7.6313]
 I^2 = 51.7% [0.0%; 86.1%]; H = 1.44 [1.00; 2.68]
Test of heterogeneity:     Q d.f. p-value
                        4.14    2  0.1263
```

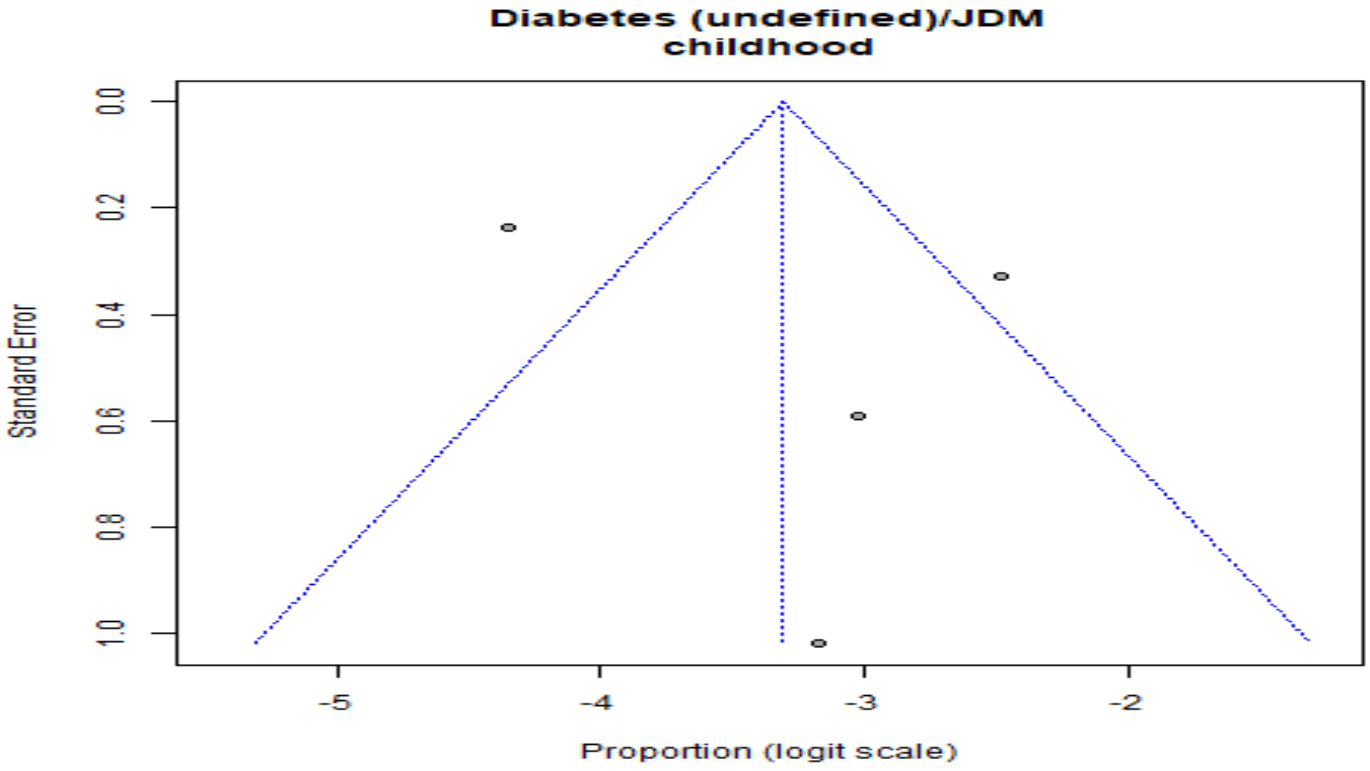
**Supplementary Figure S4.67**: Forest and Funnel Plots of studies reporting the prevalence of diabetes (undefined) in children with Juvenile Dermatomyositis

**Diabetes (undefined)**

Forest




Funnel




Summary statistics

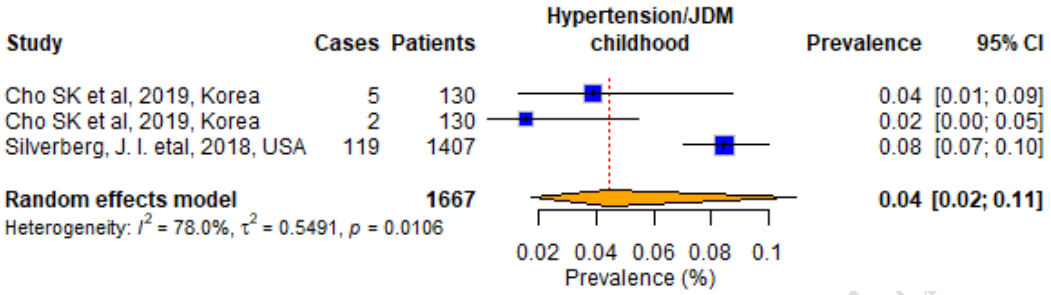
```
                                  proportion    95%-CI          %W(random)
Huber, A. M. etal, 2000, Canada       0.0462    [0.0096; 0.1290]        23.3
Okong'o et al, 2016, S Africa         0.0400    [0.0010; 0.2035]        14.1
Silverberg, J. I. etal, 2018, USA     0.0128    [0.0076; 0.0201]        32.3
Cho SK et al, 2019, Korea             0.0769    [0.0375; 0.1369]        30.3
Number of studies: k = 4; Number of observations: o = 1627; Number of events: e = 32
                     proportion           95%-CI
Random effects model     0.0352 [0.0136; 0.0879]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7032 [0.1139; 8.4433]; tau = 0.8386 [0.3376; 2.9057]
 I^2 = 86.6% [67.7%; 94.5%]; H = 2.74 [1.76; 4.25]
Test of heterogeneity:     Q d.f.  p-value
                       22.44    3 < 0.0001
```

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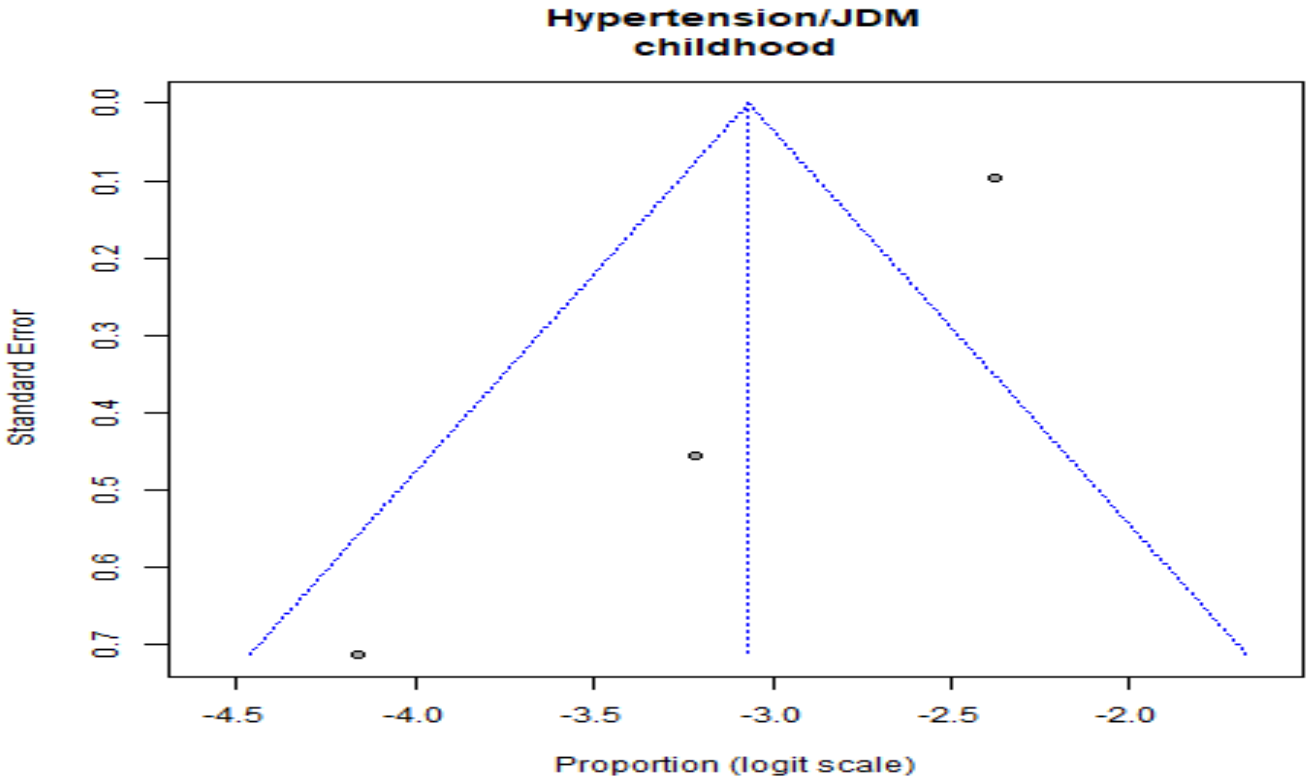
**Supplementary Figure S4.68**: Forest and Funnel Plots of studies reporting the prevalence of hypertension in children with Juvenile Dermatomyositis

**Hypertension**

Forest




Funnel



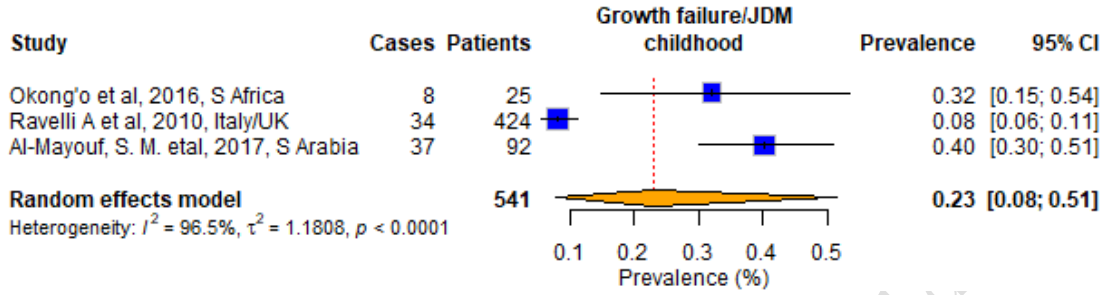

Summary statistics

```
                                proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea           0.0385    [0.0126; 0.0875]       32.5
Cho SK et al, 2019, Korea           0.0154    [0.0019; 0.0545]       23.3
Silverberg, J. I. etal, 2018, USA   0.0846    [0.0706; 0.1004]       44.1
Number of studies: k = 3; Number of observations: o = 1667; Number of events: e = 126
                     proportion           95%-CI
Random effects model     0.0444 [0.0173; 0.1095]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5491 [0.0293; 30.9540]; tau = 0.7410 [0.1711; 5.5636]
 I^2 = 78.0% [29.1%; 93.2%]; H = 2.13 [1.19; 3.83]
Test of heterogeneity:     Q d.f. p-value
                        9.10    2  0.0106
```

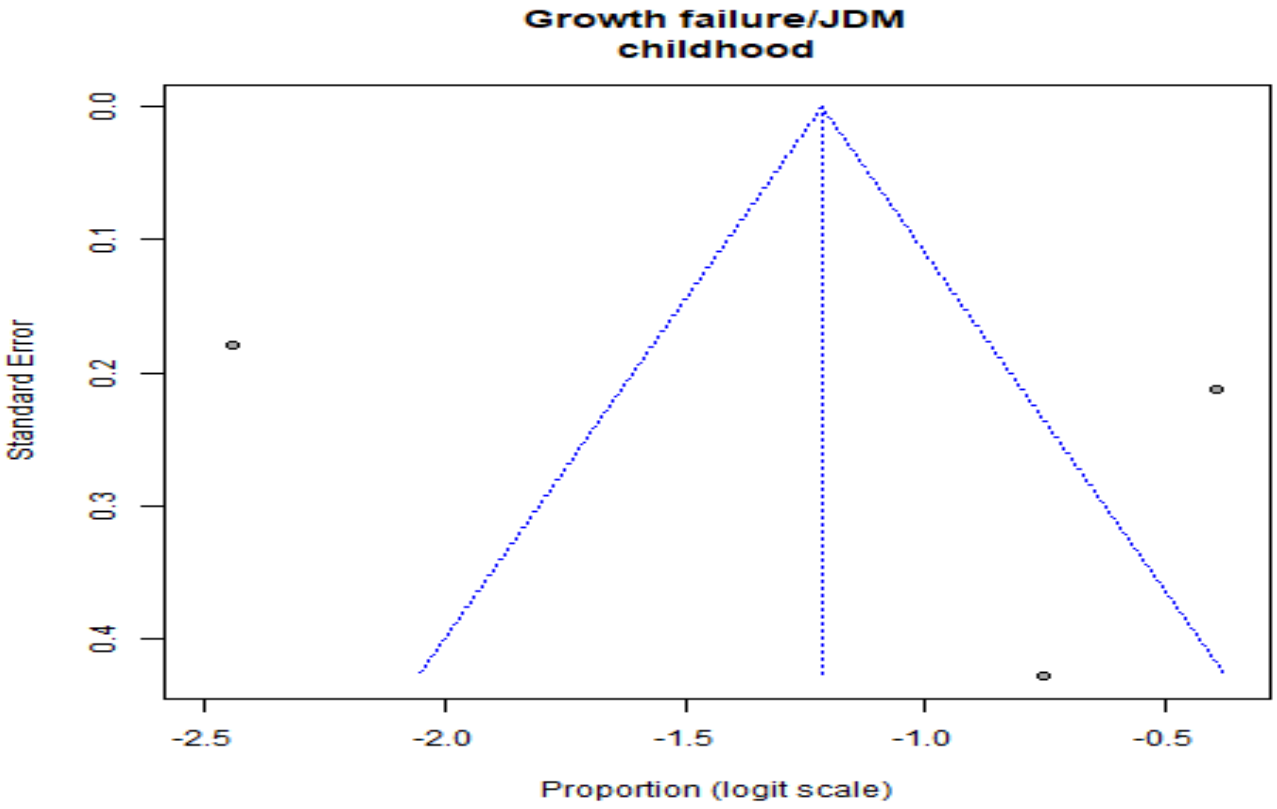
**Supplementary Figure S4.69**: Forest and Funnel Plots of studies reporting the prevalence of growth failure in children with Juvenile Dermatomyositis

**Growth failure**

Forest




Funnel




Summary statistics

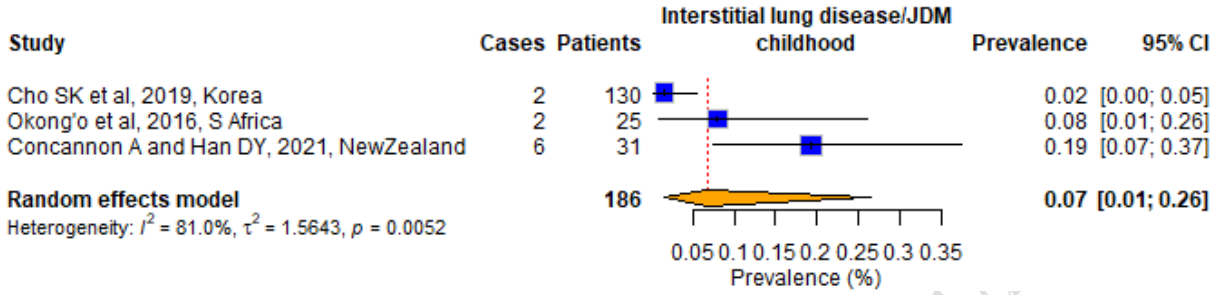
```
                                     proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa            0.3200        [0.1495; 0.5350]        30.9
Ravelli A et al, 2010, Italy/UK          0.0802        [0.0562; 0.1103]        34.7
Al-Mayouf, S. M. etal, 2017, S Arabia    0.4022        [0.3012; 0.5096]        34.4
Number of studies: k = 3; Number of observations: o = 541; Number of events: e = 79
                     proportion           95%-CI
Random effects model     0.2285 [0.0766; 0.5139]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.1808 [0.2769; 46.9921]; tau = 1.0866 [0.5262; 6.8551]
 I^2 = 96.5% [92.8%; 98.3%]; H = 5.37 [3.72; 7.74]
Test of heterogeneity:      Q d.f.  p-value
                        57.58    2 < 0.0001
```

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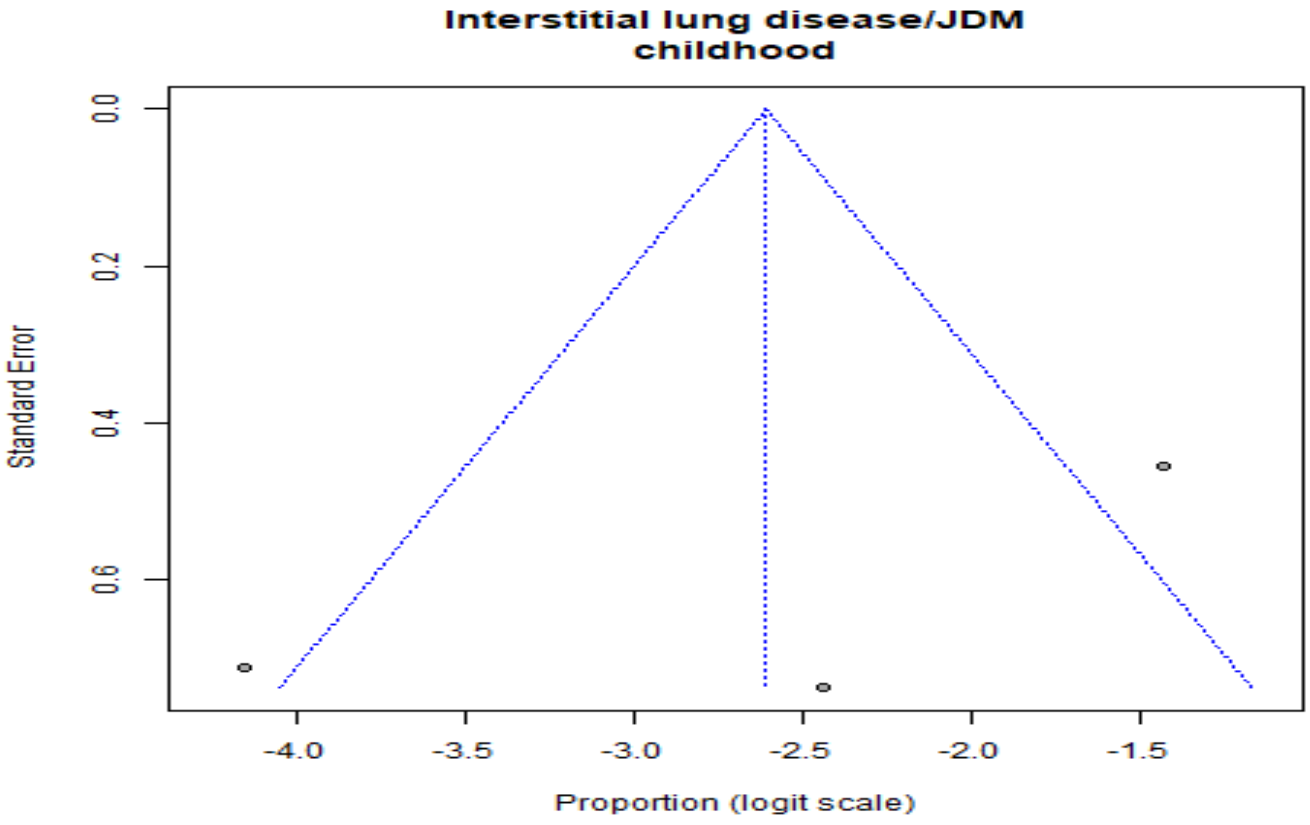
**Supplementary Figure S4.70**: Forest and Funnel Plots of studies reporting the prevalence of Interstitial lung disease in children with Juvenile Dermatomyositis

**Interstitial lung disease**

Forest




Funnel




Summary statistics

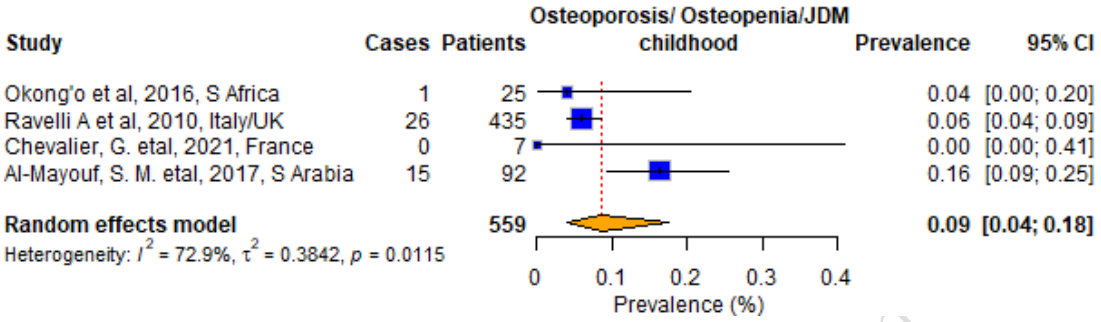
```
                                          proportion    95%-CI          %W(random)
Cho SK et al, 2019, Korea                     0.0154    [0.0019; 0.0545]        31.7
Okong'o et al, 2016, S Africa                 0.0800    [0.0098; 0.2603]        31.2
Concannon A and Han DY, 2021, NewZealand      0.1935    [0.0745; 0.3747]        37.1
Number of studies: k = 3; Number of observations: o = 186; Number of events: e = 10
                     proportion           95%-CI
Random effects model     0.0685 [0.0148; 0.2648]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.5643 [0.1487; 74.9226]; tau = 1.2507 [0.3856; 8.6558]
 I^2 = 81.0% [40.6%; 93.9%]; H = 2.29 [1.30; 4.05]
Test of heterogeneity:      Q d.f. p-value
                        10.52    2  0.0052
```

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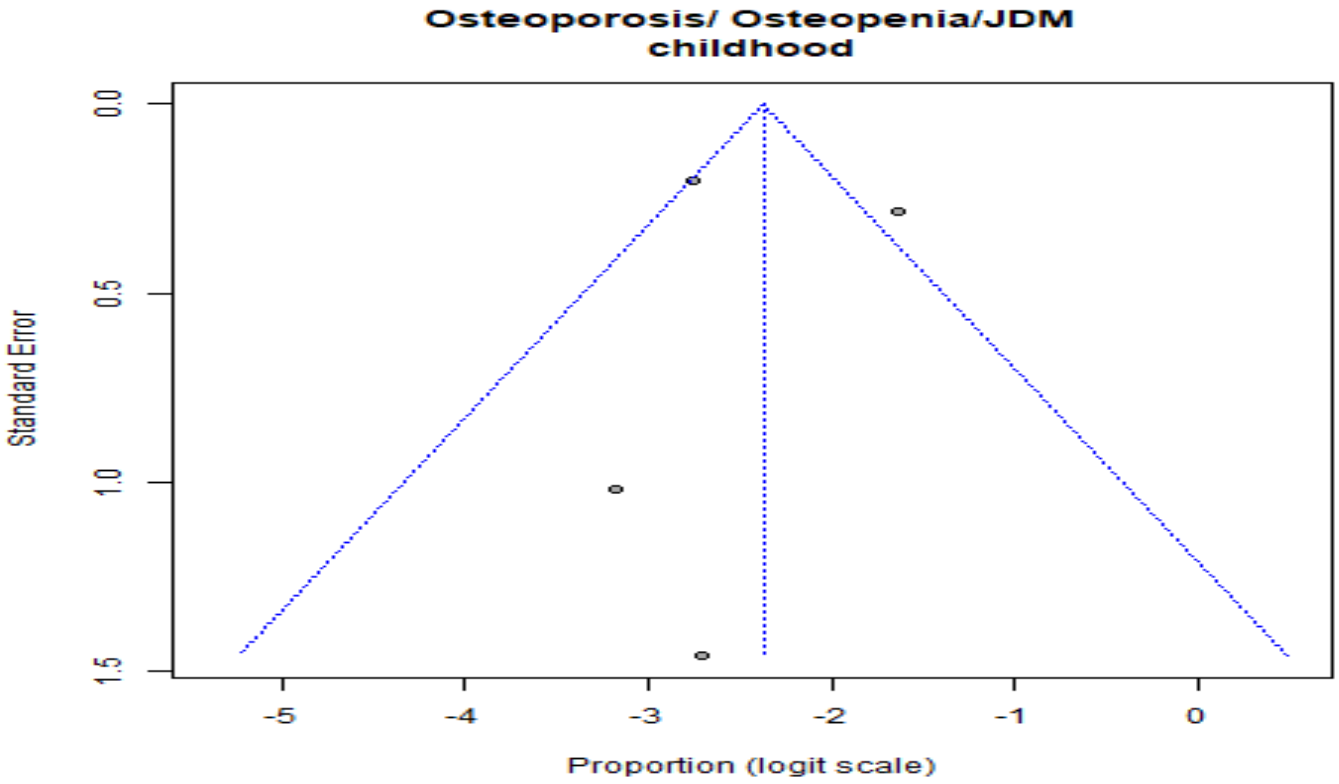
**Supplementary Figure S4.71**: Forest and Funnel Plots of studies reporting the prevalence of Osteoporosis/Osteopenia in children with Juvenile Dermatomyositis

**Osteoporosis/ Osteopenia**

Forest




Funnel




Summary statistics

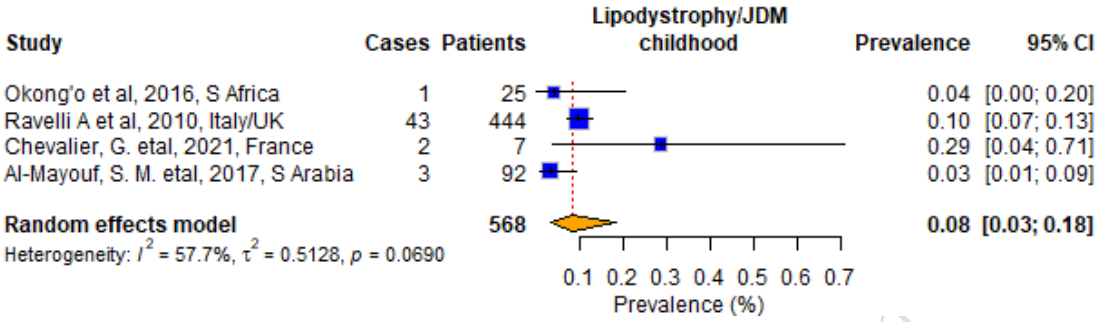
```
                                     proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa            0.0400        [0.0010; 0.2035]        12.5
Ravelli A et al, 2010, Italy/UK          0.0598        [0.0394; 0.0864]        42.0
Chevalier, G. etal, 2021, France         0.0000        [0.0000; 0.4096]         7.1
Al-Mayouf, S. M. etal, 2017, S Arabia    0.1630        [0.0942; 0.2546]        38.5
Number of studies: k = 4; Number of observations: o = 559; Number of events: e = 42
                     proportion           95%-CI
Random effects model     0.0851 [0.0391; 0.1756]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.3842 [0.0122; 5.6370]; tau = 0.6199 [0.1103; 2.3742]
 I^2 = 72.9% [23.4%; 90.4%]; H = 1.92 [1.14; 3.22]
Test of heterogeneity:      Q d.f. p-value
                        11.05    3  0.0115
```

For reproducibility only in academic research and development

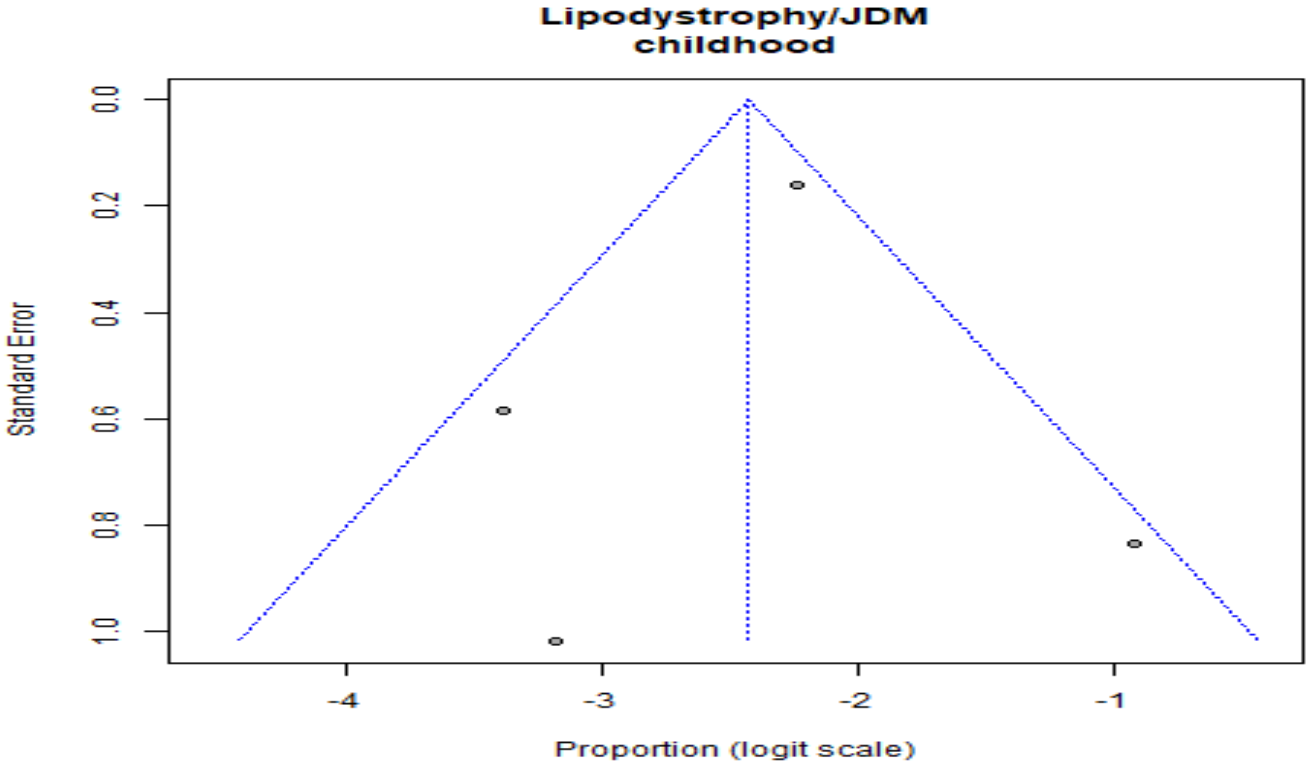
**Supplementary Figure S4.72**: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in children with Juvenile Dermatomyositis

**Lipodystrophy**

Forest




Funnel




Summary statistics

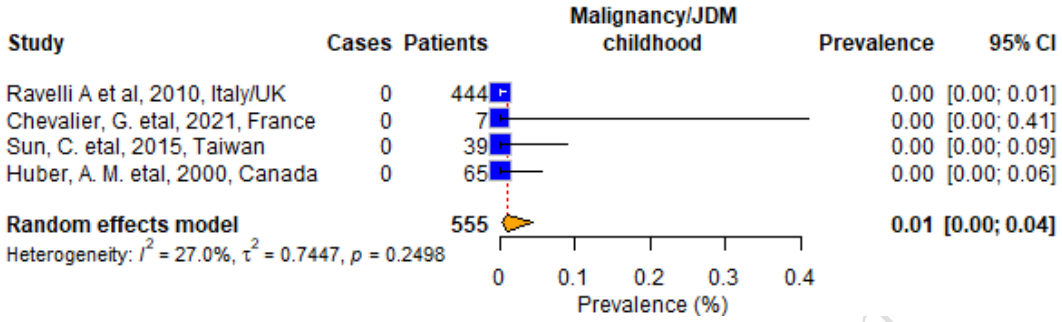
```
                                    proportion        95%-CI          %W(random)
Okong'o et al, 2016, S Africa           0.0400        [0.0010; 0.2035]        14.3
Ravelli A et al, 2010, Italy/UK         0.0968        [0.0710; 0.1282]        41.3
Chevalier, G. etal, 2021, France        0.2857        [0.0367; 0.7096]        18.4
Al-Mayouf, S. M. etal, 2017, S Arabia   0.0326        [0.0068; 0.0923]        26.0
Number of studies: k = 4; Number of observations: o = 568; Number of events: e = 49
                     proportion           95%-CI
Random effects model     0.0811 [0.0338; 0.1821]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.5128 [0.0000; 17.0078]; tau = 0.7161 [0.0000; 4.1241]
 I^2 = 57.7% [0.0%; 85.9%]; H = 1.54 [1.00; 2.67]
Test of heterogeneity:     Q d.f. p-value
                        7.09    3  0.0690
```

For reproducibility only in academic research and development

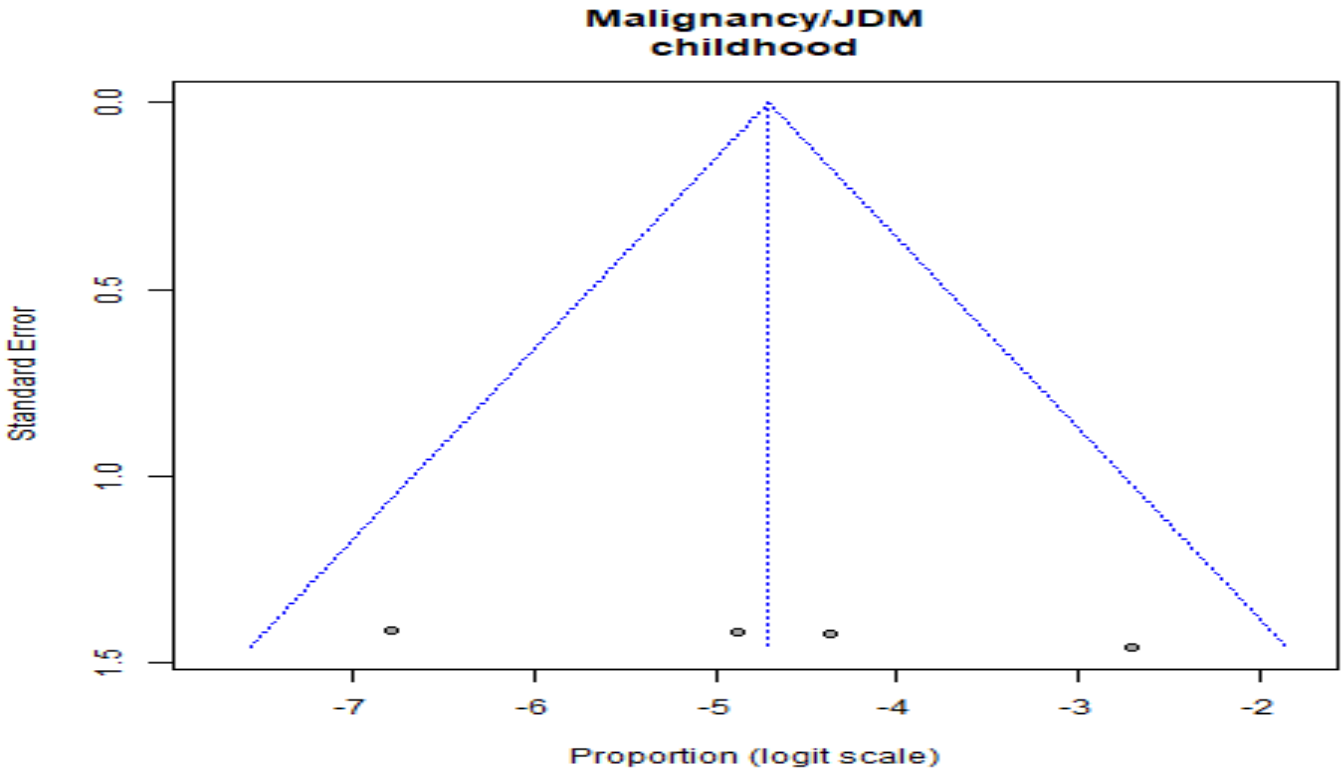
**Supplementary Figure S4.73**: Forest and Funnel Plots of studies reporting the prevalence of malignancy in children with Juvenile Dermatomyositis

**Malignancy**

Forest




Funnel




Summary statistics

```
                                proportion     95%-CI          %W(random)
Ravelli A et al, 2010, Italy/UK      0.0000     [0.0000; 0.0083]
Chevalier, G. etal, 2021, France     0.0000     [0.0000; 0.4096]
Sun, C. etal, 2015, Taiwan           0.0000     [0.0000; 0.0903]
Huber, A. M. etal, 2000, Canada      0.0000     [0.0000; 0.0552]
Number of studies: k = 4; Number of observations: o = 555; Number of events: e = 0
                     proportion           95%-CI
Random effects model     0.0089 [0.0018; 0.0442]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7447 [0.0000; 37.2127]; tau = 0.8629 [0.0000; 6.1002]
 I^2 = 27.0% [0.0%; 72.6%]; H = 1.17 [1.00; 1.91]
Test of heterogeneity:     Q d.f. p-value
                        4.11    3  0.2498
```

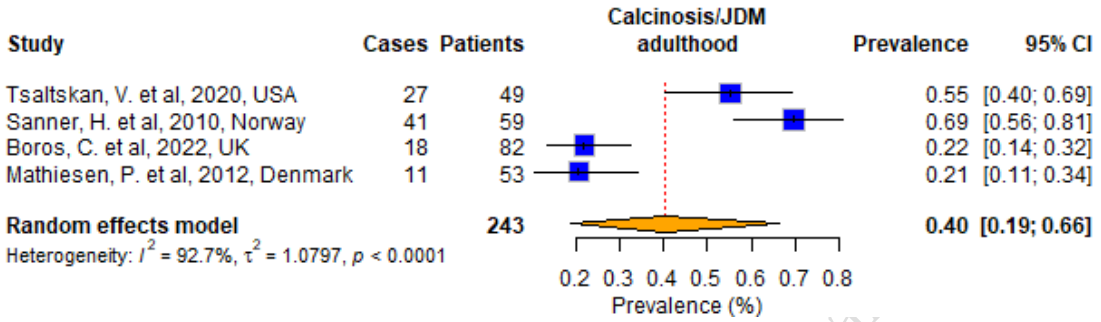
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**Adulthood/JDM**

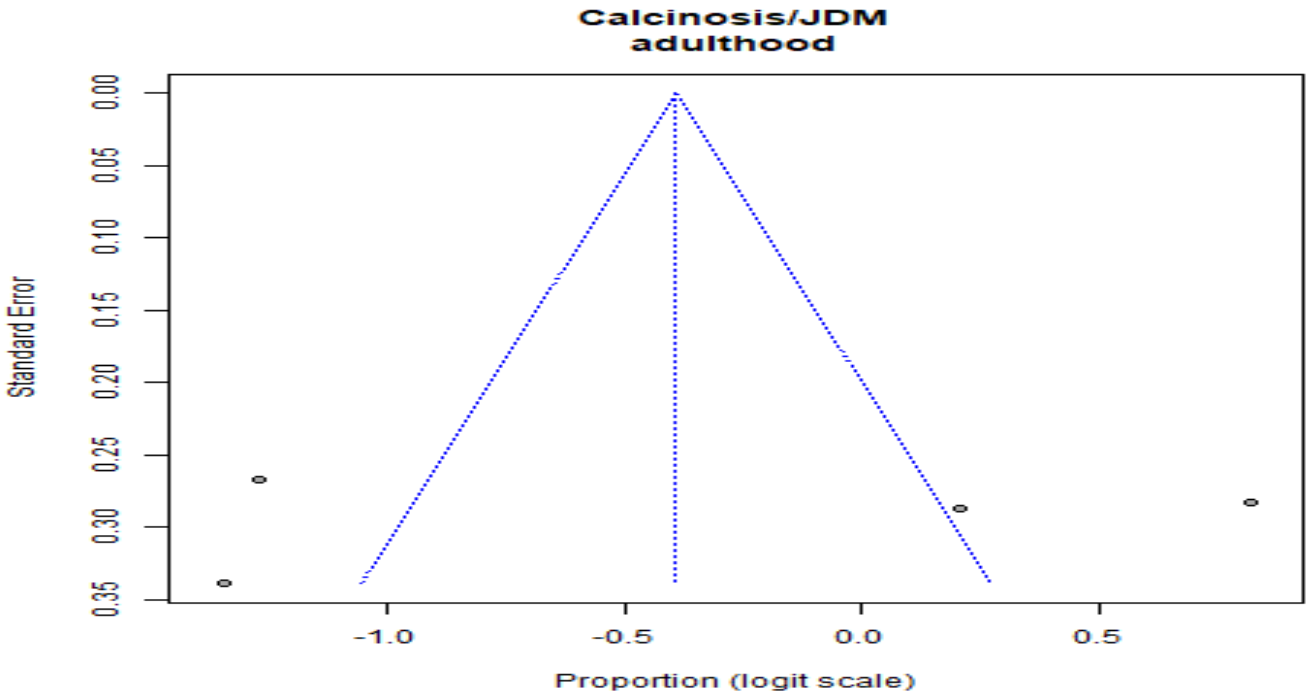
**Supplementary Figure S4.74**: Forest and Funnel Plots of studies reporting the prevalence of calcinosis in adults with Juvenile Dermatomyositis

**Calcinosis**

Forest




Funnel



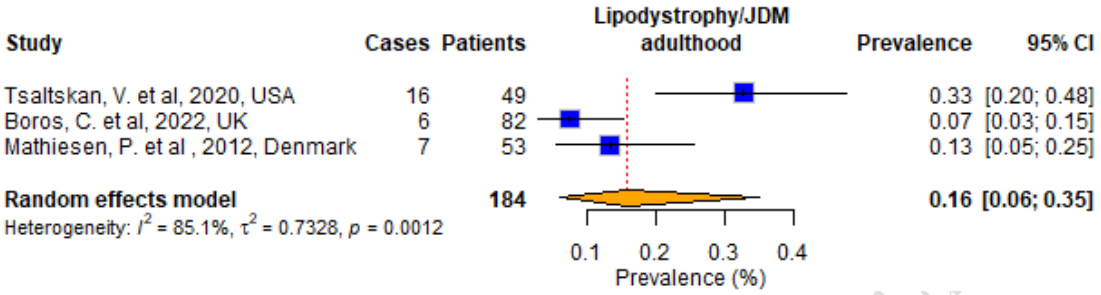

Summary statistics

```
                                  proportion   95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.5510   [0.4023; 0.6933]       25.1
Sanner, H. et al, 2010, Norway        0.6949   [0.5613; 0.8081]       25.1
Boros, C. et al, 2022, UK             0.2195   [0.1356; 0.3246]       25.3
Mathiesen, P. et al, 2012, Denmark    0.2075   [0.1084; 0.3411]       24.4
Number of studies: k = 4; Number of observations: o = 243; Number of events: e = 97
                    proportion           95%-CI
Random effects model     0.4037 [0.1902; 0.6611]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 1.0797 [0.2878; 16.1291]; tau = 1.0391 [0.5365; 4.0161]
 I^2 = 92.7% [84.6%; 96.6%]; H = 3.71 [2.55; 5.40]
Test of heterogeneity:      Q d.f.  p-value
                        41.34    3 < 0.0001
```

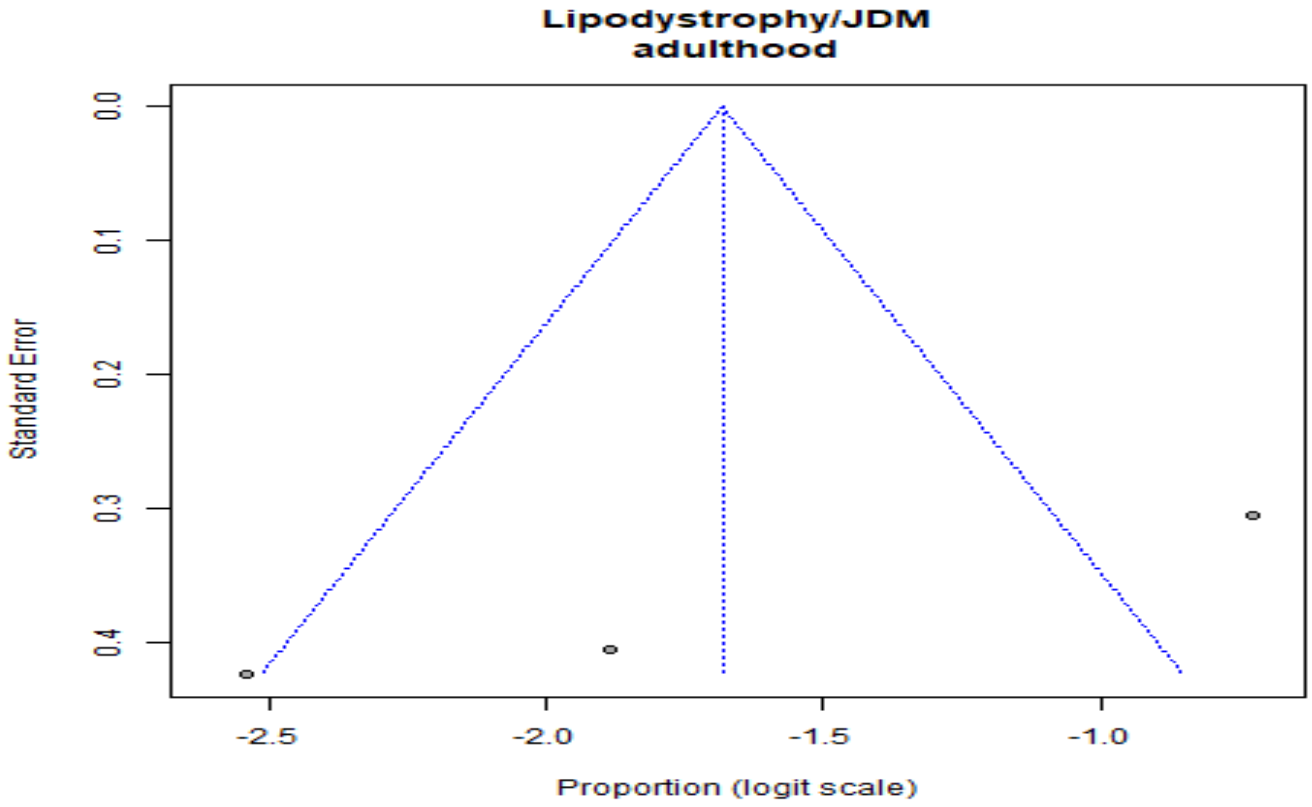
**Supplementary Figure S4.75**: Forest and Funnel Plots of studies reporting the prevalence of lipodystrophy in adults with Juvenile Dermatomyositis

**Lipodystrophy**

Forest




Funnel




Summary statistics

```
                                  proportion  95%-CI          %W(random)
Tsaltskan, V. et al, 2020, USA        0.3265  [0.1995; 0.4754]       35.4
Boros, C. et al, 2022, UK             0.0732  [0.0273; 0.1525]       32.0
Mathiesen, P. et al , 2012, Denmark   0.1321  [0.0548; 0.2534]       32.6
Number of studies: k = 3; Number of observations: o = 184; Number of events: e = 29
                     proportion           95%-CI
Random effects model     0.1567 [0.0605; 0.3491]
Quantifying heterogeneity (with 95%-CIs):  tau^2 = 0.7328 [0.1017; 33.2331]; tau = 0.8560 [0.3189; 5.7648]
 I^2 = 85.1% [55.9%; 94.9%]; H = 2.59 [1.51; 4.44]
Test of heterogeneity:      Q d.f. p-value
                        13.38    2  0.0012
```